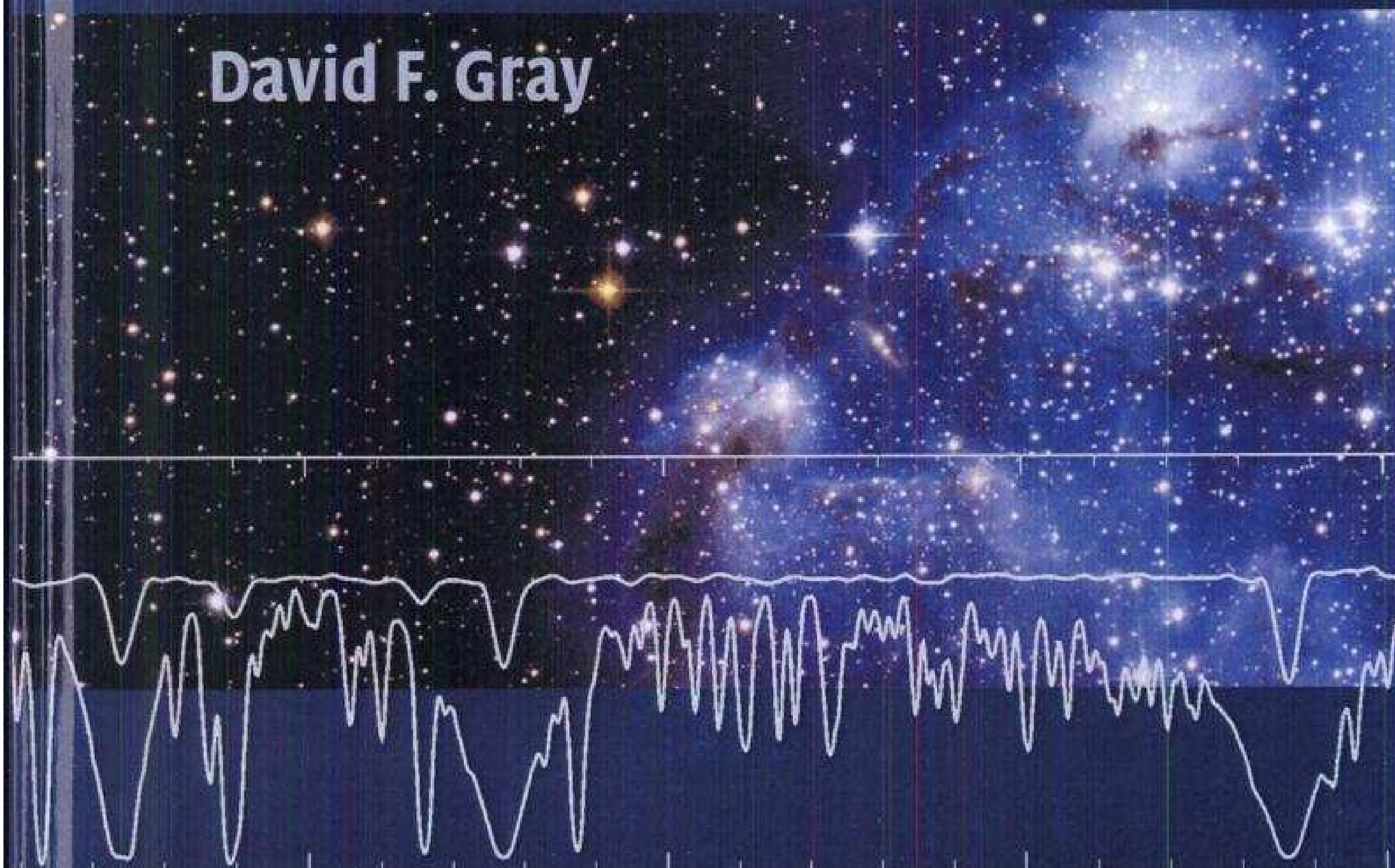
The Observation and Analysis of

Stellar Photospheres

David F. Gray



Third Edition

CAMBRIDGE

CAMBRIDGE UNIVERSITY PRESS
Cambridge, New York, Melbourne, Madrid, Cape Town, Singapore, São Paulo

Cambridge University Press
The Edinburgh Building, Cambridge CB2 2RU, UK

Published in the United States of America by Cambridge University Press, New York

www.cambridge.org
Information on this title: www.cambridge.org/9780521 851862

First published 2005

Printed in the United Kingdom at the University Press, Cambridge

A record for this publication is available from the British Library

Library of Congress in Publication data

ISBN-13 978-0-521-85186-2
ISBN-10 0-521-85186-6

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Preface to the first edition

The remarkable nature of stars is transmitted to us by the light they send. The light escapes from the outer layers of the star – called, by definition, the atmosphere. The complete atmosphere of a star can be viewed comprehensively as a transition from the stellar interior to the interstellar medium. And yet almost the whole visible stellar spectrum comes from a relatively thin part called the photosphere. Obviously we cannot disconnect the photosphere from the adjacent portions of the atmosphere, but in actual fact it is the only region we can study extensively for most stars. It is for this reason that the photosphere has taken its place as the central theme of this book.

Several books have appeared during the last decade dealing with the *theory* of stellar atmospheres. These works are for the most part excellent. It is to the material largely omitted by these books that the present treatise is directed. My students and I have felt for some time the need of a book that presents the basics of the field through the eyes of an observer and analyzer of stellar atmospheres.

An introduction to a subject, in my opinion, should be presented in a way that can be understood by a reader who has not studied the topic before. It follows that the material should be presented in as simple and straightforward a manner as possible. The Fourier transform (as covered in Chapter 2) is a unifying theme helping to accomplish this aim. Transforms lead naturally into the material on data collection, optical instruments, the instrumental profile, line absorption coefficients, velocity fields, and spectral line analysis. In addition, I have selected and developed topics that I consider to be important to those of us who look at stars and attempt to understand what we see. At the same time, I have tried to present the material in the least complicated manner. The word "complicated" is affixed to things that are difficult to understand. Complicated things consequently are often unsuitable topics for the novice. We should seek not the most general case conceivable, but the least complicated case that is serviceable (a version of the principle of minimum assumption).

The development of each of the main topics starts at an elementary level, proceeds with a discussion of the topic, and ends by pointing the direction to the more advanced literature. It should be easy to expand from this book into the areas holding an attraction for you. Realizing that astrophysics is a very dynamic field, I have documented (or otherwise made clear) the source of the material used in examples. When no source is indicated, the material is from observations or calculations of my own. The references in general have been selected because they are good illustrations of the material being discussed or because they have a basic lasting approach to the subject. I have also biased the referencing toward good starting points in the literature and toward review articles and journals to which the student is likely to have access. The references are listed at the end of each chapter and ordered according the author's name and date of publication.

The first two chapters contain preparatory material. The main theme starts in Chapter 3 with a discussion of spectroscopic tools. Generally the continuous spectrum topics are developed first, followed by the somewhat more involved subject of the line spectrum. From Chapter 14 (Chemical analysis) through to Chapter 18, the material is oriented completely toward analysis and deduction. These later chapters are closely interlinked with the preceding chapters.

The book is suitable as a text for a one-year course and as a reference to the more advanced reader.

Preface to the second edition

Wonderful growth has occurred in our understanding of stellar photospheres during the 15 years since the appearance of the first edition of "Photospheres." I have managed to retain the same chapter names and the general plan of the first edition, and many of the equation numbers are also the same. But a significant portion of the material is new or revised. A revolution in light detectors has given us hundreds of times greater efficiency in measuring stellar spectra; Chapter 4 on detectors has been re-done. The astronomical literature is burgeoning with new results on the structure of photospheres, chemical abundances, radius measurements, stellar rotation, and photospheric velocity fields. Many of these results have been incorporated in this second edition, of course. At the same time, I stayed with my original purpose of making this volume an introduction to the subject. Unhappily, this means leaving out numerous exciting topics. My book *Lectures* (Gray 1988) takes up some of these, and it is recommended as a second installment, after the material in "Photospheres" has been mastered.

More than ever, the reader should keep in mind the fundamental nature of the stellar photosphere: of interest in its own right, with marvelous and intriguing physics, yet the link between the interior and chromospheres, coronae, and interstellar surroundings, and the source of most of our basic information about stars and stellar systems.

Once again, I thank my students and colleagues for their help and patience.

Reference

Gray, D. F. 1988. *Lectures on Spectral-Line Analysis: F, G, and K Stars* (Arva, Ontario: The Publisher).

Preface to the third edition

Studies of stars continue to flourish. More detailed and sophisticated observations continue to be made, analysis tools are honed, understanding grows for more complex situations. The beauty of the stars is integrated more fully into our lives. I hope you enjoy the many revisions incorporated in this third edition of *Photospheres*: new figures, more complete data sets, re-organization of some of the material, a cleaner presentation of several topics, and some exercises and questions to help you probe the material.

I extend my thanks to J. Power for proofreading and to the many others who have guided my thoughts, given me corrections and suggestions, and contributed their efforts to knowing the stars.

1

Background

We study stellar spectra, including both lines and continua, because we are interested in the nature of the star's atmosphere. The behavior of the atmosphere is controlled by the density of the gases in it and the energy escaping through it. These in turn depend on the mass and age of the star and to a lesser extent on chemical composition and angular momentum. Stellar atmospheres are the connecting links between the observations and the rest of stellar astrophysics. In this way two philosophies arise. One is the study of the atmosphere for its own sake and the other is the use of the atmosphere as a tool to connect our observations to other parameters of interest. This book should be useful for students of both philosophies.

The topics brought together to form this chapter are background material which the reader will need. The more advanced reader can profitably skim through to Chapter 2.

What is a stellar atmosphere?

A stellar atmosphere is a transition region from the stellar interior to the interstellar medium. One way to quantify this description is to look at the change in average kinetic temperature with height as observed in the Sun. Figure 1.1 shows the solar temperature profile with the four basic sections labeled, sub-photosphere, photosphere, chromosphere, and corona. For an observer of stellar atmospheres, one concept is very important: the major portion of the visible stellar spectrum originates in the region marked "photosphere." A study of the visible-light spectrum is essentially a study of the photosphere. Higher layers are typically studied in the shorter and longer wavelengths with instruments on satellites, rockets, or balloons, and by radio

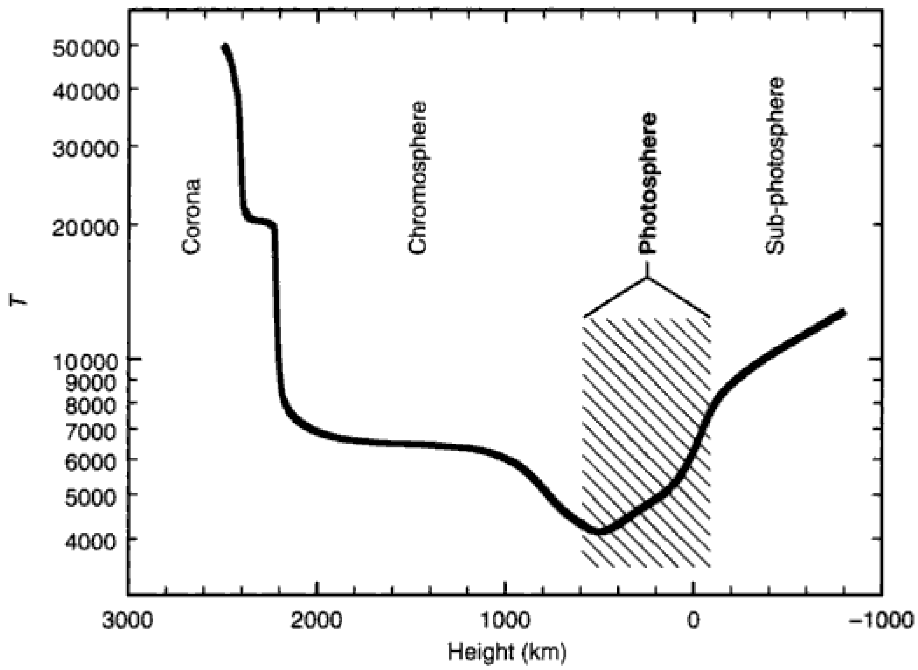


Fig. 1.1. The temperature distribution in the outer layers of the Sun is shown as a function of the geometrical depth in kilometers. The center of the Sun is toward the right. The photosphere is indicated by the shaded region. The visible-light spectrum comes from the photosphere. Adapted from Vernazza *et al.* (1973).

telescopes.

In the Sun, the photosphere is of the order of 1000 km thick, as illustrated in Fig. 1.1. The geometrical extent of the photosphere in other stars differs inversely with the surface gravity defined by

$$g = g_{\odot}\frac{\mathfrak{M}}{R^2}, \tag{1.1}$$

where $g_{\odot}$ is the surface gravity of the Sun ($2.740 \times 10^4\,\mathrm{cm/s^2}$ or $2.740 \times 10^2\,\mathrm{m/s^2}$) and the mass, $\mathfrak{M}$, and radius, R, of the star are in solar units (see Appendix A). The thickness also depends upon the opacity of the gases comprising the photosphere.

The second physical variable strongly affecting the nature of the atmosphere is its characteristic temperature. Typically the temperature drops by somewhat more than a factor of two from the bottom to the top of the photosphere (Fig. 1.1), and instead of choosing a temperature at some depth to characterize the temperature parameter, it is customary to use the effective temperature. The effective temperature is defined in terms of the total power per unit area radiated by the star,

$$\int_0^\infty \mathfrak{F}_\nu \mathrm{d}\nu = \sigma T_{\mathrm{eff}}^4, \tag{1.2}$$

where the total radiant power per unit area is given by the integral and $\sigma = 5.670\,40 \times 10^{-5}\,\mathrm{erg/(s\,cm^2\,deg^4)}$ or $5.670\,40 \times 10^{-8}\,\mathrm{W/(m^2\,deg^4)}$. Here $\mathfrak{F}_\nu$ is the flux leaving the stellar surface, which is defined formally in Chapter 5. Equation (1.2) has the form of the Stefan–Boltzmann law, which we will meet in Chapter 6, making T_{eff} the temperature of a black body having the same power output per unit area as the star. However, the distribution of power across the spectrum may differ dramatically from a black body at the same effective temperature.

The total power emitted by the star is termed the luminosity. Luminosity is related to the effective temperature through the surface area, namely,

$$L = 4\pi R^2 \int_0^\infty \mathfrak{F}_\nu \mathrm{d}\nu = 4\pi R^2 \sigma T_{\mathrm{eff}}^4, \tag{1.3}$$

for a spherical star of radius R.

Traditionally and of historic necessity, the more empirical parameters of magnitude, spectral type, and color index have been used to describe stellar surface gravity and temperature.

Spectral types

One of the most powerful ways to analyze starlight is through spectroscopy, i.e., looking at the amount of light as a function of wavelength. The portion of the stellar spectrum emitted by the photosphere is an absorption-line spectrum. The absorption lines are narrow dark regions, of which there are many, and are sometimes called Fraunhofer lines after Joseph Fraunhofer who first saw them in the solar spectrum. The relative strengths and shapes of the absorption lines are used to classify a star by effective temperature and photospheric pressure, a process referred to as assigning a spectral type. A spectral type consists of a capital letter followed by an Arabic number representing the fraction toward the next letter class, followed by a Roman numeral. The letter represents the temperature classification and the Roman numeral denotes the pressure classification. Spectrograph dispersions of $\sim 100\,\text{Å/mm}$, giving $\sim 2\,\text{Å}$ resolution, are found to be best for classification. Higher resolution resolves details of line profiles giving unwanted complications; lower resolution does not adequately separate individual lines. The traditional approach uses photographic records, called spectrograms, that are centered in the blue region of the spectrum where photographic emulsions are most sensitive, but electrical detectors work equally well or better. Line strengths and ratios are estimated visually by comparing an untyped spectrogram with standards. Establishing a grid of standard stars is a major undertaking with deep historical roots (see Huggins & Huggins 1899, Draper 1935, Garrison 1984, Keenan 1985, 1987, Hearnshaw 1986). Figure 1.2 shows a set of standard spectra in order of decreasing effective temperature. Those spectra toward the cool end of the sequence are referred to as "late-type" spectra, and those on the hot end as "early-type" spectra. In principle the spectral classification separates the observed (normal) stars into about 60 temperature steps. These range (hottest to coolest) from O2 to M8 through the standard (temperature) sequence: O, B, A, F, G, K, M with ten subdivisions per letter group. But some subclasses are rarely used. For example, the main subtypes for the cooler stars are G0, G2, G5, G8, K0, K1, K2, K3, K4, K5, M0, M1, M2, M3, M4, M5, M6, M7, and M8. In a few cases, half intervals are given full weight as a subclass, for example, O9.5 and B2.5. Statistically smooth relations exist between these main subtypes and other temperature indicators such as

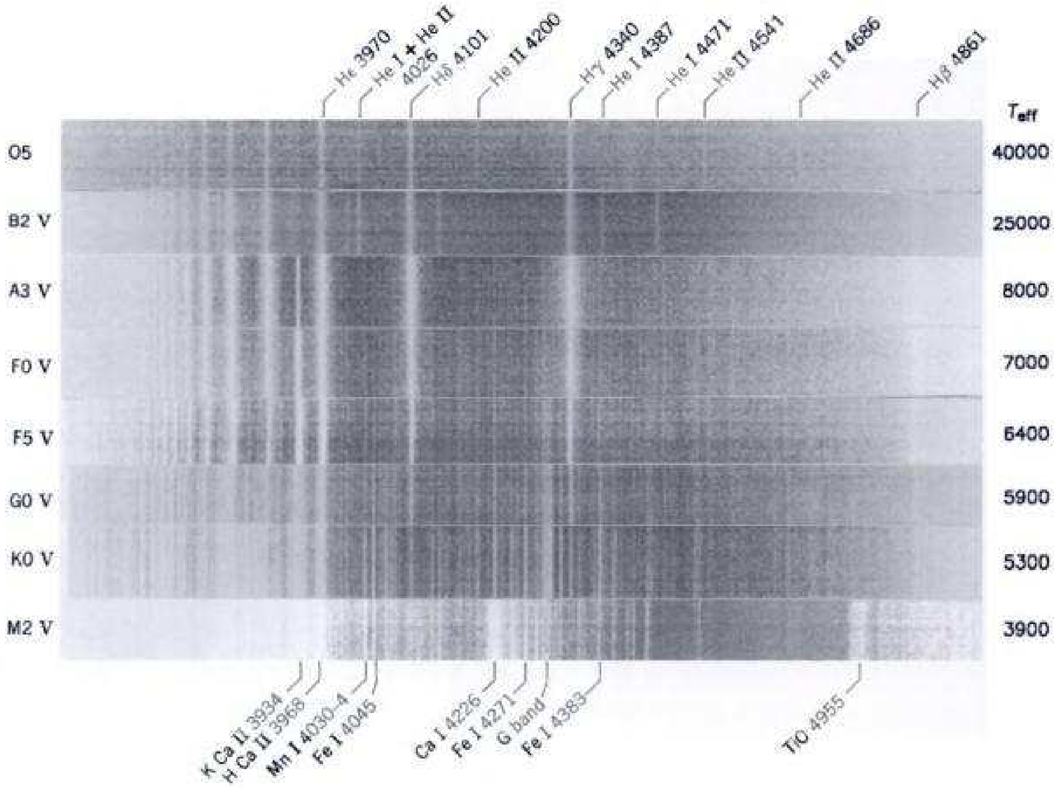


Fig. 1.2. The spectral temperature classification is illustrated by these samples of standard spectra from Abt *et al.* (1968). Notice how the hydrogen lines reach maximum strength in the A stars, high excitation lines are seen in O and B stars, while most spectral lines increase their strength with cooler temperatures.

color indices (see below). The other subclasses are sometimes used; a star classified as G1 is judged to lie between G0 and G2, for example. The spectral type of the Sun is G2 V by definition. The effective temperature associated with each spectral type is discussed in Chapter 14.

Spectral classification of very cool stars, also called brown dwarfs (see Geballe *et al.* 2002, Leggett *et al.* 2002), includes extension to classes L (2200–1300 K) and T (1300–800 K) as well as side branches: R stars parallel K stars, but show carbon-molecular bands; N stars parallel M stars, but show very strong carbon-molecular bands; S stars parallel M stars, but have the usual titanium oxide bands overshadowed by those of zirconium oxide (see Hearnshaw 1986).

Pressure sensitive features in the spectrum show markedly less variation, and only about a dozen classes are recognized. It is an observed fact that luminosity (the total power output of the star) and pressure are strongly correlated, and the pressure classification is universally referred to as the luminosity classification. The luminosity class designations in order of decreasing luminosity are 0, I, Ia, Ib, II, III, IV, and V. The names associated with these numerals are: 0 hypergiants, I supergiants, II bright giants, III giants, IV subgiants, and V dwarfs. Sometimes subdwarfs are denoted as class VI. Luminosity effects are illustrated in Fig. 1.3 and discussed in detail in Chapters 13 and 15. Sometimes the suffixes listed in Table 1.1 are used to supply additional information.

Before classification can begin, the observer must acquire spectrograms of the standard stars using the same equipment and technique intended to be used with the program stars. The first step in classification is to identify the lines

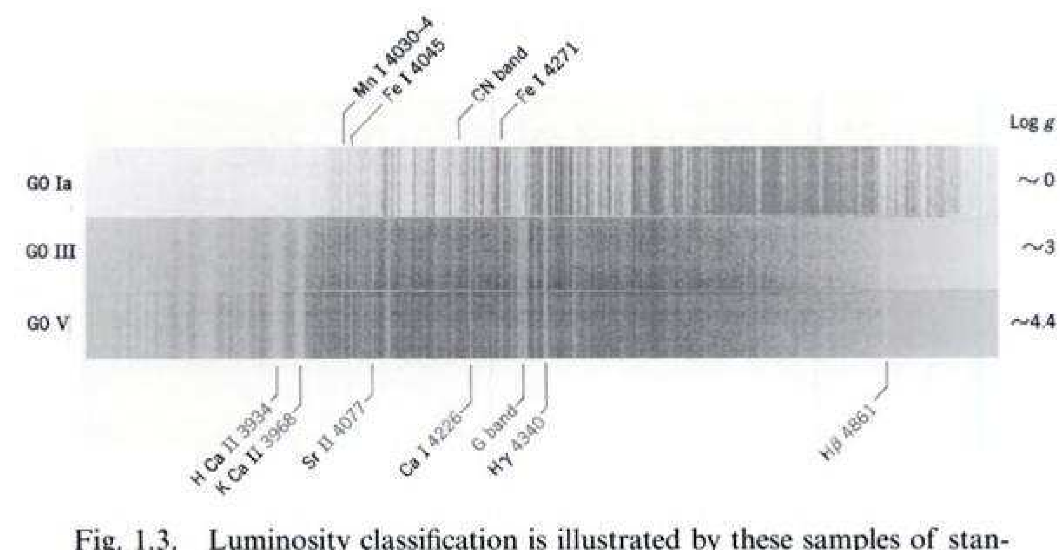


Fig. 1.3. Luminosity classification is illustrated by these samples of standard spectra from Abt *et al.* (1968). The lines are generally stronger in higher-luminosity stars. Note also specific changes as in the ratio of Sr II $\lambda 4077$ compared to H_δ, the strong line just to the right of $\lambda 4077$.

Table 1.1. *Spectral classification.*

Temperature criteria	
O4–B0	He I λ4471/He II λ4541 increasing with type He I + He II λ4026/He II λ4200 increasing with type
B0–A0	H lines increase with type He I lines reach maximum at B2 Ca II K line becomes visible at B8
A0–F5	Ca II K/H_δ increasing with type Neutral metals become stronger *G* band visible starting at F2
F5–K2	H lines decrease Neutral metals increase *G* band strengthens
K2–M5	*G* band changes appearance Ca I λ4226 increases rapidly TiO starts near M0 in dwarfs, K5 in giants
Luminosity criteria	
O9–A5	H and He lines weaken with increasing luminosity Fe II becomes prominent in A0–A5
F0–K0	Blend $\lambda\lambda$4172–9 increases with luminosity (early F) Sr II λ4077 increases with luminosity CN λ4200 increases with luminosity
K0–M6	CN λ4200 increases with luminosity Sr II increases with luminosity
Suffix notation	
e	Emission lines are present
f	He II λ4686 and/or C III λ4650 in emission; mostly for O stars
k	Ca II K line when unexpected, e.g., interstellar in hot stars
m	Metallic; metal lines are stronger than normal
n	Nebulous; lines are broad and shallow; usually high rotation
nn	Very nebulous!
p	Peculiar; spectrum is abnormal
q	Queer; unusual emission; evolved from Q novae designation (archaic)
s	Sharp; lines are sharp, usually for early-type stars with low rotation
v	Variable; spectrum changes with time
w	Wolf–Rayet bands present (archaic)
Old prefix notation	
c for supergiants; g for giants; d for dwarfs	

whose strengths serve as the criterion for classification. The required lines can be identified by their position relative to lines in the comparison spectrum or to the few prominent features of the stellar spectrum itself. Iron and titanium are commonly used comparison spectra. In the stellar spectrum we easily recognize the hydrogen lines from class O to F5, and the less conspicuous lines can be referred to them. H_β remains distinguishable as the most intense line at the long-wavelength end of the spectrogram (Fig. 1.2) as late as K0. From F5 to M, the

G band near the center of the spectrogram and the H and K lines on the short-wavelength end can be used as guideposts.

The spectrum is first classed roughly. Does it have few lines, many lines, or does it show molecular bands? This rough classification is refined in successive steps. It is generally considered unwise to attempt an immediate accurate classification. Table 1.1 gives examples of the classification criteria from the atlases of Keenan and McNeil (1976) and Morgan *et al.* (1978). The earlier atlas by Morgan *et al.* (1943) is similar (see also Johnson & Morgan 1953). Although these special features are singled out, it is the appearance of the whole spectrum that is ultimately compared with the standards. Some care must be exercised, especially with early-type stars, because high rotation (see Chapter 18) can wash out weaker classification lines. Most modern spectral types are on the Morgan–Keenan system (MK) or a close derivative of it. A review of spectral classification is given by Morgan and Keenan (1973) and there is also a symposium dealing with this topic (Garrison 1984). Additional classification information can be found in a comprehensive objective-prism survey by Houk and Cowley (1975). Early O star classification is discussed by Walborn *et al.* (2002). Most catalogues of stars list spectral types, e.g., *The Bright Star Catalogue* (Hoffleit & Warren 1996) and the *Geneva Photometric Catalogue 1988* (Rufener 1988).

Magnitudes and color indices

The brightness of a star is often expressed in magnitude units defined by

$$m = -2.5 \log \int_0^\infty F_\nu W(\nu)\, \mathrm{d}\nu + \text{constant} \tag{1.4}$$

in which the amount of light or flux of the star, F_ν, is recorded in the spectral interval specified by $W(\nu)$. The visual magnitude, often denoted by m_v or V, has a spectral window, $W(\nu)$, centered on the yellow–green wavelengths. The constant is arbitrary and usually adjusted to suit an adopted set of standard stars. Notice that the minus sign in Eq. (1.4) implies smaller magnitudes for brighter stars. The bright star Vega, for example, has a visual magnitude of 0.03 according to Hoffleit and Warren (1996). Should we wish to include radiation at all wavelengths, $W(\nu) = 1$ everywhere, and the corresponding magnitude is the bolometric magnitude. Even for this case, there remains an arbitrary zero-point constant.

Standard magnitude systems have been set up. The first was the *RGU* system of Becker (1948, Becker & Stock 1954). The most commonly encountered systems are: the *UBV* system (Johnson & Morgan 1953, Nicolet 1978) and its extension to longer wavelengths in the *RIJKLMNH* magnitudes (Johnson 1966, Johnson *et al.* 1966, Morel & Magnenat 1978, Landolt 1992), the *uvby* system

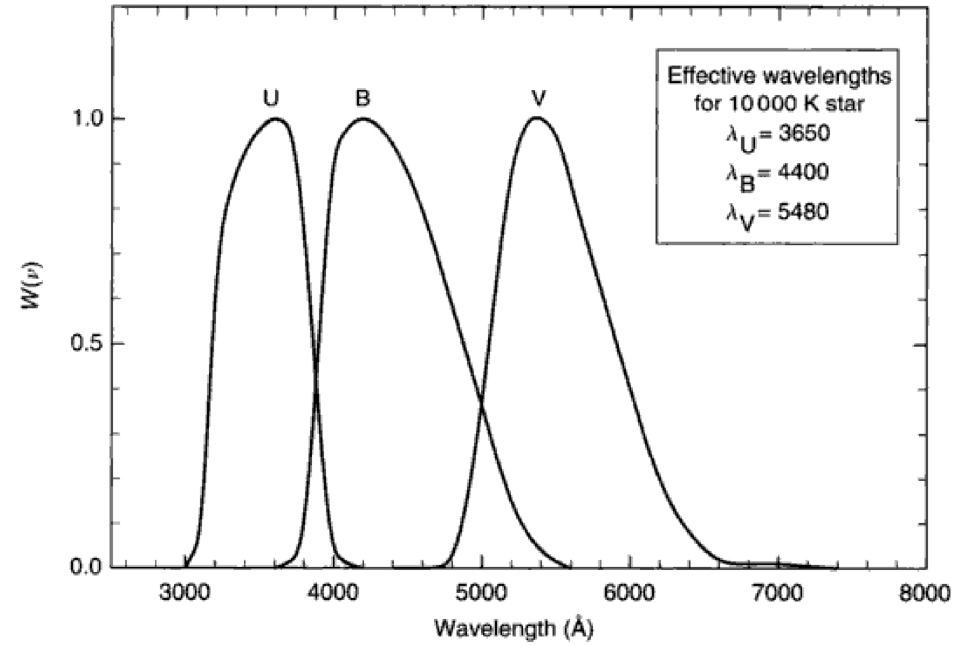


Fig. 1.4. The response functions for the *UBV* standard magnitude system are shown (from Johnson 1965). Because the bandpasses are so wide, their effective (average) wavelengths depend on the temperature of the star through the gross changes in the shape of the spectrum.

(Strömgren 1966, Hauck & Mermilliod 1980, Perry *et al.* 1987), the Geneva $UBVB_1B_2V_1G$ system (Rufener 1988, Rufener & Nicolet 1988), and the DDO system (McClure & van den Bergh 1968). The standard magnitude transmission bands are defined in terms of filter and detector response. The *UBV* response functions are shown in Fig. 1.4, for example.

By comparing the magnitudes of the same star measured in different parts of the spectrum, a measurement is made of the general shape of the spectrum. The *UBV* color indices are related to the flux and the transmission bands by

$$B - V = -2.5\log\left(\frac{\int F_\nu W_B(\nu)\,\mathrm{d}\nu}{\int F_\nu W_V(\nu)\,\mathrm{d}\nu}\right) + 0.710 \tag{1.5}$$

$$U - B = -2.5\log\left(\frac{\int F_\nu W_U(\nu)\,\mathrm{d}\nu}{\int F_\nu W_B(\nu)\,\mathrm{d}\nu}\right) - 1.093. \tag{1.6}$$

The constants by definition are adjusted to ensure that both indices are zero for an average A0 V star, and their value depends on the absolute photometry discussed in Chapter 10. The $B-V$ color index is a relative measure of the tem-

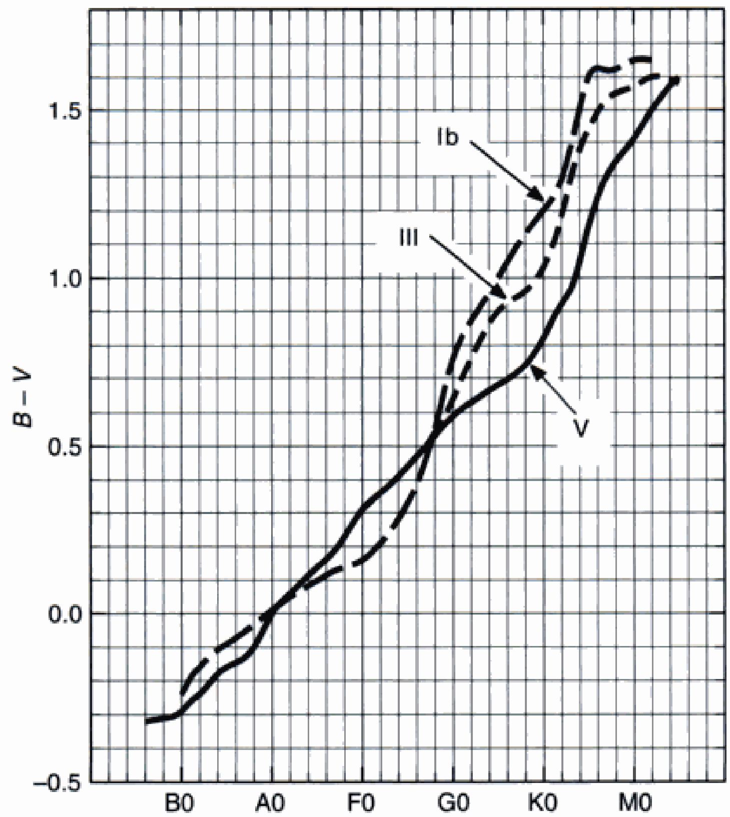


Fig. 1.5. The mean relation between the $B-V$ color index and the MK spectral type is shown based on the tabulation of FitzGerald (1970).

perature of the star through the slope of the Paschen continuum. With 0.01 magnitude precision in $B-V$, the temperature sequence can be divided into about 200 steps. Figure 1.5 relates $B-V$ and spectral type empirically. We can use this relation to estimate the $B-V$ of a star from its spectral type or vice versa.

Interstellar extinction alters the continuous energy distribution of a star, and consequently its color indices. Stars affected in this way deviate from the relation shown in Fig. 1.5. The spectral lines are unaffected by the interstellar extinction because the wavelength dependence of the extinction is negligible over the extent of an absorption line. There are, however, interstellar absorption lines, such as the calcium H and K lines and the sodium D lines, arising from the gaseous component of the interstellar material.

Interstellar extinction is a serious problem when the continuous spectrum of a star is to be analyzed because there is no really satisfactory way of correcting for the effect with the precision needed in stellar atmosphere analyses. For stars nearer than $\approx 100\,\mathrm{pc}$, little or no detectable reddening is expected. Inside this relatively small volume, the shapes of stellar continua and hence color indices such as $B-V$ are independent of distance.

The $U-B$ color index is a measure of the Balmer discontinuity and makes a useful gravity indicator for early-type stars, but it is also sensitive to temperature. Indeed, many photometric indices have been defined to sense variations in temperature, gravity (or luminosity), and global chemical abundances (Philip 1979). When the stars of interest are too faint to measure with high spectral resolution, these photometric measurements can play an important role.

The Hertzsprung–Russell diagram

The mass and age of a star, and to a lesser extent its chemical composition and rotation, determine the luminosity, radius, and temperature of the star. A log–log plot of luminosity versus temperature is called an Hertzsprung–Russell or HR diagram and can take on several forms depending on exactly which coordinates are chosen. Figure 1.6 shows one example in which instead of log luminosity, absolute visual magnitude is used, and log temperature is replaced by spectral type. Because of the multi-valued relation between spectral type and $B-V$, as in Fig. 1.5, lines of constant $B-V$ for cool stars are not vertical. Similarly, if the $B-V$ were chosen for the abscissa, lines of constant spectral type would not be vertical. Most stars are found on the main sequence (class V) because this is the longest-lasting stage, during which hydrogen is converted to helium. Stars above the main sequence, luminosity classes I–IV, are "evolved" stars. Details on stellar evolution can be found in most books on stellar interiors.

The gas laws

The photospheres of stars are hot gas. Because the gas pressure is $\lesssim 10^5$ dynes/cm^2 ($\lesssim 10^4$ N/m^2) in most photospheres, the perfect gas law is exactly what we need since it holds in the limit of low pressures. We can write this law as

$$PV = nRT, \tag{1.7}$$

where P is the gas pressure of the volume of gas V at a temperature T in Kelvin. The number of moles of gas is n and the gas constant R has a value of 8.314×10^7 erg/(mol K) or 8.314 J/(mol K). More frequently we use the gas law in a slightly different form given by dividing Eq. (1.7) by V as follows. Let the number of particles per mole ($6.022\,04 \times 10^{23}$) be denoted by N_0. Then

$$P = \frac{nN_0}{V}\frac{R}{N_0}T = NkT$$

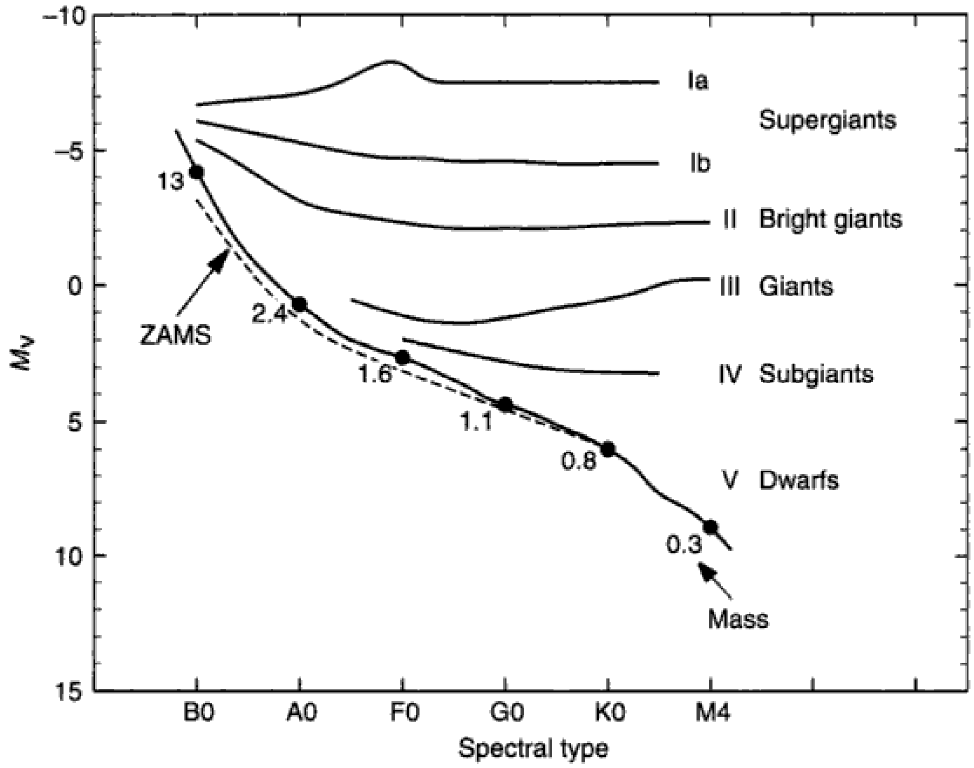


Fig. 1.6. This reference Hertzsprung–Russell diagram shows the absolute visual magnitude, a measure of the power output, plotted against spectral type, a measure of the temperature. Hotter stars are to the left and brighter ones higher. The zero-age main sequence is denoted by ZAMS. The main sequence is a mass sequence, as indicated by a few mass points. Ageing or evolutionary changes, as they are called, are very slow on the main sequence. The mean positions of stars of each of the major luminosity classes are indicated by the various lines. Old or highly evolved stars comprise the luminosity classes other than dwarfs (V).

$$= \frac{\rho}{\mu} kT, \tag{1.8}$$

where N is the number of particles per unit volume. Boltzmann's constant, k, has a value of $1.380\,66 \times 10^{-16}$erg/K ($1.380\,66 \times 10^{-9}$J/K). And as usual, ρ is the density and μ is the mean molecular weight. In fact, the partial pressures of each species in the atmosphere when summed together give the total gas pressure

$$P_g = kT \sum_j N_j. \tag{1.9}$$

One of these partial pressures is due to the free electrons in the ionized gas of the photosphere,

$$P_e = N_e kT, \tag{1.10}$$

where N_e is the number of electrons per unit volume.

The velocity distributions

Gas pressure is produced by the motions of the gas particles. The velocities of particles in a hot gas are distributed in a Gaussian or Maxwellian distribution. This is nature's "signature" of randomness stemming from the huge number of collisions taking place in the gas. The orientation of the coordinate system is arbitrary since this thermal velocity distribution is isotropic. In rectangular coordinates the fraction of particles in the velocity intervals v_x to $v_x + dv_x$, v_y to $v_y + dv_y$, and v_z to $v_z + dv_z$ is

$$\frac{dN(v_x,v_y,v_z)}{N_{\text{total}}} = \left(\frac{m}{2\pi kT}\right)^{3/2} e^{-mv^2/2kT} dv_x dv_y dv_z = \left(\frac{m}{2\pi kT}\right)^{3/2} e^{-mv^2/2kT} dv, \tag{1.11}$$

where we have used

$$v^2 = v_x^2 + v_y^2 + v_z^2.$$

The temperature of the gas is T and the particles have a mass m. Because they produce Doppler shifts, the line of sight velocities have a distribution that is an important special case for spectroscopy:

$$\frac{dN(v_R)}{N_{\text{total}}} = \left(\frac{m}{2\pi kT}\right)^{3/2} e^{-mv_R^2/2kT} dv_R, \tag{1.12}$$

where v_R is the radial (line of sight) velocity component. Translation from radial velocity into wavelength shifts ($\Delta\lambda = \lambda v_R/c$) ensures that all spectral lines have a thermal component to their broadening.

In situations where energy aspects of the gas are being considered, the speed distribution, also called the Maxwell–Boltzmann distribution, may be more appropriate. It is

$$\frac{dN(v)}{N_{\text{total}}} = \left(\frac{2}{\pi}\right)^{1/2} \left(\frac{m}{kT}\right)^{3/2} v^2 e^{-mv^2/2kT} dv. \tag{1.13}$$

That is, the fraction of particles in the speed interval (v, $v + dv$) is dN/N_{total}. The maximum of the speed distribution occurs at

$$v_1 = \left(\frac{2kT}{m}\right)^{1/2} = \text{the most probable velocity.} \tag{1.14}$$

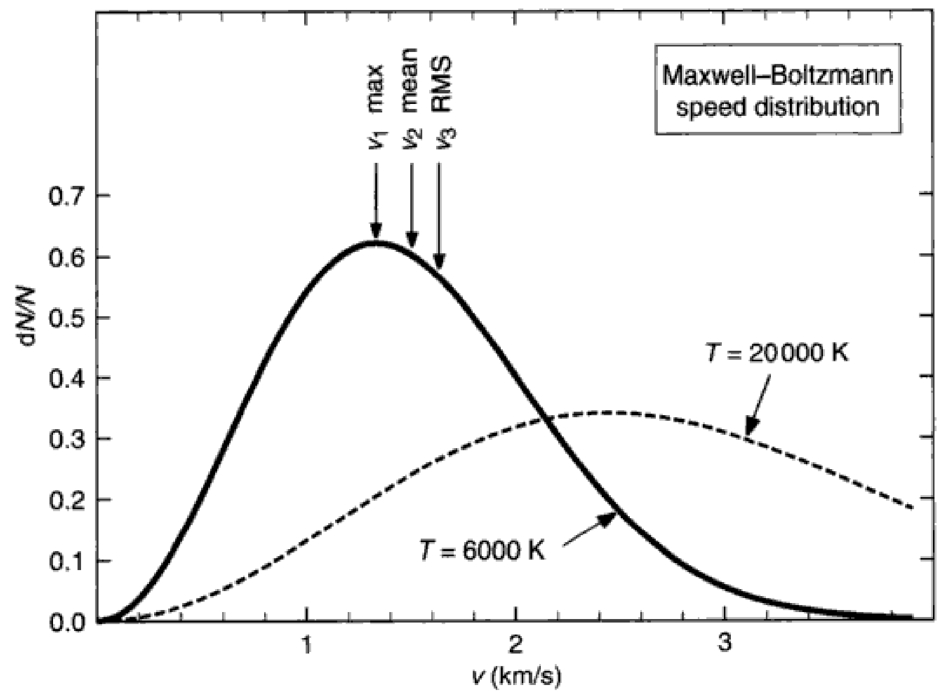


Fig. 1.7. The curve shows a Maxwell–Boltzmann speed distribution according to Eq. (1.13) for iron atoms for temperatures of 6000 and 20 000 K. The initial rise from the origin goes as v^2, while the tail at high velocities drops exponentially. The velocities v_1, v_2, and v_3 are shown for the cooler temperature.

The average velocity, v_2, is

$$v_2 = \left(\frac{8}{\pi}\frac{kT}{m}\right)^{1/2} = 1.128 v_1, \tag{1.15}$$

and the root mean square velocity, v_3, is

$$v_3 = \left(\frac{3kT}{m}\right)^{1/2} = 1.225 v_1. \tag{1.16}$$

The positions of v_1, v_2, and v_3 relative to the Maxwell–Boltzmann distribution are shown in Fig. 1.7.

Atomic excitation and ionization in thermodynamic equilibrium

The strengths of spectral lines, the strength of the continuous absorption, and the general excitation–ionization state of the gases in a star's photosphere

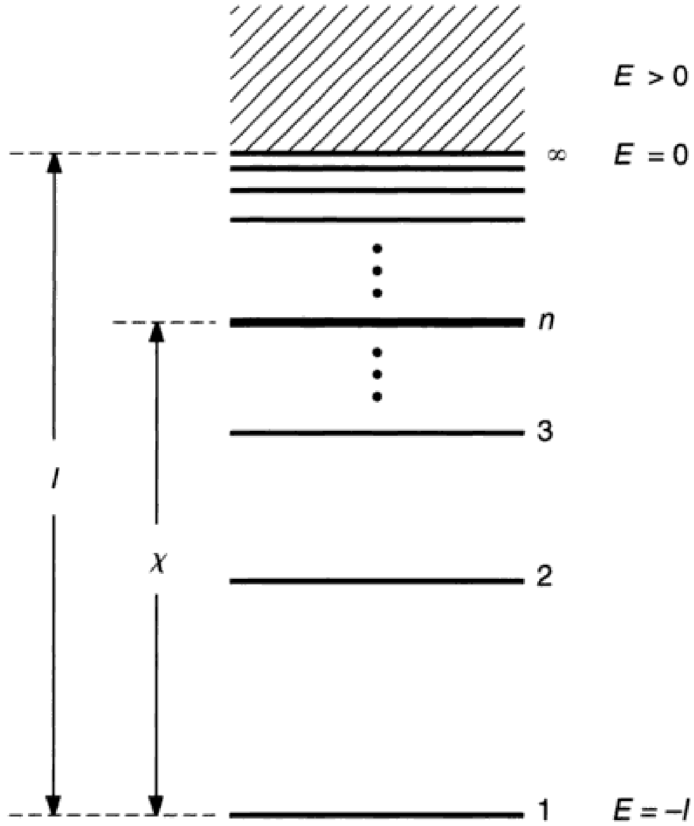


Fig. 1.8. This one-dimensional schematic energy-level diagram shows the excitation potential, χ, for an arbitrary level n and the ionization potential I. Levels are numbered upward from the ground level, $n = 1$. The actual energy of the electron, as indicated on the right, is negative for all bound states and positive in the continuum (shaded area). The energy is zero at the ionization limit, $n = \infty$.

depend on the concepts in this section. We use these equations extensively when we build models of stars and analyze their spectra. In Fig. 1.8, a schematic energy-level diagram of an atom shows the nth level among others and the continuum at the top. When an electron is raised to a level above the ground level, the atom is said to be excited. The energy above the ground level is called the excitation potential, χ. The energy difference between the ground level and the continuum is the ionization potential, I. Although all of the bound levels have negative energy, the ground level is most negative, so that χ and I are positive numbers.

The relative population of each level depends on the mechanisms for populating and depopulating the levels. The mechanisms of interest in a stellar photosphere are: (1) radiative, where a photon is absorbed and its energy used to raise the electron to a higher level; (2) collisional, where atoms bump into each other using mechanical (thermal) energy to cause excitation or de-excitation; (3) spontaneous transitions, in which de-excitation occurs without provocation accompanied by the emission of a photon; and (4) stimulated emission, a de-excitation caused by a passing photon. In some cases collisional interactions dominate radiative ones and in other cases the reverse is true. When collisions do dominate, we can calculate the populations from the equations that hold in thermodynamic equilibrium as shown below.

The fraction of atoms (or ions) excited to the nth level is proportional to the statistical weight g_n and the so-called Boltzmann factor,

$$N_n = \text{constant } g_n e^{-\chi_n/kT}.$$

The statistical weight is $2J+1$, where J is the inner quantum number (obtainable from Moore 1959). For hydrogen, $g_n = 2n^2$. The absolute temperature T is in Kelvin as usual.

The ratio of populations in two levels m and n is then

$$\frac{N_n}{N_m} = \frac{g_n}{g_m} e^{-\Delta\chi/kT}, \tag{1.17}$$

with $\Delta\chi = \chi_n - \chi_m$. Similarly, the number of atoms per unit volume in a level n expressed as a fraction of all atoms of the same species is

$$\begin{aligned}\frac{N_n}{N} &= \frac{g_n e^{-\chi_n/kT}}{g_1 + g_2 e^{-\chi_2/kT} + g_3 e^{-\chi_3/kT} + \cdots} \\ &= \frac{g_n}{u(T)} e^{-\chi_n/kT}.\end{aligned} \tag{1.18}$$

The denominator is lumped together in $u(T) = \sum g_i e^{-\chi_i/kT}$, called the partition function. In practice, the partition function is finite because the highest levels are removed by collisions (see Chapter 11). Partition functions have been calculated for most elements of interest, and a tabulation can be found in Appendix D. Partition functions for some molecules are given by Irwin (1988).

Equation (1.18) is often expressed as a power of 10,

$$\frac{N_n}{N} = \frac{g_n}{u(T)} 10^{-\log e\chi_n/kT} = \frac{g_n}{u(T)} 10^{-\theta\chi_n}, \tag{1.19}$$

with $\theta = (\log e)/(kT) = 5040/T$, where k must be in units of electron volts per

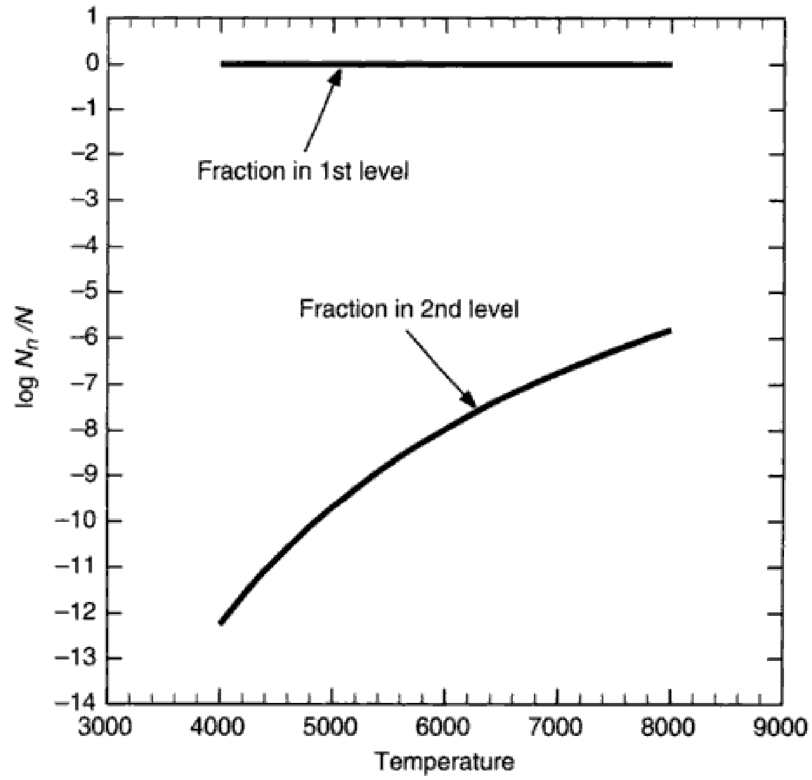


Fig. 1.9. Hydrogen excitation to the second level, $\chi = 10.20\,\mathrm{eV}$, changes by many orders of magnitude, but is so small in absolute terms that the fraction in the ground state remains sensibly at unity. At temperatures higher than these, hydrogen becomes ionized.

Kelvin to be compatible with χ_n in electron volts. Equation (1.18) or (1.19) is called the excitation or Boltzmann equation.

Figure 1.9 illustrates an important fact of excitation: the fraction excited to some level can vary by many orders of magnitude while the number in the ground state remains nearly constant.

The ionization for the collision-dominated gas can be calculated using

$$\frac{N_1}{N_0} P_e = \frac{(2\pi m_e)^{3/2}(kT)^{5/2}}{h^3} \frac{2u_1(T)}{u_0(T)} \, e^{-I/kT}, \tag{1.20}$$

where N_1/N_0 is the ratio of ions to neutrals, u_1/u_0 is the ratio of ionic to neutral partition functions, m_e is the electron mass, and h is Planck's constant. This equation was first derived by M. N. Saha and sometimes bears his name. See Struve and Zebergs (1962) and Hearnshaw (1986) for historical details. Numerically, using c.g.s. units for m, k, and h, the equation is

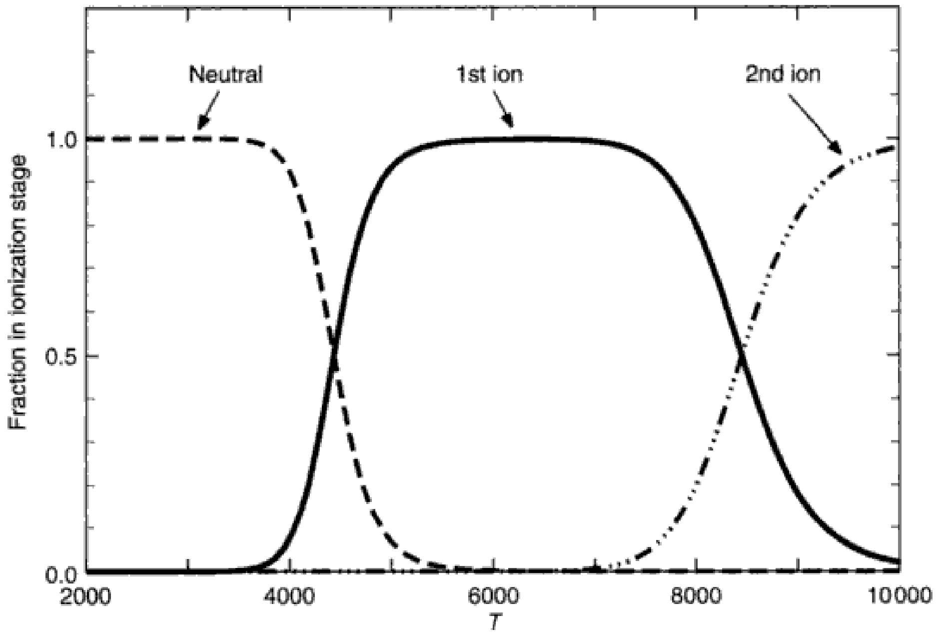


Fig. 1.10. As temperature increases, ionization occurs rather abruptly, as shown by this example using iron. An electron pressure of 1 dyne/cm^2 is used in the computation. In a stellar photosphere, elements exist mainly in just two ionization stages.

$$\log \frac{N_1}{N_0} P_e = \frac{-5040}{T} I + 2.5 \log T + \log \frac{u_1}{u_0} - 0.1762$$

or

$$\frac{N_1}{N_0} = \frac{\Phi(T)}{P_e}, \tag{1.21}$$

where

$$\begin{aligned} \Phi(T) &= 0.6665 \frac{u_1}{u_0} T^{5/2} 10^{-5040I/T} \\ &= 1.2020 \times 10^9 \frac{u_1}{u_0} \theta^{-5/2} 10^{-\theta I}. \end{aligned} \tag{1.22}$$

The ratio of doubly ionized to singly ionized particles is calculated in the same way: $N_2/N_1 = \Phi(T)/P_e$, where the I in Eq. (1.22) now stands for the ionization potential of the ion, and the partition function ratio is $u_2(T)/u_1(T)$. Derivations of these relations can be found in Aller (1963) or in statistical mechanics expositions such as that by Tolman (1938).

The transition from neutral to first ion, from first to second ion, and so on occurs fairly rapidly along the temperature coordinate as shown in Fig. 1.10. This has important implications for the roles played by any ion as a source of continuous opacity, a donor of electrons to the photospheric gas, and for the strength of its lines in the spectrum of a star.

In most calculations, Eqs. (1.19) and (1.21) are combined to find the number of atoms in a given level as a fraction of the total number of the element. For example, if we let N_n be the number in the nth level of the singly ionized atom, then

$$\begin{aligned}\frac{N_n}{N_{\text{total}}} &= \frac{N_n}{N_0 + N_1 + N_2 + \cdots}\\ &= \frac{N_n/N_1}{N_0/N_1 + 1.0 + N_2/N_1 + \cdots}.\end{aligned} \tag{1.23}$$

The ratio N_n/N_1 comes from Eq. (1.18) and each of the ratios N_0/N_1, N_2/N_1, etc. comes from Eq. (1.21). Virtually all of an element is accounted for by including three stages of ionization (as per Fig. 1.10).

Stellar catalogues, table, and atlases

A large body of useful material has been compiled over the years. Access to many of the following sources of data is necessary for successful observation and analysis of stars.

Astrophysical Quantities (Allen 1973, Cox 2000): reference volume giving astronomical data on all topics ranging from physical constants through basic stellar parameters to clusters of galaxies, and containing many useful numbers for studies in stellar photospheres.

Catalogue of Bright Stars (Hoffleit & Warren 1996): lists 9091 stars by HR = BS number. Gives HD and ADS numbers, equatorial coordinates for 1900 and 2000, precession rates, visual magnitude, color index, spectral type, proper motion, parallax, radial velocity, rotational velocity, and useful notes.

The Hipparcos and Tycho Catalogues (Hipparcos 1997): primarily astrometric data, parallaxes and proper motions, but also summarizes spectral types, magnitudes, and color indices for over 1 million stars, 118 218 from the Hipparcos satellite with parallax errors ~1 milliarcsecond (mas) and nearly ten times as many with parallax errors ~7 mas.

Catalogue of Nearby Stars (Gliese 1969): includes all stars having measured parallaxes larger than 0.045 arcsec. In addition to parallax, the tabulation gives proper motion, absolute magnitude, and galactic velocity components.

General Catalogue of Stellar Radial Velocities (Wilson 1953): lists radial velocities for 15 107 stars.

Catalogue of Rotational Velocities of the Stars (Uesugi & Fukuda 1982): $v \sin i$ measurements for 6472 stars are tabulated with references to individual measurements. Be careful here; mean values listed in the catalogue sometimes include upper limits. Most listings with $v \sin i \lesssim 15$ km/s should be treated as upper limits.

Photometric Atlas of the Solar Spectrum (Minnaert *et al.* 1940): graphical atlas covering the solar spectrum from λ3332 to λ8771; based on photographic record of the light from the center of the solar disk. The resolving power is between 100 000 and 150 000. More accurate atlases are now available, but the work remains useful for background and orientation work, especially since it is coupled with the following tabulation.

The Solar Spectrum 2935 Å to 8770 Å (Moore *et al.* 1966): gives data for 24 000 lines, including the wavelength, the equivalent width, the element identification, the excitation potential, and the multiplet number. The equivalent widths were obtained from the preceding reference.

A Multiplet Table of Astrophysical Interest (Moore 1959): multiplets are listed for about 26 000 lines of most common elements. It gives the wavelength, the intensity class, the excitation potentials, the inner quantum number J, the atomic configuration, and the multiplet number.

Lines of the Chemical Elements in Astronomical Spectra (Merrill 1958): summarizes many interesting facts about the appearance of each element in the stellar spectra, with extensive referencing to the early literature. Merrill gives useful energy-level diagrams in Appendix A.

The Solar Spectrum from λ7498 to λ12016, A Table of Measures and Identification (Swensson *et al.* 1970): gives wavelengths, elements identification, intensity scale, and transition information for 10 840 lines.

Photometric Atlas of the Solar Spectrum from λ3000 to λ10 000 (Delbouille *et al.* 1973): graphical atlas recorded photoelectrically with a double-pass monochromator at a resolving power in excess of 700 000.

A High Resolution Spectral Atlas of the Solar Irradiance from 380 to 700 Nanometers (Beckers *et al.* 1976): two volumes; a map and a numerical tabulation of the same data, a handy combination. Disk-integrated sunlight measured with a resolving power generally near 300 000 and with a signal-to-noise ratio of 200.

Solar Flux Atlas from 296 to 1300 nm (Kurucz *et al.* 1984): map of the disk-integrated sunlight using resolving power in excess of 348 000 and having a signal-to-noise ratio of several thousand.

Photometric Atlas of the Solar Spectrum from 1850 to 10 000 cm^{-1} (Delbouille *et al.* 1981): map of the infrared spectrum of the center of the solar disk with signal-to-noise ratio of several thousand and a resolution ranging from 400 000 at 1 μm to 130 000 at 5 μm.

An Atlas of the Solar Photospheric Spectrum in the Region from 8900 to 1360 cm^{-1} (7350 to 11 230 Å) (Wallace *et al.* 1993): near-infrared spectral atlas of the center of the solar disk, with a resolving power of ≈300 000, performed with a 1 m Fourier transform spectrometer.

An Atlas of the Solar Spectrum in the Infrared from 1850 to 9000 cm^{-1} (1.1 to 5 μm) (Livingston & Wallace 1991): continuation of the previous atlas to longer wavelengths.

Infrared Atlas of the Arcturus Spectrum, 0.9–5.3 μm (Hinkle *et al.* 1995): Fourier transform spectrometer observations, resolving power ≈100 000, useful line identifications.

Visible and Near Infrared Atlas of the Arcturus Spectrum 3727–9300 Å (Hinkle *et al.* 2000): cross-dispersed echelle spectrographic data, resolving power ≈130 000–200 000; also useful line identification.

A Photometric Atlas of the Spectrum of Arcturus (Griffin 1968): tracings of photographic spectra of α Boo (K2 III) from 3600 to 8825 Å with a resolving power of ≈130 000.

A Photometric Atlas of Procyon (R. & R. Griffin 1979): tracings of the α C Mi spectrum (F5 IV–V) from 3140 to 7470 Å. Sister atlas to the Arcturus atlas above, but with a better signal-to-noise ratio.

References

Abt, H. A., A. B. Meinel, W. W. Morgan, & J. W. Tapscott 1968. *An Atlas of Low-Dispersion Grating Stellar Spectra.* (Tucson: Kitt Peak Nat. Obs.).

Allen, C. W. 1973. *Astrophysical Quantities* (London: Athlone), 3rd edn.

Aller, L. H. 1963. *Astrophysics: the Atmospheres of the Sun and Stars*, 2nd edn. (New York: Ronald).

Becker, W. 1948. *Ap. J.* **107**, 278.

Becker, W. & J. Stock 1954. *Z. f. Ap.* **34**, 1.

Beckers, J. M., C. A. Bridges, & L. B. Gilliam 1976. *A High Resolution Spectral Atlas of the Solar Irradiance from 380 to 700 Nanometers*, AFGL-TR-76-0126(I), Environmental Research Papers, No. 565, Vols. I and II. (Sunspot: Sacromento Peak Obs.)

Cox, A. N., ed. 2000. *Allen's Astrophysical Quantities.* (New York: AIP, Springer-Verlag).

Delbouille, L., G. Roland, & L. Neven 1973. *Photometric Atlas of the Solar Spectrum from λ3000 to λ10,000.* (Liege: Inst. Astrophys.).

Delbouille, L., G. Roland, J. Brault, & L. Testerman 1981. *Photometric Atlas of the Solar Spectrum from 1,850 to 10,000 cm^{-1}.* (Tucson: Kitt Peak Nat. Obs.).

Draper, H. 1935. *IAU Trans.* **5**, 180.

FitzGerald, M. P. 1970. *Ast. Ap.* **4**, 234.

Garrison, R. F., ed. 1984. *The MK Process and Stellar Classification.* (Toronto: David Dunlap Obs.).

Geballe, T. R., G. R. Knapp, S. K. Leggett, *et al.* 2002. *Ap. J.* **564**, 466.

Gliese, W. 1969. *Catalogue of Nearby Stars* No. 22 (Heidelberg: Ast. Rechn.-Inst.).

Griffin, R. F. 1968. *A Photometric Atlas of the Spectrum of Arcturus, λλ3600–8825 Å.* (Cambridge: Cambridge Philosophical Soc.).

Griffin, R. & R. 1979. *A Photometric Atlas of the Spectrum of Procyon, λλ3140–7470 Å.* (Ilkey: Scholar Press), obtainable from the authors at The Observatories, Madingley Road, Cambridge CB3 0HA, UK.

Hauck, B., & M. Mermilliod 1980. *Ast. Ap. Suppl.* **40**, 1.

Hearnshaw, J. B. 1986. *The Analysis of Starlight, One Hundred and Fifty Years of Astronomical Spectroscopy*. (Cambridge: Cambridge Univ. Press).

Hinkle, K., L. Wallace, & W. Livingston 1995. *Infrared Atlas of the Arcturus Spectrum, 0.9–5.3 μm*. (San Francisco: Ast. Soc. Pacific).

Hinkle, K., L. Wallace, J. Valenti, & D. Harmer 2000. *Visible and Near Infrared Atlas of the Arcturus Spectrum 3727–9300 Å*. (San Francisco: Ast. Soc. Pacific).

Hipparcos 1997. *The Hipparcos and Tycho Catalogues*, European Space Agency SP-1200.

Hoffleit, D. & W. H. Warren, Jr. 1996. *The Bright Star Catalogue*, 5th edn. (NASA: W. H. Warren).

Houk, N. & A. P. Cowley 1975. *University of Michigan Catalogue of Two-Dimensional Spectral Types for the HD Stars*. (Washington, DC: Comnet-Publicate).

Huggins, W. & Lady Huggins 1899. *Publ. Sir Wm. Huggins Observatory*, Vol. I. (New York: W. Wesley & Son).

Irwin, A. W. 1988. *Ast. Ap. Suppl.* **74**, 145.

Johnson, H. L. 1965. *Ap. J.* **141**, 923.

1966. *Ann. Rev. Ast. Ap.* **4**, 193.

Johnson, H. L. & W. W. Morgan 1953. *Ap. J.* **117**, 313.

Johnson, H. L., R. I. Mitchell, B. Iriarte, & W. Z. Wisniewski 1966. *Comm. Lunar Planetary Lab.* **4**, 99.

Keenan, P. C. 1985. IAU Symposium 111, *Calibration of Fundamental Stellar Quantities*, ed. D. S. Hayes, L. E. Pasinetti, & A.G.D. Philip. (Dordrecht: Reidel), p. 121.

Keenan, P. C. 1987. *Pub. Ast. Soc. Pacific* **99**, 713.

Keenan, P. C. & R. C. McNeil 1976. *An Atlas of Spectra of the Cooler Stars*. (Columbus: Ohio State Univ.).

Kurucz, R. L., I. Furenlid, & J. Brault 1984. *Solar Flux Atlas from 296 to 1300 nm*. (Cambridge: Univ. Pub. Harvard), available from the National Solar Obs., Sunspot, New Mexico.

Landolt, A. U. 1992. *Ast. J.* **104**, 340.

Leggett, S. K., D. A. Golimowski, X. Fan, *et al.* 2002. *Ap. J.* **564**, 452.

Livingston, W. & L. Wallace 1991. *An Atlas of the Solar Spectrum in the Infrared from 1850 to 9000 cm^{-1} (1.1 to 5.4 μm)*, Nat. Solar Obs. Tech. Report No. 91.001. (Tucson: NSO).

McClure, R. D. & S. van den Bergh 1968. *Ast. J.* **73**, 313.

Merrill, P. W. 1958. *Lines of the Chemical Elements in Astronomical Spectra*, Pub. No. 610 (Washington: Carnegie Inst. Washington).

Minnaert, M. G. J., G. F. W. Mulders, & J. Houtgast 1940. *A Photometric Atlas of the Solar Spectrum*. (Amsterdam: Schnabel).

Moore, C. E. 1959. *A Multiplet Table of Astrophysical Interest*, National Bureau of Standards Tec. Note 36, U.S. Department of Commerce Office of Technical Services, Washington.

Moore, C. E., M. C. J. Minnaert, & J. Houtgast 1966. *The Solar Spectrum 2935 Å to 8770 Å*. (Washington: National Bureau Standards), Monograph 61.

Morel, M. & P. Magnenat 1978. *Ast. Ap. Suppl.* **34**, 477.

Morgan, W. W., P. C. Keenan, & E. Kellman 1943. *An Atlas of Stellar Spectra*. (Chicago: Univ. Chicago).

Morgan, W. W. & P. C. Keenan 1973. *Ann. Rev. Ast. Ap.* **11**, 29.

Morgan, W. W., H. A. Abt, & J. W. Tapscott 1978. *Revised MK Spectral Atlas for Stars Earlier than the Sun*. (Williams Bay: Yerkes Obs.; Chicago: Univ. Chicago;

Tucson: Kitt Peak National Obs.).
Nicolet, B. 1978. *Ast. Ap. Suppl.* **34**, 1.
Perry, C. L., E. H. Olsen, & D. L. Crawford 1987. *Pub. Ast. Soc. Pacific* **99**, 1184.
Philip, A. G. D., ed. 1979. *Problems of Calibration of Multicolor Photometric Systems*, Dudley Obs. Rept. No. 14.
Rufener, F. 1988. *Catalogue of Stars Measured in the Geneva Observatory Photometric System*, 4th edn. (Sauverny, Switzerland: Obs. Geneve).
Rufener, F. & B. Nicolet 1988. *Ast. Ap.* **206**, 357.
Strömgren, B. 1966. *Ann. Rev. Ast. Ap.* **4**, 433.
Struve, O. & V. Zebergs 1962. *Astronomy of the 20th Century*. (New York: Macmillan).
Swensson, J. W., W. S. Benedict, L. Delbouille, & G. Roland 1970. *The Solar Spectrum from λ7498 to λ12016, A Table of Measures and Identifications*, Mem. Soc. Royale des Sci. de Liége, No. 5.
Tolman, R. C. 1938. *The Principles of Statistical Mechanics*. (London: Oxford Univ.).
Uesugi, A. & I. Fukuda 1982. *Revised Catalogue of Stellar Rotational Velocities*. (Kyoto: Univ. Kyoto).
Vernazza, J. E., E. H. Avrett, & R. Loeser 1973. *Ap. J.* **184**, 605.
Walborn, N. R., I. D. Howarth, D. J. Lennon, *et al.* 2002. *Ast. J.* **123**, 2754.
Wallace, L., K. Hinkle & W. Livingston 1993. *An Atlas of the Solar Photospheric Spectrum in the Region from 8900 to 1360 cm^{-1} (7350 to 11230 Å)*, Nat. Solar. Obs. Report No. 93-001. (NSO: Tucson).
Wilson, R. E. 1953. *General Catalogue of Stellar Radial Velocities*, Publ. No. 601 (Washington: Carnegie Inst. Washington).

Questions and exercises

1. Along the main sequence, the mass, temperature, color indices, spectral type, rotation rates, the strength of convection, and to a lesser degree the surface gravity all change. What makes us think that the spectral type depends primarily on temperature rather than the other parameters?

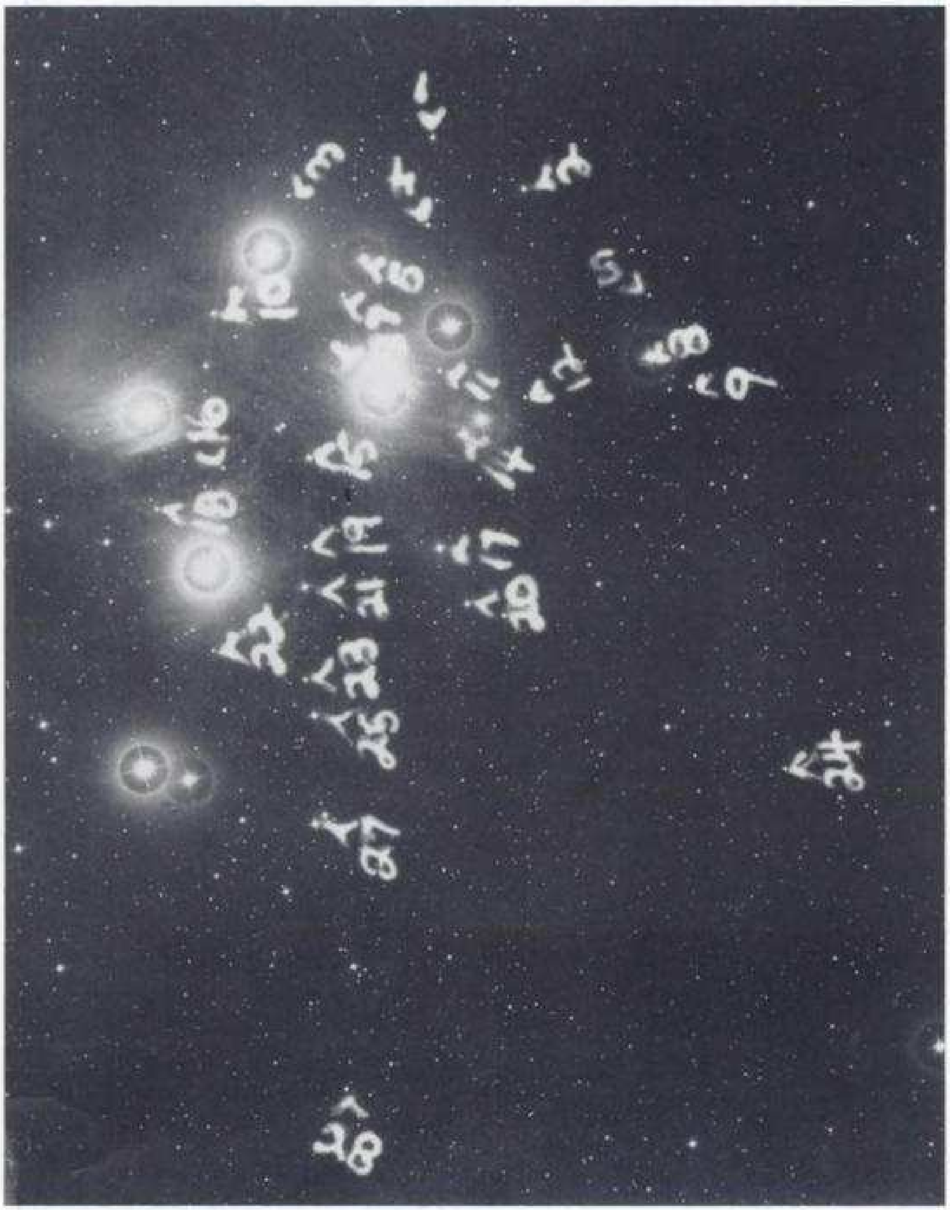
2. Develop some introductory skills in spectral classification by constructing a conceptual HR diagram for the Pleiades. Use the simplified flow diagram below to obtain approximate spectral types for the numbered stars in the objective-prism photograph. This exposure was taken with a Schmidt telescope having a 10° prism over the front end of the telescope tube, thereby turning each star image into a spectrum. You can use the line map to help you find the spectral lines used for classification. Second, use the direct image photograph to rank the numbered stars by brightness. Third, plot the ranking as ordinate against the spectral type as abscissa to make a rough HR diagram. Notice how the objective-prism technique has the advantage of recording many spectra at once, but the disadvantage of having the spectra of the brightest stars over-exposed while the spectra of the faint stars are underexposed and hard to work with.

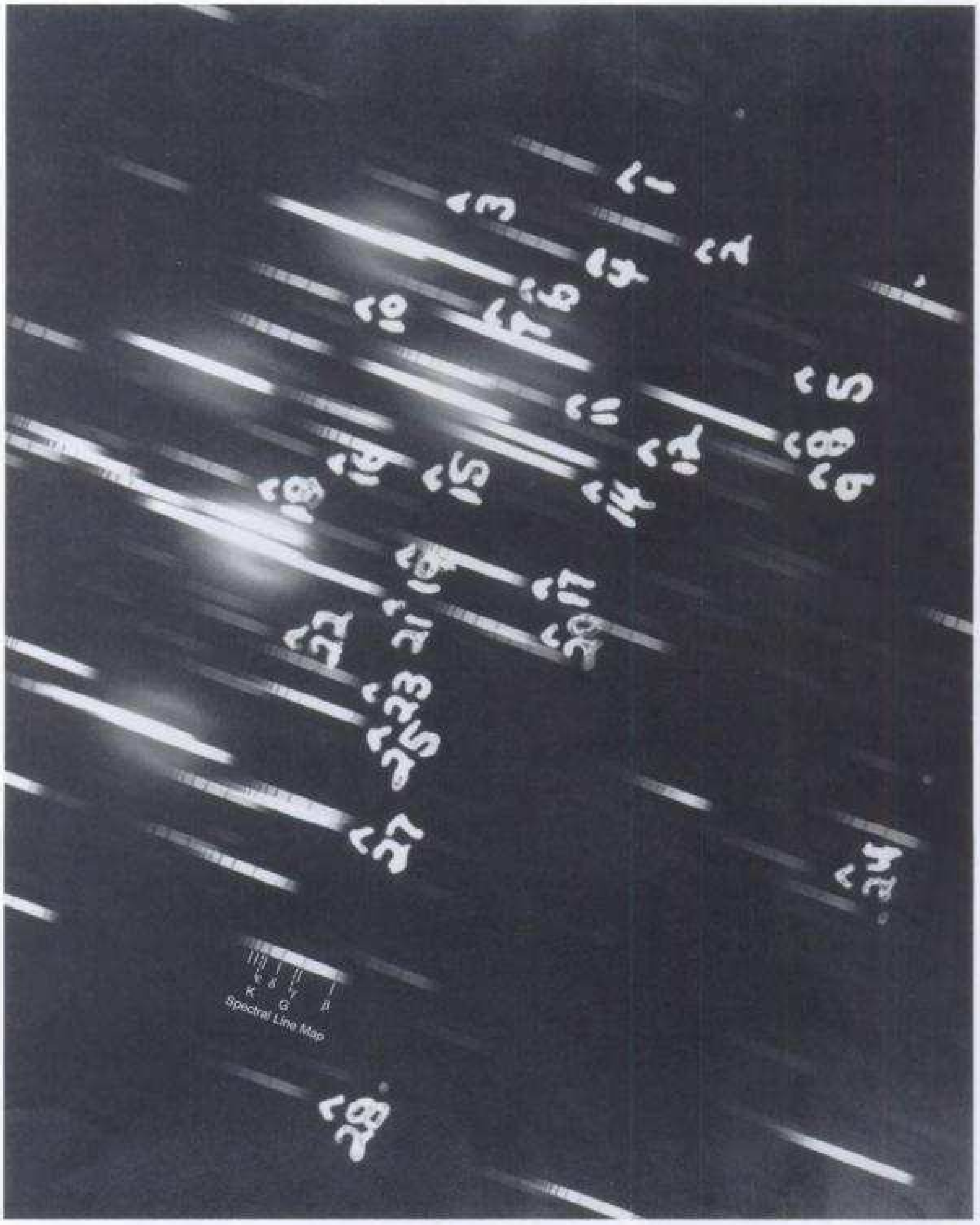
3. Show that the physical units are the same on both sides of Eq. (1.13). Then, for a pure isothermal hydrogen atmosphere having a temperature of 4000 K, calculate and graph the speed distribution using Eq. (1.13). Be particularly aware of the units of dv and how the ordinate is affected by your choice. That is, $dN(v)$ is a distribution, the graph

is really a histogram, and the bin size is dv. What is the most probably speed for your distribution? Does it depend on your choice of dv? Show on your graph the position of the most probable velocity. Compare your results with Fig. 1.7.

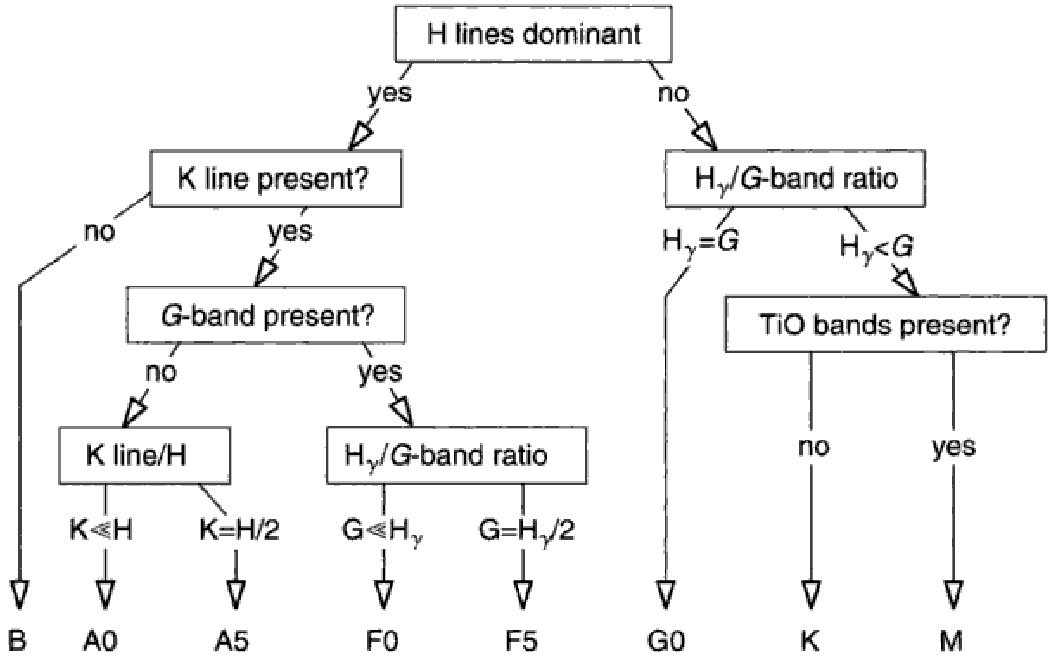
4. Add to your graph above the velocity distribution given by Eq. (1.12). What is the most probable velocity for this distribution?

5. What is the velocity of an iron atom that has enough kinetic energy to ionize hydrogen? Ionization potentials can be found in Appendix D. What gas temperature would this correspond to if this were the root-mean-square velocity? What would the temperature be if this velocity corresponded to $3v_3$, i.e., only atoms in the upper




K G β
Spectral Line Map

Spectral classification



tail of the velocity distribution did the ionizing?

6. Use Eq. (1.22) and the approach leading to Eq. (1.23) to compute N_0, N_1, and N_2 as fractions of the total number of iron particles. In this way, re-do the curves in Fig. 1.10 using an electron pressure of 1000 dynes/cm^2 instead of the value of 1 dyne/cm^2 used there. The ionization potentials for iron can be found in Appendix D. For convenience, take $u_2(T) = u_1(T) = u_0(T) = 1$. Can you explain physically why the curves change when the electron pressure is altered?

2

Fourier transforms

There is hardly a phase of modern astrophysics to which Fourier techniques do not lend some insight or practical advantage. Fourier concepts prove useful in the context of line absorption coefficients, the analysis of line profiles, spectrograph resolution, telescope diffraction, and the study of noise. In these and other applications, convolutions appear in the physics of the situation. Usually it is much easier to visualize a product of functions in place of their convolution and this can be done with Fourier transforms through the convolution theorem.

This chapter forms an introduction to Fourier transforms for those unfamiliar with them and a useful refresher for those who have studied them in past years. The treatment is highly abbreviated, but covers all the concepts used in the remainder of the book. Those wishing to learn the material in a more rigorous and extensive way are referred to the books dealing specifically with Fourier transforms, for instance, the books of Jennison (1961), Bracewell (1965), and Gaskill (1978).

The definition

The Fourier transform of a function is a specification of the amplitudes and phases of sinusoidals which, when added together, reproduce the function. Only one-dimensional functions are treated here. Expansion to two or higher dimensions can be done by the reader with modest effort.

We take a function $F(x)$ and write

$$f(\sigma) = \int_{-\infty}^{\infty} F(x) e^{2\pi i x \sigma} dx. \tag{2.1}$$

This new function, $f(\sigma)$, is the Fourier transform of $F(x)$. The variables x and σ are called a Fourier pair. The inverse transform is

$$F(x) = \int_{-\infty}^{\infty} f(\sigma) e^{-2\pi i x \sigma} d\sigma. \tag{2.2}$$

Upper- and lowercase symbols denote, respectively, the functions and their transforms. Thus $f(\sigma)$ and $F(x)$ are transforms of each other. Notice the change in sign of the exponent between Eqs. (2.1) and (2.2).

Not all functions have Fourier transforms. In the physical world of stars we simply assume the transform exists, and refer to Bracewell's book for more complex situations. A derivation of the relations (2.1) and (2.2) can also be found in a book by Kharkevich (1960).

Suppose we expand Eq. (2.1) using Euler's formula, that is, $e^{ix} = \cos x + i \sin x$. We obtain four integrals, provided we allow $F(x)$ to be a complex function, $F(x) = F_R(x) + i F_I(x)$, consisting of a real, $F_R(x)$, and imaginary, $F_I(x)$, component. Then

$$\begin{aligned} f(\sigma) = &\int_{-\infty}^{\infty} F_R(x) \cos 2\pi x\sigma \, dx + i \int_{-\infty}^{\infty} F_I(x) \cos 2\pi x\sigma \, dx \\ &+ i \int_{-\infty}^{\infty} F_R(x) \sin 2\pi x\sigma \, dx - \int_{-\infty}^{\infty} F_I(x) \sin 2\pi x\sigma \, dx. \end{aligned} \tag{2.3}$$

This equation is useful for visualizing the Fourier transform process. Consider the first integral as an example. It says: multiply the function $F_R(x)$ by a cosine with periodicity $1/\sigma$, with σ being a constant for this integration. The total area resulting from this product is the value needed. Figure 2.1 shows this process. The other three integrals in Eq. (2.3) can be visualized in a similar manner. Combining the integrals gives $f(\sigma)$ at one value of σ. Changing the frequency of the sines and cosines and repeating the process gives $f(\sigma)$ at a second point, and so on. This process is the essence of the numerical evaluation of Fourier transforms which is discussed later in the chapter.

If we further write in Eq. (2.3), $f(\sigma) = f_R(\sigma) + i f_I(\sigma)$, we see that $f_R(\sigma)$ is composed of the first and last integrals, while $f_I(\sigma)$ is given by the center two integrals. So in the general case, both $F(x)$ and $f(\sigma)$ are complex functions.

Under some physical circumstances, $F(x)$ may be a real function. One such case is when $F(x)$ is the spectrum of starlight. Since $F_I(x) = 0$, $f(\sigma)$ is given by

$$f(\sigma) = \int_{-\infty}^{\infty} F_R(x) \cos 2\pi x\sigma \, dx + i \int_{-\infty}^{\infty} F_R(x) \sin 2\pi x\sigma \, dx.$$

We see that $f(\sigma)$ is still a complex function. Only if $F(x)$ is an even function, $F_R(x) = F_R(-x)$, as well as real we find

$$f(\sigma) = \int_{-\infty}^{\infty} F_R(x) \cos 2\pi x\sigma \, dx.$$

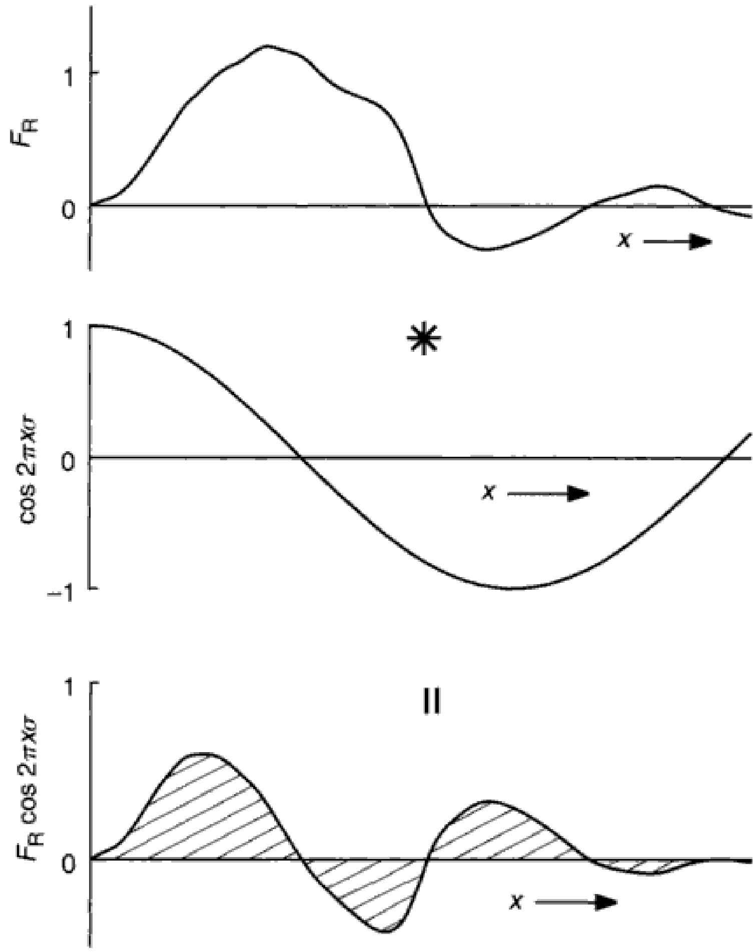


Fig. 2.1. The arbitrary real function $F_R(x)$ is multiplied by $\cos 2\pi x\sigma$ to give the bottom curve. The area is the value of the transform $f(\sigma)$ for one value of σ.

Here the evenness of $F_R(x)$ combined with the oddness of $\sin 2\pi x\sigma$ gives exactly as much negative as positive area, causing the imaginary term to be zero.

All complex numbers can be specified in rectangular or polar coordinates. In rectangular form, we have written $f(\sigma) = f_R(\sigma) + \mathrm{i} f_I(\sigma)$. We can convert this to polar form in the usual way using an amplitude and phase,

$$f(\sigma) = |f(\sigma)| \, \mathrm{e}^{\mathrm{i}\psi}, \tag{2.4}$$

where the amplitude is

$$|f(\sigma)| = [f_R^2(\sigma) + f_I^2(\sigma)]^{1/2}$$

and

$$\tan\psi = \frac{f_I(\sigma)}{f_R(\sigma)},$$

where ψ is the phase.

We conclude this section by pointing out that the zero ordinate of the transform is the area under the original function. And because the original function is the transform of the transform, its zero ordinate specifies the area under the transform. This conclusion can be reached trivially from the definitions, Eqs. (2.1) and (2.2), by setting $\sigma=0$, $f(0)=\int F(x)\,\mathrm{d}x$ and $F(0)=\int f(\sigma)\,\mathrm{d}\sigma$. These relations are useful, for example, when the Fourier transform of a spectral line is being investigated. Then the $\sigma=0$ value is the total absorption caused by the line.

Some common transforms

We now consider several simple functions and their transforms to illustrate some of the basic behavior. First consider the box shown in Fig. 2.2. It might represent graphically the light distribution in the plane of an opaque screen having a long slit of width W in it. We can denote the function by

$$B(x) = \begin{cases} 0 & \text{for} \quad -\dfrac{\mathrm{W}}{2} > x > \dfrac{\mathrm{W}}{2} \\ 1 & \text{for} \quad -\dfrac{\mathrm{W}}{2} \le x \le \dfrac{\mathrm{W}}{2}. \end{cases}$$

And immediately the Fourier transform can be written as

$$\begin{aligned} b(\sigma) &= \int_{-\infty}^{\infty} B(x)\mathrm{e}^{2\pi\mathrm{i}x\sigma}\,\mathrm{d}x \\ &= \int_{-W/2}^{W/2} \mathrm{e}^{2\pi\mathrm{i}x\sigma}\,\mathrm{d}x \\ &= \left(\frac{\mathrm{e}^{2\pi\mathrm{i}x\sigma}}{2\pi\mathrm{i}\sigma}\right)_{-W/2}^{W/2} \\ &= \frac{1}{2\pi\mathrm{i}\sigma}\left(\mathrm{e}^{2\pi\mathrm{i}W\sigma/2} - \mathrm{e}^{-2\pi\mathrm{i}W\sigma/2}\right) \\ &= W\frac{\sin\pi W\sigma}{\pi W\sigma} \\ &\equiv W\,\mathrm{sinc}\,\pi W\sigma. \end{aligned} \tag{2.5}$$

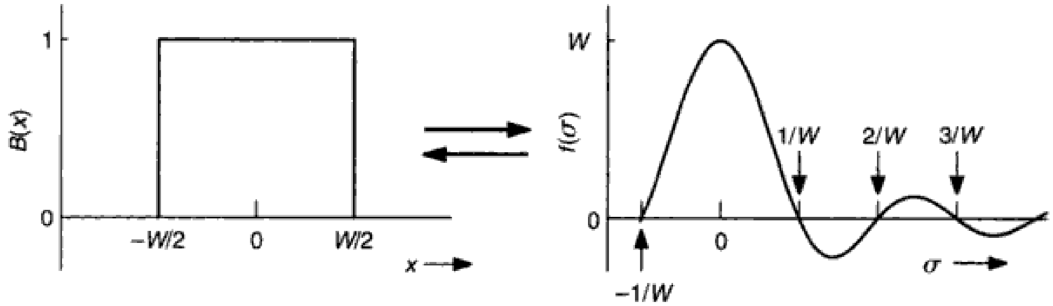


Fig. 2.2. The box of width W transforms into $f(\sigma) = \mathrm{sinc}\,\pi W\sigma$ of characteristic width $1/W$.

This sinc function is shown on the right-hand side in Fig. 2.2. The reciprocal behavior in the widths of these functions is particularly important. When W is large the transform is narrow, and vice versa. This illustrates a general behavior between all functions and their transforms. The first zeros of this transform are at $\sigma = \pm 1/W$. The other zeros are equally spaced at multiples of $1/W$ along the σ axis.

As a second function, take the unit-area Gaussian given by

$$G(x) = \frac{1}{\beta\pi^{1/2}}\,\mathrm{e}^{-x^2/\beta^2}. \tag{2.6}$$

(In statistics, the notation is slightly different, with β^2 being replaced by $2\beta^2$.) Again, we find the transform from the definition.

$$\begin{aligned} g(\sigma) &= \int_{-\infty}^{\infty} G(x)\,\mathrm{e}^{2\pi i x\sigma}\,\mathrm{d}x \\ &= \int_{-\infty}^{\infty} \frac{1}{\beta\pi^{1/2}}\,\mathrm{e}^{-x^2/\beta^2}\mathrm{e}^{2\pi i x\sigma}\,\mathrm{d}x. \end{aligned}$$

But this function is real and symmetric, so we can use the expansion as in Eq. (2.3) to obtain

$$g(\sigma) = \frac{1}{\beta\pi^{1/2}}\int_{-\infty}^{\infty} \mathrm{e}^{-x^2/\beta^2}\cos 2\pi x\sigma\,\mathrm{d}x.$$

Standard integral tables supply the necessary integration with the result that

$$g(\sigma) = \mathrm{e}^{-\pi^2\beta^2\sigma^2}. \tag{2.7}$$

In short, a Gaussian transforms into a Gaussian. If the dispersion of the original Gaussian is β, the dispersion of the transform is $1/(\pi\beta)$, again an inverse relation.

Another function that comes up, particularly in the study of spectral line profiles, is the dispersion profile defined by

$$F(x) = \frac{1}{\pi}\frac{\beta}{x^2+\beta^2}. \tag{2.8}$$

Here β is the half-half width of the function, and $F(x)$ has unit area. The transform is

$$\begin{aligned} f(\sigma) &= \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\beta}{x^2+\beta^2}\, e^{2\pi i x\sigma}\, dx \\ &= \frac{2}{\pi}\int_0^{\infty} \frac{\beta}{x^2+\beta^2} \cos 2\pi x\sigma\, dx, \end{aligned}$$

where we have jumped straight to the cosine integral since this function is real and symmetric (see Eq. 2.3). This integral is of the form

$$\int_0^{\infty} \frac{\cos mu}{1+u^2}\, du = \frac{\pi}{2}\, e^{-m}.$$

We find that

$$f(\sigma) = \begin{cases} e^{-2\pi\beta\sigma} & \text{for} \quad \sigma \geq 0 \\ e^{2\pi\beta\sigma} & \text{for} \quad \sigma < 0, \end{cases}$$

or

$$f(\sigma) = e^{-2\pi\beta|\sigma|}. \tag{2.9}$$

Now let us turn to a very useful but slightly more complicated function, the delta-function. Physically the δ-function represents an impulse. This is equivalent to saying that the physical process takes place so rapidly that it is beyond our power to resolve it. Therefore the integrated property of the impulse is what we observe, which for the δ-function is defined as unity,

$$\int_{-\infty}^{\infty} \delta(x)\, dx = 1. \tag{2.10}$$

Although we cannot resolve the impulse, we can specify its position or time of occurrence. Symbolically we write

$$\delta(x) = 0 \qquad \text{for} \quad x \neq 0,$$

which implies that $\delta(x)$ occurs at $x = 0$. If we write $\delta(x - x_1)$, then we mean the impulse occurs at $x = x_1$.

Mathematically it is convenient to think of some unit area function that can be made infinitesimal in width while maintaining unit area. This process is necessary to provide a rigorous proof of many of the properties of a δ-function. In keeping with the level of treatment in this chapter, we omit these proofs. It should be understood, however, that the very large amplitude implied in making a unit area function go to infinitesimal width is not a practical problem because either the δ-function is used in conjunction with another function or only its integral property is of concern. As an example, consider the product of the function $F(x)$ with the δ-function,

$$\delta(x-x_1)F(x) = \delta(x-x_1)F(x_1), \tag{2.11}$$

since $\delta(x-x_1)=0$ except at $x=x_1$. In other words, multiplication by the δ-function retains only one point in the original function.

Furthermore, if we integrate Eq. (2.11), we have

$$\int_{-\infty}^{\infty}\delta(x-x_1)F(x)\,\mathrm{d}x = F(x_1)\int_{-\infty}^{\infty}\delta(x-x_1)\,\mathrm{d}x = F(x_1). \tag{2.12}$$

This interesting property proves useful when the δ-function is applied in analysis. Notice also that Eq. (2.12) can be rearranged slightly and written as

$$\int_{-\infty}^{\infty}\delta(x_1-x)F(x)\,\mathrm{d}x = F(x_1).$$

This expression has the form of a convolution which is defined and discussed later in this chapter.

The transform of $\delta(x-x_1)$ is given by

$$f(\sigma) = \int_{-\infty}^{\infty}\delta(x-x_1)\,\mathrm{e}^{2\pi i x\sigma}\,\mathrm{d}x.$$

But since $\delta(x-x_1)=0$, except when $x=x_1$, we can write

$$f(\sigma) = \mathrm{e}^{2\pi i x_1\sigma}\int_{-\infty}^{\infty}\delta(x-x_1)\,\mathrm{d}x = \mathrm{e}^{2\pi i x_1\sigma}, \tag{2.13}$$

and so the amplitude of $f(\sigma)$ is unity independent of σ, but there is a linear phase term, $2\pi x_1\sigma$. These properties are shown in Fig. 2.3. This function represents the ultimate in the inverse dimension behavior: an infinitesimal width for the function gives an infinite extension in the transform. The linear phase term is shown as a dashed line with a slope of $2\pi x_1$. The relation between the x_1 shift in $\delta(x-x_1)$ and the phase term is discussed in more detail in the theorems section below (Theorem 2).

We can easily obtain the transforms of sines and cosines using δ-functions and their transforms (Eq. 2.13). Suppose, for example, we have two δ-functions

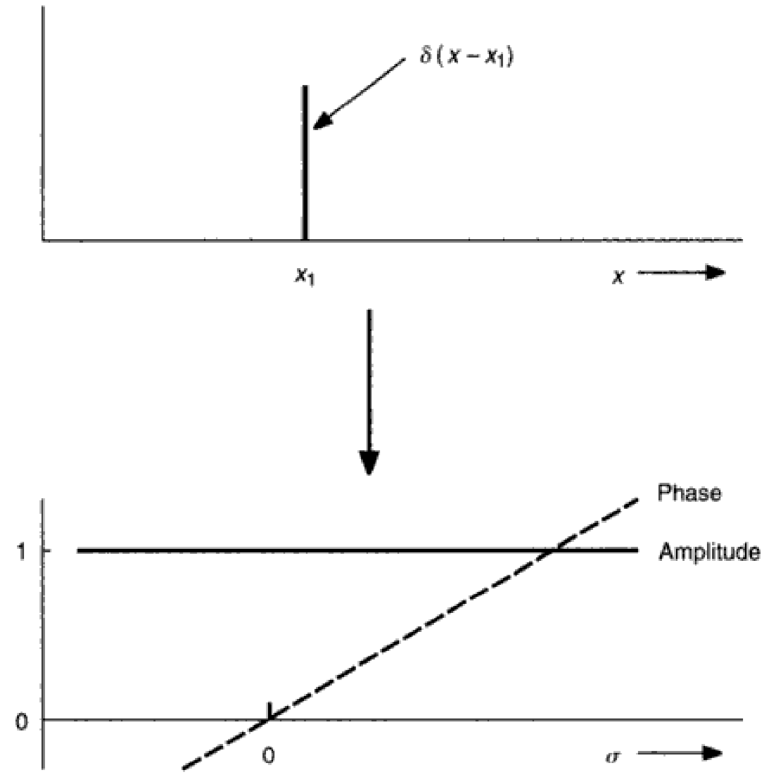


Fig. 2.3. A δ-function at $x = x_1$ transforms into the constant amplitude function with linear phase $2\pi x_1 \sigma$.

symmetrically placed about $x = 0$. That is, $F(x) = \delta(x - x_1) + \delta(x + x_1)$, and because Fourier transforms are linear (Theorem 1 below), we can write the transform as the sum of the individual transforms,

$$\begin{aligned} f(\sigma) &= e^{2\pi i x_1 \sigma} + e^{-2\pi i x_1 \sigma} \\ &= 2\cos 2\pi x_1 \sigma. \end{aligned} \tag{2.14}$$

In a similar way, $F(x) = \delta(x - x_1) - \delta(x + x_1)$, transforms to $f(\sigma) = 2i \sin 2\pi x_1 \sigma$. In short, pairs of δ-functions transform into sinusoidals (see Fig. 2.4).

Instead of two δ-functions, take an infinite array of them equally spaced at intervals of Δx. Then denoting this array by the Shah function $III(x)$, we can write

$$III(x) = \sum_{n=-\infty}^{\infty} \delta(x - n\Delta x), \tag{2.15}$$

where n is an integer. The transform of $III(x)$ is another infinite set of δ-functions, that is, $III(\sigma) = \Sigma\delta(\sigma - n/\Delta x)$ and, as always, the spacing between the peaks in

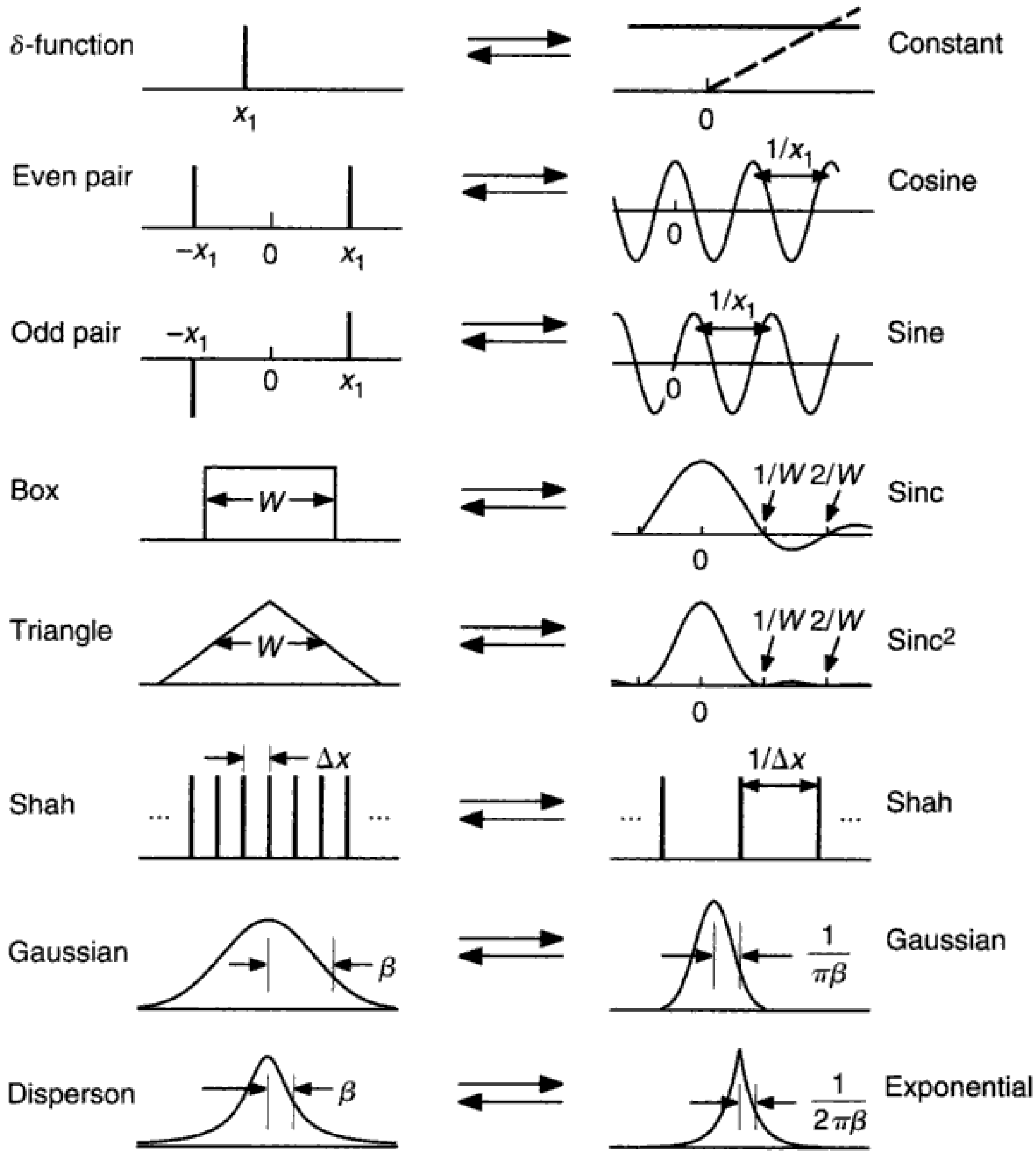


Fig. 2.4. Here is a summary of important functions and their transforms.

$III(\sigma)$ is inversely proportional to the spacing in $III(x)$. Figure 2.4 summarizes some basic functions and their transforms.

Data sampling and data windows

An important application of $III(x)$ is to convert a continuous function or distribution to a "sampled" one. We commonly measure a spectral line profile by taking data points spaced in wavelength by $\Delta\lambda$. Call these measurements $D(\lambda)$, and write

$$D(\lambda) = III(\lambda)F(\lambda). \tag{2.16}$$

where $F(\lambda)$ is the true spectrum and multiplication by $III(\lambda)$ gives the discrete sampling. We are quite accustomed to thinking of $D(\lambda)$ as the measured line profile. We can extend this concept and imagine *any* function to be synthesized from δ-functions. These δ-functions can be as closely packed as necessary for the job at hand, and they have amplitudes modulated by the true function as in Eq. (2.16). Figure 2.5 shows this process applied to spectral line profiles.

In addition to sampling data, we record data in a window. When we measure a stellar spectrum, our data extend from λ_1 to λ_2, the ends of our data window, or when we measure a variable star from a time t_1 to a time t_2, this time interval forms a window. The windows can be expressed in our data string

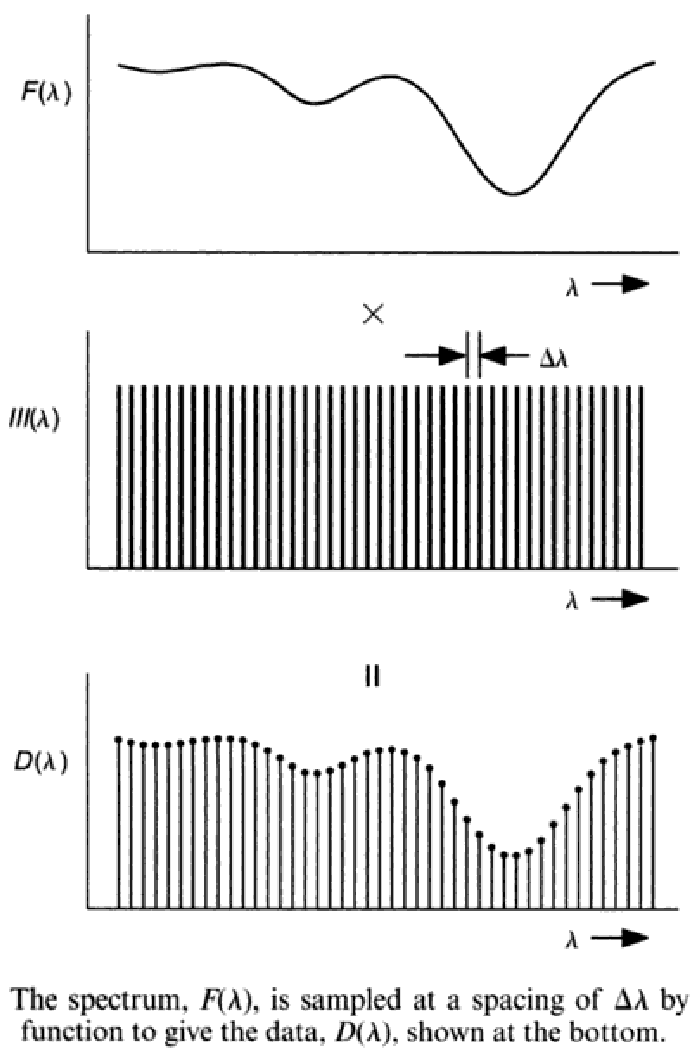


Fig. 2.5. The spectrum, $F(\lambda)$, is sampled at a spacing of $\Delta\lambda$ by the Shah function to give the data, $D(\lambda)$, shown at the bottom.

by multiplication with a box, $B(\lambda)$, or a series of boxes, having a width equal to the window limits of our observation. With this in mind, our spectral data in Eq. (2.16) become

$$D(\lambda) = B(\lambda)\,III(\lambda)F(\lambda). \tag{2.17}$$

It is $D(\lambda)$, sampled and bounded, which we record and compare with theory. Particularly important applications of these concepts arise in Chapters 12, 17, and 18.

Convolutions

Multiplication of functions, as in Eq. (2.17), clearly needs to be done. If we then ask about the Fourier transform of a product of functions, say $F(x)G(x)$, we find that it is a new process called a convolution, defined by

$$k(\sigma) \equiv \int_{-\infty}^{\infty} f(\sigma_1)g(\sigma - \sigma_1)\,\mathrm{d}\sigma_1, \tag{2.18}$$

where $f(\sigma)$ and $g(\sigma)$ are the transforms of $F(\lambda)$ and $G(\lambda)$. This integral is abbreviated by writing

$$k(\sigma) = f(\sigma) * g(\sigma),$$

with the "star" denoting convolution. The correspondence between multiplication in one Fourier domain and convolution in the other is called the *convolution theorem.*

It is relatively straightforward to look at Eq. (2.18) and understand what convolution means. $k(\sigma)$ is to be evaluated point by point, the integration being done over σ_1 with σ being constant. The function $g(\sigma_1)$ is translated by an amount σ relative to $f(\sigma_1)$, but because of the negative sign of σ_1 in the argument, $g(\sigma - \sigma_1)$ is reversed right for left. The product of these functions is formed, and the resulting net area is $k(\sigma)$ for that σ. Changing σ amounts to placing $g(\sigma - \sigma_1)$ at a new position. Tabulating $k(\sigma)$ then means sliding the reversed $g(\sigma)$ over $f(\sigma)$ and at each step recording the area under the product of the curves.

If we instead move a reversed $f(\sigma)$ over $g(\sigma)$, we obtain exactly the same result for $k(\sigma)$. The order of convolution does not matter,

$$f(\sigma) * g(\sigma) = g(\sigma) * f(\sigma). \tag{2.19}$$

Consider the simple case of the convolution of one unit amplitude box with another, as shown in Fig 2.6. This is a convenient example because the ordinates of each function are constant when they are not zero. When the boxes are

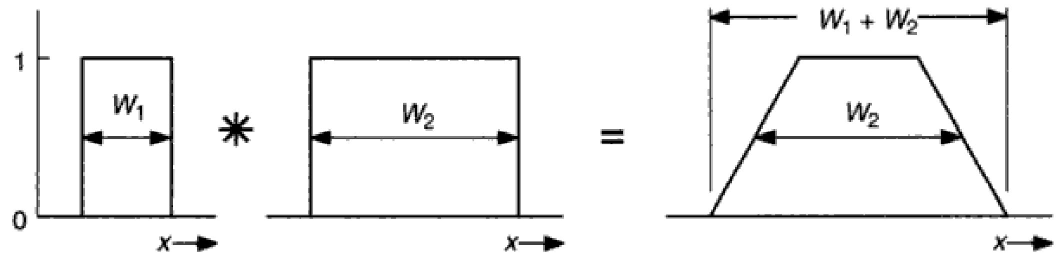


Fig. 2.6. The convolution of two boxes is a trapezoid. As the narrower box slides over the wider one, the overlapping area increases linearly until the narrower box is contained within the wider one. This interval corresponds to the width W_1. Then for the dimension $W_2 - W_1$, the common area is constant. Finally, the narrower box emerges, mirroring the initial rise.

separated, the product is zero. When the boxes overlap, the product is unity over the length of overlap. The length of overlap is proportional to the translation (A to B) until one of the boxes is contained within the other (B to C). Then the reduction of common area occurs as the boxes begin to separate (C to D), becoming zero beyond point D. A special case of interest is when the widths of the boxes are the same, so we have the convolution of two identical boxes or the convolution of the box with itself. Since the unit width box, $B(x)$, transforms into sinc $\pi\sigma$, we deduce from the convolution theorem that the triangle $B(x) * B(x)$ transforms into $\mathrm{sinc}^2\pi\sigma$. Figure 2.7 illustrates this general pattern of analysis for the specific case just cited. There are two paths by which one can get from the function to the transform. One path is usually easier to visualize or to compute than the other.

With the exception of the δ-function discussed below, all convolutions span a wider range of the variable than either of the individual functions. The convolution is a blurring process.

Convolution with a δ-function

The δ-function, $\delta(x)$, has a transform that is unity for all values of σ. Convolution of a function $F(x)$ with $\delta(x)$ implies a multiplication of their transforms, which gives $f(\sigma)$ times unity for all σ. Clearly the convolution of $F(x)$ with $\delta(x)$ results in $F(x)$ again.

However, if we instead have $\delta(x - x_1)$ convolved with $F(x)$, we find $F(x - x_1)$. We refer to Eq. (2.13), where we see that the transform of a δ-function that is not at the origin results in a phase term $e^{2\pi i x_1 \sigma}$. The product of transforms is then $f(\sigma)e^{2\pi i x_1 \sigma}$, corresponding to a translation of $F(x)$ to the position of the δ-function. This is a special case of Theorem 2 below. We conclude that convolution with a δ-function gives the function back again, but displaced to the position of the δ-function.

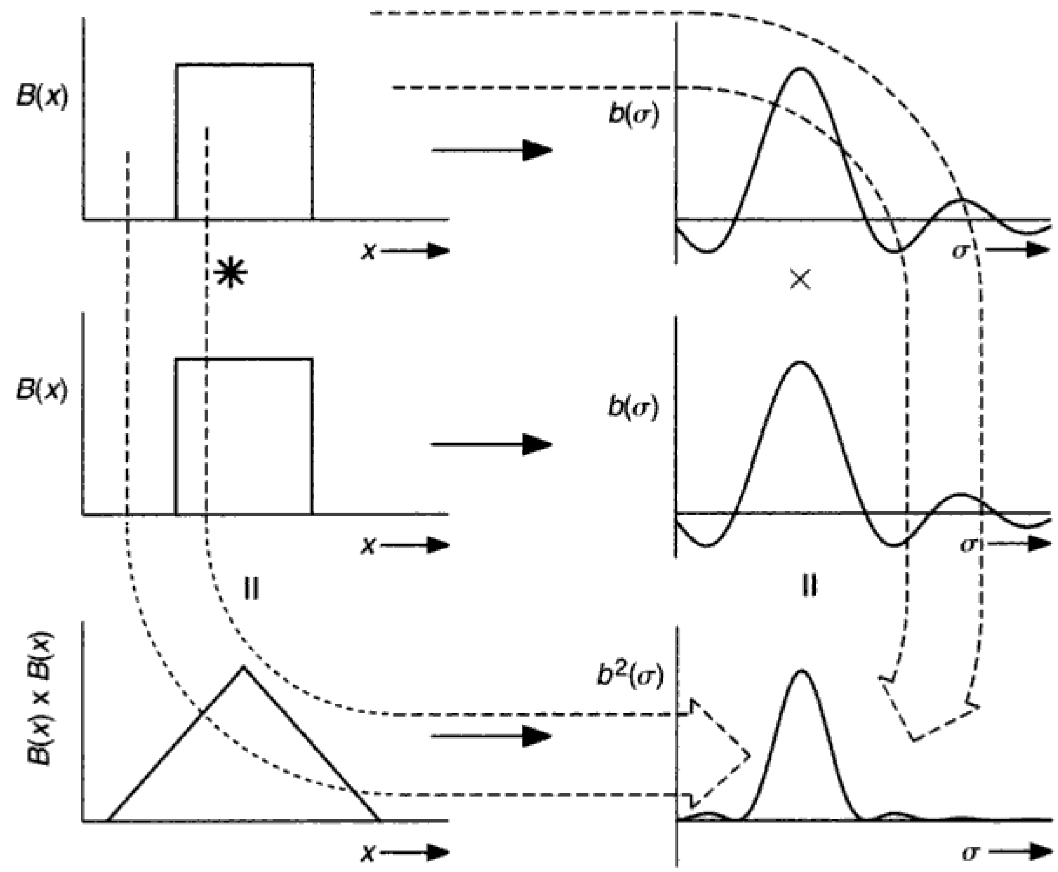


Fig. 2.7. In the x domain on the left, the convolution of a box with itself gives the triangle at the bottom. The transform of the triangle is $b^2(\sigma) = \mathrm{sinc}^2\,\pi\sigma$. An alternative route is to take the transform of the box, $B(x) \rightarrow b(\sigma) = \mathrm{sinc}\,\pi\sigma$, which is then multiplied by itself, according to the convolution theorem, to give the $\mathrm{sinc}^2\pi\sigma$ transform at the bottom right.

Convolutions of Gaussians and dispersion profiles

Equation (2.7) can be written for two Gaussians and multiplied together

$$\begin{aligned} g_a(\sigma)g_b(\sigma) &= e^{-\pi^2\beta_a^2\sigma^2}\,e^{-\pi^2\beta_b^2\sigma^2} \\ &= e^{-\pi^2\beta_c^2\sigma^2}, \end{aligned}$$

where $\beta_c^2 = \beta_a^2 + \beta_b^2$. In other words, the transform corresponds to the Gaussian

$$\begin{aligned} G_c(x) &= \frac{1}{\beta_c\pi^{1/2}}\,e^{-x^2/\beta_c^2} \\ &= G_a(x) * G_b(x). \end{aligned} \tag{2.20}$$

The convolution of two Gaussians is a third Gaussian.

In an identical way, using Eq. (2.9), one can show that the convolution of two dispersion profiles is a new dispersion profile with a half-half width given by $\beta_c = \beta_a + \beta_b$.

The convolution between a Gaussian and a dispersion function is called a Voigt function. If we let β_1 and β_2 represent the width parameters for Gaussian and dispersion profiles, then the transform of the Voigt function is

$$v(\sigma) = \mathrm{e}^{-\pi^2\beta_1^2\sigma^2}\mathrm{e}^{-2\pi\beta_2|\sigma|}. \tag{2.21}$$

The convolution of two Voigt functions leads to a product of transforms of this form. Since the exponents can be grouped together, it is trivial to show that the result is a new Voigt function with $\beta_1^2 = (\beta_1^a)^2 + (\beta_1^b)^2$ and $\beta_2 = \beta_2^a + \beta_2^b$. Voigt functions are considered in greater detail in Chapter 11.

Resolution: our blurred data

All measuring devices imprint their signature on observations we make with them. This signature is called the instrumental profile or the δ-function response, $I(x)$. To measure $I(x)$, we send through the instrument a δ-function, in practical terms, a spike that is narrow compared with the instrumental profile. (This process is applied to spectrographs in Chapter 12, where the spike is a narrow spectral line.) We then look at what comes out of the instrument, a blurred spike, and know that every δ-function sent in will be blurred this way. We recognize this as a convolution of a δ-function with the instrumental profile.

Next, we consider a more general function that will be processed through the instrument. Think of representing this function with a set of δ-functions as we did in Eq. (2.16), but now each δ-function will be blurred by the instrument. Since the convolution process is linear, the result will be the sum of all the blurred δ-functions, with each entry in the sum being displaced to the position of the corresponding δ-function. This amounts to simply convolving the function with the instrumental profile.

Therefore, we expand Eq. (2.17) to include this convolution and write

$$D(\lambda) = B(\lambda)III(\lambda)F(\lambda) * I(\lambda), \tag{2.22}$$

where $I(\lambda)$ is the instrumental profile. So not only are our collected data "boxed" and sampled, they are also blurred.

In the Fourier domain, the convolution becomes a product. In most common physical cases, $I(\lambda)$ is roughly like a Gaussian or a dispersion profile. In which case, multiplication implies stronger filtering, i.e., reduction in amplitude, toward higher Fourier frequencies. It is common for signal at high Fourier frequencies to be reduced to the noise level and consequently lost from the observations.

In the x or λ domain, the rapidly varying features have been "blurred" out by spatial integration across the dimensions of the instrumental profile.

Sampling and aliasing

Now consider Eq. (2.22) specifically with regard to the effect of the $III(x)$ sampling. The transform of Eq. (2.22) is of the form

$$d(\sigma) = b(\sigma) * III(\sigma) * f(\sigma) \times i(\sigma), \tag{2.23}$$

where $i(\sigma)$ is the transform of $I(x)$. Assume for the present that $B(x)$ is very wide so that $b(\sigma)$ is essentially an impulse. In a like manner, take $i(\sigma)$ to be unity over any σ range we discuss. Then we are essentially back to Eq. (2.16). The spacing between δ-functions in $III(\sigma)$ is $1/\Delta x$, where Δx is the data spacing in $D(x)$ or $III(x)$. The convolution of $f(\sigma)$ with $III(\sigma)$ causes $f(\sigma)$ to be reproduced over and over as shown in Fig. 2.8. We see now that if $f(\sigma)$ becomes zero for $\sigma < 0.5/\Delta x$, these replications are separated. But if not, they overlap and there is trouble. This trouble is called aliasing. The frequency

$$\sigma_N = \frac{1}{2\Delta x} \tag{2.24}$$

is called the cut-off or Nyquist frequency.

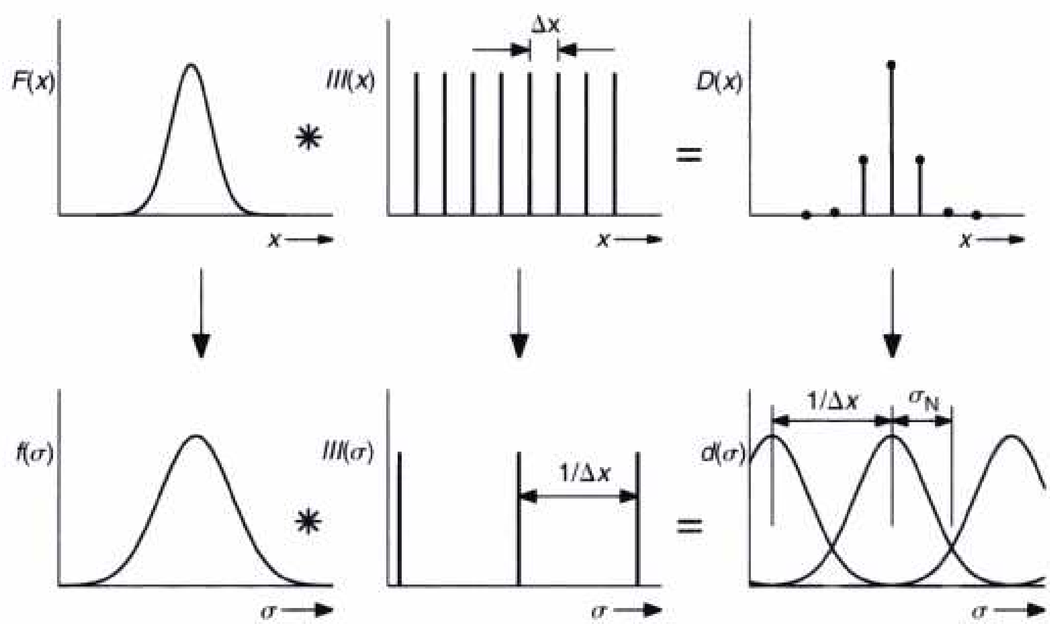


Fig. 2.8. Here is a graphical illustration of the sampling of $F(x)$ and the multiplicity of $f(\sigma)$ it causes in $d(\sigma)$. In this example, Δx is too large, causing $d(\sigma)$ to show undesirable overlapping of adjacent peaks. This is aliasing.

Generally we measure $D(x)$ and compute $d(\sigma)$, so we do not know initially if $f(\sigma)$ has significant amplitudes at $\sigma \gtrsim \sigma_N$. On the other hand, from the physics of the situation and the behavior of $f(\sigma)$ for $\sigma < \sigma_N$ we can often feel confident that $f(\sigma)$ decreases to smaller and smaller values as σ increases. In particular, multiplication with $i(\sigma)$ may force this to happen. For the case where $f(\sigma)$ does become zero for $\sigma < \sigma_N$, we can manipulate the transform $d(\sigma)$ as needed and unambiguously reconstruct $F(x)$ by taking the inverse transform of $d(\sigma)$ for $-\sigma_N \leq \sigma \leq \sigma_N$.

For the case where successive patterns overlap, there is no longer a unique solution. Many different $F(x)$ functions can be concocted that have the same transform $d(\sigma)$. These functions differ in the way the overlapped portion is divided, as shown in Fig. 2.9. This of course is the origin of the term aliasing. The low frequencies of the peak to the right are disguised as high frequencies in the peak to the left, and so on. The only real solution for such a situation is to redo the measurements with the data spacing, Δx, decreased sufficiently to separate the replications $f(\sigma)$.

Additional mixing of the $f(\sigma)$ patterns can result from the convolution of $b(\sigma)$ in Eq. (2.23). In unfavorable circumstances, $b(\sigma)$ may be sufficiently wide to result

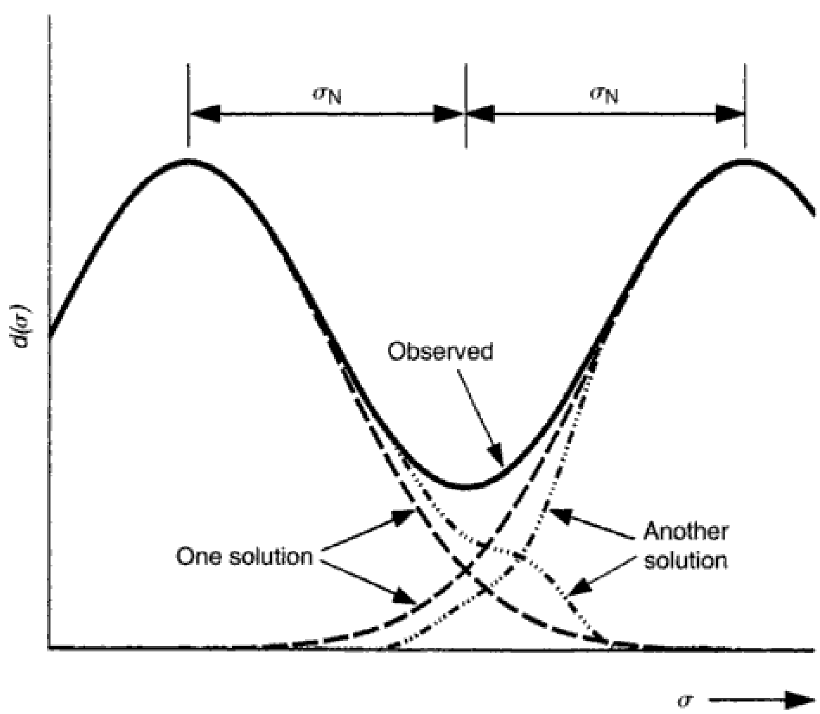


Fig. 2.9. In the aliased situation, we obtain in $d(\sigma)$ only the vector sum of the overlapping replications (————). Without additional information, it is impossible to separate the overlap in a unique way. Two arbitrary examples of separation are shown (– ··· – ··· – and – – –).

in aliasing that would otherwise not exist. The extended sidelobes of $b(\sigma) = \operatorname{sinc} \pi\sigma$ can accentuate this problem, and for this reason it may be necessary to modify the ends of $B(x)$ by bringing them smoothly to zero. The smooth wings may still be significant, and so to make certain $b(\sigma) * f(\sigma)$ is no larger than needed, there should be no large signals in $f(\sigma)$ that are unnecessary. Subtraction of the mean value of $F(x)$, for example, will often remove a large spike at $\sigma = 0$.

Application of these principles is made in Chapter 12.

Useful theorems

Physical insight as well as computational saving can be gained by application of theorems. Most of these can be proved by reverting to the defining integrals, Eqs. (2.1) and (2.2). In all cases, $f(\sigma)$ is the transform of $F(x)$ and the arrow means "transforms to."

Theorem 1. *Fourier transforms are linear.*
If

$$F(x) = F_1(x) + F_2(x) + \cdots,$$

then

$$f(\sigma) = f_1(\sigma) + f_2(\sigma) + \cdots. \tag{2.25}$$

Theorem 2. *Translation in one domain gives a phase term in the opposite domain.*

$$F(x - x_1) \to e^{2\pi i x_1 \sigma} f(\sigma). \tag{2.26}$$

Theorem 3. *Scaling the argument in one domain leads to inverse scaling in the other domain of both the argument and the amplitude.*

$$F(ax) \to \left|\frac{1}{a}\right| f(\sigma/a). \tag{2.27}$$

Theorem 4. *Differentiation in one domain is equivalent to multiplication by $2\pi i\sigma$ in the other domain.*

$$\frac{dF(x)}{dx} \to 2\pi i \sigma f(\sigma). \tag{2.28}$$

Integration is the opposite of this.

Theorem 5. *Derivatives of convolutions involve differentiation of only one or the other of the convolved functions but not both. If*

$$K(x) = F(x) * G(x),$$

then

$$\frac{dK(x)}{dx} = \frac{dF(x)}{dx} * G(x)$$
$$= F(x) * \frac{dG(x)}{dx}. \tag{2.29}$$

Theorem 6. *If $G(x)$ is normalized to unit area, then $K(x) = F(x) * G(x)$ has the same area as $F(x)$, that is,*

$$\int_{-\infty}^{\infty} K(x)\,dx = \int_{-\infty}^{\infty} F(x)\,dx. \tag{2.30}$$

Theorem 7. *If we call the centroid of a function x', where*

$$x' = \frac{\int_{-\infty}^{\infty} xF(x)\,dx}{\int_{-\infty}^{\infty} F(x)\,dx},$$

then the centroid of a convolution $K(x) = F(x) * G(x)$ is the sum of the centroids of the two functions being convolved, that is,

$$x_K = x_F + x_G. \tag{2.31}$$

Theorem 8. *Rayleigh's theorem relates the amplitudes of the function and its transform:*

$$\int_{-\infty}^{\infty} |F(x)|^2\,dx = \int_{-\infty}^{\infty} |f(\sigma)|^2\,d\sigma \tag{2.32}$$

An almost identical theorem called the power theorem is

$$\int_{-\infty}^{\infty} F(x)G^{\dagger}(x)\,dx = \int_{-\infty}^{\infty} f(\sigma)g^{\dagger}(\sigma)\,d\sigma, \tag{2.33}$$

where the dagger denotes a complex conjugate. However, in several common physical applications $F(x)$ and $G(x)$ are proportional to each other so that Eq. (2.33) reverts to Eq. (2.32). Examples of this are electric and magnetic fields in radiation, and voltage and current in resistive dissipation.

Numerical calculation of transforms

We have indicated above how some simple analytic functions can be transformed. Now consider numerical computation of the transforms, particularly for spectroscopic data. It is possible to use quadrature formulae on Eqs. (2.1)

Table 2.1. *A comparison of efficiency.*

N	Non-FFT[a]	FFT[b]	Ratio
128	32 768	2689	12
256	131 072	6147	21
512	524 288	13 830	38
1024	2 097 152	30 734	68
2048	8 388 608	67 614	124
4096	33 554 432	147 521	227
8192	134 217 728	319 629	420
16 384	536 870 912	688 432	780
32 768	2 147 483 648	1 475 212	1456

[a] The number of computations needed without the fast Fourier transform algorithm.
[b] The number of computations needed with the fast Fourier transform algorithm.

and (2.2) directly, and such a procedure may be useful in some applications, especially when the data points are spaced non-uniformly along the abscissa.

A mild revolution occurred with the invention of the fast Fourier transform, also called the Cooley–Tukey algorithm. This method of calculating transforms depends on a clever method of nesting sums and avoiding redundancy in calculating sines and cosines that the direct integration does not. The gain in efficiency over direct integration is phenomenal for lengthy calculations. Bell (1972) shows that the computation time for the classical method varies as $2N^2$, while the fast Fourier transform equivalent is $3N \log_2 N$. The ratio is $0.46N/\ln N$ for N points. Table 2.1 gives examples of this gain. The N values listed are integral powers of 2, as required by the fast Fourier transform. This requirement is easy to meet for any string of data by simply extending it to the next larger power of 2 using zeros or some other suitable constant.

The second requirement of the fast Fourier transform is that all data points be equally spaced along the abscissa. The nesting procedure does not work without even spacing. Many types of data can easily be recorded at equal steps. In other cases, treating unequally spaced data as if it were equally spaced results in negligible error.

We can write our data, bearing in mind Eq. (2.16), as

$$D(x) = D(j\Delta x) = D(j)$$

in which Δx is the step and j is an integer index running from 1 to N for N data points. The Fourier integral can then be written as the sum

$$\begin{aligned} d(\sigma) &= \sum_{-\infty}^{\infty} D(x_j)\, e^{2\pi i x_j \sigma}\, \Delta x \\ &= \sum_{1}^{N} D(j)\, e^{2\pi i (j\Delta x)\sigma}\, \Delta x. \end{aligned}$$

We can also write $\sigma = k\Delta\sigma$, where $\Delta\sigma$ is the spacing of points in $d(\sigma)$ and k is an integer. Then

$$d(\sigma) = d(k) = \sum_1^N D(j)\, e^{2\pi i jk\Delta x\Delta\sigma}\,\Delta x. \tag{2.34}$$

Further, the Nyquist frequency, $\sigma_N = 0.5/\Delta x$, is the highest frequency we can hope to work to. It is usually assumed that if there are N points defining $D(x)$, then N values of $d(\sigma)$ will be computed. Since $d(\sigma)$ will be computed equally far on both sides of $\sigma = 0$, the computations extend to $\pm N/2$ points away from zero, and the step in σ must then be

$$\Delta\sigma = \frac{\sigma_N}{N/2} = \frac{1}{\Delta x\, N},$$

or

$$\Delta\sigma\Delta x = 1/N. \tag{2.35}$$

With this substitution, Eq. (2.34) gives

$$d(k) = \sum_1^N D(j)\, e^{2\pi i jk/N}\,\Delta x. \tag{2.36}$$

It is a summation of this form that is evaluated in the fast Fourier transform techniques, and most versions also assume $\Delta x = 1$. The inverse transform corresponding to Eq. (2.36) is

$$D(j) = \sum_1^N d(k)\, e^{-2\pi i jk/N}\,\Delta\sigma. \tag{2.37}$$

The same algorithm can be used to evaluate this sum, but $\Delta\sigma$ must be explicitly taken into account using Eq. (2.35). This is equivalent to re-normalization,

$$D(j) = \frac{1}{\Delta x N}\sum_1^N d(k)\, e^{-2\pi i jk/N}. \tag{2.38}$$

One version of the fast Fourier transform is given in Appendix C as a Fortran listing.

Noise transfer between domains

Suppose we measure a function, $F(x)$, and take its transform. The measurements will contain noise, and we ask how the noise appears in the transform. Let $E(x)$ be the error or noise in these measurements such that we can write

$$F(x) = F_0(x) + E(x),$$

where $F_0(x)$ is the error-free function. Because of the linearity (Theorem 1) we can write

$$f(\sigma) = f_0(\sigma) + e(\sigma),$$

where $e(\sigma)$ is the transform of $E(x)$.

Theorem 8 tells us that

$$\int_{-\infty}^{\infty} E^2(x)\,\mathrm{d}x = \int_{-\infty}^{\infty} e^2(\sigma)\,\mathrm{d}\sigma,$$

but our measurements extend over a finite range in x, call it L, and our transform is unique only over a finite range in σ, say $\pm l/2$, due to the data sampling spacing and the attendant aliases. We can rewrite Theorem 8 as

$$\int_{x_0}^{x_0+L} E^2(x)\,\mathrm{d}x = \int_{-l/2}^{l/2} e^2(\sigma)\,\mathrm{d}\sigma, \tag{2.39}$$

where x_0 is the starting value. If we denote the average by angled brackets, the relation can also be written as

$$\langle E^2\rangle L = \langle e^2\rangle l. \tag{2.40}$$

This is the fundamental relation that says the mean square error times the length of the data is the same in both domains.

More specifically, the data we transform are usually an array of numbers. Let us also assume the fast Fourier transform is used. Then with N equally spaced data points Δx apart, Eq. (2.39) takes the form

$$\sum_1^N E^2(x_j)\,\Delta x = \sum_1^N e^2(\sigma_j)\,\Delta\sigma,$$

with $L = \Delta x N$ and $0.5l = \sigma_{\mathrm{N}} = 0.5/\Delta x$. In place of the mean square errors, we define the standard deviations

$$S_x = \left[\sum_1^N \frac{E^2(x_j)}{N-1}\right]^{1/2}$$

$$S_\sigma = \left[\sum_1^N \frac{e^2(\sigma_j)}{N-1}\right]^{1/2}.$$

The corresponding form for Eq. (2.40) is

$$S_\sigma^2\, l = S_x^2\, L, \tag{2.41}$$

or

$$\begin{aligned} S_\sigma &= S_x(L/l)^{1/2} \\ &= S_x(\Delta x L)^{1/2} \\ &= S_x \Delta x N^{1/2}. \end{aligned} \tag{2.42}$$

As might be expected, the noise in the two domains is proportional. When the measured span L is more finely divided by making the sampling interval Δx smaller, S_σ is reduced because the noise is spread over a larger span in σ.

The common practice of padding the true data array to reach an even power of 2 would alter Eq. (2.40) because the padding values would increase L or N but carry no noise. Consequently, L or N in Eq. (2.42) refers to the true data set before any modification for the fast Fourier transform calculation. It also follows that the noise contribution from a spectral line-profile measurement can be minimized by restricting the number of points to only those defining the profile itself; the measured continuum points should be replaced with their mean value.

When the noise $E(x)$ is "white," meaning independent of frequency, the noise $d(\sigma)$ is statistically constant with σ. We have seen how it is important to choose Δx small enough so that the whole transform is sensibly within $\sigma_N = 0.5/\Delta x$ (no aliasing). If σ_N is made still larger, the only non-zero amplitudes at the high frequencies are noise. In other words, it is possible to see the noise level in the σ domain (see Fig. 12.7, for example) and in some cases it is advantageous to calculate the noise in the original data by working back through Eq. (2.42) to get S_λ from S_σ.

Suppose, for instance, a spectral-line profile is defined by 16 points separated by $\Delta x = \Delta\lambda = 0.04$ Å. A transform might be computed using 64 points, 49 of which are continuum values of unity. Assuming the oversampling is sufficient to allow the noise level in the σ domain to be seen at $S_\sigma = 10^{-3}$ Å, we have $S_\lambda = S_\sigma/(\Delta\lambda N^{1/2}) = 0.6\%$ error in the original line profile data. We do a similar thing, perhaps only semiconsciously, in the λ domain. When $\Delta\lambda$ is small, that is, well below the dimension associated with signal variations, adjacent measured points should be strongly correlated. Differences in value are then due to noise. We assume white noise and estimate the noise level from the "grass," which is the high-frequency noise.

Time-series analysis

The existence of a period in physical variations can often tell us important information about a physical process. For example, spots on the surface of a star carried across the visible disk will introduce modulation in the shapes of spectral lines and the brightness of the star. A time-series analysis will give us the period of rotation and in many cases specific information concerning the size and location of the spot. More generally, the parameter we measure might be the velocity span of a line bisector, a line–depth ratio, the Zeeman broadening, the radial velocity, and so on. In this section, we focus on extracting the period(s).

In the trivial case, where the variation can be mapped more or less continuously, there will be no ambiguity, and one can establish the pattern of variability by simple inspection of the observations. Choose the first cycle as the reference and compare successive cycles with it by translating each in the time coordinate until a match with the first cycle is obtained. Call these time differences Δt_j. Then from each one we compute the period, $P_j = \Delta t_j / n_j$, where n_j is the number of cycles occurring in the time span Δt_j. Call the error associated with matching each cycle with the first cycle δt. Then the error in the period is $\delta P_j = \delta t / n_j$. In other words, we know the period better in proportion to the number of cycles elapsed. This is a very general and powerful result: precise periods come from long time series covering many cycles.

The best estimate of the true period, P, is the weighted average of the N values of P_j. Since each P_j improves in proportion to the number of cycles involved in its calculation, the weights are the square of the number of cycles, thus

$$P = \frac{\sum_{j=1}^{N} n_j^2 P_j}{\sum_1^N n_j^2} = \frac{\sum_1^N n_j \Delta t_j}{\sum_1^N n_j^2}. \tag{2.43}$$

However, in many cases the periodicity of a variation is not obvious. In such cases, a Fourier analysis can be of value. The Fourier transform of a time series is called a periodogram. Suppose we have a simple case with one frequency, as shown in Fig. 2.10. This oscillation is observed at certain times and over time windows leading to the observations shown in the lower panel of Fig. 2.10. In the time domain we can express the observations, $D(t)$ as

$$D(t) = F(t) \times W(t), \tag{2.44}$$

in which the original signal $F(t)$ is sampled by the window $W(t)$. Naturally, in the Fourier domain, the product becomes a convolution, and the window function, $w(\sigma)$, the transform of $W(t)$, is translated to the positions of the delta-functions representing the sinusoidal oscillation. The two window functions interfere with each other where they overlap; their vector components (real and imaginary) are added. We do not look for a single spike in the periodogram, but rather the central position of the window pattern (see Gray & Desikachary 1973).

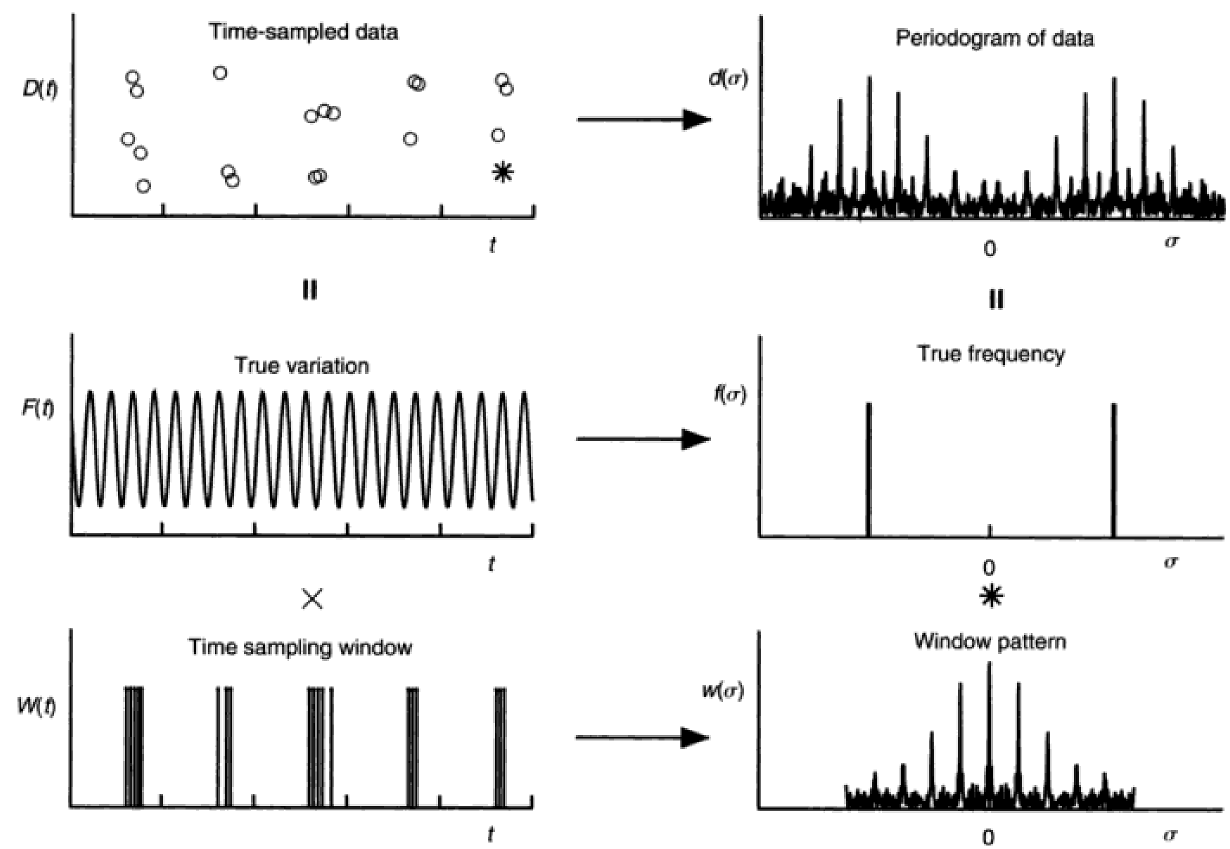


Fig. 2.10. Astronomical data is often sparsely sampled and typically gives a time series like the one shown in the upper left-hand panel. This can be thought of as the product of the true variation (the cosine wave in this example) with delta-functions at the times of sampling, as illustrated in the two lower left-hand panels. In the Fourier domain on the right, the time sampling produces a window pattern (bottom) that is convolved with the two delta-functions corresponding to the cosine wave. This explains the complicated shape of the periodogam.

If the oscillation is undersampled, as in Fig. 2.10, the window pattern will have several strong peaks, and the worse the undersampling, the closer these peaks will be in amplitude. The first peak on either side of the central peak is the first alias, the second peaks, the second alias, and so on. Because of the interference between the positive- and negative-frequency window functions, the amplitudes of the peaks can be distorted, making it hard to distinguish which of the peaks is the true central peak. Furthermore, if the true frequency is small compared with the width of the window pattern, the negative-frequency pattern will spill over into positive frequencies and the two patterns will be interleaved (an example can be seen in Gray & Hatzes 1997).

The difficulty can be compounded when the signal is more complicated than a simple sinusoid because then the harmonics, representing the more complex shape of the oscillation in the time domain, also produce displaced window patterns. Still more confusing is the case where there are several frequencies actually occurring in the physical measurements, again each bringing in at least one pair of window patterns. All such window patterns behave as vector functions. At each frequency, the periodogram signal is the vector for which the real component is the sum of the real components of all window patterns and for which the imaginary component is the sum of the imaginary components of all window patterns. The amplitude of the vector is normally the quantity plotted in periodograms. Some investigators prefer to use power rather than amplitude in the Fourier domain, which amounts to squaring the amplitude.

The reality of a signal can be judged by the visibility of the window pattern compared to the noise. It is generally not a good idea to simply focus on individual peaks in a periodogram. A formal "false-alarm probability," given by e^{-p}, where p is the power at the frequency of interest, has been used in some cases (Horne & Baliunas 1986).

References

Bell, R.J. 1972. *Introductory Fourier Transform Spectroscopy*. (New York: Academic).
Bracewell, R. 1965. *The Fourier Transform and Its Applications*. (New York: McGraw-Hill).
Gaskill, J.D. 1978. *Linear Systems, Fourier Transforms, and Optics*. (New York: Wiley).
Gray, D.F. & K. Desikachary 1973. *Ap. J.* **181**, 523.
Gray, D.F. & A.P. Hatzes 1997. *Ap. J.*, **490**, 412.
Hinkle, K., L. Wallace, J. Valenti, & D. Harmer 2000. *Visible and Near Infrared Atlas of the Arcturus Spectrum 3727–9300Å*. (San Francisco: Ast. Soc. Pacific).
Horne, J.H. & S.L. Baliunas 1986. *Ap. J.* **302**, 757.
Jennison, R.C. 1961. *Fourier Transforms and Convolutions for the Experimentalist*. (New York: Pergamon).
Kharkevich, A.A. 1960. *Spectra and Analysis*. (New York: Consultants Bureau).

Questions and exercises

1. Fourier transformation is a linear operation. What does this mean and why is it important?

2. If you know that the Fourier transform of a function has no imaginary component, what does that tell you about the function?

3. Convince yourself that Fourier decomposition really does work by constructing a box from cosines. Since the transform of a box is a sinc function, the sinc function must be the specification of the amplitudes of the cosine components comprising a box. Further, we know that pairs of delta-functions transform into a cosine. Therefore, proceed as follows. Draw a sinc function. Within the sinc envelope, draw pairs of delta-functions, symmetrically placed and evenly spaced across the main lobe and the first sidelobe (about a dozen pairs is a good choice). For each delta-function pair, plot the corresponding cosine and finally the sum of all the cosines. Be sure to scale each cosine with the ordinate of the sinc function where the delta-functions are, including the negative sign for those in the first sidelobe. Does your cosine sum resemble a box? What needs to be done to make it even more box-like? Is the zero level of your box correct, i.e., at the bottom of the box? Or did you forget to include the delta-function "pair" with no separation at the zero position of the sinc function? Is the area of your box equal to the zero ordinate of your sinc function? Did you remember that your cosine sum is an integration and is to be multiplied by an abscissa increment?

4. It is easy to see that the convolution of two boxes is a triangle. (If you have not worked this through yet, do so first.) What does the convolution of two identical isosceles triangles look like? What is the transform of this convolution?

5. Write a short computer program using Eq. (2.36) and the subroutine FOUR1, given in Appendix C, so that you can take numerical Fourier transforms. Test your program on boxes and triangles and Gaussians, for example. Try computing a transform to frequencies two or three times higher than the Nyquist frequency given in Eq. (2.24). What do you see?

6. Select two or three line profiles of Arcturus from the Hinkle *et al.* (2000) atlas CDROM. Set the continuum to zero by subtracting unity and throwing away the minus sign. Compute the transforms out to the Nyquist frequency. Does the zero ordinate of your transform correctly give the total area of the line profile (called the equivalent width)? If not, check that Δx in Eq. (2.36) has been included. Can you see the noise level at the higher frequencies? Is it consistent with Eq. (2.42)?

3

Spectroscopic tools

Observations of stellar photospheres require instruments for collecting and analyzing light. Low-resolution spectrographs are used for measuring the continuous spectrum, while high-resolution spectrographs are needed to measure the spectral lines. In this chapter, we delve into several basic aspects of spectrographs and diffraction gratings, then turn our attention briefly to interferometers, and finally to telescopes. Special features and application of spectrophotometric equipment are discussed separately in Chapter 10 (continua) and Chapter 12 (lines). Chapter 4 is about light detectors.

Spectrographs: some general relations

The basic astronomical spectrograph is shown in Fig. 3.1. It consists of an entrance slit placed at the focus of the telescope, a collimator that intercepts the divergent telescope beam, a dispersing element (prism or grating), and a camera that focuses the dispersed light onto a detector. Since the purpose of the collimator is to make the divergent beam parallel, the distance between the slit and the collimator is the focal length of the collimator, F_{coll}. Similarly, the distance between the camera and the focused spectrum is the focal length of the camera, F_{cam}. The distances between the collimator, disperser, and camera affect the detailed design and optimization of the spectrograph, but do not matter for the basics we are considering at the moment. We concentrate on plane diffraction gratings as the dispersing element since these are almost universally used in astronomical spectrographs.

In *mono*chromatic light, a single image of the entrance slit is formed in the focal plane of the camera. This image is inverted and the magnification along the direction of the slit (perpendicular to the dispersion) is the ratio of camera to collimator focal lengths, and is often less than unity. The magnification of the width of the slit depends not only on the ratio of focal lengths, but also on

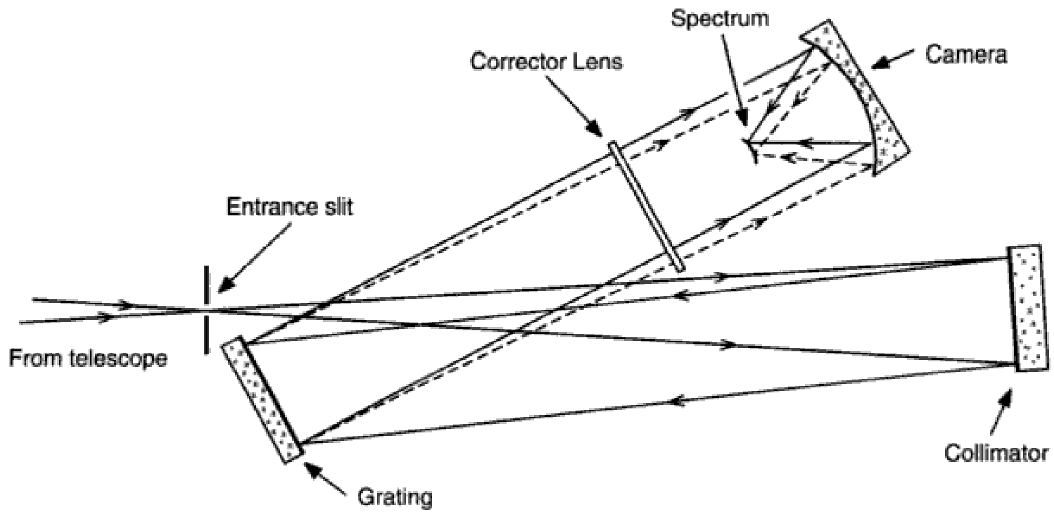


Fig. 3.1. This is the optical layout for a typical spectrograph. The stellar seeing disk formed by the telescope is focused on the entrance slit. The beam diverges inside the spectrograph until it reaches the collimator mirror and is collimated. The collimated light is returned to the grating, and from it monochromatic collimated beams are sent to the camera (only two such beams are indicated). The spectrum is focused on the curved surface. Spectrograph cameras have a wide field of view, and often need a corrector lens to attain it.

the orientation of the grating, and this is discussed later in Eq. (3.14). Only for very narrow slit settings, not normally encountered in astronomy, will the structure of the grating diffraction pattern contribute significantly to the width of the monochromatic image. When *poly*chromatic light is analyzed with the same instrument, the spectrum is composed of a continuum of monochromatic images of the entrance slit spread out across the focal plane of the camera. The chromatic purity of the light in the spectrum depends on the width, $\Delta\lambda$, of these monochromatic images, and this determines the fineness of detail we can see in the stellar spectrum. The term "chromatic resolution," or "spectral resolution," or simply "resolution," is used to describe how well the colors are separated. Generally the higher the resolution, the happier the astronomer, but there is an eternal trade-off between discerning detail in the spectrum and the faintness of the stars that can be measured. Higher resolution implies fewer photons per detector element. Quite generally, we need $\Delta\lambda$ to be smaller than the width of the structure in the spectrum we wish to observe. A quantitative discussion requires a measurement of the monochromatic image produced by the spectrograph, called the instrumental profile, and this is discussed in Chapter 12.

Now let us turn our attention to the plane diffraction grating of the type used in astronomical spectrographs.

Diffraction gratings

The plane diffraction grating has proved to be the most popular dispersing element for a number of reasons. Its luminous efficiency is higher than that of concave gratings or prisms of similar angular dispersion. The linear dispersion, Eq. (3.13) below, can be changed by merely switching cameras, whereas concave gratings have a radius of curvature that fixes the dispersion. In addition, the plane grating can be used over a wide range of angles of incidence, which makes it possible to direct the dispersed light into any of several useful directions by simply rotating the grating.

A typical plane reflection grating consists of a large number of parallel grooves ruled with a diamond tool into an aluminum or gold coating on a flat glass substrate. (Holographic gratings and volume phase holographic gratings are also making an appearance, but they will not be discussed here; see Kohl & Parkinson 1976, Barden *et al.* 2000, and Baldry *et al.* 2004.) The grooves should all have the same shape, be smooth, and be equally spaced. A microscopic view of rulings is shown in Fig. 3.2. The best results so far have been obtained on ruling engines that are servo-controlled by interferometer monitors to maintain the strict positional requirement (Fig. 3.3). In high-resolution applications, large gratings are needed. Wear on the diamond ruling tool limits the maximum ruled area. The largest gratings now available are 350 mm by 450 mm. Harrison (1973) discusses ruling engines and grating quality.

The commonly encountered gratings are replica gratings constructed by using the ruled grating as a mold. A replica is far less expensive than an original master grating, and usually has higher reflection efficiency and less scattered light.

How a grating works is most easily understood by first considering the simple transmission grating illuminated in collimated monochromatic light as in Fig. 3.4. (We will presently convert our results to the reflection gratings actually used in stellar spectrographs.) The slit width is denoted by b and the slit spacing by d. The angle of incidence, α, is measured from the grating normal, as shown in the figure. The diffraction angle, β, we choose to have the same sign as α when it is on the same side of the grating normal, as in Fig. 3.4, where α and β have the same sign. Because the rulings are so long compared to their width, we can justifiably consider the grating as a one-dimensional object. Further, let us restrict the angles α and β to the plane of the paper in Fig. 3.4. The incident plane scalar wave (for a more rigorous vector wave treatment, see Stroke 1967) can be written as

$$F(x,t) = F_0(t)e^{2\pi i(x\sin\alpha)/\lambda}, \tag{3.1}$$

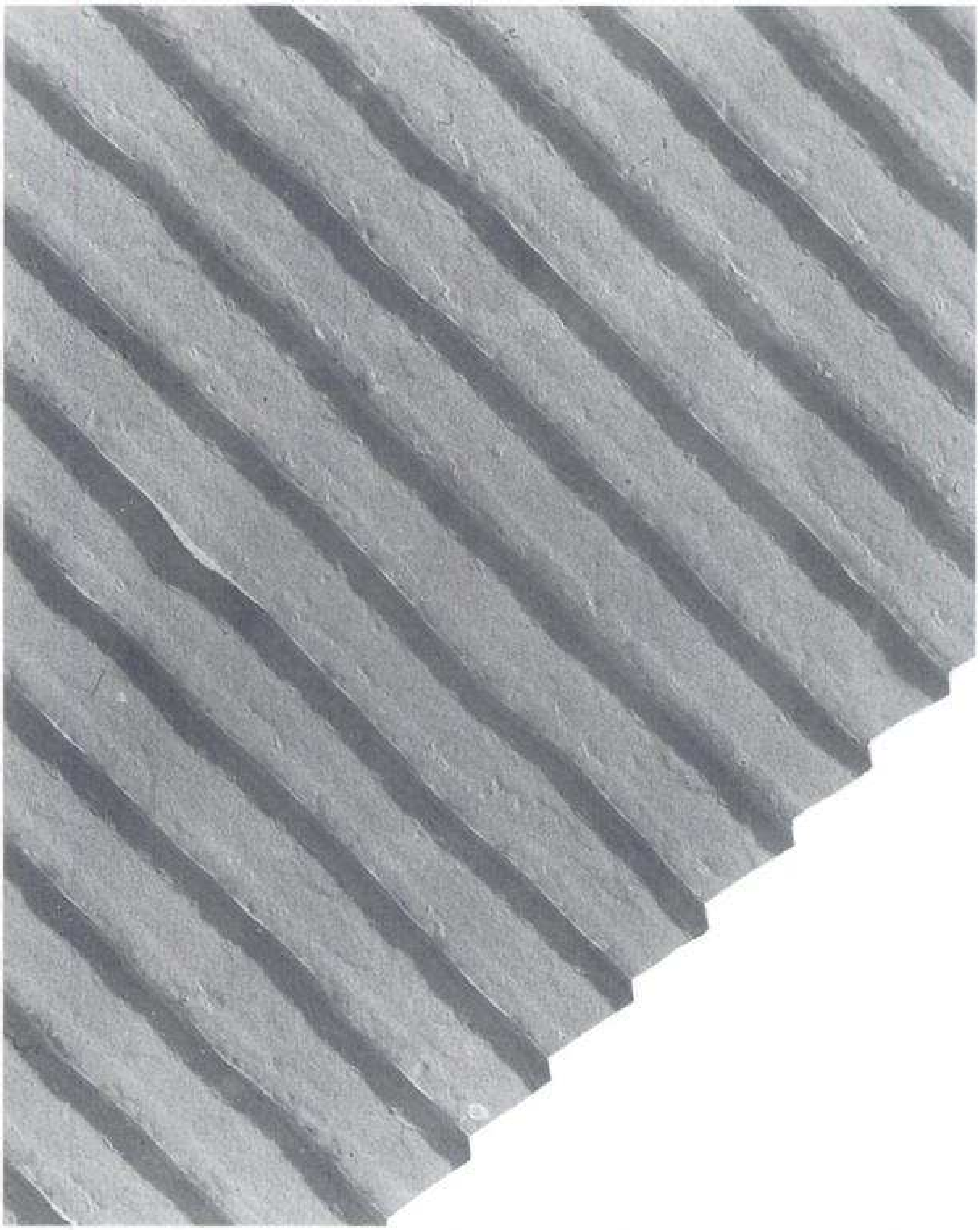


Fig. 3.2. This microscopic picture shows the rulings on a plane reflection grating having 1180 lines/mm. (Courtesy of Jarrell-Ash.)

in which $x \sin \alpha$ is a linear dimension along the direction of the wave. The time variations in the light wave occur with time scales $\sim 10^{-15}$ s and are completely undetectable during astronomical measurements. We can therefore think of $F_0(t)$ being replaced by its time average, F_0, which we take to be

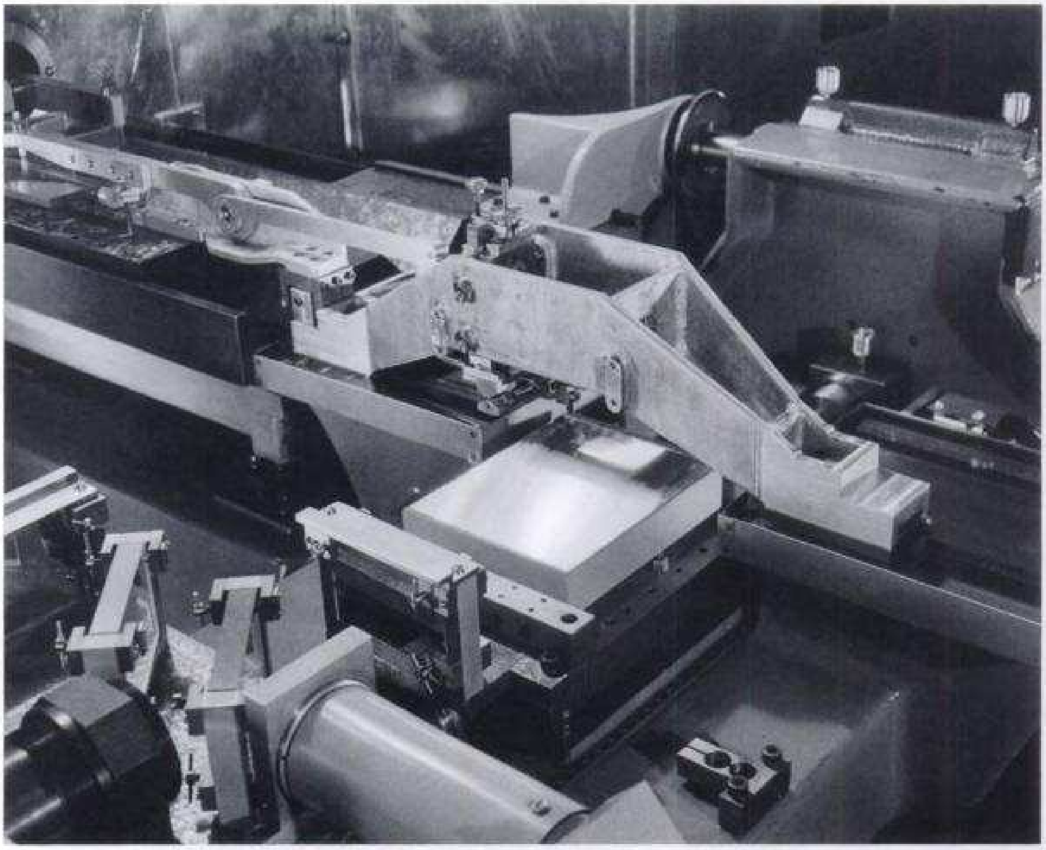


Fig. 3.3. This ruling engine has its ruling diamond poised above the corner of the grating blank. Part of the interferometer control system can be seen at the lower left. (Courtesy of Bausch and Lomb, Analytical Systems Division, Rochester, New York.)

constant. Now describe the grating transmission with the function $G(x)$ given by

$$G(x)\begin{cases} = 1 & \text{for } x \text{ values within the slits} \\ = 0 & \text{for all other } x \text{ values.} \end{cases}$$

It is the product of $F(x, t)$ with $G(x)$ that emerges from the back of the grating. The diffracted wave, $g(\beta)$, going in any arbitrary direction β will be the sum of contributions, with their proper phase shifts, from all portions of the grating. In other words, we integrate the contributions along the x coordinate.

$$g(\beta) = \int_{-\infty}^{\infty} F(x, t)G(x)\mathrm{e}^{2\pi\mathrm{i}(x\sin\beta)/\lambda}\mathrm{d}x,$$

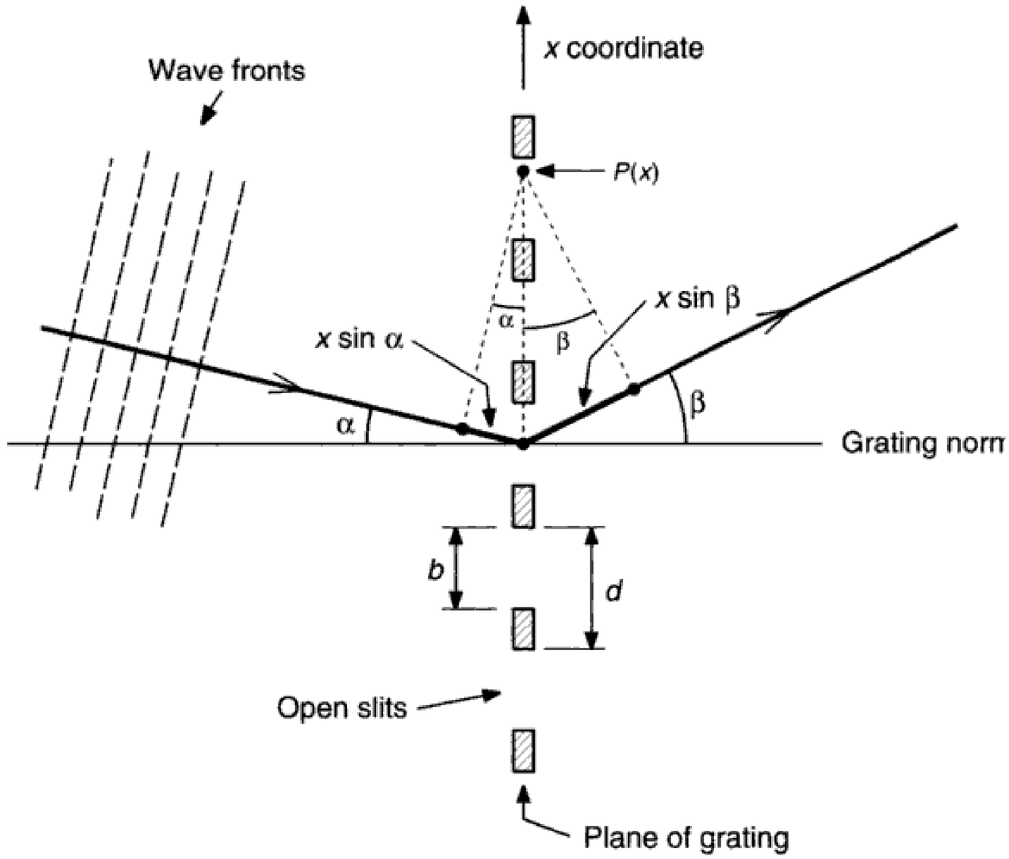


Fig. 3.4. The plane transmission grating of slit spacing d and slit width b lies in the plane perpendicular to the paper. The linear coordinate x runs across the rulings as shown. Light is incident at an angle α. The angle of diffraction is β. Both of these angles have the same sign as drawn. The dashed-line triangles show the path differences relative to the point labeled $P(x)$.

where $2\pi(x\sin\beta)/\lambda$ is the phase difference along the grating. Putting in Eq. (3.1), we have

$$g(\beta) = F_0\int_{-\infty}^{\infty} G(x)\mathrm{e}^{2\pi\mathrm{i}(x\sin\alpha + x\sin\beta)/\lambda}\,\mathrm{d}x,$$

and if we choose to measure x in units of λ while denoting $\sin\alpha + \sin\beta$ by θ, we have

$$g(\theta) = F_0\int_{-\infty}^{\infty} G(x)\mathrm{e}^{2\pi\mathrm{i}x\theta}\,\mathrm{d}x. \tag{3.2}$$

We now recognize Eq. (3.2) to be Eq. (2.1), i.e., the resultant waves behind the grating can be written as the Fourier transform of the grating transmission function, $G(x)$.

The description of $G(x)$ in terms of the Shah function and boxes is

$$G(x) = B_1(x) * III(x)B_2(x), \tag{3.3}$$

where as usual, the star denotes convolution. We take $G(x)$ to be purely real, that is, an amplitude function. A more general case allows $G(x)$ to be complex. Gratings that have $G(x)$ purely imaginary are termed phase gratings. The box $B_1(x)$ represents the transmission through a single slit (see Fig. 3.5), and so has a width b. The box $B_2(x)$ matches the total width of the grating, which we take to be W. The spacing of the δ-functions in $III(x)$ is d. The grating diffraction pattern is shown on the upper right-hand side of Fig. 3.5. It is immediately

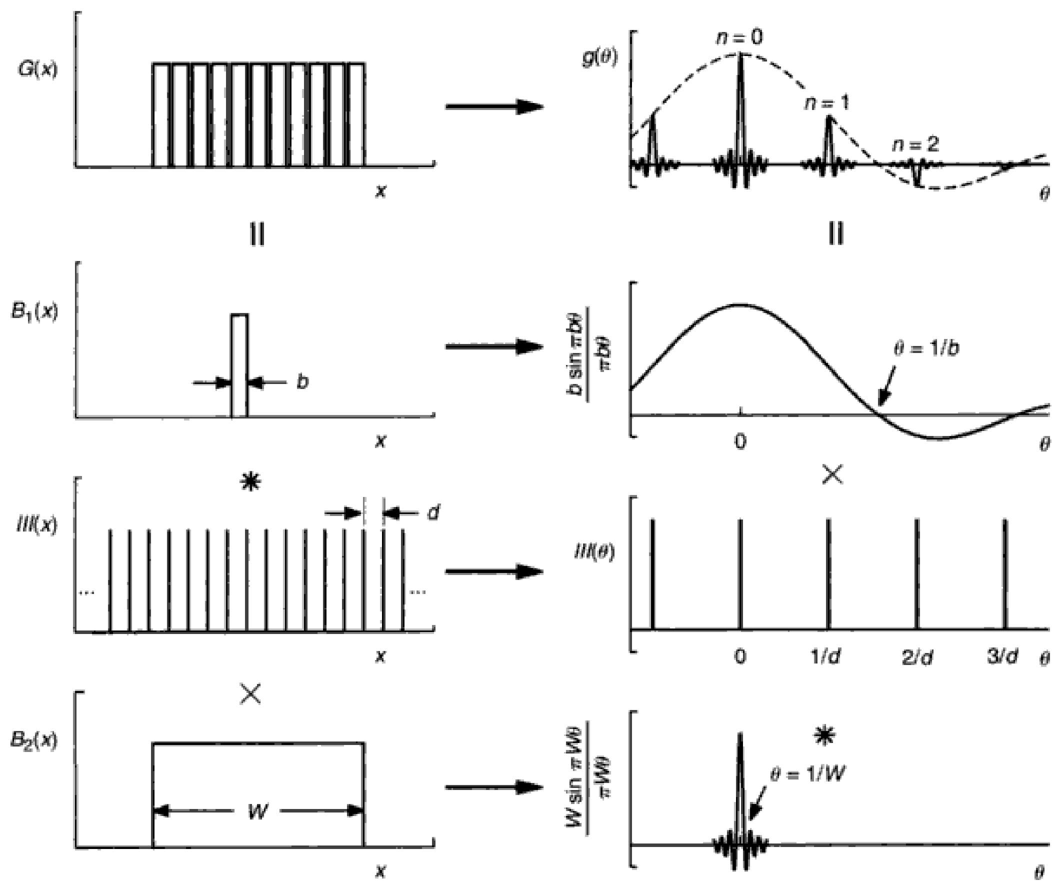


Fig. 3.5. The light distribution for a diffraction grating is shown on the upper left as $G(x)$. It is composed of the convolution of the box $B_1(x)$ of width b with $III(x)$ of spacing d multiplied by $B_2(x)$ representing the total width W of the grating. This is shown on the left-hand side of the diagram. The right-hand side shows the transforms. The amplitude diffraction pattern, $g(\theta)$, is seen to be the convolution of sinc $\pi\theta W$ with $III(\theta)$ which is multiplied by sinc $\pi\theta b$. The distribution of light in the diffraction pattern of the grating is proportional to $g^2(\theta)$.

clear that the different orders, n, arise from the many rulings (Shah function), the envelope comes from the individual slit width, b, and the full width of the grating, W, sets the width of the individual interference maxima. The amplitude diffraction pattern of the grating follows from Eqs. (3.2) and (3.3),

$$g(\theta) = III(\theta) * \frac{W \sin \pi\theta W}{\pi\theta W} \frac{b \sin \pi\theta b}{\pi\theta b},$$

where $III(\theta) = \Sigma\delta(\theta - n/d)$. The convolution with a δ-function produces a translation to the position of the δ-function, and so

$$g(\theta) = \sum_n \frac{W \sin \pi(\theta - n/d)W}{\pi(\theta - n/d)W} \frac{b \sin \pi\theta b}{\pi\theta b}. \tag{3.4}$$

The square of this wave amplitude is proportional to the light intensity in the usual way for wave optics.

In most applications we need the grating pattern as a function of β, but since $\sin\alpha + \sin\beta = \theta$ and α is fixed in any given case, $\beta = \sin^{-1}(\theta - \sin\alpha)$. Further, $d\theta = \cos\beta\, d\beta$. When β is small, such as when gratings are used in low orders, $d\theta$ and $d\beta$ are essentially the same.

Now consider the width of the interference maxima. The zeros are equally spaced in θ from the center of the maximum by $\Delta\theta = \lambda/W$ or $\cos\beta\Delta\beta = \lambda/W$. The grating (not the spectrograph) resolution is therefore proportional to its overall width.

The large maxima in $g(\theta)$ occur when $\theta - n/d$ becomes zero; that is, each maximum corresponds to a value of the order integer n. This can also be written as

$$\frac{n\lambda}{d} = \theta = \sin\alpha + \sin\beta, \tag{3.5}$$

where d and λ must be in the same units. Equation (3.5) is so fundamental in working with gratings that it is called *the* grating equation.

For gratings illuminated in polychromatic light, the collection of maxima for different wavelengths having one and the same n comprise the nth-order spectrum. The zero-order spectrum is the white light or undispersed order and falls in the $\beta = -\alpha$ direction, i.e., straight through transmission or the specular reflection. The larger the value of n, the larger β becomes for a given λ, although naturally β is restricted to range between 90° and −90°. The polychromatic light is spread out in θ in proportion to n (Eq. 3.5), and any band of colors is seen successively farther from the zero order with increasing order. The position and orientation of visible wavelengths is shown in Fig. 3.6. The overlap of wavelengths at a given θ occurs for all combinations of $n\lambda$ having

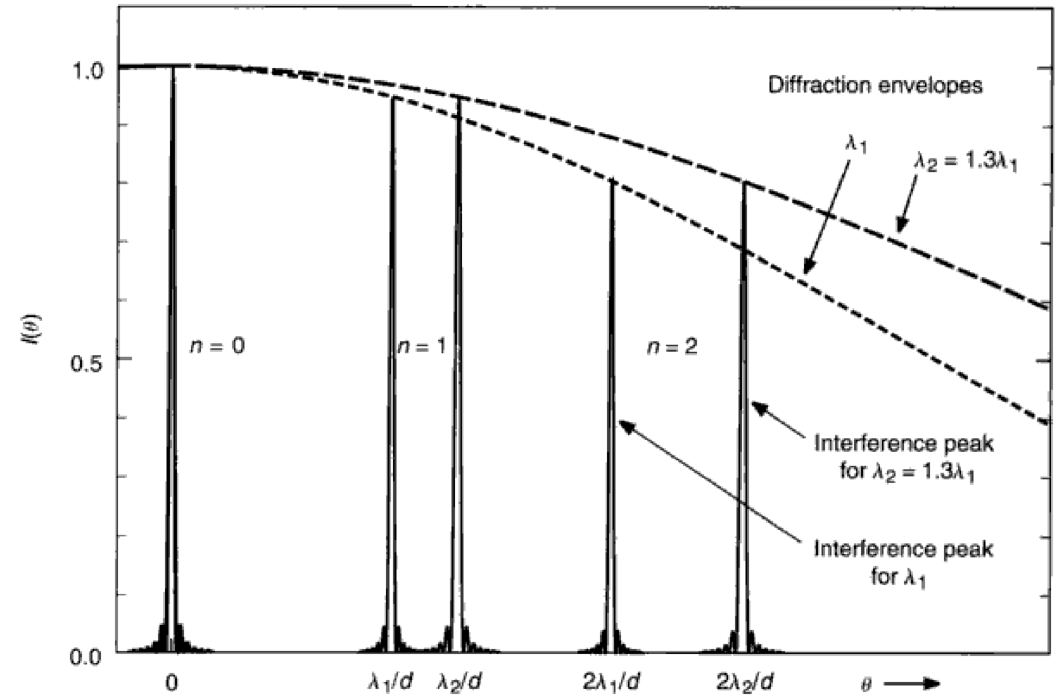


Fig. 3.6. The collection of interference peaks for a given order, n, comprise the spectrum of that order. The zero order has no dispersion. Dispersion in the second order is twice that of the first order. The heights of the interference peaks are set by the diffraction envelopes, as shown. This is the non-blazed case.

the same value – a result that follows trivially from the grating equation. Consequently, it is usually necessary to place a filter in the light beam to remove light from unwanted orders. This filter is best placed before the entrance slit of the spectrograph.

The angular dispersion of the grating follows from Eq. (3.5),

$$\frac{d\theta}{d\lambda} = \frac{n}{d} \quad \text{or} \quad \frac{d\beta}{d\lambda} = \frac{n}{d\cos\beta}. \tag{3.6}$$

Many spectrographs are designed so that the grating normal points toward the camera. Then $\beta \sim 0$, and the angular dispersion is essentially constant with wavelength. Equation (3.6) shows that the angular dispersion can be increased by observing in a higher order or by selecting a grating having smaller spacing between rulings. High-density rulings have as many as 1200 lines/mm, and values down to 300 lines/mm are typical of gratings used in low orders, e.g., $n = 1, 2$, or 3. Gratings used in much higher orders, e.g., $n \sim 200$, are called echelles. Their ruling densities are usually lower, with 79 lines/mm being a popular choice.

If we now combine the angular dispersion from Eq. (3.6), $\Delta\lambda = \Delta\theta d/n$, with the angular resolution, $\Delta\theta = \lambda/W$, given above, we find the chromatic resolution,

$$\Delta\lambda = \frac{\lambda}{W}\frac{d}{n}. \tag{3.7}$$

This is the wavelength resolution implicit in a grating of width W, ruling spacing d, working in the nth order. The ratio $\lambda/\Delta\lambda$ is called the resolving power of the grating and is independent of the wavelength. Resolving powers $\sim 10^6$ have been realized. As we shall see later in this chapter, the resolving power of the whole spectrograph is usually much less than the resolving power of the grating itself.

The blazed reflection grating

One of the objectionable features of the simple transmission grating is that the diffraction envelope has its largest maximum at $\theta = 0$, where the chromatic dispersion is zero. A large part of the light is essentially lost to this zero order. Further, as we try to use the higher orders to gain more resolution, we find the light intensity continues to decline, reaching uncomfortably low levels. To solve this problem, the diffraction envelope must be shifted relative to the interference pattern to those θ values where we wish to work. The Fourier shift theorem (Chapter 2, Theorem 2) tells us that we can accomplish this by introducing a phase term in $B_1(x)$ of Eq. (3.3). When this is done the grating is said to be blazed.

Conceptually we could blaze a transmission grating by placing a small prism over each slit, as shown in Fig. 3.7. The beam would be deviated through the angle γ, thus shifting the diffraction envelope, but of course the interference pattern is not changed because the slits would still have the same separation and would still be oriented along the same plane. This system would not be very successful for polychromatic light because of the dispersive property of the prisms (not to mention the practical problem is installing the prisms).

A reflecting surface can also give the needed phase shift and it is non-dispersive. The slits and prisms of the transmission grating are replaced by tilted "mirrors" (i.e., the rulings), with the tilt giving the phase shift. Another advantage also occurs at this point: the opaque portions between the slits are no longer needed in order to distinguish one ruling from the next, leading to increased optical efficiency. The grating cross section then makes the conceptual transition shown in Fig. 3.8. The individual reflecting surfaces are called facets. The tilt or facet angle, ϕ, is set by the manufacturer according to the shape and orientation of the diamond in the ruling engine. The front and back

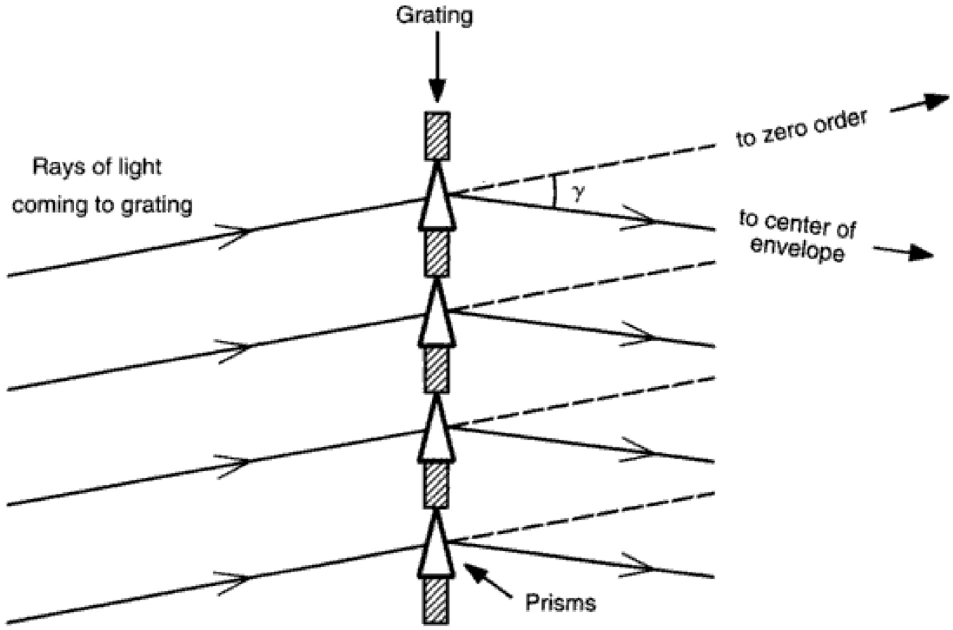


Fig. 3.7. A hypothetical transmission grating is shown with a prism in each slit. A monochromatic incident wave experiences an angular deviation of γ. The zero-order image is no longer at the center of the diffraction envelope.

sides of the facets are usually at right angles, making $b = d\cos^2\phi$. This has the effect of making the slits slightly narrower than the limiting case where $b = d$, and the diffraction envelope is correspondingly wider.

The equations derived for the transmission grating hold for the reflection grating. The angles α and β are both positive when measured on the same side of the grating normal as the facet normal. The distribution of light in the spectrum produced by a blazed grating for a white-light source can be derived in the following way. Referring back to Eq. (3.4), we see that the diffraction *envelope* of $g(\theta)$ is given by

$$b\frac{\sin \pi\theta b}{\pi\theta b} = b\frac{\sin[\pi b(\sin\alpha + \sin\beta)]}{\pi b(\sin\alpha + \sin\beta)}.$$

Tilting the reflection facet through the angle ϕ changes the phase of the reflected wave such that α is replaced by $\alpha - \phi$ and β by $\beta - \phi$. Then the envelope light distribution is proportional to

$$I(\beta) = \left[\frac{\sin\{(\pi b/\lambda)[\sin(\alpha - \phi) + \sin(\beta - \phi)]\}}{(\pi b/\lambda)[\sin(\alpha - \phi) + \sin(\beta - \phi)]}\right]^2, \tag{3.8}$$

where we have explicitly converted to b/λ in place of b measured in λ units, and we have squared the amplitude pattern to obtain the light intensity. Another

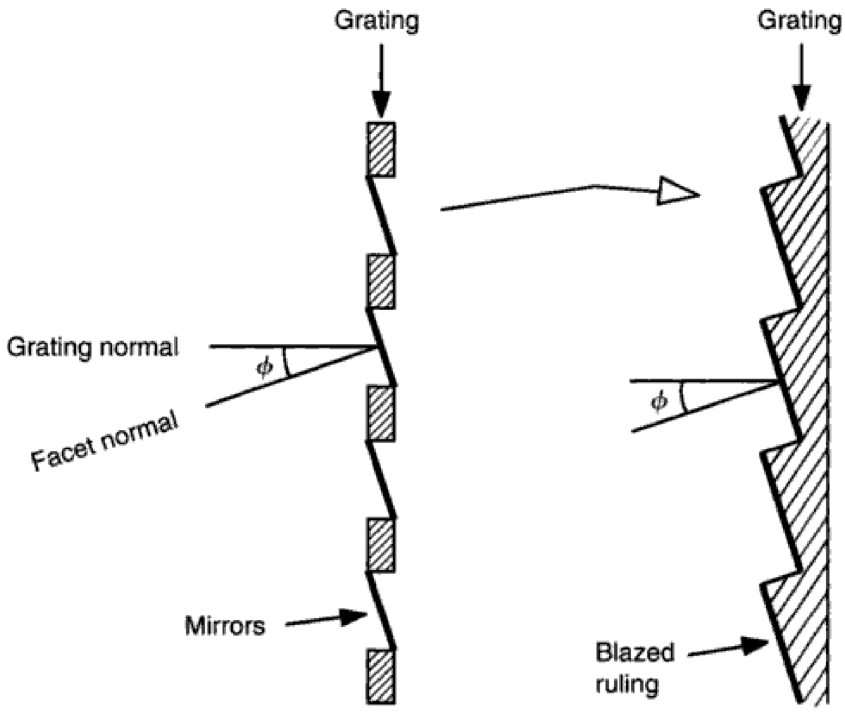


Fig. 3.8. On the left is a hypothetical grating with slits replace by mirrors. The inter-slit spacing is conceptually eliminated in the more realistic grating on the right. The facet normal makes an angle ϕ with the grating normal. The direction of positive angles is defined so that ϕ is positive.

useful form of this expression is to solve for λ in Eq. (3.5) and substitute into Eq. (3.8). After applying a half-angle formula, we obtain

$$I(\beta) = \left[\frac{\sin\{(n\pi b/d)(\cos\phi - [\sin\phi/\tan\frac{1}{2}(\alpha+\beta)])\}}{(n\pi b/d)(\cos\phi - [\sin\phi/\tan\frac{1}{2}(\alpha+\beta)])}\right]. \tag{3.9}$$

For a given grating, b, d, and ϕ are fixed so we can plot its "universal blaze curve," giving $I(\beta)$ as a function of $\alpha+\beta$ for any order n. Sample curves are shown in Fig. 3.9 for the case of $\phi = 20°$ and $b = d\cos^2\phi$. Notice how the higher orders have a blaze width that is inversely proportional to their order number. The maximum lies at $\alpha+\beta = 2\phi$, which is the configuration for specular reflection off the facet surfaces.

The blaze distribution with wavelength is easily calculated once the configuration of the spectrograph is specified. The value of λ associated with each $\alpha+\beta$ of the curves in Fig. 3.9 is calculated from the grating equation. For example, if we choose $\alpha = \beta = 20°$ and $d = 1/600$ mm, we obtain the results given in Fig. 3.10. A comparison of this scalar-wave theory to an

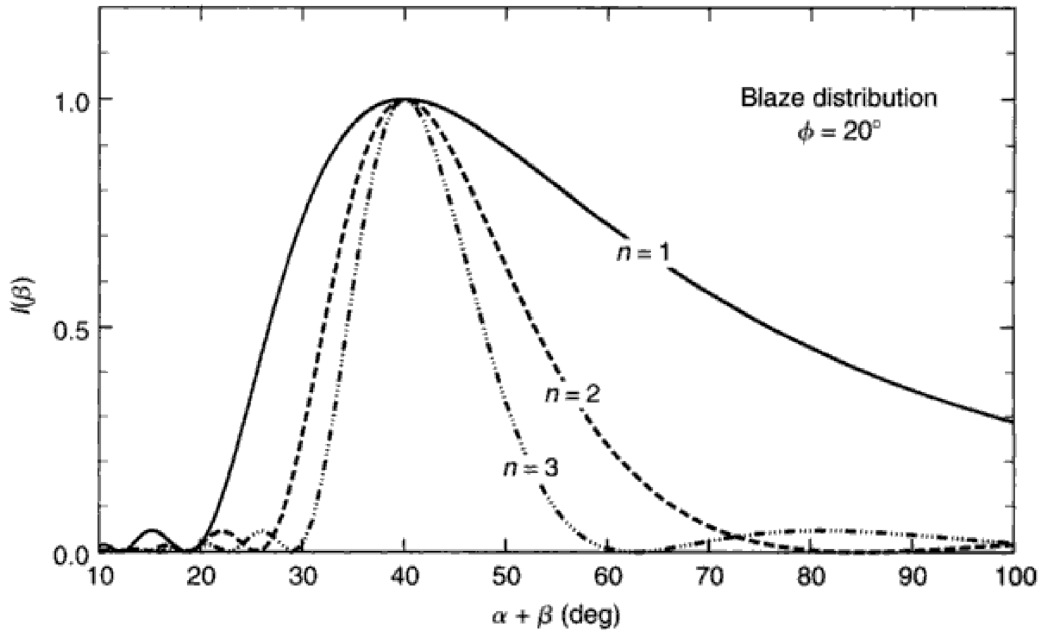


Fig. 3.9. Equation (3.9) gives this scalar-wave blaze distribution of light. The first three orders are shown for $\phi = 20°$.

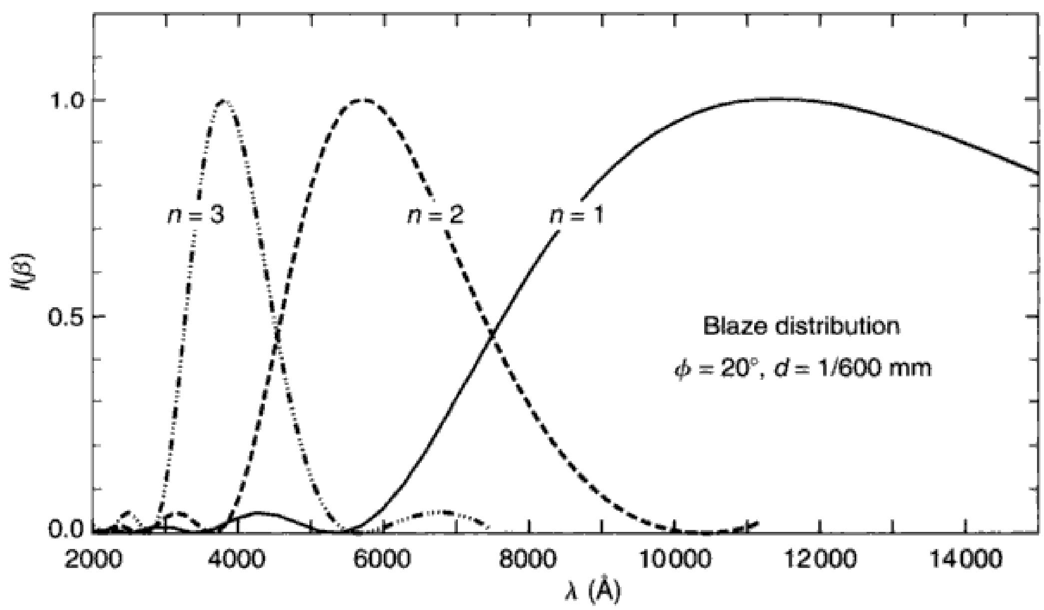


Fig. 3.10. Here the blaze distributions of Fig. 3.9 are plotted on a wavelength scale using $\alpha = \phi = 20°$ and a ruling density of 600 lines/mm.

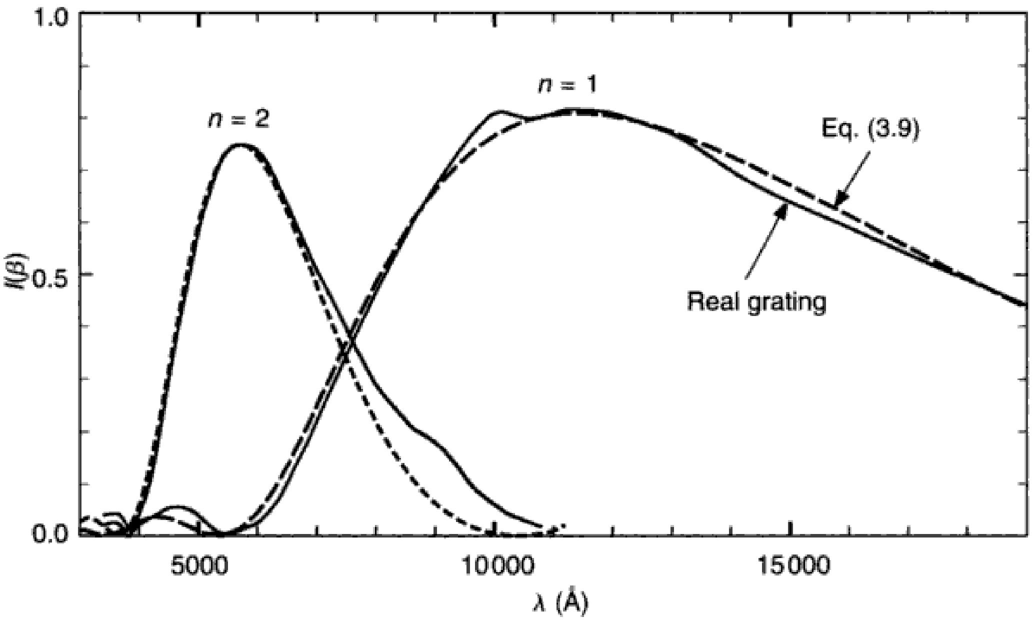


Fig. 3.11. The scalar-wave blaze equation, Eq. (3.9) (- - - - -), is compared to a real grating blaze distribution (——). Jarrell-Ash grating no. 1300. Data courtesy of Jarrell-Ash.

observed blaze distribution in unpolarized light is made in Fig. 3.11. The agreement is reasonable. However, the efficiency does depend on the vector properties of the light, namely the polarization. Radiation polarized with the electric vector perpendicular to the rulings generally shows a higher efficiency and a shift of the blaze to longer wavelengths compared to the opposite orientation, as shown in Fig. 3.12. The polarizing or analyzing property depends on wavelength and, at about 9600 Å for the grating of Fig. 3.12, the sense of the polarization reverses. We conclude that near the peak of the blaze, linear polarization of a few per cent can be expected, but several tens of per cent can arise somewhat farther from the center of the blaze. Polarization properties of gratings are further discussed by Breckinridge (1971), and polarization introduced by the entrance slit of a spectrograph is discussed by Ismail (1986).

Wavelength of the true blaze

Manufacturers of gratings give the blaze wavelength for the $\alpha = \beta$ (called the Littrow) configuration used in first order. The true blaze will appear at a shorter wavelength if the grating is employed in some other orientation. We let β_b be the diffraction angle at which the blaze is directed, λ_0 the blaze wavelength for the $\alpha = \beta$ configuration, and λ_b the wavelength of the actual blaze. The

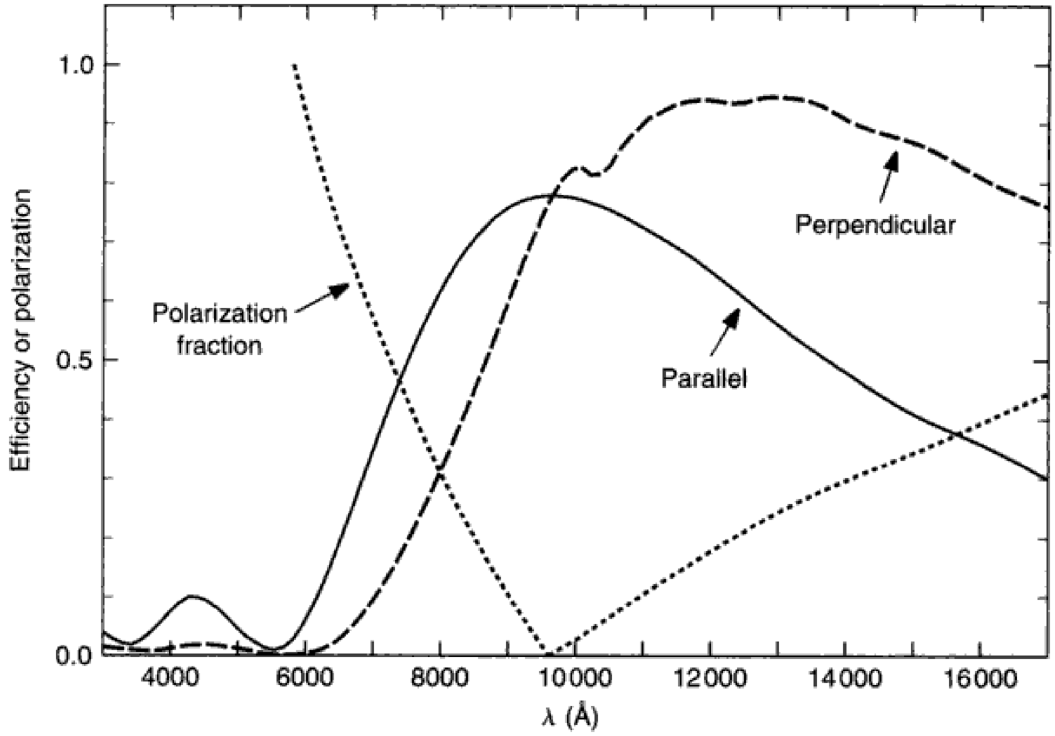


Fig. 3.12. Incident light polarized perpendicular to (- - - - -) and parallel to (———) the rulings show rather different blaze distributions. The sum of these curves is the $n = 1$ curve in Fig. 3.11. If the incident light is unpolarized, the grating will polarize it. The sense of the polarization reverses as it goes through zero (--------). Adapted from data supplied by Jarrell-Ash.

situation is shown in Fig. 3.13. Since the facet reflection gives the blaze, the law of reflection gives $\alpha - \phi = \phi - \beta_b$, or $\alpha + \beta_b = 2\phi$. The grating equation for the general case is

$$n\lambda_b/\mathrm{d} = \sin\alpha + \sin\beta_b,$$

and for the $\alpha = \beta$ case, it is

$$\begin{aligned} n\lambda_0/d &= \sin\alpha + \sin\beta_b \\ &= 2\sin\phi \\ &= 2\sin\tfrac{1}{2}(\alpha + \beta_b). \end{aligned}$$

Dividing these two expressions gives

$$\frac{\lambda_b}{\lambda_0} = \frac{\sin\alpha + \sin\beta_b}{2\sin\frac{1}{2}(\alpha + \beta_b)}.$$

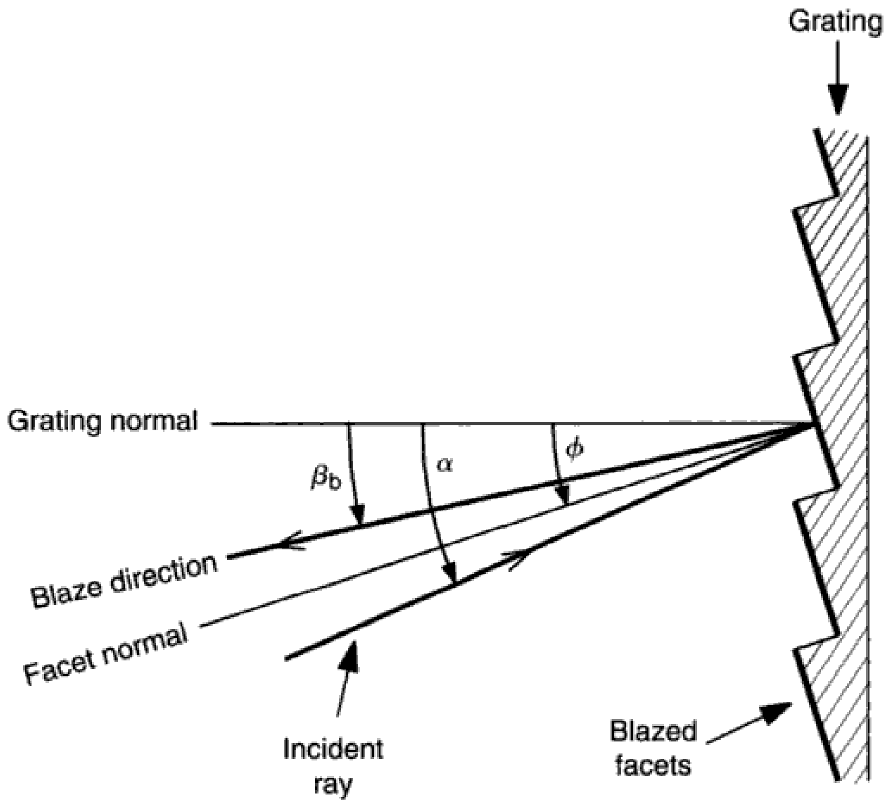


Fig. 3.13. The specular reflection at the facet determines the direction of the blaze, β_b. All the angles shown in this figure are positive because they lie on the same side of the grating normal as the facet normal.

Using a trigonometric identity, the right-hand side of the equation is replaced, giving

$$\lambda_b = \lambda_0 \cos\tfrac{1}{2}(\alpha - \beta_b). \tag{3.10}$$

For example, if the blaze is to be directed 40° from the incident light, a grating that is said to be blazed at 8000 Å will actually have its blaze at 7518 Å.

Shadowing

Most gratings can be used in two orientations. The gratings can be tilted so α is on the same side or the opposite side of the grating normal as the facet normal. These two possibilities are illustrated in Fig. 3.14. We see, upon careful inspection of the figure, that case B may involve considerable light loss, especially when the diffraction angles are large. This is shadowing. In fact, shadowing causes more harm than just light loss; it increases the level

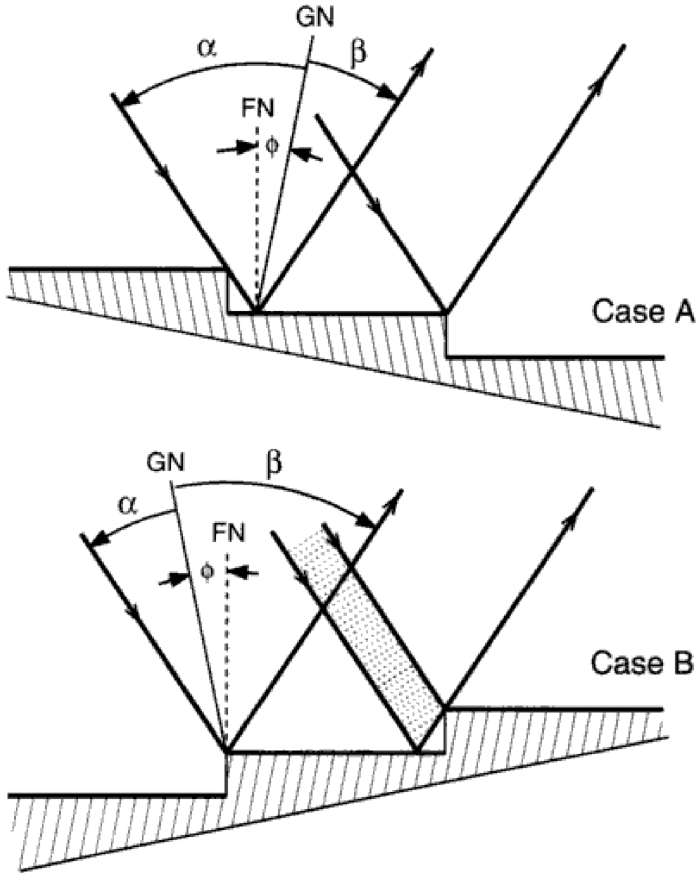


Fig. 3.14. The grating can be used in either of two orientations, α negative, as in case A in the upper part of the figure, or α positive, as in case B. Notice that the incoming and outgoing beams are in the same direction for both cases. The effective width of the facets is decreased in both cases, broadening the angular width of the blaze. In case B, the light in the shaded strip is reflected off the back of the facet and scattered into the spectrograph.

of scattered light in the spectrograph (see Chapter 12 for a discussion of scattered light). The lost light reflects off the back of the facet, then on to the face of the facet, and into the spectrograph. Shadowing is larger for those orientations of the grating where α is smaller than β. A diffracted monochromatic beam has a smaller width than the incident collimator beam, and the angular dispersion is greater for this orientation. The fraction of the beam lost to shadowing is given by

$$f = \frac{2 \tan \phi \tan(\beta - \phi)}{1 + \tan \phi \tan(\beta - \phi)}. \tag{3.11}$$

Obviously, in the Littrow configuration, both the incoming and outgoing beams will be along the facet normal, and shadowing is zero. Near the

Littrow configuration, $f \approx 2 \tan\phi \tan(\beta - \phi)$. While shadowing can readily be demonstrated for real gratings, its behavior is more complex than the foregoing geometrical arguments indicate, mainly because of diffraction effects.

Grating ghosts

Spurious lines are generated by various imperfections in the gratings. One class, the ghosts, arises from periodic errors in the position of the rulings. These errors can usually be traced back to flaws in the ruling engine, such as minute deviations in the pitch of the carriage-advance screw used to space the rulings. The multiple line pattern of the type shown in Fig. 3.15 appears in place of each single emission line in the spectrum. Insight into the ghost phenomenon can be gained by looking at the grating in the light of a ghost. We find that the grating appears to be striped (Fig. 3.16), that is, the light forming the ghost comes from a series of bright stripes. Each of these stripes corresponds to one cycle of the periodic error in the rulings. The set of stripes forms a ghost grating. The grating aperture distribution of Eq. (3.3) should then be replaced by

$$G'(x) = G(x)H(x),$$

where $H(x)$ is the periodic modulation from the ghost grating. In analogy with Eq. (3.3), we write

$$H(x) = B_3(x) * III(x)\, B_2(x),$$

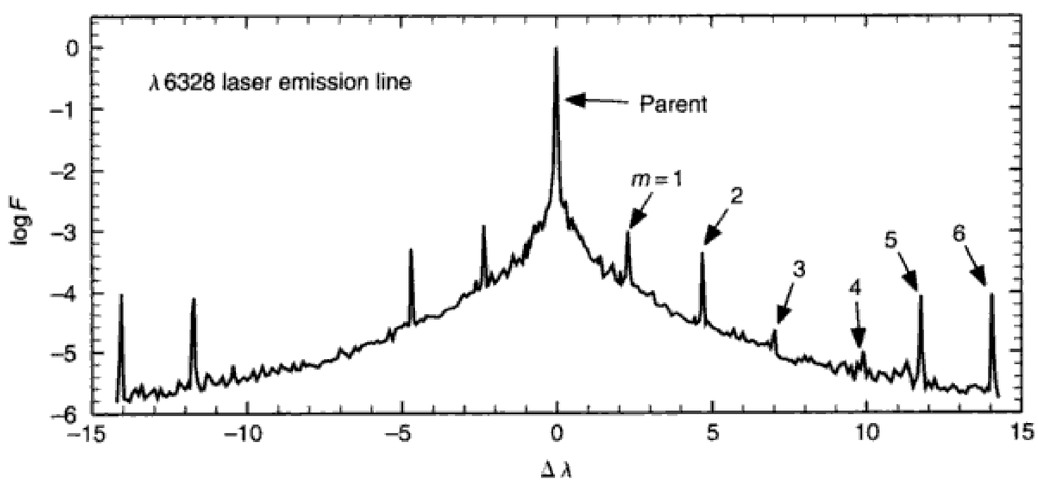


Fig. 3.15. Ghosts are weak spurious lines symmetrically placed on each side of the parent line. In this example, some of them are labeled with their order number. Note the logarithmic ordinate. Adapted from Griffin (1968).

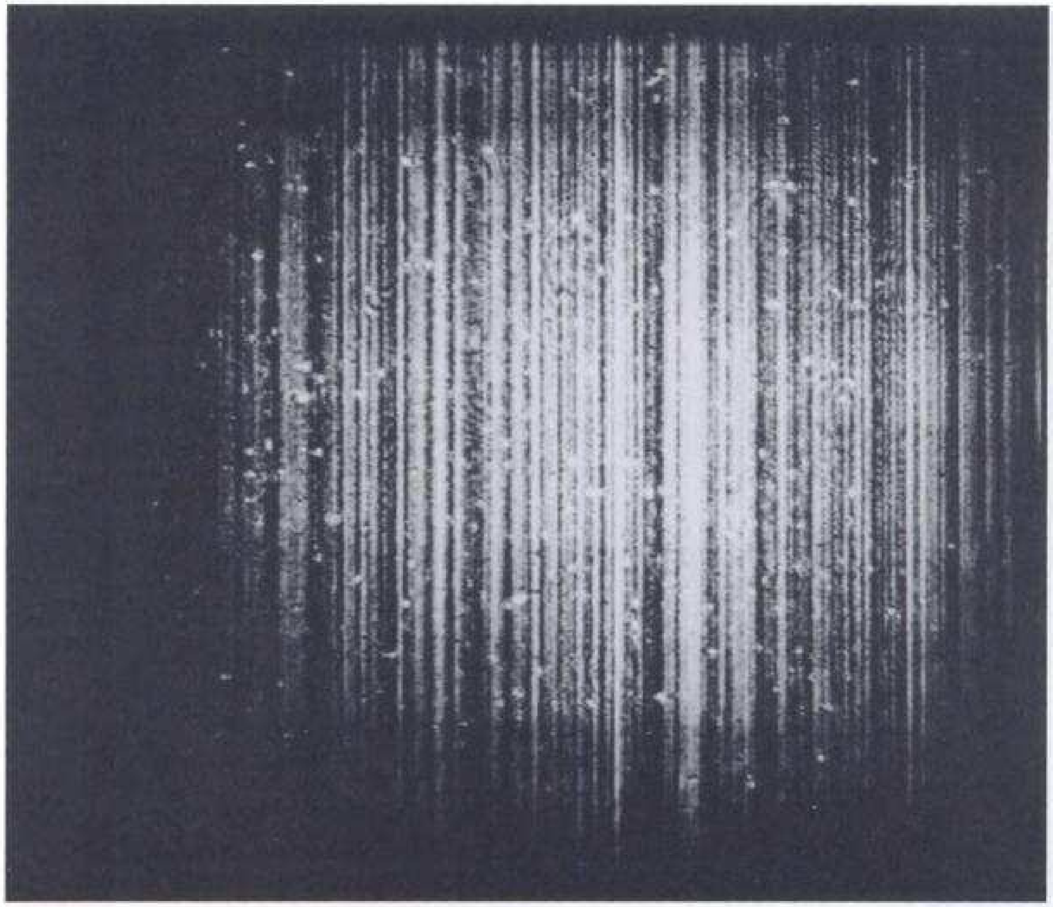


Fig. 3.16. A photograph of a large grating in the light of the $m = 1$ ghost shows the ghost grating as a series of stripes superimposed across the face of the main grating. Courtesy of J. B. Breckinridge.

where $B_3(x)$ describes the illumination across one stripe of the ghost grating, $B_2(x)$ is the box corresponding to the full width of the grating, and $III(x)$ is a new Shah function, $\Sigma\delta(x - mD)$, with δ-functions spaced D (the periodic error spacing) apart with m being the order integer.

The amplitude spectrum formed is then

$$g'(\theta) = g(\theta) * h(\theta)$$

where $g(\theta)$ is given by Eq. (3.4) and $h(\theta)$ is the spectrum from the coarse ghost grating. The convolution brings the spectrum of this grating to the position of each line in $g(\theta)$, thus giving the ghosts. The dispersion of the ghost grating is low because of the large spacing D, and in fact, we can write

$$\Delta\theta_g = \frac{m\lambda_g}{D}$$

by analogy with Eq. (3.5). Since $\Delta\theta_g = 0$ at the center of the parent line according to the convolution above, $\Delta\theta_g$ is a measure of the angular separation of the mth ghost from the parent line. The ghost spacing in wavelength follows from Eq. (3.7):

$$\Delta\lambda_g = \frac{d}{n}\Delta\theta_g = \frac{d}{n}\frac{m}{D}\lambda_g. \tag{3.12}$$

This is the ghost spacing equation. The ghosts associated with each real line are found at the positions given by letting m run through the $\pm$ integers starting at $m = \pm 1$. The spacing can be quite large. Take $d = 1/600$ mm, $n = 1$, $\lambda = 6000$ Å, and $D = 1$ mm (which is typical of ruling engine screw pitch). We find $\Delta\lambda_g = \pm 10$, ± 20 Å, and so on.

The color of light comprising the ghost images is the same as that of the parent line. This follows from the fact that the different order ghosts (m) are really different orders of the same λ formed by the ghost grating. Equation (3.12) gives the spacing of the ghosts, but certainly does not imply that the photons in the ghost spectrum have changed their wavelength from that of the parent line.

When more than one periodic error occurs in the rulings, the interference of the two ghost diffraction patterns can cause ghosts very distant from the parent line. These are often called Lyman ghosts. Those near the parent and positioned according to Eq. (3.12) are called Rowland ghosts. The distant ghost still has the color of the parent. Visual inspection can be used to identify them on this basis.

The run of ghost intensity with m is usually irregular. Often the $m = \pm 1$ ghosts are the largest, and the general trend is for decreasing strength with increasing m, which must follow from general considerations of the diffraction envelope. The irregular behavior (as seen in Fig. 3.15) arises because there is more than one periodic error in the rulings and the interference between them reduces the intensity of some of the ghosts. Non-symmetry in pairs of ghosts results if the ghost grating shows blaze characteristics.

Rowland (1893) and Meyer (1934) give a derivation of the ghost strength dependence with grating order,

$$I_g = I_p(\pi m e/d)^2,$$

where I_g and I_p are the intensity of the ghost and parent line, respectively, and e is the amplitude of the Fourier component of the ruling error under consideration and is expressed as a fraction of the average ruling spacing d. Ghosts are likely to be a bother in high orders since they increase quadratically with order, but for a "good" grating they are less than 10^{-4} as strong as the parent line in first order. Newer gratings have weaker ghosts because of the precise interferometer control of the ruling engine steps, and holographic gratings are essentially free of ghosts (see Kohl & Parkinson 1976), although holographic gratings are difficult to blaze. Ghost strength has also been considered by Calatroni and Garavaglia (1973, 1974).

A second class of grating imperfections gives rise to spurious lines called satellites. Looking in the light of the satellite feature reveals that certain portions of the grating surface are distorted. In effect, one has a little grating superimposed on the main grating but not in the same plane. Satellites are usually much closer to the parent line than the ghost spacing. They are not measurable in the best gratings. Poor mounting of the grating can distort it and produce satellite lines.

The third type of undesirable behavior is called Wood's anomaly (Wood 1935, Breckinridge 1971). It consists of a relatively large discontinuity introduced into an otherwise smooth continuous spectrum. Fortunately they generally lie away from the center of the blaze.

We now turn our attention from gratings back to spectrographs more generally.

Dispersion, slit magnification, and spectral resolution

Spectral resolution is the scientifically important parameter; it specifies the finest details one can see in the spectrum. But first, we start with the dispersion.

The angular dispersion produced by the dispersing element, when combined with the focal length of the camera, determines how spread out the spectrum will be in the focal plane of the camera. If two collimated beams having wavelengths λ and $\lambda + \Delta\lambda$ leave the grating at angles differing by $\Delta\beta$, then, as shown in Fig. 3.17, the linear separation in the spectral image is

$$\Delta x = f_{\mathrm{cam}} \Delta\beta,$$

or

$$\frac{\mathrm{d}\beta}{\mathrm{d}x} = \frac{1}{f_{\mathrm{cam}}},$$

and it follows immediately that the linear dispersion in angstroms per millimeter is given by

$$\begin{aligned}\frac{\mathrm{d}\lambda}{\mathrm{d}x} &= \frac{\mathrm{d}\lambda}{\mathrm{d}\beta}\frac{\mathrm{d}\beta}{\mathrm{d}x} \\ &= \frac{1}{f_{\mathrm{cam}}}\frac{1}{\mathrm{d}\beta/\mathrm{d}\lambda} \\ &= \frac{1}{f_{\mathrm{cam}}}\frac{d\cos\beta}{n},\end{aligned} \tag{3.13}$$

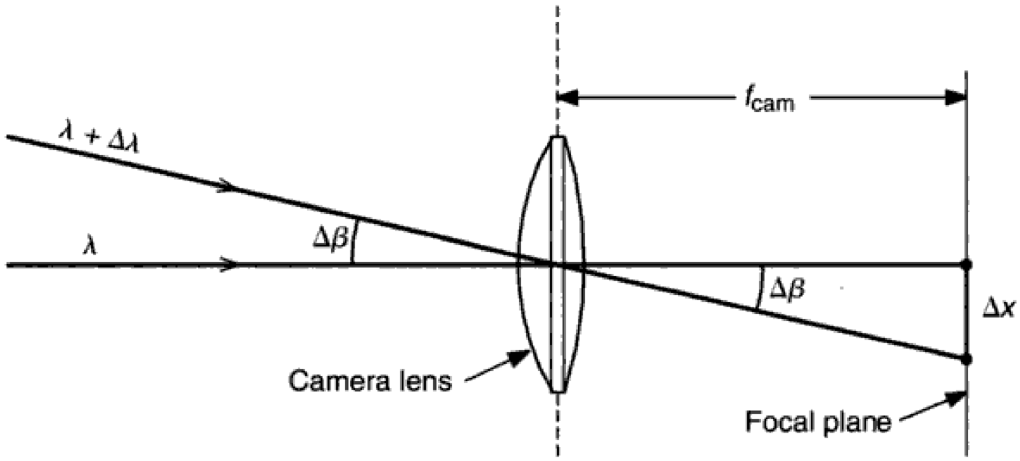


Fig. 3.17. The camera in a spectrograph focuses the collimated monochromatic beams from the grating to positions differing by Δx for the difference in angle $\Delta\beta$, corresponding to two wavelengths differing by $\Delta\lambda$. The scale of the image in the focal plane is proportional to the focal length of the camera. If the camera is a mirror, the same diagram holds true if it is folded about a line through the lens, as shown.

where $d\beta/d\lambda$ is the angular dispersion given by the grating (Eq. 3.6). Note that high dispersion is a small number of angstroms per millimeter and vice versa. It is common for a spectrograph to have several cameras in order to supply several linear dispersions.

Linear dispersion and resolution are often connected. The connection occurs (only) when something about the detector "graininess" is tacitly understood. In years past, when photographic plates were used, one often said "high dispersion" while really meaning "high resolution." With the wide range of detectors now in use, the two terms have become de-coupled and can mean very different things. It is the resolution, not the dispersion, that is scientifically important. Let us proceed in that direction.

The width of the image of the entrance slit in millimeters as seen in the spectrum depends on the focal lengths of the collimator and camera and on the orientation of the grating. Let W' be the width of the entrance slit, $f_{\rm coll}$ the collimator focal length, and $f_{\rm cam}$ the camera focal length. The angular size of the entrance slit as seen from the collimator (or the grating) is

$$\Delta\alpha = W'/f_{\rm coll}.$$

From the grating equation (Eq. 3.5), we know that

$$\Delta\beta = -\frac{\cos\alpha}{\cos\beta}\Delta\alpha,$$

so

$$\Delta\beta = -\frac{\cos\alpha W'}{\cos\beta f_{\mathrm{coll}}}$$

is the angular size of the monochromatic image of the entrance slit. If we call the slit image width w, then (as in Fig. 3.17 with $w = \Delta x$)

$$\begin{aligned} w &= \Delta\beta f_{\mathrm{cam}} \\ &= -\frac{\cos\alpha}{\cos\beta}\frac{f_{\mathrm{cam}}}{f_{\mathrm{coll}}}W'. \end{aligned} \tag{3.14}$$

The magnification of the width of the entrance slit is w/W'. As mentioned earlier in the chapter, the magnification along the slit is just $f_{\mathrm{cam}}/f_{\mathrm{coll}}$.

The focal length of a spectrograph camera is chosen to give a suitable scale for the spectrum. Associated with each detector is some characteristic "graininess" or pixel size. For photographic plates, it is the graininess of the emulsion, typically 10–20 μm. For Si-diode arrays, it is the width of the diodes, characteristically 15–30 μm, depending on the array. The spectrum should be scaled so that w in Eq. (3.14), spans two to three resolution elements in the detector. Spreading the spectrum out more than this gives oversampling. A small amount of oversampling is scientifically advantageous, but *any* oversampling is paid for immediately and painfully with longer exposure times or brighter limiting magnitudes. A lower limit of two resolution elements across w corresponds to the Nyquist sampling limit (Eq. 2.24). Lower dispersion than this will produce undersampling; information will be lost because each detector pixel will be integrating information over wavelength spans exceeding w. In effect, the detector will then limit the spectrograph resolution, a situation that should not normally be allowed.

The spectral resolution, occasionally referred to as spectral purity, is the most important parameter one can specify to describe the capabilities of a spectrograph. When the width of the entrance-slit image dominates the broadening of the instrumental profile, which is the usual case, it determines the spectral resolution. (Optical aberrations are discussed by Schroeder 1974.) Convert the image width, w, to a wavelength interval by combining Eqs. (3.14) and (3.13),

$$\begin{aligned} \Delta\lambda &= w\frac{\mathrm{d}\lambda}{\mathrm{d}x} \\ &= -\frac{\cos\alpha}{\cos\beta}\frac{f_{\mathrm{cam}}}{f_{\mathrm{coll}}}W'\frac{1}{f_{\mathrm{cam}}}\frac{d\cos\beta}{n} \\ &= -\cos\alpha\frac{W'}{f_{\mathrm{coll}}}\frac{d}{n}. \end{aligned} \tag{3.15}$$

This is the important relation that ties together the grating parameters d and n with the spectrograph parameters W' and f_{coll}. We see that the camera focal length has canceled out; the resolution is independent of the camera focal length. In general, we say that the resolution is higher when $\Delta\lambda$ is smaller, and a smaller $\Delta\lambda$ value is what we want in order to measure the shapes of spectral lines more accurately.

We can choose a grating with high ruling density (small d) or one blazed to work in high orders (large n). Blazing for high orders implies steep facet angles and therefore deep grooves, making it difficult to manufacture high ruling densities.

The other factor we control is W'. Making W' small gives higher resolution, but at the cost of light loss – a cardinal sin. Naturally image slicers are a wise choice at this point. (Image slicers recycle the light intercepted by the slit jaws so that most of the seeing disk gets into the spectrograph; see Chapter 12.) We also notice in Eq. (3.15) that it is the *angular* size of the entrance slit, as seen from the collimator that is involved. In other words, resolution is gained equally well by increasing f_{coll}. This is exactly what is done in practice, and it is the reason why coude spectrographs are large (see Chapter 12). Recall how (Fig. 3.1) the beam diverges with the f-ratio of the telescope as it enters the spectrograph. The expansion of the beam is arrested and the beam size is fixed by the collimator. The limit to extending f_{coll} is set by the maximum available grating size. In some cases mosaics of gratings are used to partly get around this limitation.

High-resolution ($\Delta\lambda/\lambda > 10^5$) spectroscopy on big telescopes is seriously affected by limited grating size. A big telescope has a correspondingly big effective focal length (or else an equally troublesome small f-ratio). For example, doubling the size of the telescope means a stellar seeing disk of twice the diameter, so the spectrograph slit needs to be twice as wide to let in the same fraction of the image. Then to preserve the spectral resolution, the collimator must be twice as far from the entrance slit, so the beam has doubled its diameter by the time it is collimated. With telescope apertures of 3.4 m, four-element mosaic gratings are used to gain sufficient beam size. High resolution can be attained on telescopes of still larger aperture only with great difficulty. Excellent atmospheric seeing is an important advantage in this regard.

Figure 3.18 illustrates how details of line profiles are readily seen with a spectral resolution of $\approx 100\,000$. Some high-resolution spectrographs are described in the literature (e.g., Gray 1986, 1997, Vogt *et al.* 1994, Pilachowski *et al.* 1995).

Echelle spectrographs

In recent years cross-dispersed echelle spectrographs have been used effectively for certain types of work (e.g., Chaffee & Schroeder 1976, Weiss *et al.* 1981,

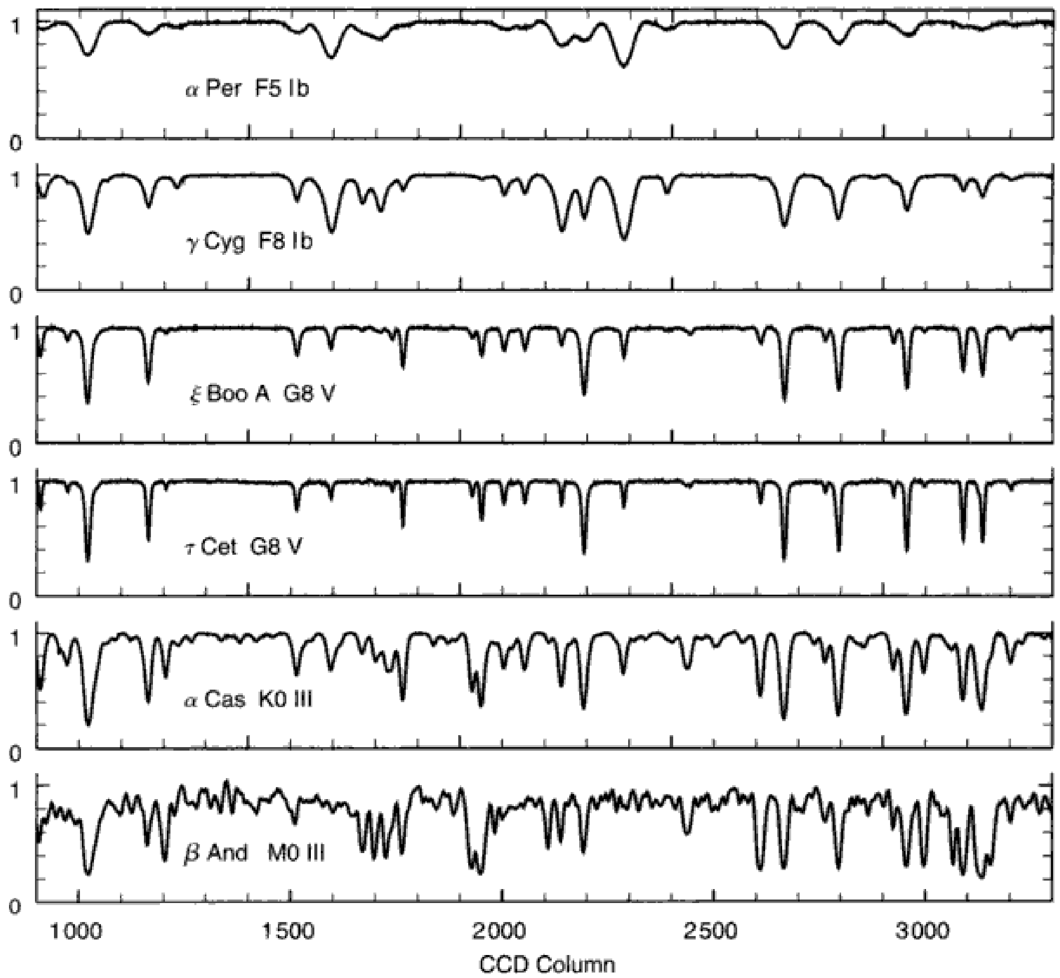


Fig. 3.18. These coude-spectrograph measurements, normalized to unit continuum, have a spectral resolution of ≈100 000. The noise can just barely be discerned on some of the exposures. The differences in widths and shapes of the spectral lines are easy with spectra like these. Many very weak lines can also be seen, especially for the cooler stars. These graphs show a 23 Å portion centered at λ6245. Data from the Elginfield Observatory, University of Western Ontario.

Vogt 1987, Robinson 1988, Epps & Vogt 1993, Alencar *et al.* 2001, Montes *et al.* 2001, Ryan *et al.* 2001, Soderblom *et al.* 2001, Lacy *et al.* 2002). The leverage offered by the echelle grating is its high angular dispersion, $d\beta/d\lambda$, resulting from working in very high orders, $n \sim 100$, for example. As a result, the camera focal length can be small and the physical size of the spectrograph can be kept small enough to use at the Cassegrain focus, as long as modest resolving powers ($\lesssim 40\,000$) are not exceeded. The wavelength span of the blaze distribution varies inversely with the order number, giving rise to a spectrum that is effectively cut into dozens of small pieces. A cross-dispersing

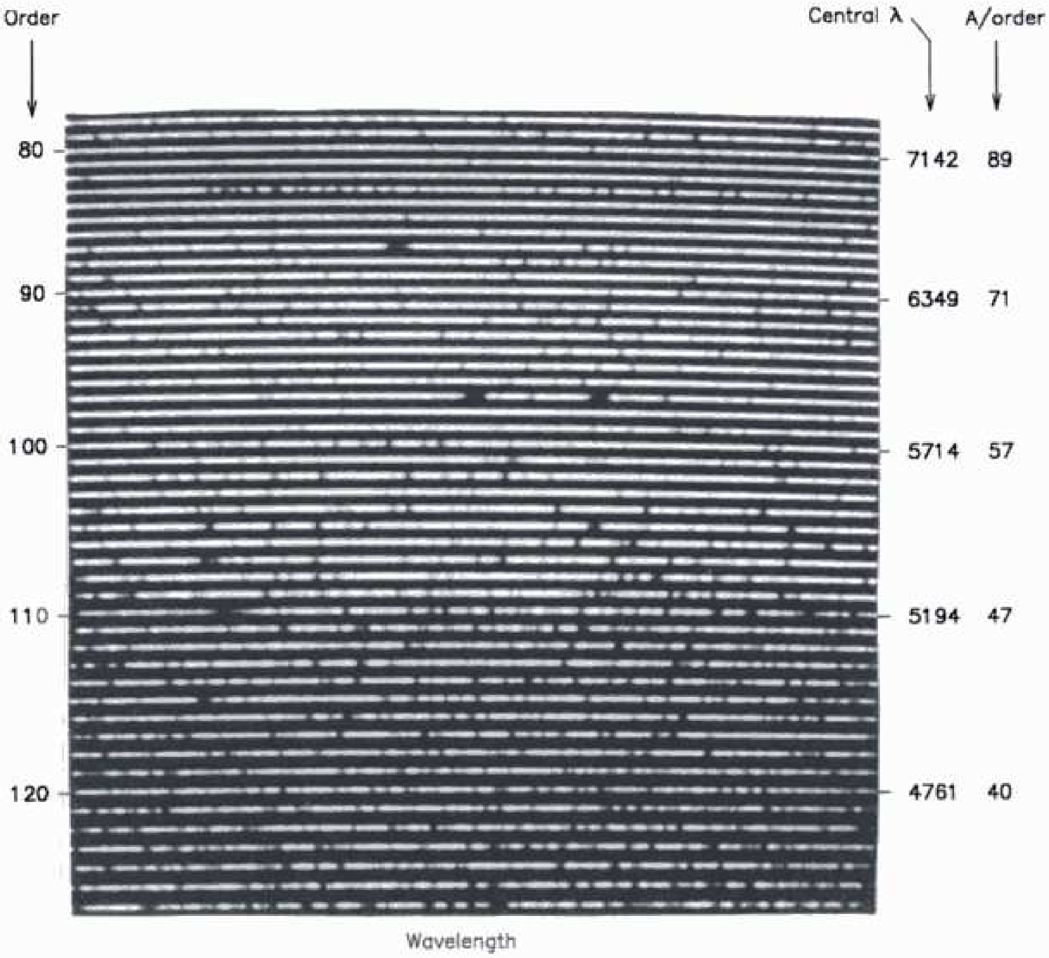


Fig. 3.19. This echellogram shows the central portions of 49 orders spanning a range of wavelength from 4500 to 7300 Å. The field is limited by the size of the detector. The full coverage by the spectrograph is 3600 to 9500 Å. The resolving power is 30 000, and the star (A4, V = 12.3) is in the globular cluster M71. The strong line in order 87 is H_α, and the two in order 97 are the Na D lines. Details can be seen in Vogt (1987). Photograph courtesy of S. S. Vogt.

grating or prism is used to separate and stack the orders, and one must have a two-dimensional detector to record them (see Fig. 3.19).

The scientific advantages of the echelle spectrograph lies mainly in the wide wavelength coverage obtained in a single exposure. This is important for chemical abundance studies, for instance, where numerous and widely separated lines must be measured. The price paid is limited accuracy. The narrow blaze distribution causes steep variations in the apparent continuum with wavelength. Accurate measurements of profiles of strong lines, such as the hydrogen Balmer lines in hotter stars, is exceedingly difficult. The fact that thousands of angstroms of spectrum are allowed into the spectrograph leads to relatively high levels of scattered light, typically several per cent. (With low-order gratings, a bandpass filter is used to exclude the light not being measured.) Reductions are complicated by the slant of the spectrum resulting from the cross dispersion; the spectrum does not run perpendicular to the spectral lines. There is also the question of how strong the cross dispersion should be. If the adjacent orders are close together, optical aberrations will extend light from each order into those adjacent to it on the detector – a nasty contamination. There may also be insufficient room for the traditional comparison spectrum needed for absolute wavelength calibration. But if the orders are spaced well apart, fewer of them will be recorded because detectors have a limited area. When observing extended sources, such as nebulae or emission shells around stars, be prepared to see the lines extending across adjacent orders. The strong variation of brightness across the blaze and sometimes with order number (owing to the changing efficiency of the cross-disperser) make for widely ranging signal-to-noise ratios on each exposure, a significant impediment to precise measurement. Because of various field errors, the instrumental profile (Chapter 12) is a function of the position in the field.

Cross-dispersed echelle spectrographs are powerful tools, and should be used when very wide wavelength coverage is deemed important enough to sacrifice the ultimate possible accuracy. Otherwise lower-order spectrographs, i.e., $n \lesssim 20$, have the advantage.

Multi-object spectroscopy

The advent of two-dimensional light detectors has also allowed tens of stars to be measured simultaneously in a single order. The light from many star images is piped from the focal plane of the telescope via fiber optics to a stacked position along the entrance slit of the spectrograph. Each row on the light detector then corresponds to a spectrum of a different star. This technique is most useful in low-resolution, modest signal-to-noise situations. Positioning the input ends of the fibers is a considerable technical challenge (Taylor & Douglas 1995). Additional information can be found in Barden *et al.* (1993), Allington-Smith *et al.* (2000), Doyon *et al.* (2000), Parry *et al.* (2000) and Conti *et al.* (2001).

Spectra from interferometers

Fourier spectroscopy, the technique of using an interferometer to measure the spectrum, offers certain advantages for astronomical measurements including high optical throughput and high spectral resolution (see Connes 1961, 1969, 1970, Beer *et al.* 1972, Bell 1972, Marschall & Hobbs 1972, Brault 1982). Interferometers

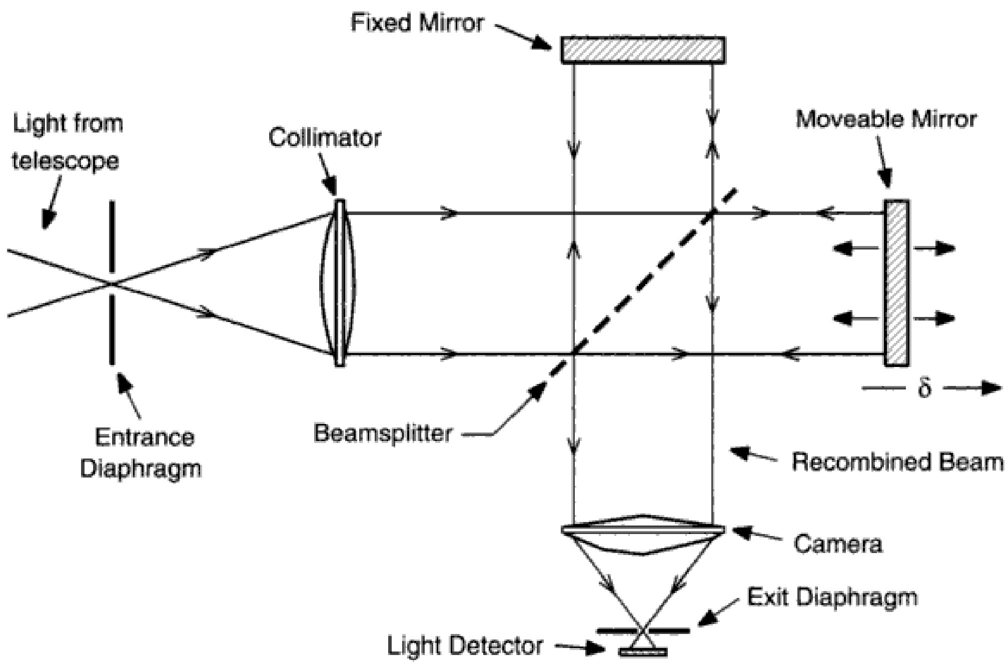


Fig. 3.20. A basic Michelson interferometer uses a beamsplitter to create two beams. The path length and therefore the phase of the horizontal beam is changed by moving the mirror on the right through the path difference δ. The interference pattern, formed when the beams are recombined, is focused on the detector by the camera lens.

measure the Fourier transform of the spectrum. Several types of interferometers have been used, notably the Michelson and Fabry–Perot instruments. In the Michelson interferometer shown in Fig. 3.20, the path difference between the two arms is δ (the mirror moves half this distance). The moveable mirror introduces a phase term $e^{2\pi i\delta\sigma}$ for the monochromatic light of wave number $\sigma = 1/\lambda$. The state of interference in the exit diaphragm varies sinusoidally with δ as the successive multiples of λ and $\lambda/2$ in path difference give maxima and minima. Take $B(\sigma)$ to be the spectrum of the light and $b(\delta)$ to be the signal recorded at the detector. These functions, for the monochromatic example, are shown in Fig. 3.21. We recognize the two functions $B(\sigma)$ and $b(\delta)$ as Fourier transforms if we take $B(-\sigma) = B(\sigma)$ and $b(-\delta) = b(\delta)$ In this way $B(\sigma)$ and $b(\delta)$ are symmetric.

In polychromatic light, we sum the monochromatic contributions and write

$$\begin{aligned} b(\delta) &= \int_{-\infty}^{\infty} B(\sigma)\, e^{2\pi i\delta\sigma}\, d\sigma \\ &= \int_{-\infty}^{\infty} B(\sigma) \cos 2\pi\delta\sigma\, d\sigma. \end{aligned} \tag{3.16}$$

The cosine integral follows since $B(\sigma)$ is symmetric and real.

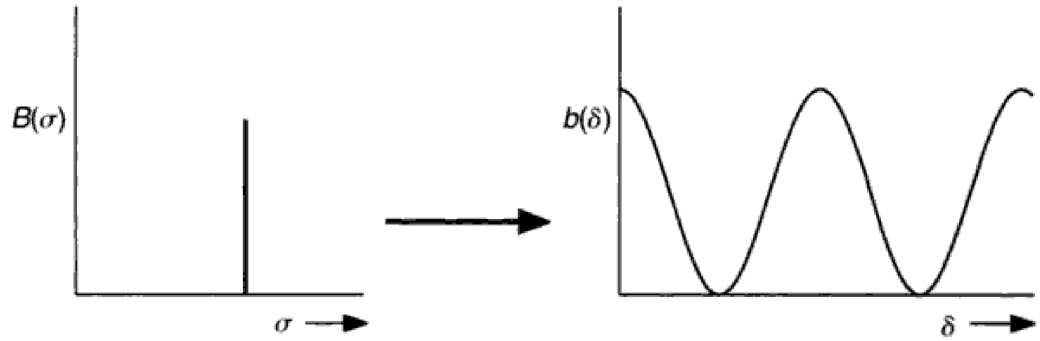


Fig. 3.21. The monochromatic spectrum on the left gives the cosine output on the right.

The interferogram, $b(\delta)$, is the detector output as a function of path difference. The spectrum of the light source is the Fourier transform of the interferogram. Some of the first useful measurements of infrared stellar spectra were obtained by Mertz (1965b). An example of a more modern interferogram and the resulting spectrum of α Per (F5 Ib) are shown in Figs. 3.22 and 3.23. At small path differences, all wavelengths are nearly in phase, and large amplitudes appear in $b(\delta)$. The phases rapidly become non-coherent as δ increases, and the interference between different Fourier components reduces the amplitudes to lower values. There is, however, a great deal of meaningful structure right out to the maximum path difference, as illustrated in Fig. 3.22.

The maximum path difference, δ_m, determines the resolution afforded by the interferometer in exact analogy with the width of the diffraction grating, as described by Eq. (3.3); that is, the full interferogram, which in principle extends from $-\infty$ to $+\infty$, is actually restricted to $\pm\delta_m$. This amounts to multiplying $b(\delta)$ in Eq. (3.16) by a box of width $2\delta_m$. The spectrum computed from $b(\delta)$ is then convolved with sinc $2\pi\delta_m\sigma$, which is the instrumental profile. Its characteristic width, and therefore the resolution of the interferometer, is $\Delta\sigma = 1/\delta_m$. Because it is possible to work over path differences of 2 m, interferometers can produce substantially higher resolution than diffraction gratings, the largest of which is less than 0.5 m in width. Non-trivial technical difficulties in measuring δ are overcome by using reference laser interferometers to measure the positions of the optical components (see Maillard & Michel 1982).

In a grating spectrograph, we saw how the spectral resolution could only be kept high by using a small entrance slit (Eq. 3.21). The slit allowed into the spectrograph only a small fraction of the light actually available in the seeing disk. An interferometer has no entrance slit; the whole stellar seeing disk can be measured without loss in resolution. This offers a significant efficiency gain

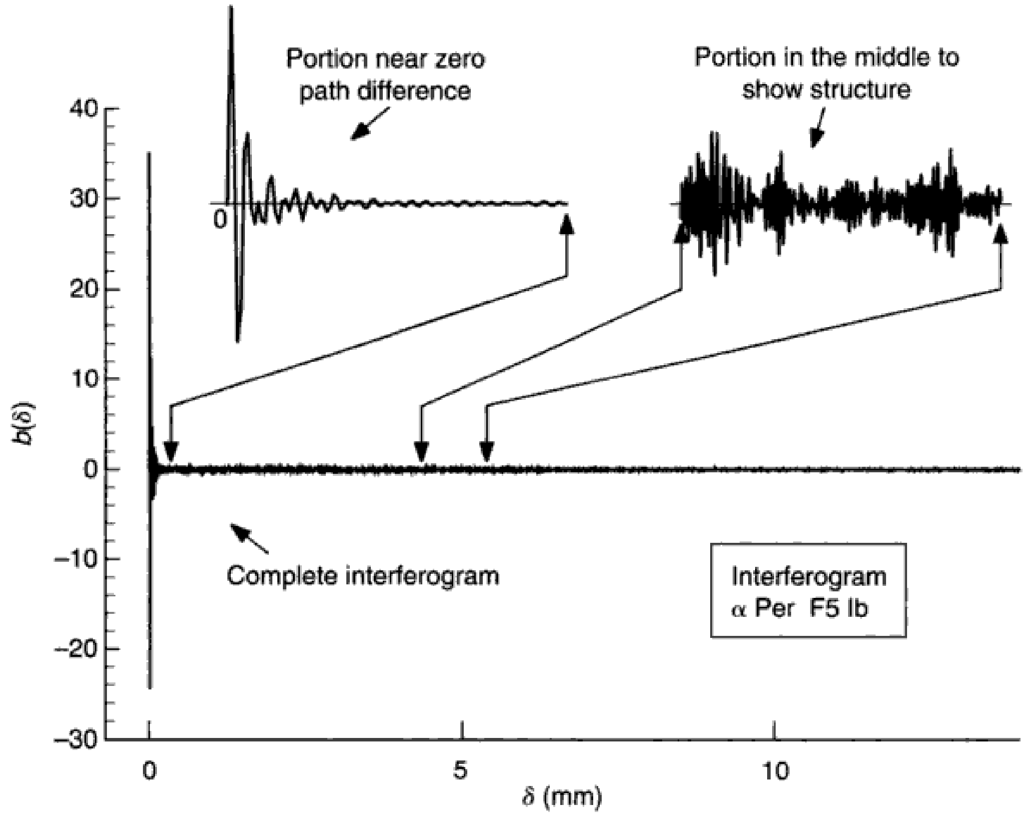


Fig. 3.22. A characteristic interferogram shows large amplitudes near zero path difference where the phase differences are still small. The signal out to large path differences, although of smaller amplitude, contains the high Fourier frequency information of the wavelength domain. Two portions are enlarged to show detail. Data courtesy of D. D. Sasselov and J. B. Lester.

over slit spectrographs. On the other hand, image slicers (discussed in Chapter 12) offset a significant portion of this advantage. Additional information can be found in the detailed work of Jaquinot (1960), Vanasse and Sakai (1967), Schnopper and Thompson (1974), or Ridgway and Brault (1984).

Numerous other advantages are offered by interferometers, including low levels of scattered light, an absolute wavelength calibration to $\lesssim 100$ m/s, a very well determined and controllable instrumental profile, an "adjustable" resolving power (δ_m), versatile wavelength coverage, and insensitivity to non-linearities in the detectors.

Unfortunately, interferometers are not as fast as conventional spectrographs used with multi-element array detectors, nor are they as easy to work with. Numerous comparisons have been made in the literature between interferometers and scanning monochromators. In the usual case, where photon noise dominates the total noise, scanning the spectrum or scanning the interferogram produces

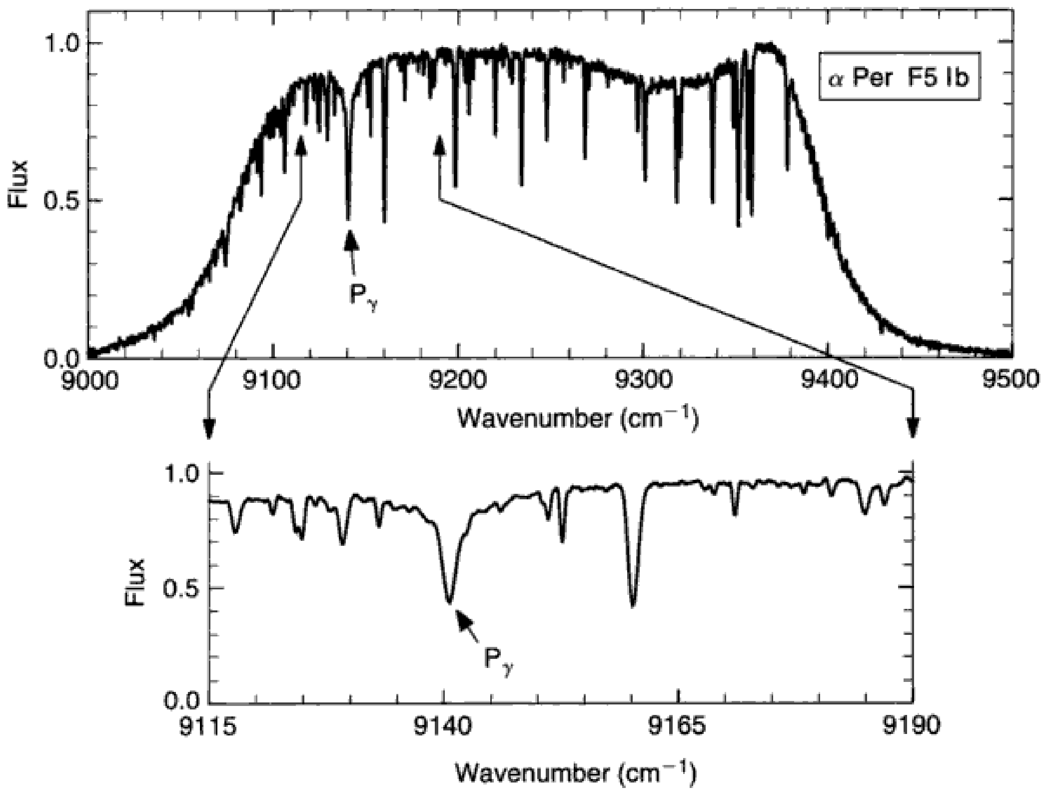


Fig. 3.23. The wavelength-domain reconstruction of the interferogram in Fig. 3.22 is shown. In the upper portion, the overall shape is the transmission curve of the bandpass filter used to isolate the measured portion of the spectrum. Division by an exposure of a continuous-spectrum lamp can be used to remove this shape and normalize to the continuum. The lower part shows an expansion around the Paschen gamma line to better illustrate the details of the line profiles. The spectral resolving power is about 27 000.

equal results (although subtle differences in noise distribution do occur; e.g., see Mertz 1965a). But one would not normally use a scanning spectrometer anymore, since detector arrays can now measure thousands of spectral elements simultaneously. The analogous device for the interferometer is to have a post-interference dispersing device to measure hundreds of narrow-band interferograms simultaneously. This type of multiplexing has yet to be put into common use.

In a few cases, notably in the far infrared, detector technology has not yet progressed to the low-noise domain. When the detector noise exceeds the photon noise, the interferometer has the added (original) multiplex advantage, called Fellgett's advantage, gaining in speed relative to the scanning monochromator by the number of resolution elements recorded.

Attaining a photometric precision of better than about 1% requires considerable skill. But very high values can be attained, e.g., the solar atlas by Kurucz *et al.* (1984) where the signal-to-noise ratio is ~2000 (see also Marschall & Hobbs 1972 and Hinkle *et al.* 1995).

Aspects of telescopes

The main function of the telescope for studies of stellar photospheres is to collect light for spectral analysis. The size and design of a telescope must be compatible with the spectroscopic equipment coupled to it. The focal ratio of the telescope, for instance, must match the focal ratio of the collimator in a spectrograph. The telescope focal ratio is controlled by the figure on the secondary mirror, and often telescopes are constructed with interchangeable secondary mirrors to accommodate several types of instruments and to give the long focal length needed for coude operation (see Fig. 3.24). The *f*-ratios of Cassegrain foci typically range from 7 to 20; values of 20–40 are more common for the coude focus. The effective focal length is defined to be the focal ratio (the angular convergence of the beam at the focus) multiplied by the aperture of the telescope, and it may be much larger than the physical distance from the primary mirror to the focus. The scale in the focal plane of the telescope, just as with the discussion leading to Eq. (3.13), is given by

$$S = \frac{1}{F_{\text{eff}}}\ \text{rad/mm} = \frac{206\,265}{F_{\text{eff}}}\ \text{arcsec/mm}, \tag{3.17}$$

where the effective focal length, F_{eff}, is given in millimeters. Typical Cassegrain-focus scales are 10–30 arcsec/mm, while at the coude, values of 2–6 arcsec/mm are common. A stellar seeing disk of 2 arcsec would then be ~0.5 mm across at a typical coude focus.

Seeing is a major consideration for ground-based instruments. The actual distribution of light in the stellar seeing disk is sharply peaked toward the center and often shows a large but faint outer halo reaching to several seconds of arc. The general distribution of light in seeing disks is discussed by Moffat (1969), Piccirillo (1973), Young (1974), Diego (1985), and Racine (1989). Physical causes of seeing are reviewed by Coulman (1985). Seeing is often degraded as the telescope size increases owing to the larger number of atmospheric cells encompassed by the larger aperture (Dyck & Howell 1983, O'Byrne 1988; see also Chapter 14). All of these factors are of interest in considering Eq. (3.15) and the need to optimize throughput while preserving resolving power. Image slicers (see Chapter 12) can partially compensate for poor seeing. Locating telescopes in sites having good seeing is just as important for spectroscopy as for direct imaging.

Although the light gathering power of a telescope increases with the square of the aperture, the throughput into a slit spectrograph grows only *linearly* with

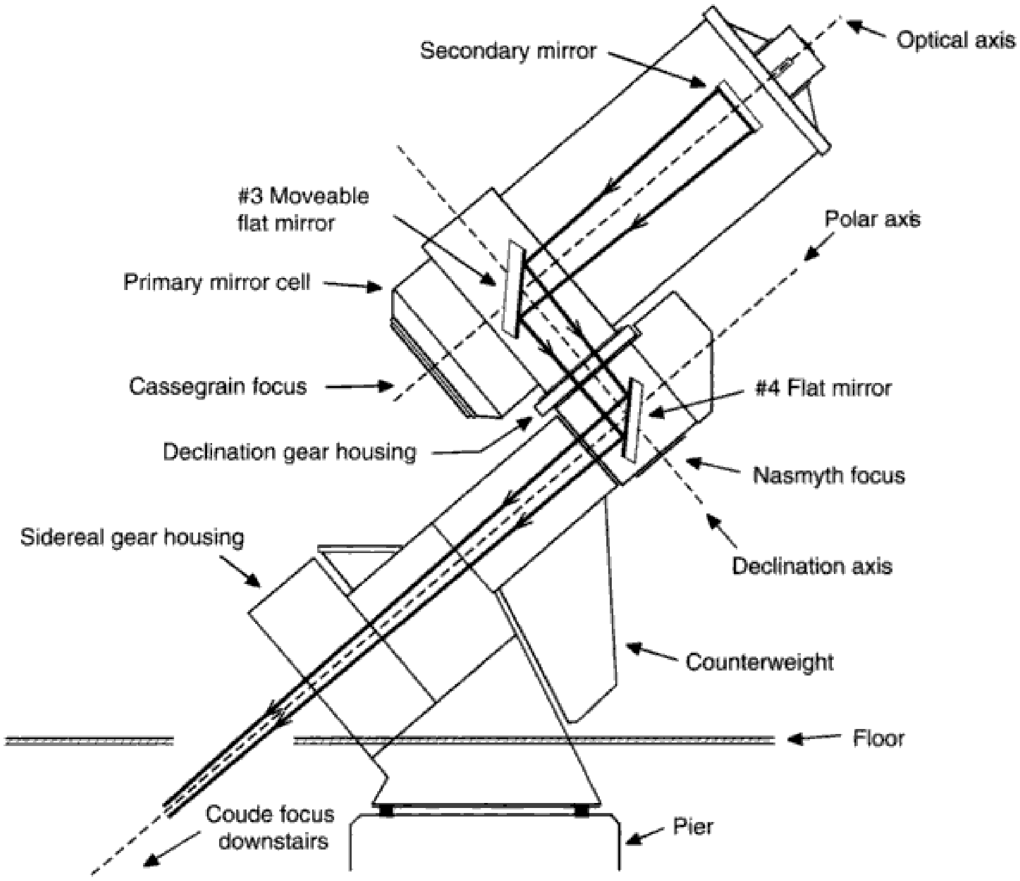


Fig. 3.24. Many telescopes allow the beam to be brought to any of several foci. The arrangement for coude focus is shown. The beam is shown from the secondary mirror onward. The #3 flat mirror is placed where the telescope optical axis intersects the declination axis, bringing the beam through the hollow declination axle. The #4 mirror lies at the intersection of the declination and polar axes, and directs the beam through the hollow polar axle to the coude focus on the floor below the telescope. In this way the focus remains fixed and independent of the direction in which the telescope is pointing. When a Cassegrain focus is needed, the #3 flat mirror inside the telescope tube is moved to the side and the secondary mirror is changed to give a shorter focal length. The light can also be brought to the Nasmyth focus on the declination axis opposite the polar axis from the telescope tube.

aperture once the maximum grating size has been reached. This follows from the discussion after Eq. (3.15).

Adaptive optics can correct some of the effects of seeing. By introducing real-time corrections to seeing distortions in the wave front, images can be sharpened and the throughput enhanced. Artificial laser guide stars have proved useful (e.g., Hackenberg & Bonaccini 2000). These systems are still relatively costly; nor are they free of problems (see Coudé du Foresto *et al.* 2000, Bloemhof *et al.* 2001).

Aluminum has been the traditional reflecting surface for telescope and spectrograph mirrors, but enhanced and over-coated silver offer throughput advantages. Seasoned aluminum typically has a reflectivity of 85% over much of the visible window, while the reflectivity of over-coated silver is about 97%. If one is working at a coude focus with, say, five mirrors to get the light to the spectrograph slit, and another three mirrors to get the light to the detector, there is potentially a gain of as much as a factor of three.

Polarization introduced by the telescope is small at the Cassegrain focus but quite significant at the coude focus. The oblique reflections needed to bring the light to the coude focus (notice the #3 and #4 mirrors in Fig. 3.24) produce about 6 or 7% polarization. How these contributions combine depends on the direction in which the telescope is pointed, so the net effect varies with time and the star.

Fiber optics are an effective alternative to the usual coude-relay mirrors. A fiber feed can also be used to bring the starlight to a Cassegrain-type spectrograph place on the floor instead of having it mounted on the telescope. This eliminates flexure and sagging of the spectrograph that always occurs when it is mounted on a telescope owing to the changing orientation of gravity as the telescope is pointed to different parts of the sky. Fiber optics bring some challenges. Some fibers suffer from transmission inefficiency, especially at shorter wavelengths. Sometimes the *f*-ratio of the telescope has to be modified (with a lens in the beam) to use the fiber optics effectively, and keeping all rays within the same *f*-ratio cone at the output end of the fiber can be difficult. The interested reader should consult Hubbard *et al.* (1979), Mandel (1987), Felenbok and Guerin (1988), Furenlid and Cardona (1988), Barden *et al.* (1981), and Coudé du Foresto *et al.* (2000).

References

Alencar, S.H.P., C.M. Johns-Krull, & G. Basri 2001. *Ast. J.* **122**, 3335.
Allington-Smith, J.R., R. Content, G.N. Dodsworth, *et al.* 2000. *SPIE* **4008**, 1172.
Baldry, I.K., J. Bland-Hawthorn, & J.G. Robertson 2004. *Pub. Ast. Soc. Pacific* **116**, 403.
Barden, S.C., L.W. Ramsey, & R.J. Truax 1981. *Pub. Ast. Soc. Pacific* **93**, 154.
Barden, S.C., T. Armandroff, P. Massey, *et al.* 1993. ASP Conf. Proc. 37, *Fiber Optics in Astronomy II*, ed. P. M. Gray. (San Francisco: Ast. Soc. Pacific), p. 185.
Barden, S.C., J.A. Arns, W.S. Colburn, & J.B. Williams 2000. *Pub. Ast. Soc. Pacific* **112**, 809.
Beer, R., R.B. Hutchison, R.H. Norton, & D.L. Lambert 1972. *Ap. J.* **172**, 89.
Bell, R.J. 1972. *Introductory Fourier Transform Spectroscopy*. (New York: Academic).

Bloemhof, E.E., R.G. Dekany, M. Troy, & B.R. Oppenheimer 2001. *Ap. J.* **558**, L71.
Brault, J.W. 1982. *Phil. Trans. Roy. Soc. London*, Ser. A, **307**, 503.
Breckinridge, J.B. 1971. *Appl. Optics* **10**, 286.
Calatroni, J.A. & M. Garavaglia 1973. *Appl. Optics* **12**, 2298.
1974. *Appl. Optics* **13**, 1009.
Chaffee, F.H. & D.J. Schroeder 1976. *Ann. Rev. Ast. Ap.* **14**, 23.
Connes, P. 1961. *Rev. Optics* **40**, 45, 116, 171, 231.
1969. *Theory and Observation of Normal Stellar Atmospheres*, ed. O. Gingerich. (Cambridge: MIT Press), p. 323.
1970. *Ann. Rev. Ast. Ap.* **8**, 209.
Conti, G., E. Mattaini, L. Chiappetti, *et al.* 2001. *Pub. Ast. Soc. Pacific* **113**, 452.
Coudé du Foresto, V., M. Faucherre, N. Hubin, & P. Gitton 2000. *Ast. Ap. Suppl.* **145**, 305.
Coulman, C.E. 1985. *Ann. Rev. Ast. Ap.* **23**, 19.
Diego, F. 1985. *Pub. Ast. Soc. Pacific* **97**, 1209.
Doyon, R., S. Thibault, D. Nadeau, & P. Vallee, 2000. *SPIE* **4008**, 1103.
Dyck, H.M. & R.R. Howell 1983. *Pub. Ast. Soc. Pacific* **95**, 786.
Epps, H. & S.S. Vogt 1993. *Appl. Opt.* **32**, 6270.
Felenbok, P. & J. Guerin 1988. *Instrumentation for Ground Based Optical Astronomy*, ed. L. B. Robinson. (New York: Springer-Verlag), p. 260.
Furenlid, I. & O. Cardona 1988. *Pub. Ast. Soc. Pacific* **100**, 1001.
Gray, D.F. 1986. *Instrumentation and Research Programmes for Small Telescopes*, I.A.U. Symposium No. 118, ed. J. B. Hearnshaw & P. L. Cottrell. (Dordrecht: D. Reidel), p. 401.
1997. *Instrumentation for Large Telescopes*, ed. J. M. Rodriguez Espinosa, A. Herrero, & F. Sanchez. (Cambridge: Cambridge University Press), p. 173.
Griffin, R.F. 1968. *A Photometric Atlas of the Spectrum of Arcturus.* (Cambridge: Cambridge Phil. Soc.).
Hackenberg, W.K. & D. Bonaccini 2000. *SPIE* **4007**, 258.
Harrison, G.R. 1973. *Appl. Optics* **12**, 2039.
Hinkle, K., L. Wallace, & W. Livingston 1995. *Infrared Atlas of the Arcturus Spectrum, 0.9–5.3 μm.* (San Francisco: Ast. Soc. Pacific).
Hubbard, E.N., J.R.P. Angel, & M. S. Gresham 1979. *Ap. J.* **229**, 1074.
Ismail, M.A. 1986. *Ap. Space Sci.* **122**, 1.
Jaquinot, P. 1960. *Rept Progress Phys.* **23**, 267.
Kohl, J.L. & W.H. Parkinson 1976. *Ap. J.* **205**, 599.
Kurucz, R.L., I. Furenlid, J. Brault, & L. Testerman 1984. *Solar Flux Atlas from 296 to 1300 nm.* (Sunspot: National Solar Observatory).
Lacy, J.H., M. J. Richter, T.K. Greathouse, D.T. Jaffe, & Q. Zhu 2002. *Pub. Ast. Soc. Pacific* **114**, 153.
Maillard, J.P. & G. Michel 1982. *Instrumentation for Astronomy with Large Optical Telescopes*, I.A.U. Coll. 67, ed. C.M. Humphries. (Dordrecht: Reidel), p. 213.
Mandel, H. 1987. *The Impact of Very High S/N Spectroscopy on Stellar Physics*, I.A.U. Symp. 132, ed. G. Cayrel-de-Strobel & M. Spite. (Dordrecht: Reidel), p. 9.
Marschall, L.A. & L.M. Hobbs 1972. *Ap. J.* **173**, 43.
Mertz, L. 1965a. *Transformations in Optics.* (New York: Wiley).
1965b. *Ast. J.* **70**, 548.
Meyer, C. F. 1934. *The Diffraction of Light, X-Rays, and Material Particles.* (Chicago: Univ. Chicago), Appendix F.
Moffat, A.F.J. 1969. *Ast. Ap.* **3**, 455.

Montes, D., J. López-Santiago, M. J. Fernández-Figueroa, & M. C. Gálvez 2001. *Ast. Ap.* **379**, 976.
O'Byrne, J. W. 1988. *Pub. Ast. Soc. Pacific* **100**, 1169.
Parry, I. R., C. D. Mackay, R. A. Johnson, *et al.* 2000. *Optical and IR Telescope Instrumentation and Detectors*, ed. M. Iye & A. Moorwood, *SPIE* **4008**, 1193.
Piccirillo, J. 1973. *Pub. Ast. Soc. Pacific* **85**, 278.
Pilachowski, C., H. Dekker, K. Hinkle, *et al.* 1995. *Pub. Ast. Soc. Pacific* **107**, 983.
Racine, R. 1989. *Pub. Ast. Soc. Pacific* **101**, 437.
Ridgway, S. T. & J. W. Brault 1984. *Ann. Rev. Ast. Ap.* **22**, 291.
Robinson, L. B., ed. 1988. *Instrumentation for Ground-Based Optical Astronomy*. (New York: Springer-Verlag).
Rowland, H. A. 1893. *Philos. Mag.* **35**, 397.
Ryan, S. G., T. Kajino, T. C. Beers, *et al.* 2001. *Ap. J.* **549**, 55.
Schnopper, D. J. & R. I. Thompson 1974. *Methods of Experimental Physics: Astrophysics*, Part A, ed. N. Carleton. (New York: Academic), p. 491.
Schroeder, D. J. 1974. *Methods of Experimental Physics: Astrophysics*, Part A, ed. N. Carleton. (New York: Academic), p. 463.
Soderblom, D. R., B. F. Jones, & D. Fischer 2001. *Ap.J.* **563**, 334.
Stroke, G. W. 1967. *Handb. Phys.* **29**, 426.
Taylor, K. & N. G. Douglas 1995. I.A.U. Coll. 149, *ASP Conf. Ser.* 71, p. 33.
Vanasse, G. A. & H. Sakai 1967. *Progr. Opt.*, **6**, 258.
Vogt, S. S. 1987. *Pub. Ast. Soc. Pacific* **99**, 1214.
Vogt, S. S., S. L. Allen, B. C. Bigelow, *et al.* 1994. *SPIE* **2198**, 362.
Weiss, W. W., M. Barylak, J. Hron, & J. Schmiedmayer 1981. *Pub. Ast. Soc. Pacific* **93**, 787.
Wood, R. W. 1935. *Phys. Rev.* **48**, 928.
Young, A. T. 1974. *Ap. J.* **189**, 587.

Questions and exercises

1. Sketch qualitative graphs of the diffraction patterns of four hypothetical transmission gratings of infinite width having $b = d/3$, $b = d/2$, $b = 2d/3$, and $b = d$, where b is the slit width and d is the slit spacing. Explain what you see.

2. List the reasons why reflection gratings are better than transmission gratings for most astronomical spectrographs.

3. What do we mean by the term "blazed grating?" How are gratings blazed? Why does blazing not change the angular position of both the diffraction envelope *and* the interference pattern?

4. Suppose you are designing a reflection grating having 300 lines/mm to work in the fifth order at 6000 Å. What facet angle, ϕ, is needed to position the blaze for optimum efficiency? Assume $\alpha = \beta$ for 6000 Å light. For this grating, what is the angular separation of 6000 Å light from 6500 Å light as the beams leave the grating?

5. Draw a schematic diagram of a coude spectrograph using transmission optics for clarity. Show and label all relevant components. Show two rays of starlight entering your spectrograph. Draw the path of these rays as they pass through the

spectrograph, ending at the light detector. Assume that the starlight is composed of only two wavelengths. Indicate how the rays of light behave after leaving the grating, where the spectrum is formed, and the orientation of the wavelength scale.

6. Define spectral resolving power. Explain what factors determine it and how one goes about optimizing it for stellar observations.

7. Consider a coude spectrograph with two cameras, one having a focal length of 2080 mm, and the other 669 mm. The focal length of the collimator is 6960 mm. Suppose the grating, which has 316 lines/mm, is used in the ninth order at 6250 Å. Assume for convenience that $\alpha = \beta = 60°$. What linear dispersions (in Å/mm) are achieved for these two cameras? If the entrance slit is 200 μm (0.200 mm) wide, what spectral resolution (in Å) can one expect for these two cameras?

4
Light detectors

Light detectors have the important job of converting stellar photons into recordable signals. Remarkable and wonderful developments in detectors have occurred over the last two decades, resulting in dramatic increases in speed and improved signal-to-noise ratios. Most notable for stellar-photospheric studies are the silicon array detectors such as the charge-coupled devices (CCDs) and self-scanned arrays. To make quality observations of stellar spectra, the observer must be cognizant of the basic detector parameters: quantum efficiency, spectral response, linearity, noise, and spatial resolution. These properties can be defined for any detector, and the reader can extend the concepts of this chapter to the detector of choice for the job to be done.

Detectors can be grouped into "integrating" and "pulse-counting" types. In the first class, photons are accumulated for some integration time, and then the total signal is measured. Examples are (unintensified) silicon-diode detectors and the photographic process. In the second class, electrical pulses for individual photons are recorded. Examples include photomultipliers and various kinds of electron image tubes. Integrating-silicon devices tend to be used for higher signal-to-noise ratio work and pulse-counting ones for lower signal-to-noise ratios. A high signal-to-noise ratio is needed for much of our work on stellar photospheres.

Along with the detector itself, we often have electronics. Digital and logic components control the detector, issuing integration start pulses, clock sequencing to route the signal through amplifiers and pulse shapers, switches to start and stop charge integrators, and so on. Analog electronics are involved in amplifying the signal, integrating the charges from each sensing element, and in supplying highly stable reference voltages. In most cases, the analog signal is digitized by an analog-to-digital converter and fed into a computer where further manipulation and analysis is performed on what the detector originally produced from the star's photons.

Quantum efficiency and spectral response

Quantum efficiency is defined as

$$q(\lambda) = \frac{\text{number of photons detected}}{\text{number of photons incident}}, \tag{4.1}$$

and is to be thought of as the probability of a photon causing a recordable event. It is very important to have q as large as possible, especially when dealing with the low light levels encountered in astronomy. The size of the output is proportional to $q(\lambda)$, making it of fundamental importance.

The spectral response is a statement of the change in sensitivity of a detector with wavelength. Specifying $q(\lambda)$ is one way of defining the spectral response. The responsivity is sometimes given instead. It specifies, as a function of wavelength, the charge generated by the detector per unit optical energy into the detector. Suppose the radiation incident on the detector has a strength of J joules and consists of N photons. Then

$$J = Nhc/\lambda$$

in which h is Planck's constant, λ is the wavelength of the light, and c is the velocity of light. The charge generated in the detector is

$$Q = q(\lambda)eN,$$

where e is the electronic charge. Combining these two equations gives the responsivity

$$\begin{aligned} r &= \frac{Q}{J} = \frac{e\lambda}{hc} q(\lambda) \\ &= 8.066 \times 10^{-5} \lambda q(\lambda)\,\mathrm{C/J} \end{aligned} \tag{4.2}$$

for λ in angstroms. Si-diodes maintain high quantum efficiency well into the red, whereas photomultipliers and photographic detectors usually peak in the blue. Typical curves are compared in Fig. 4.1. Amazingly high quantum efficiencies are reached by silicon detectors, typically 40% at their peak, but up to ~90% with special thinning of the silicon layer. Notice the stark contrast with photographic values. Photocathodes lie in between, and can be improved somewhat with multiple-reflection devices (Oke & Schild 1968, Gunter *et al.* 1970).

Linearity

When the output of a detector is exactly proportional to the amount of incident radiation, we have a linear detector. Linearity is important because it

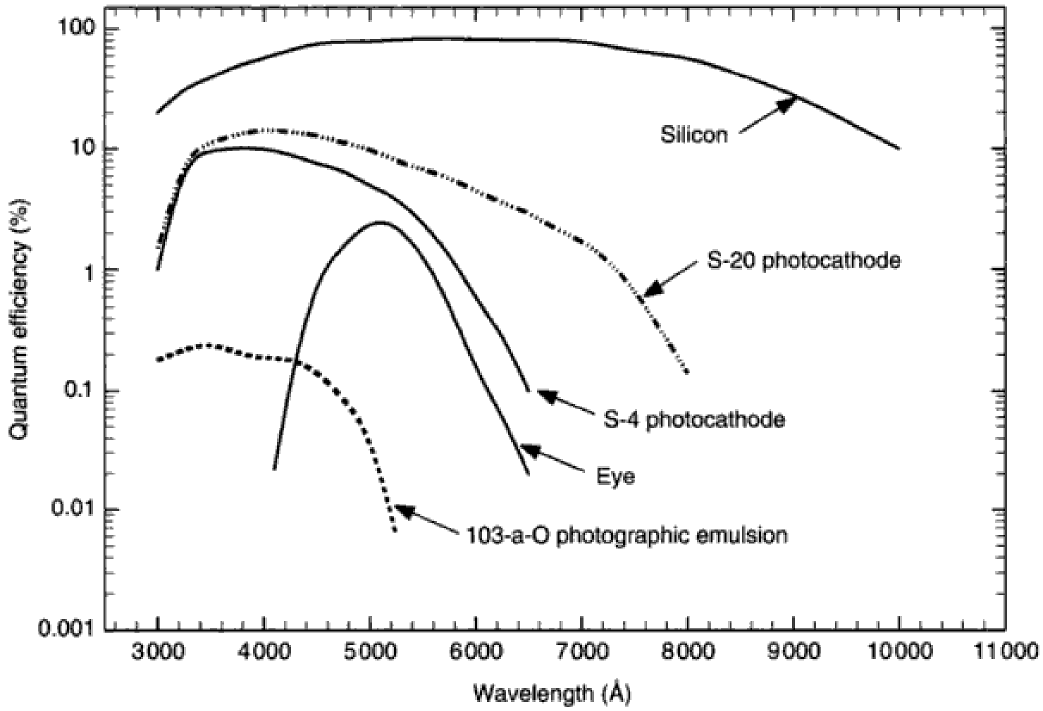


Fig. 4.1. The percentage quantum efficiency is shown for a few characteristic detectors. Variations of as much as a factor of 2 can occur between detectors having the same nominal detecting surface. The 103-a-O photographic emulsion was commonly used in astronomy in the "photographic" era. The S-4 cathode in the P21 photomultiplier was used for the *UBV* photometric system. The advent of the S-20 surface greatly improved measurements in the red. The curve shown for silicon is for a thinned CCD with a peak quantum efficiency exceeding 80%.

means that the ratio of the outputs from two measurements is equal to the ratio of the numbers of photons received during the measurements. For example, a spectral line profile can be specified directly as the detector output as a function of wavelength across the profile. When linearity is lacking, a response calibration must be used to map the detector output into something proportional to the number of photons it saw. Not only is this an inconvenience, it also results in loss of accuracy.

Linearity in detectors depends in the first instance on a photoelectric effect. A fraction, $q(\lambda)$, of the incident photons will produce an electron–hole pair in a Si-diode, or an electron will be emitted from the surface of a photocathode. As long as the energy of the photons exceeds the work function (characteristic of the material), the number of emissions will be statistically proportional to the number of incident photons. The "statistical" aspect is usually ignored under the assumption that even for relatively faint sources the actual number of photons will be large.

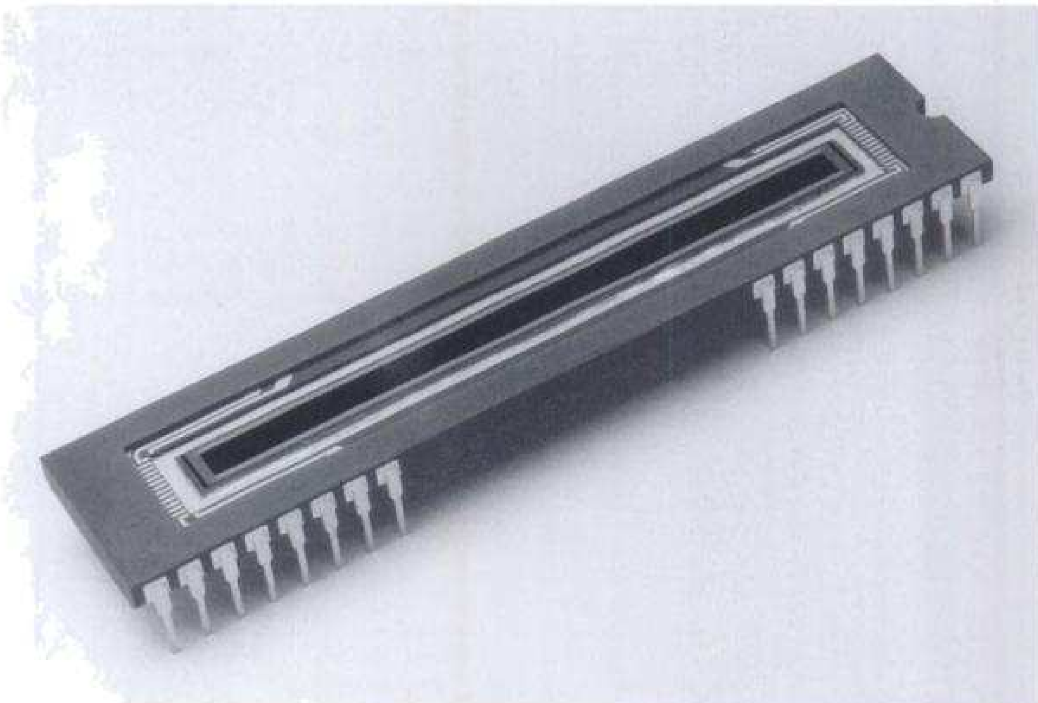


Fig. 4.2. This is a Reticon self-scanned array having a single row of 2048 diodes, each 15 microns wide by 0.8 mm long. Photograph courtesy of EG&G Reticon. Reticons have largely been replaced by CCDs.

Linearity depends in the second instance on the electronics involved in handling the charge after detection, including the manipulation of charge across other diodes, as in the case of CCDs, and through electrical switches to output lines, as well as amplification and digitization of the signal. Linearity must be maintained at each step to give us what we need. For example, Vogt *et al.* (1978) conclude that self-scanned arrays can be linear over more than four orders of magnitude.

Silicon-diode arrays

Silicon devices, like those illustrated in Figs. 4.2 and 4.3 rely on the formation of an electron–hole pair when the photon is absorbed in a silicon pixel or diode that has been charged to ~ 5 V. In effect, each pixel of the array acts like a small charged capacitor. Charges are moved onto the pixels prior to exposure to starlight, and the signal is read from them by square-wave clock signals that either turn switches on and off or that alter the barrier potentials in the material surrounding the pixel. In a self-scanned array, with pixels arranged like a picket fence and so is one dimensional, this process amounts to connecting diodes sequentially to the output or "video" line, so that a series of signal

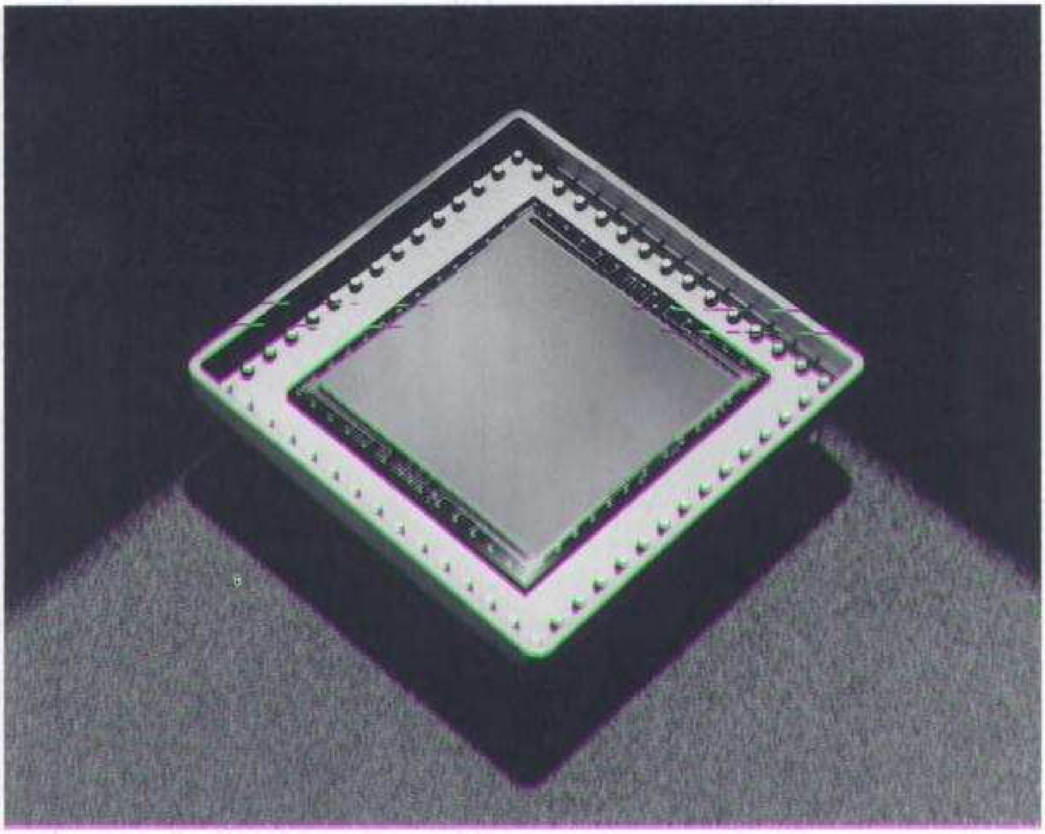


Fig. 4.3. CCDs typically have a two-dimensional array of diodes, each $\sim 20 \times 20\ \mu m^2$. This Tektronix device has 1024 diodes in each dimension. Photograph courtesy of Tektronix, Inc.

pulses comes out in phase with the clocking frequency. Readout rates of ~ 10 kHz are typical. This gives a window of $\sim 100\ \mu s$ per diode to process and digitize the signal pulse from each diode. In a spectrograph, when the spectrum is spread along a self-scanned array, the total number of charge pulses equals the number of spectral elements. CCDs, by contrast, are two-dimensional devices, and the height of the spectrum (perpendicular to dispersion) usually covers several rows of pixels. The first row is clocked out through an amplifier built into the CCD chip. The charges of subsequent rows are forced to migrate down in row number by altering the potential barriers between rows. How well this is accomplished is termed the "charge-transfer efficiency." Typical values range from 0.999 50 to 0.999 999. Although these numbers seem very good, after transfer over 1024 or 2048 rows, the losses can be significant, i.e., 0.999 900 to the 2048th power is only 81%.

Eventually all of these rows are added to give the one-dimensional spectrum. This can be done electrically on the CCD, a technique that is useful when the signal is exceedingly weak and fighting to stay above the noise, or after the readout, during computer manipulation of the data. On-chip binning typically produces far fewer numbers, making readout much faster and requiring less storage space. However, for higher signal levels, the on-chip amplifiers are likely to saturate with the larger binned signals, ruining the data. Cosmic ray hits are also better cleaned from unbinned data (see below).

After the signal leaves the detector, various types of operational amplifiers are used to toughen and strengthen the charge pulses, convert them to voltage pulses, and shape and integrate the pulses. A sensitive and stable amplifier, a "preamp," having high electronic gain is placed as close to the array as possible. The need for high sensitivity and stability is obvious. The high gain and close proximity is a matter of noise suppression. The noise of an analog electrical system varies inversely as the gain of the *first stage* of amplification. Numerous other electrical design factors need to be carefully engineered to keep the readout noise low. Sampling the electrical baseline just before and just after the signal pulse is a typical technique.

Digitization of the signal is the final process before the exposure is stored in computer-readable form. Analog-to-digital (A/D) conversion devices are readily available, but their accuracy, linearity, range, and speed can vary widely. Since the final numbers are generated on a scale set by the A/D conversion, we often see output expressed in ADU, i.e., analog-to-digital units. Translating this to numbers of photons requires a calibration of a gain-conversion factor for the system as a whole. Silicon uses 3.65 eV/photon to produce an electron–hole pair. The conversion factor for a preamp might be ~ 0.2 V generated for each MeV of energy in. There is also likely to be an amplifier in the system with a gain, call it G_{amp}, of between 1 and 50. A typical 12-bit A/D converter can count to $2^{12} = 4096$ steps over a 0–10 V range, giving 2.44×10^{-3} V/ADU. The conversion factor between photons and ADU for this case would be

$$\begin{aligned} u_0 &= \frac{2.44 \times 10^{-3}\ \text{V/ADU}}{3.65\ \text{eV/photon} \times 0.2\ \text{V/MeV} \times G_{\text{amp}}} \\ &= 3.3 \times 10^3 / G_{\text{amp}} \\ &\approx 100\ \text{photons/ADU}, \end{aligned} \tag{4.3}$$

if G_{amp} has a typical value of 30.

Pixel-to-pixel sensitivity can vary by several per cent, even on a very good array. The standard procedure for removing this variation is to divide the stellar exposure by a "flat field" exposure. This is relatively painless, since it involves a short exposure of a continuous spectrum, usually from an incandescent lamp, over the same wavelength span recorded for the star. The signal-to-noise ratio in the flat field should be made several times higher than the value for the stellar exposure so that very little increase in noise results from the flat-field division.

Fringing can cause problems with the flat-field process. Silicon becomes progressively transparent for $\lambda \gtrsim 8000$ Å, and reflections from the front and back of the thin silicon diode interfere; the thinner the silicon, the worse the problem. Since the exact pattern of the fringes depends precisely on the path of the light, small errors in telescope collimation, image slicer adjustment, positioning of the incandescent lamp, the distribution of rays within the field, vignetting, polarization, etc. all conspire to create slightly different fringing for stars in different parts of the sky and for the flat-field lamp. Consequently, the flat-field division does not necessarily remove the fringes and can even enhance them. Fortunately fringing is not always a problem, especially at visible wavelengths and with the thicker silicon chips.

More technical information can be found in Jacoby (1990) and Howell (1992).

Background and cosmic rays

Common to all detectors is an output generated in the absence of illumination. Only the output above the background constitutes real signal, and the linearity of a detector is effectively destroyed if the background is not carefully treated.

Thermal leakage at room temperature can be high enough to preclude the use of some detectors. Fortunately thermal leakage falls exponentially with temperature, and it is universal practice to refrigerate detectors for this reason. Dry ice and liquid nitrogen are common refrigerants used to bring the temperatures down to ~ -30 to $-150\,°$C, depending on the needs of the detector. Frequently, small vacuum chambers are used to house cooled detectors. After simple evacuation by a mechanical pump, vacuum generated with silica gel or Xelite cooled by the refrigerant can readily reduce the pressure to $\sim 10^{-5}$ mb, thereby supplying enough thermal insulation to prevent condensation and reduce the drain on the cooling system. Integration times of several hours are routinely possible with silicon detectors cooled in this manner. Thermal emission from photomultipliers can usually be brought to less than one electron per second per square millimeter of cathode area.

Cooling also affects the spectral response. Generally there is a decline in red response for cooler temperatures. The effect can be dramatic, amounting to as much as a factor of 10 for a 100 °C change in temperature (Young 1963, Vogt *et al.* 1978). Wide spectral coverage, as in stellar energy distribution measurements, therefore requires temperature stabilization to better than 1 °C.

Some detectors, self-scanned arrays in particular, have sizable switching transients that can be stabilized and reduced by careful adjustment of the readout clocks, but not eliminated. This output is referred to as the fixed pattern signal. It then becomes part of the background electrical signal that must be subtracted from both the dark and the stellar exposures. With CCDs, the electrical background is called a "bias" exposure. Generally one must subtract a bias exposure from the dark exposure in order to scale the dark exposure to the exposure time of the star before the dark can be subtracted from the stellar exposure.

Cosmic rays pepper a typical CCD during any exposure. The rate and density depend on the location on the Earth, altitude, and shielding. At mid-latitudes, at sea-level, a 2 h exposure will show as many as 1 event/mm^2. If not too much on-chip binning has been done, these hits are easy to identify, and interpolation between neighboring pixels can be used to patch over the affected pixel or pixels.

Noise in the measurements

The origins of noise are many. We can usually class them into one of the three categories: (1) photon noise in the starlight; (2) photon noise in the sky background; and (3) equipment noise. The combined fluctuations in output due to the inevitable sources of noise in categories 1 and 2 represents a theoretical minimum against which we compare our equipment. It is possible to come within a few per cent of this minimum under favorable conditions.

Photon noise arises from the irregular times of arrival of the photons. The fluctuations in photon rate are described by Bose–Einstein statistics. For a thermal source of characteristic temperature T, the mean-square fluctuation, ΔN photons/s, is

$$\Delta N = N^{1/2}\left(1 + \frac{1}{e^{h\nu/kT} - 1}\right),$$

and for non-thermal sources, a slightly different relation may hold (Fellgett 1955, Smith *et al.* 1957). Since $h\nu$ is appreciably greater than kT in most cases, we choose to neglect these deviations from classical Poisson statistics and write

$$\Delta N = N^{1/2}. \tag{4.4}$$

This same relation holds for photons coming from the star and those coming from the sky background. Usually the equipment noise is also assumed to follow Poisson statistics. In this case, we think of N as the equivalent number of detected photons that would cause the same noise.

We can combine the main sources of noise as follows. Take N to be the total number of *counts* (notice these are the detected photons) recorded during an integration. Of this number, L counts come from the star and B come from the equipment, sky, and/or other background sources. In practice the signal is obtained from

$$L = N - B,$$

and the statistical uncertainty in L, call it ΔL, is the square root of the quadratic sum of the individual errors, giving

$$\Delta L = (\Delta N^2 + \Delta B^2)^{1/2}$$

or, if we use Eq. (4.4) for Poisson statistics,

$$\Delta L = (N + B)^{1/2}.$$

The signal-to-noise ratio is then

$$\begin{aligned} s/n &= L/\Delta L \\ &= L/(N + B)^{1/2} \\ &= L/(L + 2B)^{1/2}. \end{aligned} \tag{4.5}$$

This equation is fundamental. There are two asymptotic cases of interest. First, consider the regime of small background where $B \ll L$ (and hence $N \approx L$). Then

$$s/n \approx L^{1/2}$$

This is the domain of pure signal-photon noise and represents the ideal, minimum noise, case. Detecting 10^4 photons gives a s/n of 100 or a 1% error. Assuming a constant photon flux with time, the situation improves with the square root of the integration time.

The opposite regime, where $L \ll B$ (making $N \approx B$) implies a small signal on top of a large background – a bad situation to be in. Now Eq. (4.5) gives

$$s/n = L/(2B)^{1/2},$$

and we should strive to increase L by every available means: a larger telescope, better throughput, higher quantum efficiency, and to decrease B: a better detector, darker sky, and so on. Notice that s/n is proportional to L in this regime, which is better than the $L^{1/2}$ dependence of the photon-noise-limited regime we are trying to reach.

In all cases, the integration time needed to maintain a fixed s/n varies inversely with the brightness of the source, since it is the number of counted photons that matter, i.e., L, the product of the detection *rate* times the exposure time. If it took 1 h to reach a signal-to-noise ratio of 500 for a sixth magnitude star, it will take a little more than 2.5 h to do the same for a seventh magnitude star.

Choosing a detector to maximize *s/n*

An observer is frequently faced with a choice among detectors. The paramount concern is which detector will give the best signal-to-noise ratio, i.e., maximize the information we wish to collect. Generalize Eq. (4.5) from the last section by explicitly including the quantum efficiency, $q(\lambda)$, and detector noise for each specific candidate. Let l be the number of incident photons, such that the counted number is $L = q(\lambda)l$. For simplicity, assume the background photons are negligible compared to the noise of the detector, which is often true for spectroscopy, and let r be the electrical ("readout") noise of the detector, making ΔB of the previous section equal to r. Then Eq. (4.5) becomes

$$s/n = \frac{q(\lambda)l}{(q(\lambda)l + r^2)^{1/2}}. \tag{4.6}$$

We see that in principle there are only two aspects of candidate detectors that are involved, the quantum efficiency and the readout noise. The relative importance of each is seen in Fig. 4.4, where two detectors are compared. One of them has a pleasantly high quantum efficiency of 90%, but a readout noise of 400 equivalent photons. The other has a desirable low readout noise of 10 photons, but a quantum efficiency of only 20%. The photon-noise domain for each detector is represented by the asymptotic approach to a slope of 1/2, and

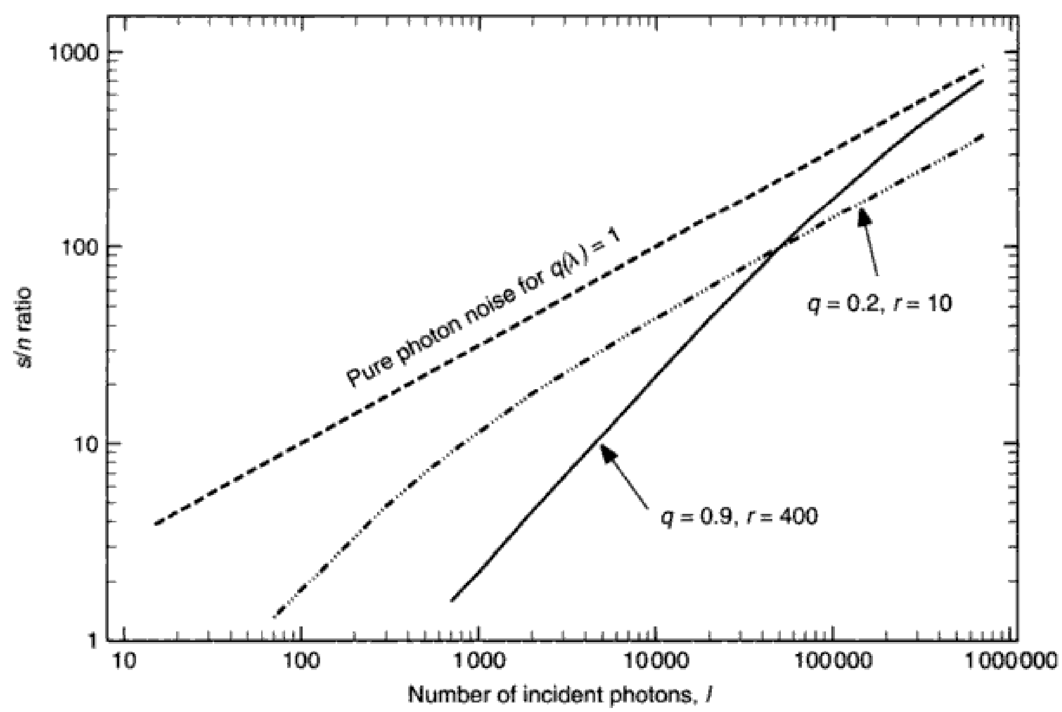


Fig. 4.4. Signal-to-noise ratio increases with the number of incident photons according to Eq. (4.6). At low values, both detectors are in the readout-noise-limited domain, and s/n rises linearly with the number of incident photons. When the slope makes a transition from unity to one-half, the photon-noise domain has been entered. The theoretical limit is shown for comparison (-----). For signal-to-noise ratios greater than about 100, the higher quantum efficiency of the noisier detector is the dominant factor.

the second detector enters this domain even for small signals, i.e., those where $L = q(\lambda)l$ exceeds $\sim 10^3$ photons. The higher curve is the one to be preferred, but the curves cross, and so the choice depends on the s/n level we hope to work at. If we need $s/n \gtrsim 100$, which we usually do in photospheric studies, the high quantum efficiency turns out to be the more crucial parameter, and we choose the first detector even though it has higher readout noise.

A readout noise of ~ 1 photon has been achieved, blurring the division between integrating and pulse-counting detectors.

Dynamic range and well capacity

Even a linear detector with high quantum efficiency will be of little use unless it can work over a substantial range of source brightness. The ratio of the smallest resolvable signal to the full range over which the signal can run is called the dynamic range. A self-scanned array, for example, might have a readout noise of 400 equivalent photons, corresponding to a charge of 400 electrons. Each diode in the array can carry ~ 2 picoCoulomb or 1.2×10^7 electron charges. The dynamic range is then the ratio $1.2 \times 10^7 \div 400 = 3.0 \times 10^4$. In practice such a large dynamic range may be limited by the analog-to-digital converter, which typically has a range of only 4096 to 1 (12 bit A/D converter) or 16384 to 1 (14 bit A/D converter). Dynamic ranges as small as 100 are encountered with some detectors. They are inconvenient to use and incapable of working to high signal-to-noise ratios.

The charge capacity of one pixel is called the well depth. For the self-scanned array in the example used above, the diode size is $15 \times 500\,\mu\text{m}^2$, and the well depth of 12 million charges is enormous. Many other detectors, especially CCDs, have much smaller pixels, typically 15–30 μm square, and they are often thinner as well. The volume available for electron–hole pair formation is correspondingly smaller, and well depths of a few hundred thousand or less is the result. We need to watch the limitation implied on the maximum signal-to-noise attainable per exposure. Since the photon noise is given by $L^{1/2}$, a full well capacity of only 10^4 for example, would limit the precision to $s/n \lesssim 100$ per exposure. In a real case, one must not work too close to the limit of the well capacity because a fully depleted well gives a saturated and useless exposure. So in practice, this device will not give signal-to-noise ratios in excess of 50–70 with a single exposure. In order to push to higher precision, one would then have to add together many exposures. This is acceptable only if we are working in the photon-noise-limited domain. The noise penalty when working in the readout-noise-limited domain is equal to the square root of the number of readouts. Obviously one long and complete integration is preferred in such cases.

Measuring the noise of a detector

Astronomers are interested in the global performance of a detector. Many different phenomena often contribute to the noise of a detector, and while understanding what these sources are crucial for engineering reductions in noise, the size of the total noise is what counts for the stellar measurements.

Perhaps the most basic approach is to make repeated measurements of darkness. The distribution of these measurements (for each individual pixel) will scatter around some mean level, and the amount of scatter is a measure of the noise of that pixel. Figure 4.5 indicates what this looks like. The standard deviation of the distribution in this example is 1.06 ADU, and is the readout noise associated with this pixel. The mean level itself, or bias, is part of the fixed pattern signal that can be subtracted away. The actual measurements are performed in analog-to-digital converter units, and a conversion

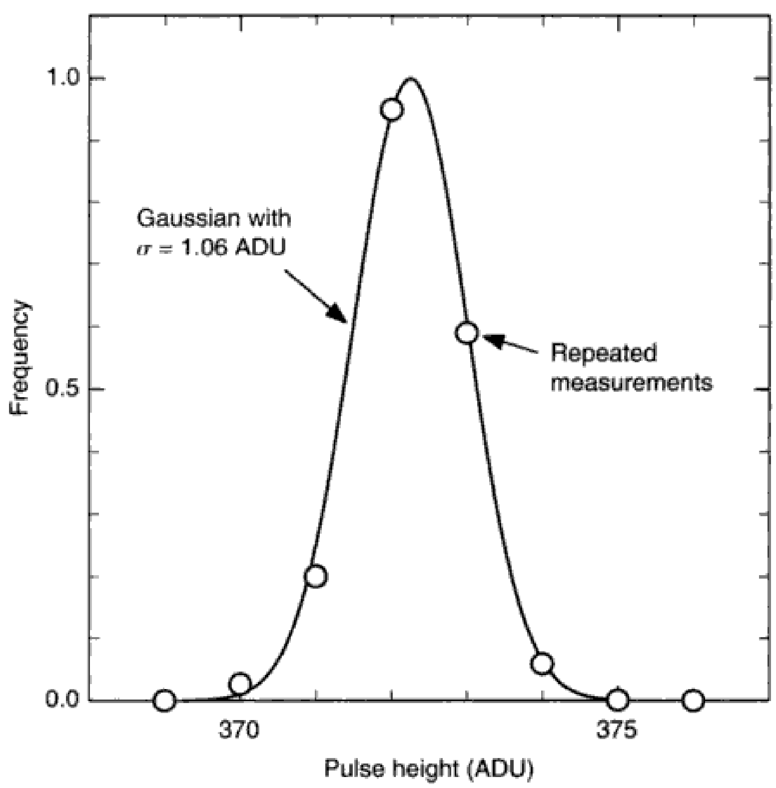


Fig. 4.5. The width of the pulse-height distribution for a single pixel, like this one, is a direct measure of the noise of the detector. The frequency distribution has been normalized to unit height, but the original measurements are the number of values at the pulse heights shown as found from repeated exposures. A Gaussian function is a good match to the observations and tells us that the noise is 1.06 ADU.

factor such as u_0 from Eq. (4.3), is needed to express the pulse width in equivalent photons.

A variation of this method is to calibrate u_0, and consequently r, through the photon noise via Eq. (4.6) rather than using Eq. (4.3). For sufficiently high light levels, photon noise will dominate. As the photon flux is reduced, the noise will deviate from the $L^{1/2}$ behavior according to Eq. (4.6), that is, in a manner depending explicitly on the value of the readout noise, r. In Eq. (4.6), express the counts in ADU using u_0 to denote the conversion factor in photons/ADU,

$$s/n = \frac{u_0 L_{\mathrm{ADU}}}{(u_0 L_{\mathrm{ADU}} + u_0^2 r_{\mathrm{ADU}}^2)^{1/2}} = \frac{L_{\mathrm{ADU}}}{\left(L_{\mathrm{ADU}}/u_0\right) + r_{\mathrm{ADU}}^2} \tag{4.7}$$

Measure the readout noise in ADU by taking repeated exposures of darkness to obtain r_{ADU}. This is essentially what was done in the previous method, but now we take the mean over the whole array. Then we take two identical exposures of a flat continuous spectrum from a lamp and divide them. Without noise, the ratio would be exactly the same number for all pixels. With noise, there is scatter about the mean number, and the root-mean-square of this scatter combined with the known mean lamp signal level, L_{ADU}, gives the left-hand side of Eq. (4.7). Armed with L_{ADU} and r_{ADU} to use on the right-hand side of the equation, we find u_0. The readout noise in equivalent photons is the measured r_{ADU} times u_0. Still better, run a series of lamp exposures at various brightness levels and fit the results into a grid calculated from Eq. (4.7) for assumed values of u_0, as illustrated in Fig. 4.6.

In the case of true photon-counting equipment, the noise is explicitly accounted for in Eq. (4.5) through the background counts B.

Spatial resolution

The ability of a detector to distinguish between closely spaced images diminishes as the image separation decreases. In order to speak of spatial resolution at all, we presuppose the detector is a multi-element device such as a phosphor on a television screen, a silicon-diode array, or a photographic plate. Each detector has a "graininess" or pixel size that filters the signal impressed upon it. Think of an image composed of many Fourier frequencies; the broad contrast variations cover a large area and are described by low spatial frequencies, while the sharp features need the high spatial frequencies to describe them.

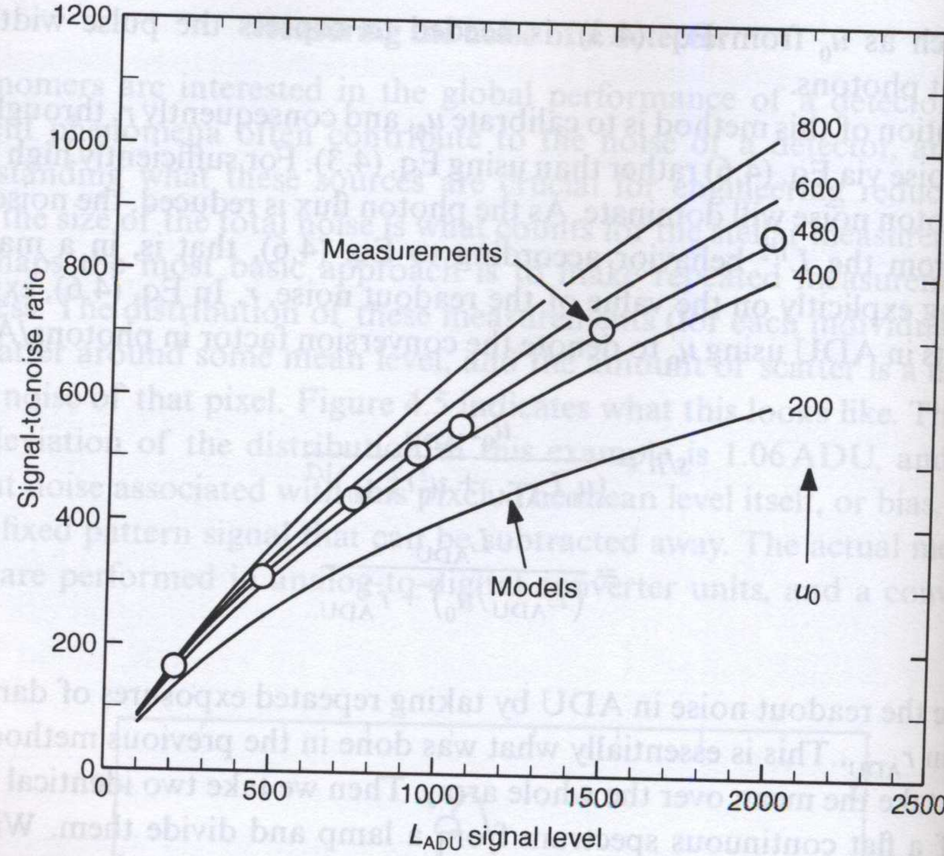


Fig. 4.6. The change of signal-to-noise ratio as a function of signal level can be used to measure the readout noise of the detector. The detector in this example has a readout noise of about 480 equivalent photons/ADU.

Fourier components having wavelengths smaller than the graininess of the detector are lost. But the transition from the recorded information to the lost information is continuous. A specification of the fraction of the Fourier amplitude that is actually recorded as a function of spatial frequency tells us exactly how this transition occurs.

Measurements can be made directly in the spatial frequency domain by placing in front of the detector a mask having a sinusoidal variation in density along one coordinate. The amplitude of the modulation as recorded by the detector will be diminished according to how much that spatial frequency is attenuated by the lack of resolution in the detector. Measurements on many masks, covering a wide range of spatial frequency, allows the filtering properties, or modulation transfer function (MTF), of the detector to be mapped out. Many common test patterns are made of black and white stripes rather than a sinusoidal variation in density. The fundamental periodicity of the stripe pattern is the last Fourier component to fade however, so that both types of patterns give the same result. A typical mask is shown in Fig. 4.7. Modulation transfer functions usually show a monotonically decreasing amplitude with increasing spatial frequency. (An example is shown in Fig. 4.8.) In lieu of a full description, some manufacturers specify a single point on the curve, such as 50 lines/mm at 30% contrast. But this is truly minimal information since the shape of the transfer functions can vary widely.

One can measure what amounts to the instrumental profile of the detector by illuminating it with a point or line image much smaller than the pixel size, i.e., a δ-function. The output of the detector for this arrangement is the Fourier transform of the modulation transfer function.

The instrumental profile associated with the detector alone is usually of limited interest to the astronomer. Instead, we need the full instrumental profile of the spectrograph, which includes the contribution from the detector. This is discussed in Chapter 12.

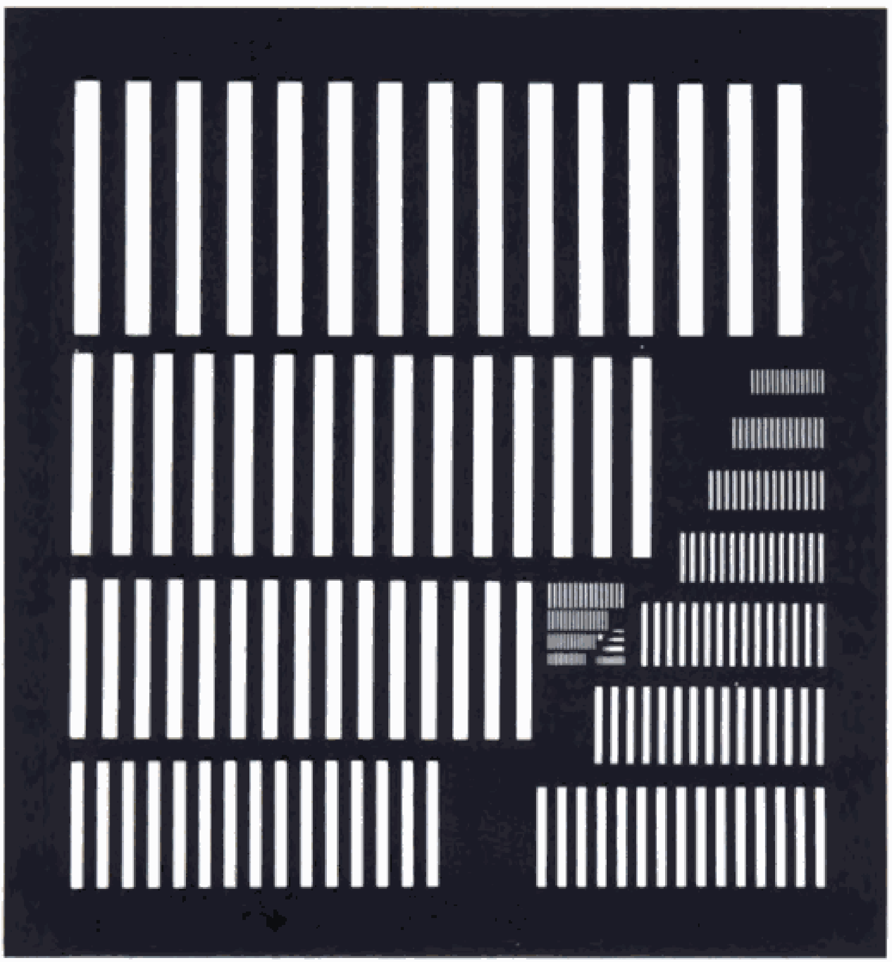


Fig. 4.7. Spatial resolution can be tested using a bar pattern like this one. Courtesy of The Ealing Corporation.

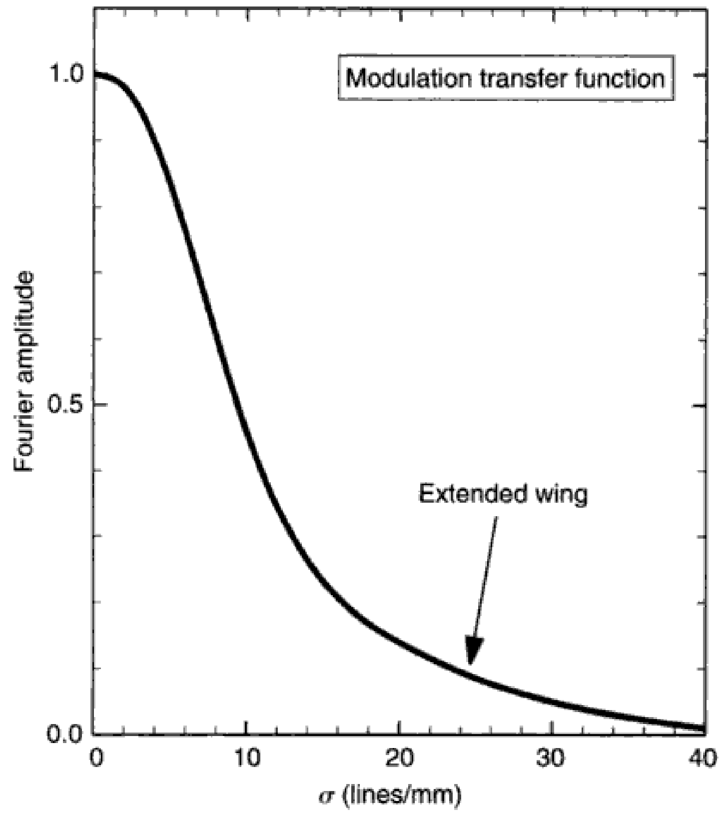


Fig. 4.8. A three-stage image tube produced this MTF.

Photomultiplier tubes

The photomultiplier is the classic pulse-counting detector (see Fig. 4.9). Although it has many desirable properties, including linearity and a quantum efficiency of ~20%, it remains a single-element detector and therefore is ~N times less efficient than an array detector having N elements. Electrons are emitted from the cathode when a photon is absorbed. This electron is multiplied by 10^6–10^9 times by cascade through a dynode chain which operates at relatively high voltage. Unlike silicon detectors, photomultipliers can be destroyed by excessive illumination.

The dynode chain is a near-perfect amplifier (Lallemand 1962, Engstrom 1963), and yields pulses lasting ~10^{-8} s at the output (anode). These pulses can be counted with appropriate amplifiers and pulse-height discriminators (Morton 1968) up to rates of 10^7/s, although coincidence corrections need to be applied for the higher count rates. Photomultipliers can also be used in a direct current mode in which the current from the tube is measured instead of individual pulses. More details can be found in the first edition of *Photospheres* (Gray 1976), Schonkeren (1970), and Young (1974).

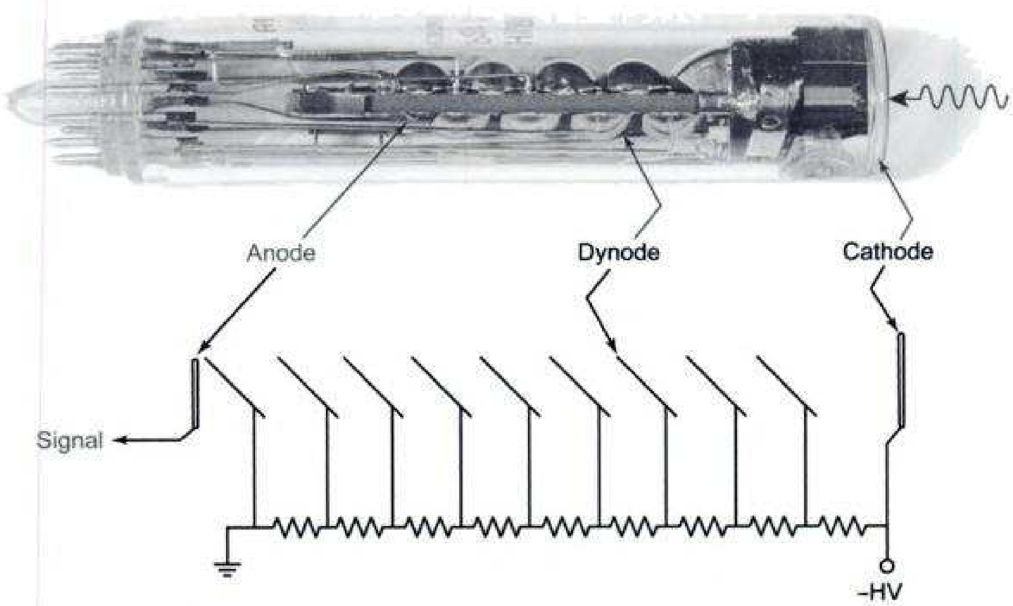


Fig. 4.9. The internal structure of a photomultiplier can be seen here. The light strikes the light-sensitive cathode on the right where the initial photo-electron is emitted. Several electrons are generated at each dynode as electrons from the previous stage are accelerated through ~100 V between dynodes. The high voltage is distributed along the dynodes by the resistor string.

The photographic plate

The wide use of photography prior to the 1970s has now been almost completely superseded by electronic detectors. Photographic processes suffer from a non-linear response, low quantum efficiency (see Fig. 4.1), and other effects detrimental to precision spectroscopy. The reader is referred to Chapter 4 of the first edition of this book (Gray 1976), Wright *et al.* (1964), or Marx (1988) for details. The one advantage still held by photography is the huge number of pixels per exposure. Technology is currently struggling to produce CCDs with a few million elements. By contrast, a typical 40×40 cm^2 photographic plate, can have $\sim 10^9$ detection elements. Unfortunately for photography, there is now only limited use for vast amounts of low-precision data.

References

Engstrom, R. W. 1963. *Phototubes and Photocells*, RCA Tech. Manual PT-60, pp. 6, 28.

Fellgett, P. B. 1955. *Vistas in Astronomy*, Vol. 1, ed. A. Beer. (London: Pergamon), p. 475.

Gray, D. F. 1976. *The Observation and Analysis of Stellar Photospheres*, 1st edn. (New York: Wiley), Chapter 4.

Gunter, W. D., Jr., G. R. Grant, & S. A. Shaw 1970. *Appl. Optics* **9**, 251.
Howell, S. B., ed. 1992. *Astronomical CCD Observing and Reduction Techniques*, Ast. Soc. Pacific Conf. Ser. **23**.
Jacoby, G. H., ed. 1990. *CCDs in Astronomy*, Ast. Soc. Pacific Conf. Ser. **8**.
Lallemand, A. 1962. *Astronomical Techniques*, ed. W. A. Hiltner. (Chicago: Univ. Chicago), p. 126.
Marx, S., ed. 1988. *Astrophotography*. (Berlin: Springer-Verlag).
Morton, G. A. 1968. *Appl. Optics* **7**, 1.
Oke, J. B. & R. E. Schild 1968. *Appl. Optics* **7**, 617.
Schonkeren, J. M. 1970. *Photomultipliers*, Phillips Application Book. (Eindhoven: Phillips), pp. 6, 28, 116.
Smith, R. A., F. E. Jones, & R. P. Chasmar 1957. *The Detection and Measurement of Infra-red Radiation*, 1st edn. (Oxford: Oxford Univ. Press), p. 209.
Vogt, S. S., R. G. Tull, & P. Kelton 1978. *Appl. Optics* **17**, 574.
Wright, K. O., E. K. Lee, T. V. Jacobson, & J. L. Greenstein 1964. *Pub. Dom. Ast. Obs. Victoria* **12**, No. 7.
Young, A. T. 1963. *Appl. Optics* **2**, 51.
1974. *Methods of Experimental Physics: Astrophysics: Part A: Optical and Infrared*, Vol. 12, *Astrophysics*, ed. N. Carleton. (New York: Academic). p. 1.

Questions and exercises

1. What is the photon noise in a spectrum in which 100 total photons have been recorded per resolution element? How would this change if 100 000 photons were recorded instead? What factors might you consider changing that would enable you to go from 100 to 100 000 photons.

2. Two light detectors have the following parameters: detector 1, peak quantum efficiency is 0.92, readout noise is 150 equivalent photons; detector 2, peak quantum efficiency is 0.40, readout noise is one equivalent photon. Which detector is more effective for faint sources and low signal-to-noise ratio work? Is the opposite one then preferred for bright sources and high signal-to-noise work? Explain. At what signal-to-noise ratio do these two detectors perform equally well?

5

Radiation terms and definitions

We must have a carefully defined terminology to properly describe light and its interaction with the photospheric material. This short chapter introduces these basic quantities. We consider the distribution of photons with wavelength, angle, position, and time by specifying measurable macroscopic quantities in the limiting case where these quantities become infinitesimal. The meaning of this limit is taken in a practical sense, with infinitesimal meaning smaller than our resolving power for the variable under consideration. In this way we can ignore physical complications due to the uncertainty principle or the atomic "lumpiness" of the gas.

Specific intensity

Consider the radiating surface depicted in Fig. 5.1. This surface might be a physical boundary or it could be an imaginary surface inside a radiating gas. We denote a small portion of this surface by ΔA. The specific intensity, I_ν, is defined by

$$\begin{aligned} I_\nu &= \lim \frac{\Delta E_\nu}{\cos\theta\,\Delta A\,\Delta\omega\,\Delta t\,\Delta\nu} \\ &= \frac{\mathrm{d}E_\nu}{\cos\theta\,\mathrm{d}A\,\mathrm{d}\omega\,\mathrm{d}t\,\mathrm{d}\nu}. \end{aligned} \tag{5.1}$$

The limit is taken as all the small quantities, ΔA, $\Delta\omega$ the increment of solid angle, Δt the increment in time, and $\Delta\nu$ the increment in the spectral band, diminish toward zero. Of course, ΔE_ν diminishes to zero as well. In this way, we define the specific intensity at a "point" on the surface, at a given time, in a direction θ, at a frequency ν in the spectrum. The adjective "specific" in front of the word intensity is often omitted. This same physical quantity goes under the name of "brightness" in radio astronomy.

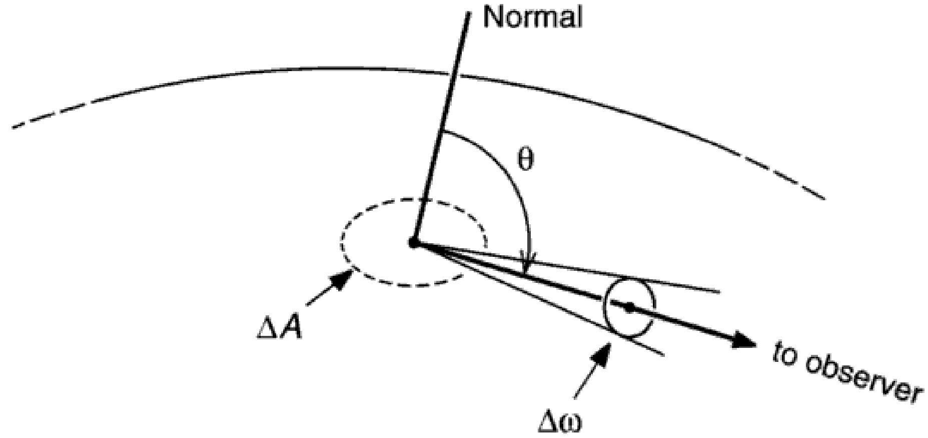


Fig. 5.1. The geometrical portion of the definition of specific intensity is illustrated here. The increment of area, ΔA, is seen foreshortened by $\cos\theta$, where θ is the angle of view from the normal to the surface. An increment of solid angle, $\Delta\omega$, is shown.

One has a choice of units for specific intensity. Equation (5.1) uses frequency units with $\Delta\nu$ in the denominator. We could equally well define I_λ by replacing $\Delta\nu$ with the wavelength increment $\Delta\lambda$. The relation between these two spectral distributions is

$$I_\nu \, \mathrm{d}\nu = I_\lambda \, \mathrm{d}\lambda, \tag{5.2}$$

which follows directly from Eq. (5.1). The cgs units of I_ν are erg/(s cm^2 rad^2 Hz)= erg/(rad^{2}cm^2), where Hz is used to denote cycles per second. The traditional cgs units of I_λ are erg/(s cm^2 rad^2 Å). The corresponding mks units are W/(m^2 rad^2 Hz) and W/(m^2 rad^2 Å).

The two spectral distributions, I_ν and I_λ, have different shapes for the same spectrum. Characteristic features such as intensity maxima appear at different wavelengths. The solar spectrum, for example, has a maximum in the blue (~4500 Å) for I_λ, but the maximum is in the near infrared (~8000 Å) for I_ν. This curious state of affairs can be understood by looking at the relation between spectral bands for the two cases. Since $c = \lambda\nu$, $\mathrm{d}\nu = -(c/\lambda^2)\,\mathrm{d}\lambda$. The minus sign refers to the direction of the coordinates; as λ increases, ν decreases. We see that equal intervals in λ correspond to very different intervals in ν across the spectrum. With increasing λ, a constant $\mathrm{d}\lambda$ (the I_λ case) corresponds to a smaller and smaller $\mathrm{d}\nu$. These smaller $\mathrm{d}\nu$ intervals obviously contain diminishing energy compared to the constant $\mathrm{d}\nu$ intervals associated with I_ν. The prism and grating spectrographs approximately portray I_ν and I_λ, respectively, since the dispersion varies approximately as λ^{-2} for the prism, and is nearly constant with λ for the grating (Eq. 3.6).

In the computation of model photospheres (Chapter 9), I_ν is obtained from the transfer equation, which we meet in Chapter 7. The study of this computation comprises the heart of the *theory* of stellar atmospheres. Some of the remainder of this chapter and portions of several others are preparation for obtaining I_ν for given boundary conditions and a specified geometry.

The mean intensity, J_ν, is defined as the directional average of the specific intensity,

$$J_\nu = \frac{1}{4\pi}\oint I_\nu \,\mathrm{d}\omega, \tag{5.3}$$

where the circle on the integral indicates that the integration is performed over the whole unit sphere centered on the point of interest.

Flux

Flux is a measure of the net energy flow across an area ΔA, over a time Δt, in a spectral range $\Delta\nu$ as these quantities are taken to the infinitesimal limit. The only directional significance is whether the energy crosses ΔA from the top or from the bottom. This total net energy is the sum of all the ΔE_ν values we would need to describe the same radiation using the specific intensity terminology, and we therefore denote it by $\Sigma\Delta E_\nu$. We write the definition of flux as

$$\begin{aligned}\mathfrak{F}_\nu &= \lim \frac{\sum \Delta E_\nu}{\Delta A\,\Delta t\,\Delta\nu} \\ &= \frac{\oint \mathrm{d}E_\nu}{\mathrm{d}A\,\mathrm{d}t\,\mathrm{d}\nu},\end{aligned} \tag{5.4}$$

where again the circle on the integral indicates a complete integration over all directions. We can relate flux and intensity using Eq. (5.1) to substitute for $\mathrm{d}E_\nu/(\mathrm{d}A\,\mathrm{d}t\,\mathrm{d}\nu)$ to obtain

$$\mathfrak{F}_\nu = \oint I_\nu \cos\theta \,\mathrm{d}\omega, \tag{5.5}$$

where θ is shown in Fig. 5.1. (In theoretical stellar atmospheres, a quantity called "astrophysical flux" is often defined. It is equal to the real flux, $\mathfrak{F}_\nu$, divided by π.) When the radiation is isotropic, $\mathfrak{F}_\nu = 0$. In this sense the flux is a measure of the flow of radiation and the anisotropy of the radiation field.

If we look at a point on the physical boundary of a radiating sphere, we can write, using Eq. (5.5)

$$\begin{aligned}\mathfrak{F}_\nu &= \int_0^{2\pi} \mathrm{d}\phi \int_0^{\pi} I_\nu \sin\theta\cos\theta\,\mathrm{d}\theta \\ &= \int_0^{2\pi} \mathrm{d}\phi \int_0^{\pi/2} I_\nu \sin\theta\cos\theta\,\mathrm{d}\theta \\ &\quad + \int_0^{2\pi} \mathrm{d}\phi \int_{\pi/2}^{\pi} I_\nu \sin\theta\cos\theta\,\mathrm{d}\theta \\ &= \mathfrak{F}_\nu^{\mathrm{out}} + \mathfrak{F}_\nu^{\mathrm{in}},\end{aligned}$$

that is, the flux leaving the surface plus the flux entering the surface. In the case at hand, the second term is zero. If, in addition, there is no azimuthal dependence for $\mathfrak{F}_\nu$, we have for the spectrum leaving the star

$$\mathfrak{F}_\nu = 2\pi \int_0^{\pi/2} I_\nu \sin\theta\cos\theta\,\mathrm{d}\theta. \tag{5.6}$$

We compute the theoretical spectrum of a star using this equation.

The simplest configuration occurs when I_ν in Eq. (5.6) is independent of direction over one hemisphere ($0 \le \theta \le \pi/2$). Then

$$\mathfrak{F}_\nu = \pi I_\nu, \tag{5.7}$$

since the integral of $\sin\theta\cos\theta$ gives a value of $\frac{1}{2}$.

The student who is meeting these definitions for the first time will do well to reflect on the difference between I_ν and $\mathfrak{F}_\nu$. For example, I_ν is independent of distance from the source, whereas $\mathfrak{F}_\nu$ obeys the standard inverse square law. Specific intensity can only be measured directly if we can resolve the radiating surface; otherwise we necessarily measure $\mathfrak{F}_\nu$. It is $\mathfrak{F}_\nu$ we measure for most stars for this reason. These differences arise because of the solid angle $\Delta\omega$ appearing in Eq. (5.1) but not in Eq. (5.4).

To elaborate slightly, consider an angularly large and uniformly bright source at a distance r being measured with the telescope and detector shown in Fig. 5.2. The small area marked A_{D} represents a resolution element of the detector. The lens (or the telescope) images the source on the detector, and the source area A_{S} matches A_{D}. The solid angle over which radiation is coming to A_{D} is A_{S}/r^2 or A_{D}/F^2. The surface in Fig. 5.1 corresponds to the lens. We clearly are measuring I_ν because it is the radiation per solid angle that the detector records. Now, as r increases, the source diminishes in angular size, but the area A_{D} is still fully illuminated and the output signal from this detector element

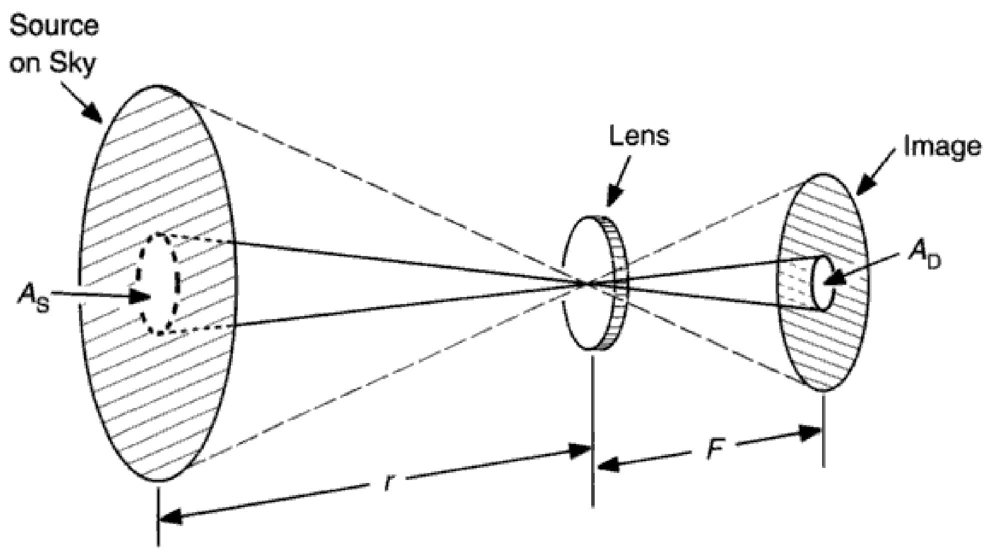


Fig. 5.2. A portion of the uniformly bright source, A_S, corresponds to the detector resolution element, A_D. The focal length of the lens is F. When A_D is only a portion of the full image, the image is resolved and we measure I_ν. As the distance to the source, r, increases, the image of the source becomes smaller until it is completely contained within A_D. We no longer resolve the image. As r increases still further, the whole image becomes smaller than the pixel size and only the flux, $\mathfrak{F}_\nu$, can be measured.

remains constant since the source is uniformly bright over its surface. Eventually, as r increases, the whole source will subtend an area equal to and then less than A_S/r^2. When this happens, the source is no longer resolved, and there is a transition from intensity measurement to flux measurement.

The *K* integral and radiation pressure

It is found to be useful to define, in addition to the mean intensity,

$$J_\nu = \frac{1}{4\pi}\oint I_\nu \, \mathrm{d}\omega$$

and the flux integral

$$\mathfrak{F}_\nu = \oint I_\nu \cos\theta \, \mathrm{d}\omega,$$

a third equation using the second moment of $\cos\theta$, that is,

$$K_\nu = \frac{1}{4\pi}\oint I_\nu \cos^2\theta \, \mathrm{d}\omega. \tag{5.8}$$

Physically this integral is related to the radiation pressure.

Electromagnetic radiation has momentum equal to its energy divided by the velocity of light, c. Consider photons transferring momentum to a solid wall in a manner analogous to the usual discussion of particles in kinetic gas theory. The component of momentum normal to the wall taken per unit time and area is the pressure,

$$\mathrm{d}P_\nu = \frac{1}{c}\frac{\mathrm{d}E_\nu \cos\theta}{\mathrm{d}t\,\mathrm{d}A},$$

where θ is the angle the photon trajectory forms with the normal to the wall (also as in Fig. 5.1). Casting this in terms of specific intensity gives

$$\mathrm{d}P_\nu = \frac{I_\nu}{c}\cos^2\theta\,\mathrm{d}\nu\,\mathrm{d}\omega.$$

The total radiation pressure comes with integration over all directions and over the whole spectrum. Thus

$$\begin{aligned} P_\mathrm{R} &= \frac{1}{c}\int_0^\infty \oint I_\nu \cos^2\theta\,\mathrm{d}\omega\,\mathrm{d}\nu \\ &= \frac{4\pi}{c}\int_0^\infty K_\nu \mathrm{d}\nu, \end{aligned} \tag{5.9}$$

when the definition of K_ν is used.

A frequently used special case assumes that I_ν is independent of direction so that it can be factored out of the directional integration in Eq. (5.9). Upon integration of $\cos^2\theta$, one finds

$$P_\mathrm{R} = \frac{4\pi}{3c}\int_0^\infty I_\nu \mathrm{d}\nu. \tag{5.10}$$

In certain applications, a temperature is defined using Eqs. (5.7) and (6.4) in the next chapter to give

$$\pi\int_0^\infty I_\nu\,\mathrm{d}\nu = \sigma T^4,$$

and the radiation pressure is recast as

$$P_\mathrm{R} = \frac{4\sigma}{3c}T^4. \tag{5.11}$$

In most applications, when Eq. (5.11) is used in place of Eq. (5.10), T is usually taken to be the kinetic gas temperature. Radiation pressure becomes quite important in early-type stars. In solar-type and cooler stars, it can be safely neglected (see Table 9.1 for numerical values).

The absorption coefficient and optical depth

Consider radiation in a specified direction shining on a thin layer of material which itself is so cool that it does not radiate measurably. The intensity of light is found experimentally to be diminished upon passage through the layer by an amount $\mathrm{d}I_\nu$, where

$$\mathrm{d}I_\nu = -\kappa_\nu \rho I_\nu \, \mathrm{d}x. \tag{5.12}$$

Here ρ is the density in mass per unit volume and the layer thickness, $\mathrm{d}x$, has units of length. The absorption coefficient, κ_ν, has units of area per mass, and is therefore a mass absorption coefficient.

There are two physical processes contributing to κ_ν. The first is true absorption where the photon is destroyed and the energy thermalized. The second is scattering where the photon is deviated in direction and removed from the solid angle being considered. So, the absorption coefficient is really an extinction coefficient.

The radiation sees neither $\kappa_\nu \rho$ nor $\mathrm{d}x$ alone. Rather, it is the combination of the two over some path length L given by

$$\tau_\nu = \int_0^L \kappa_\nu \rho \, \mathrm{d}x \tag{5.13}$$

that is important. Here τ_ν is the optical depth (unitless) and x is the geometrical depth (in meters or centimeters, for example). Using optical depth, Eq. (5.12) can be written as

$$\mathrm{d}I_\nu = -I_\nu \, \mathrm{d}\tau_\nu.$$

The solution is $I_\nu = I_\nu^0 \exp(-\tau_\nu)$, the usual simple extinction law. This is a "passive" situation where no emission occurs and is the simplest example of the radiative transfer equation (the main topic of Chapter 7) and its solution.

The emission coefficient and the source function

We treat the emission much the same as we did the absorption. The increment of radiation emitted from an increment of thickness $\mathrm{d}x$ in a specified direction is defined as

$$\mathrm{d}I_\nu = j_\nu \rho \, \mathrm{d}x. \tag{5.14}$$

Here ρ and $\mathrm{d}x$ are the same as for absorption, and the emission coefficient j_ν has units of $\mathrm{erg/(s\,rad^2\,Hz\,g)}$ or $\mathrm{J/(s\,rad^2\,Hz\,kg)}$.

Again the physical processes contributing to j_ν are: (1) real emission, the creation of photons, and (2) scattering of photons into the direction being considered. Scattering in this context rarely refers to diffraction and the like, but is absorption followed immediately by re-emission of a photon from the same atomic transition.

The ratio of emission to absorption has the same units as I_ν, and can be thought of as the specific intensity "emitted" at some "point" in a hot gas. It turns out to be convenient to use this ratio when discussing the transfer of radiation, and it is given a special name, "the source function," and is defined by

$$S_\nu = j_\nu/\kappa_\nu. \tag{5.15}$$

The physics of calculating S_ν can be complicated. Some of these complications are treated in Chapter 13 where spectral lines are considered, for it is with the line radiation that they really become important. For now we consider the two extreme cases of pure isotropic scattering and pure absorption as examples of the source function.

Pure isotropic scattering

In the pure-isotropic-scattering case, all the "emitted" energy is due to photons being scattered into the direction under consideration. We imagine the state of affairs as depicted in Fig. 5.3. The contribution, dj_ν, to the emission from the solid angle $d\omega$ is proportional to $d\omega$ and to the absorbed energy $\kappa_\nu I_\nu$. It is isotropically re-radiated so the fraction per unit solid angle is $1/4\pi$. In this way we can write

$$dj_\nu = \kappa_\nu I_\nu d\omega/4\pi.$$

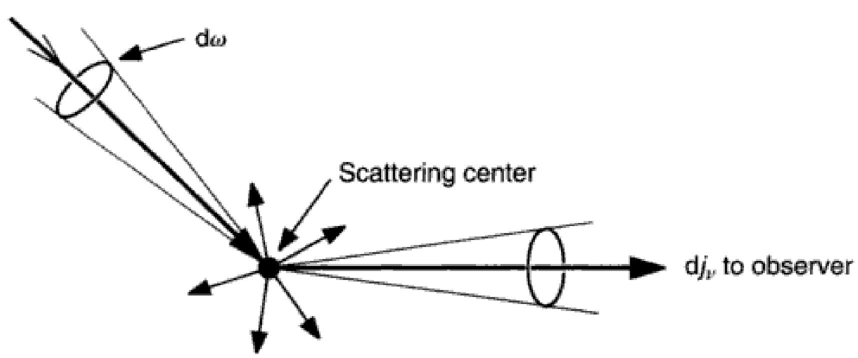


Fig. 5.3. The amount of radiation scattered toward the observer is the sum of the contributions from all increments of solid angle like the one shown, $d\omega$. The arrows around the scattering center indicate that the radiation from $d\omega$ is scattered in all directions. Only the radiation scattered into the infinitesimal solid angle around the direction toward the observer contributes to j_ν.

Since j_ν and κ_ν are both expressed per unit density, the factor of ρ on both sides has been omitted. We obtain all the contributions to j_ν by integrating over the solid angle

$$j_\nu = \frac{1}{4\pi}\oint \kappa_\nu I_\nu \,\mathrm{d}\omega.$$

But κ_ν is usually independent of ω, so

$$S_\nu = \frac{j_\nu}{\kappa_\nu} = \frac{1}{4\pi}\oint I_\nu \,\mathrm{d}\omega$$

or

$$S_\nu = J_\nu.$$

In a word, the source function (or specific intensity "emitted") for the case of pure isotropic scattering is the mean intensity. For this case, the source function depends completely on the radiation field and not at all on the local thermal emission.

Pure absorption

We have the case of pure absorption when all the absorbed photons are destroyed and all the emitted photons are newly created with a distribution governed by the physical state of the material. Common usage of the term has narrowed its meaning to that physical state of the material called thermodynamic equilibrium. Emission from a gas in thermodynamic equilibrium is described by the laws of the black-body radiator, the topic of the next chapter. The source function for this case is given by Planck's radiation law,

$$S_\nu = \frac{2h\nu^3}{c^2}\frac{1}{\mathrm{e}^{h\nu/kT}-1}. \tag{5.16}$$

This is the specific intensity emitted by a gas of temperature T. According to this equation, the source function depends only on the frequency of the radiation and the temperature of the material. The black-body source function is often denoted as B_ν instead of S_ν.

The Einstein coefficients

When we deal with spectral lines or bound–bound transitions, we can describe the probabilities for spontaneous emission in terms of the atomic constants. Consider the spontaneous transition between an upper level, u, and a lower level, l, separated by an energy $h\nu$. We assume the emission is isotropic. Then the probability that the atom will emit its quantum of energy within a time $\mathrm{d}t$ and in a solid angle $\mathrm{d}\omega$ is $A_{ul}\,\mathrm{d}t\,\mathrm{d}\omega$. The proportionality constant, A_{ul}, is the Einstein probability coefficient for spontaneous emission. If there are N_u excited atoms per unit volume, the contribution of spontaneous emission to the emission coefficient is

$$j_\nu\rho = N_u A_{ul} h\nu. \tag{5.17}$$

Additional emission is induced if a radiation field is present that has photons corresponding to the energy difference between the levels u and l. The induced emission is proportional to the amount of radiation, and each newly emitted photon shows phase coherence and a direction of propagation that is the same as that of the inducing photon. This process of stimulated emission is sometimes referred to as a process of negative absorption for this reason. The probability for stimulated emission producing a quantum of energy within the time interval $\mathrm{d}t$ in the solid angle $\mathrm{d}\omega$ is $B_{ul}I_\nu\,\mathrm{d}t\,\mathrm{d}\omega$, where the directional dependence of the specific intensity becomes very important since it fixes the directional dependence of these new photons. B_{ul} is the Einstein probability coefficient for stimulated emission.

The true absorption probability is defined in the same way, and the proportionality constant is denoted by B_{lu}. Notice how these Einstein coefficients have been tied to I_ν and j_ν. Their units are therefore per steradian, and differ by 4π from definitions using energy density.

The mass absorption coefficient for this bound–bound transition can be expressed in terms of these B values by considering the radiation absorbed per unit path length from the intensity beam I_ν, namely,

$$\kappa_\nu\rho I_\nu = N_l B_{lu} I_\nu h\nu - N_u B_{ul} I_\nu h\nu,$$

where N_l is the population of the lower level per unit volume. I_ν can be canceled in the above equation, giving

$$\kappa_\nu\rho = N_l B_{lu} h\nu - N_u B_{ul} h\nu. \tag{5.18}$$

The amount of reduction in absorption from the second term turns out to be only a few per cent in the visible spectrum (refer to Table 8.1). The numerical calculation of κ_ν is the topic of Chapter 8 for continuous absorption and Chapter 11 for line absorption.

Questions and exercises

1. Experiment with some numerical manipulation of spectra to help understand their nature as a distribution (histogram) of photons. For example, start out by making a graph of Eq. (5.16). What units does $I_\nu = S_\nu$ have? As with any histogram, one has some choice in the binning interval. What binning interval have you chosen? How would your graph change if the binning interval were widened by a factor of 10? Now use the basic relations in Eq. (5.2) to convert I_ν to I_λ. Make a graph of I_λ versus λ. Compare this new graph with the I_ν graph. Do they peak at the same wavelength? If not, why not? On the I_λ graph, plot I_ν as a function of λ. Why are these two plots not the same?

2. Prove that the flux from an isotropically radiating surface (upon which there is no incident radiation from the outside) is related to the specific intensity by $\mathfrak{F}_\nu = \pi I_\nu$. Then calculate the flux for a completely isotropic surface (as might be encountered deep inside a star).

6

The black body and its radiation

Think of a container that is completely closed except for a very small hole in one wall. Any light entering the hole has a very small probability of finding its way out again, and eventually will be absorbed by the walls of the container or the gas inside the container. The predicament of the photon is illustrated in Fig. 6.1. We have constructed a perfect absorber – all the light that enters the small hole is absorbed inside. Of course, there is a very small probability that the photon will find the hole and get out. This probability is related to the size of the hole relative to the area of the walls and their roughness and reflection coefficient. So our perfect absorber is not truly perfect, but clearly can be made close enough to perfect that we cannot measure the difference. This type of container, or more specifically the hole, is called a black body.

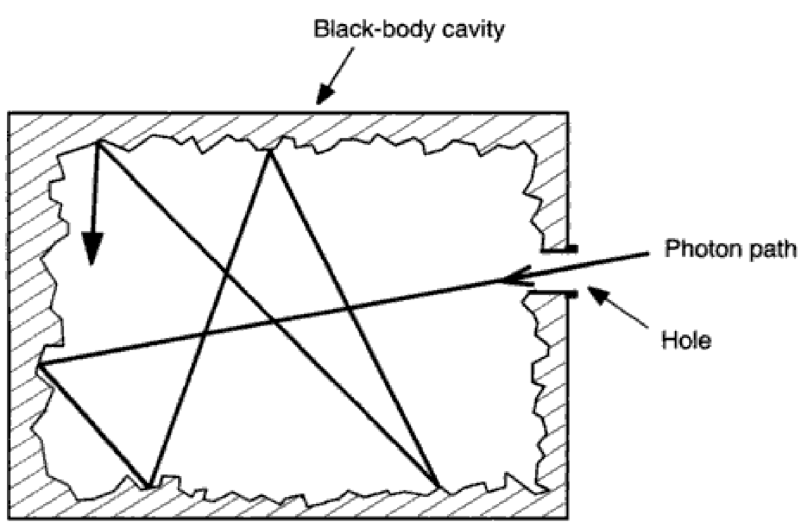


Fig. 6.1. The black-body cavity acts like a perfect absorber because light that enters the small opening in the cavity is almost certain to be absorbed inside.

The importance of a black body for stellar photospheres comes when we reverse the situation and heat the container; the walls emit photons, filling the inside with radiation. Each of these photons is re-absorbed inside the container, the small hole being negligible. Lengthy arguments can be put forward to support what seems obvious to the physically minded, namely that an equilibrium condition exists inside the chamber provided the temperature of the walls remain constant and uniform. Each physical process is balanced by its inverse. The container is said to be in thermodynamic equilibrium. A small fraction of the internal radiation leaks out the hole, and we can measure its spectrum, but this leakage is so small that the thermodynamics of the container is essentially maintained.

Strange as it may seem, stellar photospheres share some properties with such a container. The basic condition for the black body as an emitting source is that a negligibly small fraction of the radiation escapes. At the bottom of a stellar photosphere, the optical depth to the surface is high enough to prevent the escape of most photons. They are re-absorbed not far from where they are emitted. There are nearly as many photons headed into the star as there are headed outward. The material is then close to thermodynamic equilibrium, and we expect the radiation laws of a black body to describe the radiation in these deep layers of a stellar photosphere. Higher layers deviate increasingly from the black-body case as the leakage (the "hole") becomes more significant. There is a continuous transition from near perfect local thermodynamic equilibrium (LTE) deep in the photosphere to complete non-equilibrium (non-LTE) high in the atmosphere.

Observed relations

The observed black-body spectrum is found to be continuous, isotropic, and unpolarized. The intensity of the continuum depends only on wavelength and the temperature of the black body. Two fundamental empirical laws were found before the full nature of the spectrum was understood. The first of these is a scaling relation, discovered by Wien in 1893, and can be written in frequency or wavelength units,

$$I_\nu = \nu^3 f(\nu/T) \quad \text{or} \quad I_\lambda = \frac{c^4}{\lambda^5} f(c/\lambda T), \tag{6.1}$$

where c denotes the velocity of light and f is a unique function that can be tabulated from the measurements. Take $u = \nu/T$ and write

$$I_\nu = T^3 u^3 f(u).$$

We see that the frequency dependence of I_ν is always the same, $u^3f(u)$, but the strength scales with the cube of the temperature of the black body. The maximum of $u^3f(u)$ always occurs at the same u since $f(u)$ is a unique function. The observations show that the maximum I_ν occurs at $u = 5.8789 \times 10^{10}$ Hz/K, or

$$\begin{aligned} \lambda'_{\mathrm{m}} T &= 0.50995\,\mathrm{cm\,K} \\ &= 5.0995 \times 10^{7}\,\text{Å K}, \end{aligned} \tag{6.2}$$

where λ'_{m} is the wavelength at which the frequency distribution I_ν has its maximum. The corresponding relation for I_λ is

$$\begin{aligned} \lambda_{\mathrm{m}} \mathrm{T} &= 0.289\,78\,\mathrm{cm\,K} \\ &= 2.8978 \times 10^{7}\,\text{Å K}. \end{aligned} \tag{6.3}$$

The relation (6.1) is called "Wien's law," although the special case in Eqs. (6.2) and (6.3) often goes by the same name. The fact that the maximum appears at a different place in the spectrum for frequency and wavelength units was discussed near the beginning of Chapter 5.

The second law, the Stefan–Boltzmann law, says that the total power output is specified purely by the temperature according to

$$\int_0^\infty \mathfrak{F}_\nu \, \mathrm{d}\nu = \sigma T^4, \tag{6.4}$$

where $\mathfrak{F}_\nu$ is the flux and σ is the Stefan–Boltzmann constant of value $\sigma = 5.6705 \times 10^{-5}$ erg/(s cm^{2}K^4) or 5.6705×10^{-8} W/(m^{2}K^4). The Stefan–Boltzmann relation follows directly from Wien's law. Again let $u = \nu / T$. Then $I_\nu = T^3u^3f(u)$ and

$$\begin{aligned} \int_0^\infty I_\nu \, \mathrm{d}\nu &= \int_0^\infty T^3u^3f(u)T\,\mathrm{d}u \\ &= T^4 \int_0^\infty u^3f(u)\,\mathrm{d}u \\ &= \text{constant}\, T^4. \end{aligned} \tag{6.5}$$

As shown in the next section, the constant has a value of σ/π. Thermodynamic derivations of these two laws can be seen in Richtmyer *et al.* (1955), for example.

The shapes of the spectra are illustrated in Fig. 6.2. At low frequencies the curves are found to have the form

$$I_\nu = \frac{2kT\nu^2}{c^2} = \frac{2kT}{\lambda^2} \tag{6.6}$$

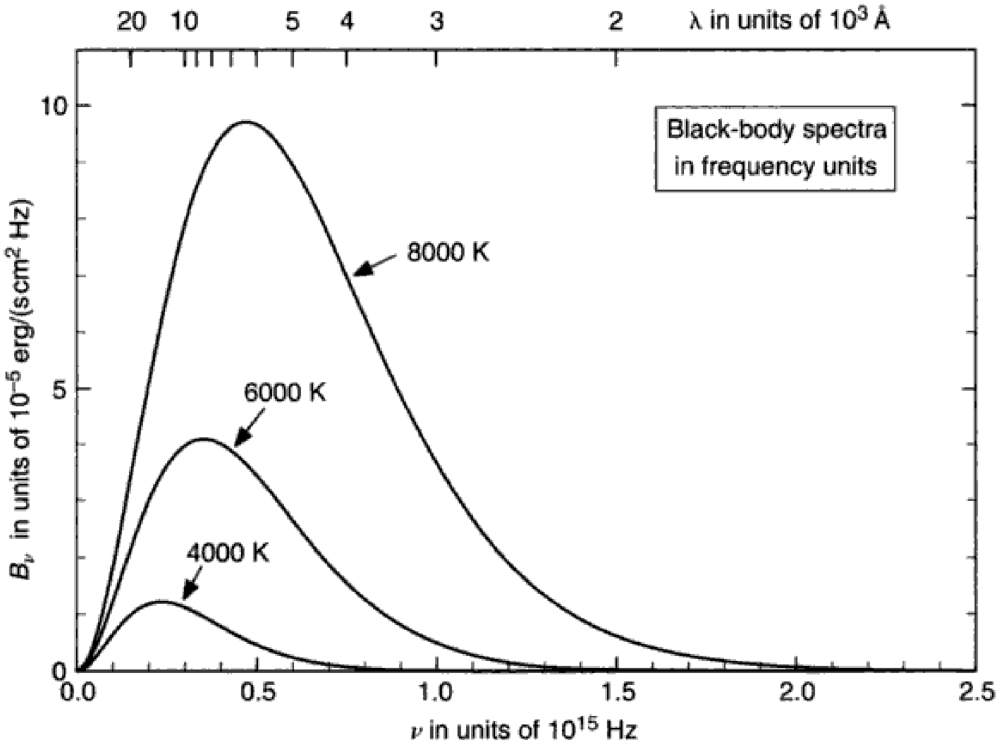


Fig. 6.2. The specific intensity emitted by a black body is shown for the three temperatures indicated. The scaling of Wien's law (Eq. 6.1) can be seen in the shape of the curves. The area under the curves obeys the Stefan–Boltzmann law, Eq. (6.4).

called the Rayleigh–Jeans approximation. At high frequencies

$$I_\nu = \text{constant}\, \nu^3 e^{-\text{constant}\, \nu/T}.$$

The name "Wien approximation" is associated with this equation.

Planck's radiation law

The history of the black body and its fundamental role in starting quantum theory is well known. We omit the historical details here except to point out that any derivation of the relation describing black-body spectra, as in Fig. 6.2, relies on the quantum approach.

We derive Planck's law using a two-level atom. (See Born 1957 and Cox & Giuli 1968 for alternative derivations.) The upper level has a population N_u, the lower N_l, and since these atoms are in a container in a state of thermodynamic equilibrium, their populations are related by Eq. (1.17),

$$\frac{N_u}{N_l} = \frac{g_u}{g_l} e^{-h\nu/kT},$$

where the energy difference between the levels is $h\nu = \chi_u - \chi_l$. The equilibrium condition implies that all the ways for the electron to go from u to l must be balanced by return paths.

The number of spontaneous emissions per second per unit solid angle per unit volume is $N_u A_{ul}$, where A_{ul} is the Einstein coefficient defined in the last section of Chapter 5. Similarly, the rate of stimulated emission per unit solid angle and volume is $N_u B_{ul} I_\nu$, and I_ν must be taken at ν appropriate to the transition. Finally, absorption pumps electrons upward, so for this rate we write $N_l B_{lu} I_\nu$. In addition to radiative transitions we could have collision-induced transitions, but in equilibrium there are as many up as there are down, and they cancel out of the equilibrium equation. And so we have

$$N_u A_{ul} + N_u B_{ul} I_\nu = N_l B_{lu} I_\nu.$$

If we now solve this for I_ν, we find

$$I_\nu = \frac{A_{ul}}{B_{lu}(N_l/N_u) - B_{ul}} \tag{6.7}$$

which is just the ratio of emission to absorption as in Eq. (5.15), that is, Eq. (6.7) has the form of a source function. If we also substitute for the population ratio according to the excitation equation above, Eq. (6.7) becomes

$$I_\nu = \frac{A_{ul}}{(g_l/g_u) B_{lu} e^{h\nu/kT} - B_{ul}}.$$

But we know that this expression must revert to the Rayleigh–Jeans approximation (Eq. 6.6) in the limit of small ν. Therefore we expand the exponential, keeping only the first-order term, and find

$$I_\nu \approx \frac{A_{ul}}{(g_l/g_u) B_{lu} - B_{ul} + (g_l/g_u) B_{lu} h\nu/kT}$$

for $h\nu/kT \ll 1$. This can equal $2kT\nu^2/c^2$ only if

$$B_{ul} = B_{lu} g_l/g_u \quad \text{and} \quad A_{ul} = \frac{2h\nu^3}{c^2} B_{ul}. \tag{6.8}$$

The result of substituting these relations back into Eq. (6.7) is

$$I_\nu = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/kT} - 1}. \tag{6.9}$$

Equation (6.9) is Planck's radiation law. It is also common to denote this expression as $B_\nu(T)$, meaning the black body source function. It gives the proper fit to the observed curves in Fig. 6.2 and has the correct limiting forms of Eqs. (6.6) and the Wien approximation. It has the functional form giving the observed relations expressed in Eq. (6.1). We can also check the maximum in the usual way by setting the derivative equal to zero. Use $x = h\nu/kT$ and obtain

$$x = \frac{3(e^x - 1)}{e^x}.$$

Solution by iteration gives $x = 2.8214$, or $\lambda'_m T = 0.50995$ cm K in accord with Eq. (6.2).

Integration of Planck's law leads to the Stefan–Boltzmann law in the following way. From Eq. (5.5), the flux is given by

$$\mathfrak{F}_\nu = \oint I_\nu \cos\theta \, d\omega,$$

and in this case, I_ν is Planck's law. Since no significant radiation is entering the hole in the black-body chamber and the escaping radiation is isotropic, we have $\mathfrak{F}_\nu = \pi I_\nu$ according to Eq. (5.7). It then follows that

$$\begin{aligned}\int_0^\infty \mathfrak{F}_\nu \, d\nu &= \pi \int_0^\infty \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/kT} - 1} \, d\nu \\ &= \frac{2\pi h}{c^2}\left(\frac{kT}{h}\right)^4 \int_0^\infty \frac{x^3}{e^x - 1} \, dx,\end{aligned}$$

where $x = h\nu/kT$. Evaluating the integral gives $\pi^4/15$ and

$$\int_0^\infty \mathfrak{F}_\nu \, d\nu = \frac{2\pi^5 k^4}{15h^3c^2} T^4 = \sigma T^4.$$

So the observed Stefan–Boltzmann law of Eq. (6.4) is retrieved.

Finally, notice that the relations (6.8) are between fundamental atomic constants. These relations must hold independent of the physical situation the atoms are encountering, and therefore are perfectly general relations. More details follow in Chapter 11.

Numerical values of black-body radiation

Tabulations of black-body functions, such as those of McDonald (1955) or Pivovonsky and Nagel (1961), are useful in some situations, but often it is quicker and more convenient to compute from Eq. (6.9), especially when B_ν is being used as the source function in a model photosphere calculation. Thus in frequency units

$$\begin{aligned}B_\nu(T) &= \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/kT}-1}\\ &= \frac{1.474\,501\times 10^{-47}\nu^3}{e^{4.799\,239\times 10^{-11}\nu/T}-1}\frac{\text{erg}}{\text{s cm}^2\text{ rad}^2\text{ Hz}}\\ &= \frac{3.972\,895\times 10^8}{\lambda^3}\frac{1}{e^{1.438\,78\times 10^8/\lambda T}-1}\frac{\text{erg}}{\text{s cm}^2\text{ rad}^2\text{ Hz}} \quad \text{for } \lambda \text{ in Å}\\ &= \frac{1.474\,501\times 10^{-50}\nu^3}{e^{4.799\,239\times 10^{-11}\nu/T}-1}\frac{\text{W}}{\text{m}^2\text{ rad}^2\text{ Hz}},\end{aligned} \tag{6.10}$$

or in wavelength units,

$$\begin{aligned}B_\lambda(T) &= \frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/\lambda kT}-1}\\ &= \frac{1.1910\,427\times 10^{27}/\lambda^5}{e^{1.438\,775\times 10^8/\lambda T}-1}\frac{\text{erg}}{\text{s cm}^2\text{ rad}^2} \quad \text{for } \lambda \text{ in Å}\\ &= \frac{1.1910\,427\times 10^{24}/\lambda^5}{e^{1.438\,775\times 10^8/\lambda T}-1}\frac{\text{W}}{\text{s m}^2\text{ rad}^2} \quad \text{for } \lambda \text{ in Å}\end{aligned} \tag{6.11}$$

where we have used $h = 6.626\,0688\times 10^{-27}$ erg s (or 10^{-34} J s), $k = 1.380\,65\times 10^{-16}$ erg/K (10^{-23} J/K), and $c = 2.997\,924\,58\times 10^{10}$ cm/s (10^8 m/s) is the velocity of light in a vacuum. If the black body is used in air, c should be decreased from its vacuum value by the index of refraction of air, $n \approx 1.000\,28$ (see Eqs. 12.18 and 12.19). Likewise, all wavelengths in air will differ from their vacuum values: $\lambda = \lambda_0/n$, where λ_0 is the vacuum value. These differences may not be negligible, amounting to $\approx 1.0\%$ at 3000 Å from 0.3% at 10 000 Å at the gold-point temperature of 1337.6 K.

The black-body as a radiation standard

For photometric work, the black body radiator is used as a primary standard of radiation against which many types of secondary standards are calibrated. The usefulness of the black body in this context stems from the simple unique dependence of the radiation on temperature.

A black body suitable for this type of work is built with a cavity in the center surrounded by electrical heating coils. Inside is a small amount of pure metal, typically gold, and in one wall there is a small hole from which the black body radiation is emitted. The apparatus is used by slowly heating the gold until it melts. The system is then allowed to cool to the freezing point of the metal, where there is a thermally stable plateau lasting several minutes. It is during this interval that the black body is used for photometry. Operation at a metallic freezing point results in good reproducibility and stability. The temperature of freezing gold is 1337.6 K. The black body can be operated at other temperatures by using other metals: platinum freezes at 2045 K; copper at 1358 K. The more inert the metal, the less it becomes contaminated with impurities. The freezing temperature is a function of the purity of the metal.

The precision of photometric calibrations can be limited by improper construction or operation of the black body. Uniform heating of the cavity walls is imperfect. The effects of the finite hole size compared with the size of the cavity and the type of wall have been calculated by DeVos (1954). Useful details of black-body operation can be found in the paper of Kostkowski and Lee (1962). The most fundamental limit though is the uncertainty in the physical temperature scales. Bless *et al.* (1968) and Griem (1964) find the ultraviolet calibration uncertain, relative to the standard wavelength λ5556 radiation, by several per cent because of uncertainty in the actual temperature within the cavity.

Black bodies are rather awkward to look at with an astronomical telescope. Standard lamps, calibrated against black bodies, are usually used instead. The stellar calibrations are discussed in Chapter 10.

References

Bless, R. C., A. D. Code, & D. J. Schroeder 1968. *Ap. J.* **153**, 545.
Born, M. 1957. *Atomic Physics*, 6th edn. (New York: Hafner).
Cox, J. P. & R. T. Giuli 1968. *Stellar Structure*. (New York: Gordon and Breach), p. 108.
DeVos, J. C. 1954. *Physica* **20**, 669.
Griem, H. R. 1964. *Plasma Spectroscopy*. (New York: McGraw-Hill), p. 257.
Kostkowski, H. J. & R. D. Lee 1962. U.S. National Bureau of Standards Monograph **41**.
McDonald, J. K. 1955. *Pub. Dominion Ap. Obs. Victoria* **10**, 127.
Pivovonsky, M. & M. R. Nagel 1961. *Tables of Black Body Radiation Functions*. (New York: Macmillan).
Richtmyer, F. K., E. H. Kennard, & T. Lauritsen 1955. *Introduction to Modern Physics*, 5th edn. (New York: McGraw-Hill).

Questions and exercises

1. Explain what we mean by a "black body," and why it is relevant to stellar photospheres. Explicitly address the question of how a closed container, like the one shown in Fig. 6.1, can have anything in common with a stellar photosphere that is gaseous and without walls.

2. Starting with Planck's law, Eq. (6.10), convert to the Rayleigh–Jeans approximation applicable at long wavelengths.

3. The Stefan–Boltzmann law, Eq. (6.4), is used to define the effective temperature of a star, as done in Eq. (1.2). This in turn is related to the power output and the radius of the star in Eq. (1.3). Using these relations, compare the following two stars. The brighter star has 11.3 times the luminosity of the fainter one. The more distant star lies at 21.3 pc. The effective temperature of the brighter star is 1.4 times that of the fainter star. Assuming these are main sequence stars, what is the ratio of their radii? If the lesser star is the Sun, estimate the temperature, radius, luminosity, and mass of the other star. You may wish to consult the data given in Appendix B.

4. How much power actually comes out of a stellar surface? Use the Stefan–Boltzmann law to find the answer for two stars of effective temperature 4000 and 40 000 K. If these stars were at the distance of the Sun from us, how much power would they give per square meter on Earth?

5. Estimate the temperature of a slowly rotating planet in a circular orbit around a star. Call the effective temperature of the star T_{eff}, the radius of the star R, the radius of the planet a, and the radius of the orbit r. Assume an equilibrium situation exists, e.g., the temperature of the planet is constant, implying that the power absorbed from the star is equal to the power re-radiated by the planet. Why does the planet's temperature not depend on the size of the planet? What complications to your calculation occur if the planet has an atmosphere?

7

Radiative and convective energy transport

The dominant mechanism of energy transport through the surface layers of a typical star is radiation, i.e., photons. Transport by convection is often important below the surface, but rarely carries a significant fraction of the flux in the photosphere. Conduction comes into play in extreme cases such as white dwarfs. So it is radiative transfer that is our main focus here. It is in the domain of radiative transfer where the physical parameters of the material comprising the star are coupled to the spectrum we see and measure. We start by setting up the differential equation describing the flow of radiation through an infinitesimal volume. The integration of the equation can then be accomplished for the geometry of the situation. Unfortunately the step from a differential to an integral equation is not a physical solution to the problem because the integrand still depends on the atomic excitation of the material, which itself depends on the temperature of the material, and the radiation field in the material. Both the thermal (collisional) and the non-thermal (radiative) excitation vary with depth in the photosphere. In most applications of the theory to real stars, a numerical model of the star's photosphere is formed from which the integrand and then the spectrum of the star can be calculated. This chapter, along with Chapters 8 and 9, develops the tools for this modeling.

The transfer equation and its formal solution

Consider radiation traveling in a direction s. The change in specific intensity, dI_ν, over an increment of path length, ds, is the sum of the losses, expressed in the absorption coefficient κ_ν, and gains, expressed in the emission coefficient j_ν,

$$dI_\nu = -\kappa_\nu \rho I_\nu \, ds + j_\nu \rho \, ds.$$

The absorption and emission coefficients, as well as the source function, S_ν, used below, were defined in Chapter 5. Now divide the equation by $\kappa_\nu \rho\, ds$, which we call $d\tau_\nu$ in accord with Eq. (5.13), and write

$$\frac{dI_\nu}{d\tau_\nu} = -I_\nu + j_\nu/\kappa_\nu,$$

or using the definition of the source function, Eq. (5.15),

$$\frac{dI_\nu}{d\tau_\nu} = -I_\nu + S_\nu. \tag{7.1}$$

This is the (differential) equation of radiative transfer.

The integration follows from a standard integrating-factor scheme. Noticing that the variable τ_ν appears alone in Eq. (7.1), we are led to try a solution of the form

$$I_\nu(\tau_\nu) = f e^{b\tau_\nu}, \tag{7.2}$$

where f is a function to be determined. Substitution into Eq. (7.1) gives

$$\begin{aligned}\frac{dI_\nu}{d\tau_\nu} &= fb e^{b\tau_\nu} + e^{b\tau_\nu}\frac{df}{d\tau_\nu} \\ &= bI_\nu + e^{b\tau_\nu}\frac{df}{d\tau_\nu}.\end{aligned}$$

or

$$bI_\nu + e^{b\tau_\nu}\frac{df}{d\tau_\nu} = -I_\nu + S_\nu.$$

We therefore make the first terms on each side of the equation equal by setting $b = -1$, and then equating the second terms gives

$$e^{-\tau_\nu}\frac{df}{d\tau_\nu} = S_\nu$$

or

$$f = \int_0^{\tau_\nu} S_\nu e^{t_\nu}\, dt_\nu + c_0$$

with t_ν serving as a dummy variable. Then Eq. (7.2) becomes

$$I_\nu(\tau_\nu) = e^{-\tau_\nu}\int_0^{\tau_\nu} S_\nu(t_\nu) e^{t_\nu}\, dt_\nu + c_0 e^{-\tau_\nu}. \tag{7.3}$$

The integration constant c_0 must be $I_\nu(0)$, as seen by setting $\tau_\nu = 0$.

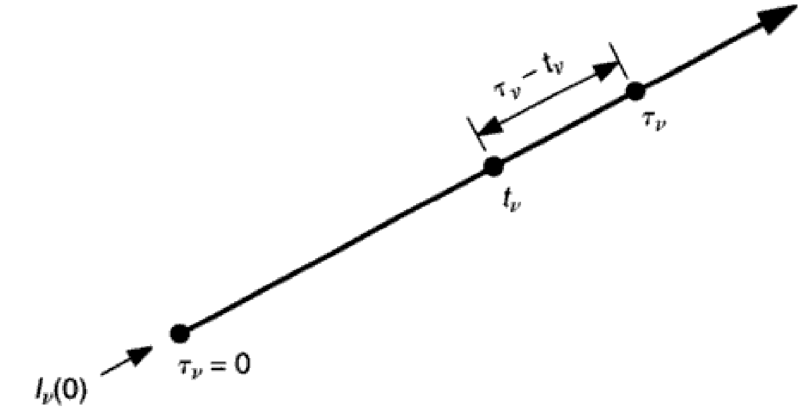


Fig. 7.1. Radiation along the line at the point τ_ν is composed of the sum of intensities, S_ν, originating at the points t_ν along the line but suffering extinction according to the optical-depth separation $\tau_\nu - t_\nu$. Any radiation incident at $\tau_\nu = 0$, called $I_\nu(0)$, suffers extinction by a factor of $e^{-\tau_\nu}$.

It is instructive to rewrite Eq. (7.3), bringing the $e^{-\tau_\nu}$ inside the integral,

$$I_\nu(\tau_\nu) = \int_0^{\tau_\nu} S_\nu(t_\nu) e^{-(\tau_\nu - t_\nu)} \, dt_\nu + I_\nu(0) e^{-\tau_\nu}. \tag{7.4}$$

Figure 7.1 illustrates the line along which radiation is being considered and the positions of the variables for this equation. In particular, at a point τ_ν, the original intensity, $I_\nu(0)$, has suffered an exponential extinction $e^{-\tau_\nu}$, and in a similar way, the intensity generated at any point t_ν, which we denote as $S_\nu(t_\nu)$, undergoes an exponential extinction $e^{-(\tau_\nu - t_\nu)}$ before being summed at the point τ_ν.

Equation (7.4) is the basic integral form of the transfer equation. If we can solve this equation, we have the answer we expect from the theory. However, in order to actually perform the integration, $S_\nu(t_\nu)$ must be a known function. In some situations this is a complicated function. In others it may be simple, as shown in the examples of Chapter 5. In the case of local thermodynamic equilibrium (LTE), $S_\nu(T) = B_\nu(T)$, the Planck function, and then knowing T as a function of x or τ_ν amounts to a solution of the transfer equation.

The transfer equation for different geometries

Equation (7.4) expresses I_ν as a function of optical depth along a line. In many applications, particularly in studies of interstellar material, this equation can be applied directly since the observation of a point on the celestial sphere corresponds to observation along such a line. On the other hand, for stellar photospheres, it is conventional to define the optical depth relative to the star

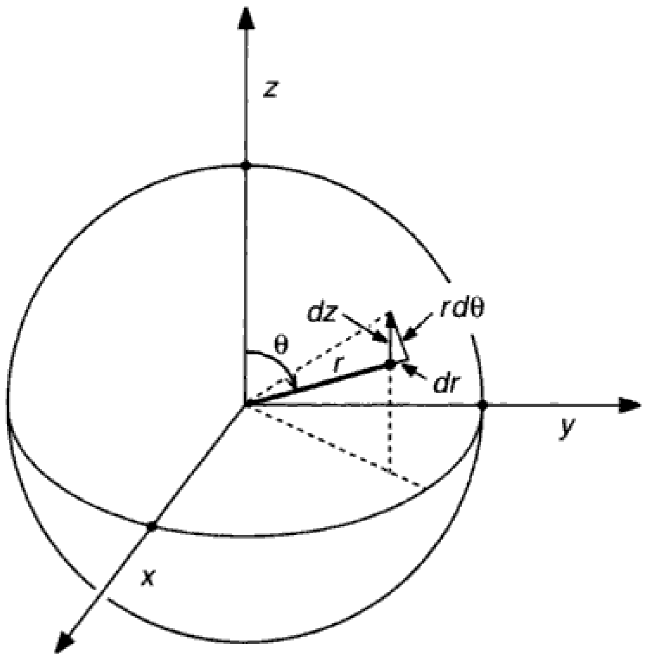


Fig. 7.2. The geometry is shown for the conversion of line-of-sight to polar coordinates centered on the star. The arrow on dz points toward the observer.

along a stellar radius, and not along the line of sight. The appropriate projection factor must then be incorporated.

Assume spherical stars, thereby making spherical coordinates appropriate as depicted in Fig. 7.2. The z axis is chosen toward the observer. Because we are concerned with the geometry, we write the transfer equation in the form

$$\frac{\mathrm{d}I_\nu}{\kappa_\nu \rho \mathrm{d}z} = -I_\nu + S_\nu.$$

In general $\mathrm{d}I_\nu/\mathrm{d}z$ in these coordinates can be written as

$$\frac{\mathrm{d}I_\nu}{\mathrm{d}z} = \frac{\partial I_\nu}{\partial r}\frac{\mathrm{d}r}{\mathrm{d}z} + \frac{\partial I_\nu}{\partial \theta}\frac{\mathrm{d}\theta}{\mathrm{d}z}, \tag{7.5}$$

where we assume I_ν has no ϕ, i.e., azimuthal, dependence. From the figure we see that

$$\mathrm{d}r = \cos\theta \mathrm{d}z$$
$$r\mathrm{d}\theta = -\sin\theta \mathrm{d}z.$$

By substitution of these expressions, the transfer equation becomes

$$\frac{\partial I_\nu}{\partial r}\frac{\cos\theta}{\kappa_\nu \rho} - \frac{\partial I_\nu}{\partial \theta}\frac{\sin\theta}{\kappa_\nu \rho r} = -I_\nu + S_\nu. \tag{7.6}$$

This form of the equation is used in stellar interiors and in the calculation of very thick stellar atmospheres such as those found in supergiants. Conveniently, the geometrical thickness of the photosphere in most stars is very small compared to the stellar radius. The solar photosphere is ~700 km thick, or ~0.1% of the solar radius (refer to Fig. 1.1). Under these conditions, a plane parallel approximation can be made, namely, θ does not depend upon z so there is no second term in Eq. (7.6). Then

$$\cos\theta \frac{\mathrm{d}I_\nu}{\kappa_\nu \rho\, \mathrm{d}r} = -I_\nu + S_\nu.$$

Custom has been to adopt a new geometrical depth variable, x, for the plane geometry case defined by $\mathrm{d}x = -\mathrm{d}r$. Adopting this convention and writing $\mathrm{d}\tau_\nu$ for $\kappa_\nu \rho\, \mathrm{d}x$ gives the basic form of the transfer equation used in the central arena of stellar atmospheres,

$$\cos\theta \frac{\mathrm{d}I_\nu}{\mathrm{d}\tau_\nu} = I_\nu - S_\nu. \tag{7.7}$$

The optical depth defined in this way is measured along x and not along the line of sight which is at some angle, θ, as shown in Fig. 7.3. This amounts to replacing τ_ν in Eq. (7.1) by $-\tau_\nu \sec\theta$. The negative sign arises from choosing $\mathrm{d}x = -\mathrm{d}r$. The integrated form corresponding to Eq. (7.4) is then

$$I_\nu(\tau_\nu) = -\int_c^{\tau_\nu} S_\nu(t_\nu) \mathrm{e}^{-(t_\nu - \tau_\nu)\sec\theta} \sec\theta\, \mathrm{d}\tau_\nu. \tag{7.8}$$

The integration limit, c, replaces the $I_\nu(0)$ integration constant of Eq. (7.4) because the boundary conditions are clearly different for radiation going in ($\theta > 90°$) and coming out ($\theta < 90°$). In the first case, we start at the outer boundary, where $\tau_\nu = 0$, and work inward, so when $I_\nu = I_\nu^{\mathrm{in}}$, $c = 0$. In the second case, we consider radiation at the depth τ_ν and deeper until no more radiation can be seen coming out. In other words, when $I_\nu = I_\nu^{\mathrm{out}}$, $c = \infty$. Therefore, the full intensity at the position τ_ν on the line of sight through the photosphere is

$$\begin{aligned} I_\nu(\tau_\nu) &= I_\nu^{\mathrm{out}}(\tau_\nu) + I_\nu^{\mathrm{in}}(\tau_\nu) \\ &= \int_{\tau_\nu}^{\infty} S_\nu \mathrm{e}^{-(t_\nu - \tau_\nu)\sec\theta} \sec\theta\, \mathrm{d}t_\nu \\ &\quad - \int_0^{\tau_\nu} S_\nu \mathrm{e}^{-(t_\nu - \tau_\nu)\sec\theta} \sec\theta\, \mathrm{d}t_\nu. \end{aligned} \tag{7.9}$$

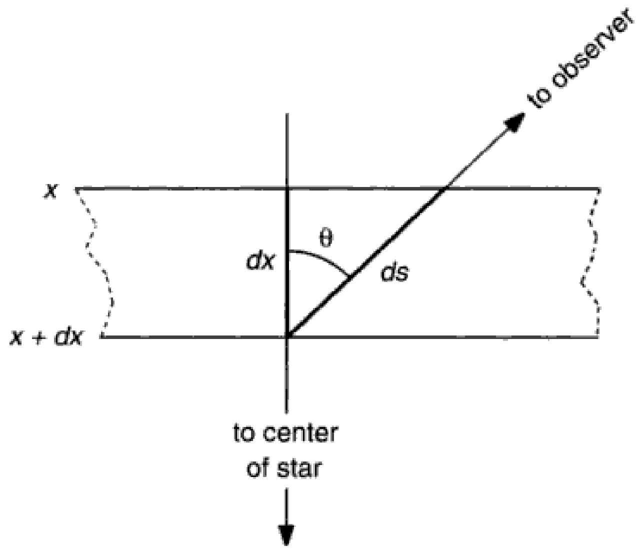


Fig. 7.3. In the plane-parallel approximation, the geometry is simpler than in the previous figure. The increment of path length along the line of sight is $ds = dx \sec\theta$.

The mathematically minded will usually add that one must require $S_\nu e^{-\tau_\nu}$ to go to zero as τ_ν goes to infinity to ensure that the first integral exists. Real stars apparently know how to meet this condition without effort.

An important special case of Eq. (7.9) occurs at the stellar surface. In this case

$$I_\nu^{\rm in}(0) = 0$$
$$I_\nu^{\rm out}(0) = \int_0^\infty S_\nu e^{-t_\nu \sec\theta} \sec\theta \, dt_\nu, \tag{7.10}$$

where we assume that the radiation from other stars, galaxies, and so forth is completely negligible compared to the star's own radiation. Equation (7.10) is the expression we need to compute the spectrum. It is especially relevant to solar work where intensity measurements can be made as a function of θ. For most stars, we must still integrate $I_\nu^{\rm out}$ over the star's disk, i.e., we observe the flux.

The flux integral

Expand the defining equation for flux (Eq. 5.5) in spherical polar coordinates and assume no azimuthal dependence in I_ν. Then

$$\begin{aligned}\mathfrak{F}_\nu &= 2\pi \int_0^\pi I_\nu \cos\theta \sin\theta \, d\theta \\ &= 2\pi \int_0^{\pi/2} I_\nu^{\rm out} \cos\theta \sin\theta \, d\theta + 2\pi \int_{\pi/2}^{\pi} I_\nu^{\rm in} \cos\theta \sin\theta \, d\theta.\end{aligned}$$

Using the definitions of I_ν^{out} and I_ν^{in} from Eq. (7.9) gives

$$\begin{aligned}\mathfrak{F}_\nu = {} & 2\pi\int_0^{\pi/2}\int_{\tau_\nu}^{\infty} S_\nu e^{-(t_\nu-\tau_\nu)\sec\theta}\sin\theta\, dt_\nu\, d\theta \\ & - 2\pi\int_{\pi/2}^{\pi}\int_0^{\tau_\nu} S_\nu e^{-(t_\nu-\tau_\nu)\sec\theta}\sin\theta\, dt_\nu\, d\theta,\end{aligned}$$

or, if S_ν is isotropic

$$\begin{aligned}\mathfrak{F}_\nu = {} & 2\pi\int_{\tau_\nu}^{\infty} S_\nu \int_0^{\pi/2} e^{-(t_\nu-\tau_\nu)\sec\theta}\sin\theta\, d\theta\, dt_\nu \\ & - 2\pi\int_0^{\tau_\nu} S_\nu \int_{\pi/2}^{\pi} e^{-(t_\nu-\tau_\nu)\sec\theta}\sin\theta\, d\theta\, dt_\nu .\end{aligned} \tag{7.11}$$

If we now let $w = \sec\theta$ and $x = t_\nu - \tau_\nu$, the first inner integral can be written as

$$\int_0^{\pi/2} e^{-(t_\nu-\tau_\nu)\sec\theta}\sin\theta\, d\theta = \int_1^{\infty}\frac{e^{-xw}}{w^2}dw. \tag{7.12}$$

Integrals of this form are well known. They are called exponential integrals. Their properties are discussed later in this chapter. For now we simply denote them by

$$E_n(x) = \int_1^{\infty}\frac{e^{-xw}}{w^n}dw, \tag{7.13}$$

realizing that $E_n(x)$ is a monotonically diminishing function of x (see Fig. 7.4 below). Using the exponential integral notation, Eq. (7.11) becomes

$$\mathfrak{F}_\nu(\tau_\nu) = 2\pi\int_{\tau_\nu}^{\infty} S_\nu E_2(t_\nu-\tau_\nu)dt_\nu - 2\pi\int_0^{\tau_\nu} S_\nu E_2(\tau_\nu - t_\nu)dt_\nu. \tag{7.14}$$

To obtain $E_2(\tau_\nu - t_\nu)$ in the second integral, it is necessary to make the substitution $w = -\sec\theta$ and $x = \tau_\nu - t_\nu$ in the inner integral of the second term of Eq. (7.11). The limit as θ goes from $\pi/2$ to π is approached with negative values of $\cos\theta$, so w goes to ∞, not $-\infty$.

Equation (7.14) is the basic relation we seek. It is analogous to Eq. (7.9) for intensity. The theoretical stellar spectrum is $\mathfrak{F}_\nu$ at $\tau_\nu = 0$, that is,

$$\mathfrak{F}_\nu(0) = 2\pi\int_0^{\infty} S_\nu(t_\nu)E_2(t_\nu)dt_\nu. \tag{7.15}$$

This says that the surface flux is composed of the sum of the source function at each depth multiplied by an extinction factor, $E_2(t_\nu)$, appropriate to that depth below the surface, and the sum is taken over all depths contributing a significant amount of radiation at the surface. Notice that $\mathfrak{F}_\nu$ is defined per unit area. The total radiation at the frequency ν is $4\pi R^2\mathfrak{F}_\nu$, where R is the radius of the star.

In the process of deriving Eq. (7.15), we assumed explicitly that S_ν was isotropic and implicitly that κ_ν (or τ_ν) is also isotropic. For many purposes this is perfectly reasonable. In others it is not. In particular, when we deal with Doppler shifts of spectral lines arising from photospheric velocity fields (Chapter 17) or from rotation of the stars (Chapter 18), isotropy no longer holds; κ_ν depends on θ, and we are forced to replace Eq. (7.15) with explicit integrations of $I_\nu(\theta, \phi)$ over the apparent disk of the star. Explicit disk integrations are also required when modeling stars with surface features such as starspots.

The mean intensity and *K* integrals

We can easily derive expressions for J_ν and K_ν as functions of τ_ν, and obtain expressions similar to Eq. (7.15) for flux. The result is

$$J_\nu(\tau_\nu) = \tfrac{1}{2}\int_{\tau_\nu}^{\infty} S_\nu E_1(t_\nu - \tau_\nu)\,\mathrm{d}t_\nu + \tfrac{1}{2}\int_0^{\tau_\nu} S_\nu E_1(\tau_\nu - t_\nu)\,\mathrm{d}t_\nu \tag{7.16}$$

and

$$K_\nu(\tau_\nu) = \tfrac{1}{2}\int_{\tau_\nu}^{\infty} S_\nu E_3(t_\nu - \tau_\nu)\,\mathrm{d}t_\nu + \tfrac{1}{2}\int_0^{\tau_\nu} S_\nu E_3(\tau_\nu - t_\nu)\,\mathrm{d}t_\nu. \tag{7.17}$$

The details of the derivation are left to the reader. Notice the analogy with Eq. (5.8).

Exponential integrals

Let us repeat the equation for the exponential integrals, $E_n(x)$, of the nth order,

$$E_n(x) = \int_1^{\infty} \frac{\mathrm{e}^{-xw}}{w^n}\,\mathrm{d}w. \tag{7.18}$$

The value at the origin is found directly by setting $x = 0$, leaving

$$E_n(0) = \int_1^\infty \frac{\mathrm{d}w}{w^n} = \frac{1}{n-1}.$$

The first three exponential functions are shown in Fig. 7.4, where it can be seen that $E_1(0)$ is infinite, $E_2(0)$ is unity, and $E_3(0)$ is one-half.

Differentiation leads to a useful relation,

$$\frac{\mathrm{d}E_n}{\mathrm{d}x} = \int_1^\infty \frac{1}{w^n} \frac{\mathrm{d}}{\mathrm{d}x} \mathrm{e}^{-xw} \mathrm{d}w = -\int_1^\infty \frac{\mathrm{e}^{-xw}}{w^{n-1}} \mathrm{d}w,$$

or

$$\frac{\mathrm{d}E_n}{\mathrm{d}x} = -E_{n-1}. \tag{7.19}$$

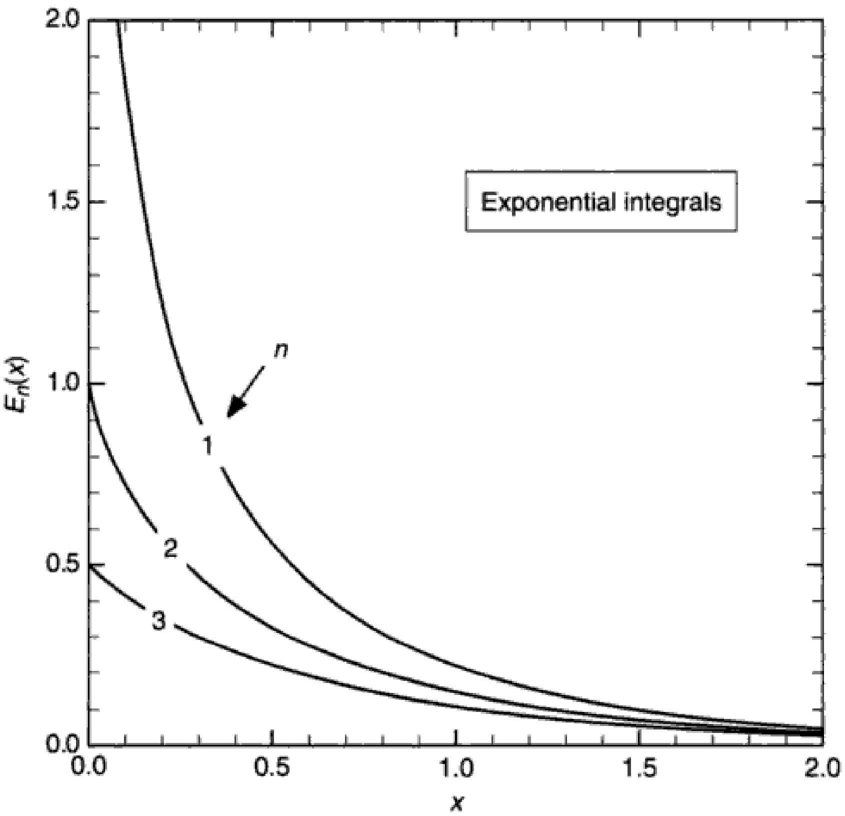


Fig. 7.4. Here are plots of the first three exponential integrals, $E_1(x)$, $E_2(x)$, and $E_3(x)$. At $x = 0$, E_1 is unbounded, E_2 is unity, and E_3 is 0.5. At large x, all the functions behave the same.

One can also show that the recurrence formula

$$nE_{n+1}(x) = \mathrm{e}^{-x} - xE_n(x) \tag{7.20}$$

holds. This expression is particularly useful for generating numerical values of $E_{n+1}(x)$ from $E_n(x)$. Abramowitz and Stegun (1964) give the following approximations suitable for computer calculation:

$$\begin{aligned} E_1(x) = &-\ln x - 0.57721566 + 0.99999193x - 0.24991055x^2 \\ &+ 0.05519968x^3 - 0.00976004x^4 + 0.00107857x^5 \quad \text{for } x \leq 1, \end{aligned}$$

and

$$E_1(x) = \frac{x^4 + a_3x^3 + a_2x^2 + a_1x + a_0}{x^4 + b_3x^3 + b_2x^2 + b_1x + b_0} \frac{1}{x\mathrm{e}^x} \qquad \text{for} \quad x > 1,$$

where

$$\begin{aligned} a_3 &= 8.573\,328\,7401, & b_3 &= 9.573\,322\,3454, \\ a_2 &= 18.059\,016\,9730, & b_2 &= 25.632\,956\,1486, \\ a_1 &= 8.634\,760\,8925, & b_1 &= 21.099\,653\,0827, \\ a_0 &= 0.267\,773\,7343, & b_0 &= 3.958\,496\,9228. \end{aligned}$$

These polynomials fit E_1 with an error of less than 2×10^{-7}. Then from Eq. (7.20), the higher-order exponential integrals can be computed.

The asymptotic behavior is given by the expansion

$$\begin{aligned} E_n(x) &= \frac{1}{x\mathrm{e}^x}\left[1 - \frac{n}{x} + \frac{n(n+1)}{x^2} - \cdots\right] \\ &\approx \frac{1}{x\mathrm{e}^x}. \end{aligned}$$

This is a very sharp decline with x, and lends confidence to the convergence of the integral in Eq. (7.15). Notice that the asymptotic form is ultimately independent of n.

Radiative equilibrium

Radiative equilibrium is an expression of conservation of energy. The central point when computing purely theoretical model atmospheres is to ensure that radiative equilibrium is enforced. Conservation of energy must apply to the flow of energy upward through the stellar photosphere, assuming that there are no sources or sinks of energy within the photosphere; the energy generated in the core of the star is simply flowing outward to the outer boundary. The general condition expressing the lack of sources and sinks is that the divergence of the flux be zero everywhere in the photosphere. However, virtually all the calculations we consider are for the plane parallel geometry. The divergence condition is then

$$\frac{\mathrm{d}}{\mathrm{d}x}\mathfrak{F}(x) = 0 \quad \text{or} \quad \mathfrak{F}(x) = \mathfrak{F}_0, \qquad \text{a constant,} \tag{7.21}$$

where $\mathfrak{F}(x)$, or $\mathfrak{F}(\tau_\nu)$, is the total energy flux in erg/(s cm^2) or W/m^2 as a function of depth. When all the energy is carried by radiation, we have

$$\mathfrak{F}(x) = \int_0^\infty \mathfrak{F}_\nu(\tau_\nu)\,\mathrm{d}\nu = \mathfrak{F}_0 \tag{7.22}$$

with $\mathfrak{F}_\nu$ given in Eq. (7.14). This special case is called radiative equilibrium, and then Eq. (7.21) is the first condition for radiative equilibrium. Although the shape of $\mathfrak{F}_\nu$ can be expected to change very significantly with depth, its integral remains invariant.

If convection plays a significant role as a mode of energy transport, then the convective flux, $\Phi(x)$, must be included,

$$\Phi(x) + \int_0^\infty \mathfrak{F}_\nu(x)\,\mathrm{d}\nu = \mathfrak{F}_0. \tag{7.23}$$

Equation (7.23) is a more general expression of flux constancy, and it might have to be expanded still further if other sources of energy transport are significant, such as conduction, acoustic waves, Alfvén waves, or magnetic-field dissipation.

Suppose now we take Eq. (7.14) and combine it with Eq. (7.22). This leads immediately to the relation

$$\int_0^\infty \left[\int_{\tau_\nu}^\infty S_\nu E_2(t_\nu - \tau_\nu)\,\mathrm{d}t_\nu - \int_0^{\tau_\nu} S_\nu E_2(\tau_\nu - t_\nu)\,\mathrm{d}t_\nu\right]\mathrm{d}\nu = \frac{\mathfrak{F}_0}{2\pi}. \tag{7.24}$$

This is referred to as Milne's second equation; the other two Milne equations will be considered shortly. In essence, it says that in the case of radiative equilibrium, the solution of the transfer equation is found when an $S_\nu(\tau_\nu)$ is known that satisfies this equation.

The other two radiative equilibrium conditions that are commonly used stem from the transfer equation, which we write as

$$\cos\theta\frac{\mathrm{d}I_\nu}{\mathrm{d}x} = \kappa_\nu\rho I_\nu - \kappa_\nu\rho S_\nu. \tag{7.25}$$

First, an integration over solid angle gives

$$\frac{\mathrm{d}}{\mathrm{d}x}\oint I_\nu \cos\theta \, \mathrm{d}\omega = \kappa_\nu \rho \oint I_\nu \mathrm{d}\omega - \kappa_\nu \rho \oint S_\nu \, \mathrm{d}\omega,$$

assuming κ_ν is independent of direction. If S_ν is also independent of direction, and we substitute for the first and second integrals the definitions of flux and mean intensity, the equation becomes

$$\frac{\mathrm{d}\mathfrak{F}_\nu}{\mathrm{d}x} = 4\pi\kappa_\nu \rho J_\nu - 4\pi\kappa_\nu \rho S_\nu. \tag{7.26}$$

Second, an integration over frequency gives

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_0^\infty \mathfrak{F}_\nu \mathrm{d}\nu = 4\pi\rho\int_0^\infty \kappa_\nu J_\nu \mathrm{d}\nu - 4\pi\rho\int_0^\infty \kappa_\nu S_\nu \mathrm{d}\nu. \tag{7.27}$$

In radiative equilibrium, the left-hand side is zero, leaving the relation

$$\int_0^\infty \kappa_\nu J_\nu \mathrm{d}\nu = \int_0^\infty \kappa_\nu S_\nu \mathrm{d}\nu. \tag{7.28}$$

In this particular form of the radiative equilibrium condition, the value of the flux constant does not appear.

Another Milne equation follows from Eq. (7.28) by using the expression for J_ν in Eq. (7.16),

$$\int_0^\infty \kappa_\nu \left[\tfrac{1}{2}\int_{\tau_\nu}^\infty S_\nu E_1(t_\nu - \tau_\nu)\mathrm{d}t_\nu + \tfrac{1}{2}\int_0^{\tau_\nu} S_\nu E_1(\tau_\nu - t_\nu)\mathrm{d}t_\nu - S_\nu\right]\mathrm{d}\nu = 0. \tag{7.29}$$

As with Eq. (7.24), a solution is in hand when we have an $S_\nu(\tau_\nu)$ value that satisfies this relation.

Equation (7.26) leads to another interesting result. In very deep layers, where the optical depth at all frequencies is large, we can be assured that, compared to the other terms in the equation, $\mathrm{d}\mathfrak{F}_\nu/\mathrm{d}x$ is negligible. We conclude that $J_\nu = S_\nu$ in these deep layers.

The final radiative equilibrium condition is obtained in a similar way. Multiply Eq. (7.25) by $\cos\theta$ before integrating over solid angle and frequency with the result

$$\int_0^\infty \frac{\mathrm{d}K_\nu}{\mathrm{d}\tau_\nu}\mathrm{d}\nu = \frac{\mathfrak{F}_0}{4\pi}. \tag{7.30}$$

The corresponding Milne equation follows by substituting K_ν from Eq. (7.17),

$$\int_0^\infty \frac{\mathrm{d}}{\mathrm{d}\tau_\nu}\left[\tfrac{1}{2}\int_{\tau_\nu}^\infty S_\nu E_3(t_\nu-\tau_\nu)\,\mathrm{d}t_\nu+\tfrac{1}{2}\int_0^{\tau_\nu} S_\nu E_3(\tau_\nu-t_\nu)\,\mathrm{d}t_\nu\right]\mathrm{d}\nu=\frac{\mathfrak{F}_0}{4\pi}. \quad (7.31)$$

The three Milne equations (7.29), (7.24), and (7.31), are not independent. $S_\nu(\tau_\nu)$ that is a solution of one will be the solution of all three.

The flux constant, $\mathfrak{F}_0$, is often expressed in terms of an effective temperature (Eq. 1.2), i.e., $\mathfrak{F}_0=\sigma T_{\mathrm{eff}}^4$. When model photospheres are constructed using flux constancy as a condition to be fulfilled by the model, the effective temperature becomes one of the fundamental parameters characterizing the model.

In the theory of stellar atmospheres, much of the technical effort goes into iteration schemes using Milne's and related equations of radiative equilibrium to find the source function, $S_\nu(\tau_\nu)$. Refer to Kourganoff (1963), Mihalas (1978), Kalkofen (1987), and Adelman *et al.* (1996). The "grey case," discussed in the next section is often a starting point for such iterations.

The grey case

One simplification of the equations of radiative equilibrium deserves special comment because of the historic spot it occupies and because it can serve as a starting point in many iterative calculations. The simplification is to assume that κ_ν is independent of frequency, hence the name grey. Electron scattering is the only opacity source relevant to stellar atmospheres that is grey, and it is usually a minor contributor to κ_ν (see Chapter 8). The grey case is, consequently, not very realistic.

Integrate the basic transfer equation (7.7) over frequency. Denote $\int_0^\infty I_\nu\mathrm{d}\nu$ by I, $\int_0^\infty S_\nu\mathrm{d}\nu$ by S, $\int_0^\infty J_\nu\mathrm{d}\nu$ by J, $\int_0^\infty \mathfrak{F}_\nu\mathrm{d}\nu$ by $\mathfrak{F}$, and $\int_0^\infty K_\nu\mathrm{d}\nu$ by K, Then

$$\cos\theta\frac{\mathrm{d}I}{\mathrm{d}\tau}=I-S,$$

where $\mathrm{d}\tau=\kappa\rho\,\mathrm{d}x$ and κ is the "grey" absorption coefficient. This is a vastly simpler situation because the total radiation is described by a single transfer equation, whereas in the general case, we had an infinite number – one for each frequency.

This type of simplification runs through the radiative equilibrium equations and Milne's equations. Thus the equivalent of the radiative equilibrium conditions, Eqs. (7.22), (7.28), and (7.30), are

$$\mathfrak{F}=\mathfrak{F}_0$$

$$J = S$$
$$\frac{\mathrm{d}K}{\mathrm{d}\tau} = \frac{\mathfrak{F}_0}{4\pi}. \tag{7.32}$$

One of the first solutions was the approximate scheme of Eddington (1926). He assumed, as a first approximation, hemispherically isotropic outward and inward specific intensity,

$$I(\tau) = \begin{cases} I^{\mathrm{out}}(\tau) & \text{for} \quad 0 \le \theta \le \pi/2 \\ I^{\mathrm{in}}(\tau) & \text{for} \quad \pi/2 \le \theta \le \pi. \end{cases}$$

Then I^{out} and I^{in} can be factored out of the spatial integrals, and the mean intensity is

$$J(\tau) = \oint I \frac{\mathrm{d}\omega}{4\pi} = \tfrac{1}{2} I^{\mathrm{out}} \int_0^{\pi/2} \sin\theta \, \mathrm{d}\theta + \tfrac{1}{2} I^{\mathrm{in}} \int_{\pi/2}^{\pi} \sin\theta \, \mathrm{d}\theta$$
$$= \tfrac{1}{2}[I^{\mathrm{out}}(\tau) + I^{\mathrm{in}}(\tau)].$$

In a similar manner,

$$\mathfrak{F}(\tau) = \pi[I^{\mathrm{out}}(\tau) - I^{\mathrm{in}}(\tau)]$$

and

$$K(\tau) = \tfrac{1}{6}[I^{\mathrm{out}}(\tau) + I^{\mathrm{in}}(\tau)].$$

From Eq. (7.32) $S(\tau) = J(\tau)$, giving

$$S(\tau) = J(\tau) = 3K(\tau). \tag{7.33}$$

Then by integrating the third equation in (7.32), we have

$$K(\tau) = \frac{\mathfrak{F}_0}{4\pi}\tau + \frac{\mathfrak{F}_0}{6\pi},$$

where the integration constant is evaluated at $\tau = 0$ using Eq. (7.33). Finally, placing this expression for K in Eq. (7.33), we arrive at the solution, Eddington's approximate solution for the grey case,

$$S(\tau) = \frac{3\mathfrak{F}_0}{4\pi}\left(\tau + \tfrac{2}{3}\right). \tag{7.34}$$

In this simple case, the source function varies linearly with optical depth. This same behavior is not terribly far from the truth in more rigorous solutions.

Using the frequency-integrated form of Planck's law, Eq. (6.5), write $S(\tau) = (\sigma/\pi)T^4(\tau)$ and $\mathfrak{F}_0 = \sigma T_{\text{eff}}^4$, so Eq. (7.34) in LTE becomes

$$T(\tau) = \left[\tfrac{3}{4}\left(\tau + \tfrac{2}{3}\right)\right]^{1/4} T_{\text{eff}}. \tag{7.35}$$

Notice that at $\tau = \frac{2}{3}$ the temperature is equal to the effective temperature, and that $T(\tau)$ scales in proportion to the effective temperature. Equation (7.34) also shows that $\mathfrak{F}(0) = \pi S\left(\frac{2}{3}\right)$, i.e., the surface flux is π times the source function at an optical depth of $\frac{2}{3}$, similar to Eq. (5.7).

Complete and rigorous solutions of the grey case (Chandrasekhar 1957) lead to a solution differing only slightly from Eq. (7.34). It is usually written in the form

$$S(\tau) = \frac{3\mathfrak{F}_0}{4\pi}[\tau + q(\tau)]$$

or

$$T(\tau) = \left\{\tfrac{3}{4}[\tau + q(\tau)]\right\}^{1/4} T_{\text{eff}}, \tag{7.36}$$

where $q(\tau)$ is a slowly varying function ranging from 0.577 at $\tau = 0$ to 0.710 at $\tau = \infty$.

The usefulness of the grey case is small when it comes to interpreting real stellar spectra. It is worthwhile recalling Eddington's (1926) comment concerning the grey case:

> This, however, is a lazy way of handling the problem and it is not surprising that the result fails to accord with observation. The proper course is to find the spectral distribution of the emergent radiation by treating each wave-length separately using its own proper value of j and κ.

Convective transport

Photospheric convection probably exists in most stars, but the rapid outward decline in density and opacity through the photosphere causes convection to be inefficient and contribute only a small fraction of the energy transported. Stars on the cool half of the Hertzsprung–Russell diagram have convective envelopes that extend up into the bottom of the photosphere. For these stars, convection carries enough energy to lower the temperature gradient, as we shall see in Chapter 9. Convection functions because some rising (or falling) volume of gas possesses an excess (or deficit) of heat compared to its surroundings. Such a cell or "energy bucket" must have sufficient optical depth across its dimensions to prevent complete radiative loss of the excess energy to the surroundings. It is immediately clear that convection cannot convey much flux in the optically thin upper photosphere, which is not to say that these layers are not full of motions stemming from convection.

The flux carried by convection can be expressed in terms of the average cell characteristics: the temperature excess, ΔT, of the cell over its terminal surroundings, the specific heat at constant pressure, C_p, the cell density, ρ, and the upward velocity, v, of the cell,

$$\Phi = \rho C_p v \Delta T . \tag{7.37}$$

Both rising and falling cells are involved here with ΔT and v changing sign for the two cases. In the phenomenon of solar granulation, we see the top of the convective cell pattern. This region is at the bottom of the photosphere where $\rho \sim 10^{-7}$ g/cm^2. The observed ΔT and v are of the order of 100 K and a few km/s. Since hydrogen is the dominant gas, $C_p \approx 2.5R \approx 8 \times 10^7$ erg/(g K), where R is the gas constant. Therefore $\Phi \approx 2 \times 10^8$ erg/(s cm^2) compared to $\approx 10^{11}$ erg/(s cm^2) of radiative flux. Convection is a small contributor to the energy flow.

On the other hand, convection plays a major role in shaping line profiles through the Doppler shifts and temperature inhomogeneities it causes. The momentum of convective cells is not dissipated when the optical depth of the cells becomes small, and substantial mass motion can exist above those photospheric layers that are actually producing the convection (sometimes called "overshoot"). In general, there will be a correlation between velocity direction and temperature differences, e.g., upward and hotter versus downward and cooler, so that the Doppler-shift distribution (number of photons per unit Doppler shift) is not symmetrical. The resulting asymmetries and shifts of spectral lines are the main observational tools for studies of stellar granulation (Chapter 17).

Convection also affects stellar atmospheres by generating noise. Acoustic waves may be a major source of power for the lower chromospheres of stars. In a parallel way, convection produces motions in the magnetic field lines that extend through the photosphere into the upper atmosphere. Some of these motions may generate Alfvén waves (waves in the magnetic field lines) which propagate energy upward to power stellar coronae.

Convection is an efficient mixing mechanism, producing chemical homogeneity in photospheres where convection is vigorous. Convection zones inside a star are also mixed by the motions. It is worth noting that the time scale for energy to be conveyed through a convective envelope is weeks to months compared to radiation-transfer time scales of 10^5–10^6 years.

Conditions for convective flow

A hot convective cell must be buoyed upward at each level if it is to continue to be driven upward. Since the density of the photosphere is less in higher layers, and since the cell can be expected to stay in pressure equilibrium with its surroundings, it expands as it rises. We calculate the density change to see if the expanded cell will continue to rise, or if it will lose its buoyancy force and stop convection.

The classical Schwarzschild (1906) criterion is obtained by assuming the convective cell behaves adiabatically – no energy leakage whatsoever from the rising cell (an approximation more suitable to dense stellar interiors than stellar photospheres). Let γ denote the ratio of specific heats and P the total pressure. Then $P\rho^{-\gamma}$ = constant. In order to have convection, the density of the cell must decrease at least as rapidly as the average photospheric density. In other words, to have convection the following must hold:

$$\frac{1}{\gamma} = \left(\frac{\mathrm{d}\log\rho}{\mathrm{d}\log P}\right)_{\text{cell}} > \left(\frac{\mathrm{d}\log\rho}{\mathrm{d}\log P}\right)_{\text{average}}. \tag{7.38}$$

In many calculations, we prefer to use temperature rather than density in these gradients. Using the gas law in the form $P = (\rho/\mu)kT$, where μ is the mean molecular weight, we write

$$\left(\frac{\mathrm{d}\log\rho}{\mathrm{d}\log P}\right)_{\text{average}} = 1 + \frac{\mathrm{d}\log\mu}{\mathrm{d}\log P} - \frac{\mathrm{d}\log T}{\mathrm{d}\log P}.$$

Placing this into inequality (7.38) gives

$$\left(\frac{\mathrm{d}\log T}{\mathrm{d}\log P}\right)_{\text{average}} > 1 - \frac{1}{\gamma} + \frac{\mathrm{d}\log\mu}{\mathrm{d}\log P}. \tag{7.39}$$

as the condition needed for convection.

Radiative transport will always be present; convection can be on or off depending on criterion (7.39) at the position being considered. One way to meet criterion (7.39) is for the gradient on the left-hand side to become large. Large gradients are caused by high opacity, a situation often found in the layers just below the photosphere. Highly opaque layers constrict the outward flow of heat from below. The star is faced with having to expand or, if relation (7.39) is satisfied, it can use convection as a "relief valve."

Convection will also occur if the adiabatic gradient – the right-hand side of relation (7.39) – is reduced. In situations where hydrogen and helium do not change their ionization stage and the gases are chemically homogeneous, $\mathrm{d}\log\mu/\mathrm{d}\log P = 0$; the value of γ for a monatomic gas is $\frac{5}{3}$. The Schwarzschild criterion is then $\mathrm{d}\log T/\mathrm{d}\log P > 0.4$ for convection. Polyatomic gases have γ approaching unity as the degrees of freedom increase with the complexity of the molecules. Therefore convection is encouraged in cool stars where molecules are numerous.

The value of γ may also be decreased from the monatomic value of $\frac{5}{3}$ by the presence of radiation. From the radiation pressure equation, Eq. (5.11), and the first law of thermodynamics, one can show (Chandrasekhar 1957, for example) that the radiation trapped in the convective cell obeys an adiabatic gas law with $\gamma = \frac{4}{3}$.

Should the transition from ionized to neutral hydrogen occur in the photosphere, μ will increase by about a factor of 2. The gradient $\mathrm{d}\log\mu/\mathrm{d}\log P$ is negative, which, according to inequality (7.39) encourages convection. Recombination of the electron and proton releases 13.6 eV of energy per atom as well. The ionization energy heats the gas in the adiabatic cell, causing an increase in volume, again encouraging convection. The ionization is a new degree of freedom, and the effect can be incorporated by redefining γ as done by Unsöld (1955) and Krishna Swamy (1961). Hydrogen and helium ionization zones are standard features of stellar envelope models.

The mixing-length formulation

The simple treatment of convection is called the mixing-length formulation because one is left with a free parameter, the mixing length, that needs to be chosen by guesswork. Most modeling of photospheres and interiors has relied on a dimensional formulation based on Eq. (7.37) to calculate the convective heat transfer, a scheme dating back several decades (Biermann 1933, 1938, Öpik 1950, Vitense 1953). If $\Delta\rho$ is the average difference in density of the convective cell compared to its surroundings, then mechanical energy conservation dictates that $\frac{1}{2}\rho v^2 = g\Delta\rho\ell$, where ℓ is the mixing length, i.e., the distance over which the cell survives. Assuming pressure equilibrium and constancy of the mean molecular weight, $\Delta\rho/\rho = \Delta T/T$, and energy conservation then gives $v = g^{1/2}\ \ell^{1/2}(\Delta T/T)^{1/2}$. Combined with Eq. (7.37), the convective flux can then be written as

$$\Phi = \rho C_p \left(\frac{g\ell}{T}\right)^{1/2} \Delta T^{3/2}.$$

Expressed in terms of temperature gradients,

$$\Phi = \rho C_p \left(\frac{g}{T}\right)^{1/2} \ell^2 \left[\left(\frac{\mathrm{d}T}{\mathrm{d}x}\right)_{\mathrm{cell}} - \left(\frac{\mathrm{d}T}{\mathrm{d}x}\right)_{\mathrm{average}}\right]^{3/2}.$$

The value one uses for the mixing length, ℓ, is arbitrary, i.e., not supplied by "theory." In stellar interior calculations, an empirical calibration is possible because the radius of the model star depends on ℓ, and values of ≈ 1.5 pressure scale heights are typically found. In the case of stellar photospheres, no such simple calibration exists. The solar limb darkening depends weakly on the strength of the convection (see Chapter 9), and is best measured near the opacity minimum at 1.6 μm. Otherwise we must trust to the condition of flux constancy and use Eq. (7.23). The value of the mixing length may also depend on any magnetic field in the material and on the rotation of the star. Several relevant papers can be seen in the conference proceedings edited by Spiegel and Zahn (1977).

Some evidence has been presented (Defouw 1970a, b) that convection may occur even when the radiative gradient is smaller than the adiabatic gradient. There is also observational evidence that convection may occur in the photospheres of hot stars even though theory tells us there is no convective envelope underneath (Gray 1988, 1989, Gray & Nagel 1989).

Significant improvements in handling the full non-linear hydrodynamical equations of more rigorous calculations have been made for solar-type photospheres and the layers immediately below the photospheres (e.g., Nordlund 1980, 1982, 1985, Nordlund & Dravins 1990, Asplund *et al.* 1999, Stein & Nordlund 2000). The convective transport remains small, but the effects on the shapes and shifts of spectral lines is highly significant and we turn our attention to these matters in Chapter 17.

References

Abramowitz, M. & I. A. Stegun, eds. 1964. *Handbook of Mathematical Functions*, National Bureau of Standards Applied Math Series **55**. (Washington, DC: National Bureau of Standards), p. 227.

Adelman, S. J., F. Kupka, & W. W. Weiss, eds. 1996. *M.A.S.S. Model Atmospheres and Stellar Spectra. 5th Vienna Workshop*, Ast. Soc. Pacific Conf. Ser. (San Francisco: Ast. Soc. Pacific).

Asplund, M., Å. Nordlund, R. Trampedach, & R. F. Stein, 1999. *Ast. Ap.* **346**, L17.

Biermann, L. 1932. *Z. f. Ap.* **5**, 117.

1938. *Ast. Nachrichten* **264**, 395.

Chandrasekhar, S. 1957. *An Introduction to the Study of Stellar Structure*. (New York: Dover), p. 55.

Defouw, R. J. 1970a. *Solar Phys.* **14**, 42.

1970b. *Ap. J.* **160**, 659.

Eddington, A. S. 1926. *The Internal Constitution of the Stars*, 1959 printing. (New York: Dover), pp. 322, 325.

Gray, D. F. 1988. *Lectures on Spectral-Line Analysis: F, G, and K Stars*. (Arva, Ontario: The Publisher).

1989. *Pub. Ast. Soc. Pacific* **101**, 832.

Gray, D. F. & T. Nagel 1989. *Ap. J.* **341**, 421.
Kalkofen, W., ed. 1987. *Numerical Radiative Transfer*. (Cambridge: Cambridge University Press).
Kourganoff, V. 1963. *Basic Methods in Transfer Problems*. (New York: Dover).
Krishna Swamy, K. S. 1961. *Ap. J.* **134**, 1017.
Mihalas, D. 1978. *Stellar Atmospheres*, 2nd edn. (San Francisco: Freeman).
Nordlund, Å. 1980. *Stellar Turbulence*, I.A.U. Coll. No. 51, ed. D. F. Gray & J. L. Linsky. (Heidelberg: Springer-Verlag), p. 213.
1982. *Ast. Ap.* **107**, 1.
1985. *Solar Phys.* **100**, 209.
Nordlund, Å. & D. Dravins 1990. *Ast. Ap.* **228**, 155.
Öpik, E. J. 1950. *Mon. Not. Roy. Ast. Soc.* **110**, 559.
Schwarzschild, K. 1906. *Akad. Wiss. Goett. Nachr.*, **1**, 41.
Spiegel, E. A. & J.-P. Zahn, eds. 1977. *Problems of Stellar Convection*, I.A.U. Coll. 38. (Heidelberg: Springer-Verlag).
Stein, R. F. & Å. Nordlund 2000. *Solar Phy.* **192**, 91.
Unsöld, A. 1955. *Physik der Sternatmosphären*. (Berlin: Springer-Verlag), p. 227.
Vitense, E. 1953. *Z. f. Ap.* **32**, 135.

Questions and exercises

1. What are Milne's equations and what are they used for?

2. Find an analytical expression for I_ν at the surface of a star when $S_\nu = a + b\tau_\nu$, i.e., express $I_\nu(\tau_\nu = 0)$ in terms of the constants a and b. The answer is the most elementary form for the limb darkening of a star.

3. In Eq. (7.15) we have an expression for computing the spectrum of a star. Within this expression we see the exponential integral E_2. What conditions are necessary to legitimately use Eq. (7.15) and when must a more rigorous expression be used instead? What is the more rigorous path?

8

The continuous absorption coefficient

As a precursor to calculating the transfer of radiation through a model stellar photosphere, we now look at the continuous absorption coefficient. The wavelength dependence of the continuous absorption coefficient shapes the continuous spectrum emitted by the star: more absorption, less light. The strength of spectral lines also depends on the continuous absorption; more continuous absorption means a thinner photosphere with fewer atoms to make spectral lines. Consequently, we must know κ_ν in order to match model-computed spectra to real stellar spectra. However, before we can compute the theoretical spectrum, we need to compute the model on which it is based, and that too can require κ_ν. Specifically, if we invoke radiative equilibrium to find $T(\tau_0)$, then κ_ν is needed from the very beginning. Furthermore, it then becomes particularly important to know κ_ν in those spectral regions carrying most of the flux. In hot stars, this means the ultraviolet region, in cool stars, the infrared region. We escape this step if $T(\tau_0)$ is determined empirically, say by scaling the solar temperature distribution; the main need for κ_ν reverts to the computation of the spectrum. Since scaled temperature distributions are emphasized in this book, κ_ν in the visible window is the main focus of this chapter.

The origins of continuous absorption

The total continuous absorption coefficient is the sum of the absorption resulting from many physical processes. These are considered here one by one. All of them can be classed into one of two categories. The first is an ionization process where a bound–free transition occurs, and the second is a so-called free–free transition, where one charge is accelerated upon passing close to another charge. The remaining possibility is a bound–bound transition which, of course, gives a spectral line. Spectral lines are not normally included in κ_ν, but when there are many overlapped and crowded lines, they act much like the continuous absorption processes. Most of the continuous opacity is due to hydrogen in some form or other. This is a direct result of the overwhelming dominance of hydrogen in the chemical composition. Reviews of continuous absorption have been given by Vitense (1951), Pecker (1965), and Gustafsson *et al.* (1975).

The atomic absorption coefficient, with units of area per absorber, we call α. It is possible to express α as a function of ν or λ, but since it is not a distribution in the sense that I_ν and I_λ are, we shall not use a ν or λ subscript on α. The wavelength versus frequency units question does not arise for α. On the other hand, it is frequently useful, particularly when dealing with spectral lines (Chapter 11) to describe the amount of power abstracted in the frequency interval $d\nu$ from a unit intensity (I_ν) beam. Such a power is denoted by $\alpha d\nu$. This quantity is a distribution and has units, for example, of erg/(s cm^2 rad^2 Hz) per absorber. The corresponding quantity in wavelength units is $\alpha d\lambda$, which is abstracted from a unit I_λ beam. The relation between them is the usual c/λ^2, thus $\alpha d\nu = (c/\lambda^2)\alpha d\lambda$. The continuous absorption coefficient per neutral hydrogen atom is denoted by κ. Like α, κ is not a distribution. At the end of the chapter, κ is converted from units of area per absorber to area per mass, denoted by κ_ν. The ν subscript here simply reminds us of the important frequency dependence of the continuous absorption coefficient.

Let us not dwell on the atomic theory but instead move directly to the question of how to calculate κ_ν. In each case, we start with α and then multiply by the number of absorbers per hydrogen atom.

The stimulated emission factor

The absorption processes we are about to consider are affected by stimulated emission (see the discussion following Eq. 5.17). Since stimulated emission amounts to negative absorption, the calculated absorption coefficient has to be reduced accordingly. A simple allowance for stimulated emissions follows from Eqs. (5.18), (6.7), and (1.16),

$$\begin{aligned}\kappa_\nu \rho &= N_l B_{lu} h\nu - N_u B_{ul} h\nu \\ &= N_l B_{lu} h\nu\left(1 - \frac{N_u B_{ul}}{N_l B_{lu}}\right) \\ &= N_l B_{lu} h\nu(1 - e^{-h\nu/kT}).\end{aligned} \tag{8.1}$$

Table 8.1. *Stimulated emission factors.*

λT (cm K)	T for $\lambda = 5000$ Å	$1-e^{-h\nu/kT}$	% decrease
0.2	4000	0.999	0.08
0.4	8000	0.973	2.8
0.6	12000	0.909	10.0
0.8	16000	0.834	19.8
1.0	20000	0.763	31.1

In other words, the absorption actually produced is lower by a factor of $1-e^{-h\nu/kT}$ because of stimulated emissions. It is convenient to write this as $1-10^{-\chi_\lambda\theta}$, where $\chi_\lambda = 1.2398 \times 10^4/\lambda$ for λ in Å. This factor need only be applied to the stated absorption coefficients and we do so below in Eq. (8.18), for example. Table 8.1 shows the size of the numbers that are involved. While it is true that we have derived Eq. (8.1) only for the case of bound–bound absorption, Milne (1924) showed that the same factor also holds with continuous absorption. When the validity of Eq. (1.16) is in doubt, a calculation more detailed than Eq. (8.1) may be needed. Such added complications are sometimes necessary when dealing with spectral lines.

Neutral hydrogen

Neutral hydrogen is the major source of absorption for temperatures exceeding about 8000 K. Both the bound–free and the free–free absorption of hydrogen are important. Bound–free transitions, i.e., ionizations, can occur from any level to the continuum. The wavelengths of the spectral lines, the bound–bound transitions, are given by

$$\frac{1}{\lambda} = R\left(\frac{1}{n^2} - \frac{1}{m^2}\right), \tag{8.2}$$

in which $R = 1.0968 \times 10^5 \,\mathrm{cm}^{-1}$ or $R = 1.0968 \times 10^{-3}$ Å^{-1} is the Rydberg constant and m and n are principal quantum numbers. Fixing n at 2 and letting m run from 3 to infinity gives the Balmer line series at visible wavelengths. To produce continuous absorption, the electron must be lifted beyond the "last" orbit. The minimum energy bound–free case is therefore given by $m = \infty$. In this case, the photon energy is just sufficient to ionize the atom, and we can write $h\nu = hc/\lambda = hcR/n^2$, where h denotes Planck's constant. Photons of wavelength shorter than any given ionization threshold therefore impart a kinetic energy

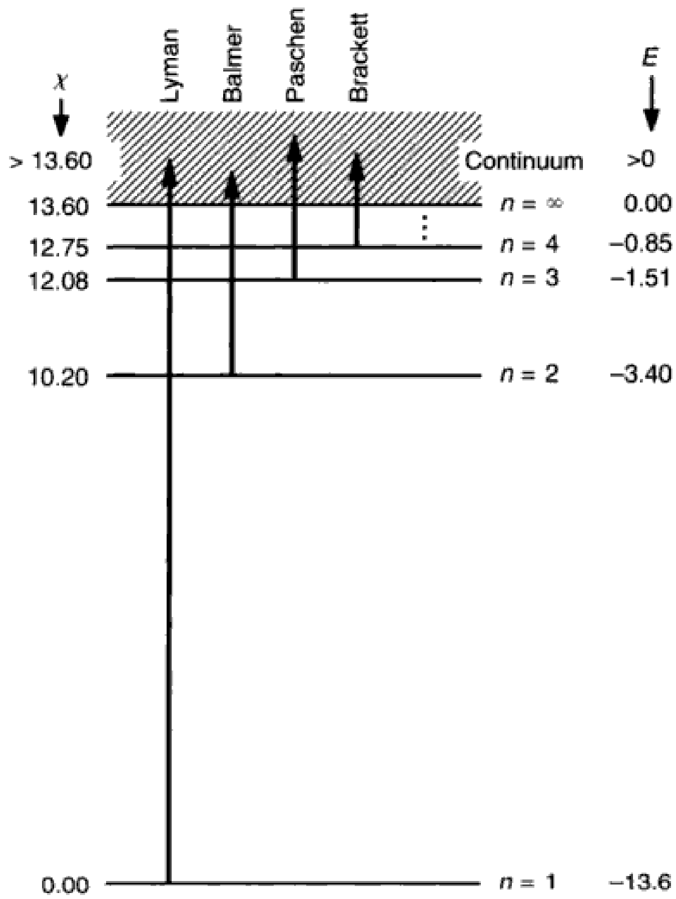


Fig. 8.1. This simplified energy-level diagram for hydrogen shows the quantum numbers, n, binding energy, E, and the excitation potential, χ, in electron volts for the first four levels and the continuum. Transitions from a bound level into the continunm comprise the bound–free absorption.

to the freed electron equal to the energy of the photon minus hcR/n^2. The excitation potential, χ, is defined to be the energy above the ground level, and the continuum is I electron volts above the ground level (hRc/I^2). This is shown graphically in Fig. 8.1. We can then express the excitation potential for the bound levels as

$$\begin{aligned}\chi &= I - hRc/n^2 \\ &= hRc - hRc/n^2 \\ &= 13.60(1-1/n^2)\ \mathrm{eV}.\end{aligned} \tag{8.3}$$

The absorption coefficient corresponding to the bound–free transitions of the type shown in Fig. 8.1 was original derived by Kramers (1923) and modified by Gaunt (1930). Their result can be written as

$$\begin{aligned}\alpha_{\rm bf}({\rm H}) &= \frac{32}{3^{3/2}}\frac{\pi^2 e^6}{h^3}\frac{R}{n^5\nu^3}g_{\rm bf}\\ &= \frac{32}{3^{3/2}}\frac{\pi^2 e^6}{h^3 c^3}R\frac{\lambda^3}{n^5}g_{\rm bf}\\ &= \alpha_0 g_{\rm bf}\frac{\lambda^3}{n^5}\ {\rm cm}^2 \text{ per neutral H atom}\end{aligned} \tag{8.4}$$

with $\alpha_0 = 1.0449\times 10^{-26}$ for λ in angstroms. In this expression, the electron charge has a value $e = 4.803\times 10^{-10}$ esu, and $g_{\rm bf}$ is the Gaunt factor (not to be confused with the statistical weight g_n), which brings Kramer's original equation into agreement with the quantum mechanical results. It is of the order of unity.

There are two Gaunt factors, one for the bound–free absorption, Eq. (8.4), and another for the free–free absorption discussed at the end of this section. Tabulations are given by Karzas and Latter (1961). However, the general expressions based on the work of Menzel and Pekeris (1935) are adequate for most purposes. They are

$$g_{\rm bf} = 1 - \frac{0.3456}{(\lambda R)^{1/3}}\left(\frac{\lambda R}{n^2} - \frac{1}{2}\right) \tag{8.5}$$

and

$$\begin{aligned}g_{\rm ff} &= 1 + \frac{0.3456}{(\lambda R)^{1/3}}\left(\frac{\lambda kT}{hc} + \frac{1}{2}\right)\\ &= 1 + \frac{0.3456}{(\lambda R)^{1/3}}\left(\frac{\log e}{\theta\chi_\lambda} + \frac{1}{2}\right),\end{aligned} \tag{8.6}$$

where λR is unitless and $\chi_\lambda = h\nu = 1.2398\times 10^4/\lambda$ with λ in angstroms is the photon energy.

The behavior of $\alpha_{\rm bf}({\rm H})$ is depicted in Fig. 8.2. At the ionization limits, where the kinetic energy of the released electron is nil, $\lambda = n^2/R$, so $\alpha_{\rm bf}({\rm H})$ at these limits is proportional to n. The Lyman limit is at 912 Å, the Balmer at 3746 Å, etc. As a point of reference, note that the $n = 1$ Bohr radius of the classical hydrogen atom is 0.53 Å, its area is 8.8×10^{-17} cm^2, which is not far from the value of $\alpha_{\rm bf}({\rm H})$ at the 912 Å ionization limit. The cross section for absorption increases as λ^3 for ionization from any given level. Although the λ^3 dependence is very rapid, it is not so rapid that $\alpha_{\rm bf}({\rm H})$ for level n becomes negligible when considering $\alpha_{\rm bf}({\rm H})$ for level $n-1$. The overlap is shown in Fig. 8.2. This means we must sum over several $\alpha_{\rm bf}({\rm H})$ for several values of n. In addition, we have to take into account the populations in each level.

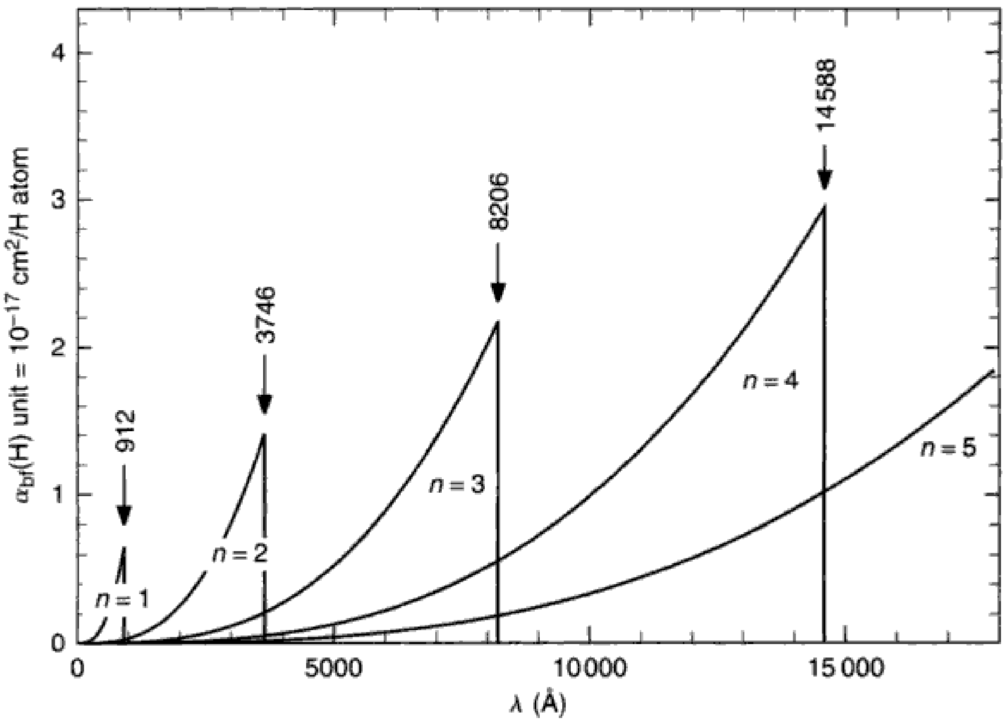


Fig. 8.2. The bound–free absorption coefficient for hydrogen increases with n.

We use Eq. (1.17),

$$\frac{N_n}{N} = \frac{g_n}{u_0(T)} e^{-\chi/kT},$$

where N_n/N is the number of hydrogen atoms excited to level n per neutral hydrogen atom, $g_n = 2n^2$ is the statistical weight, $u_0(T) = 2$ is the partition function, and χ is the excitation potential given by Eq. (8.3). The absorption coefficient in cm^2 per neutral H atom for all continua starting at n_0 is

$$\begin{aligned}
\kappa(H_{bf}) &= \sum_{n_0}^{\infty} \frac{\alpha_{bf}(H) N_n}{N} \\
&= \alpha_0 \sum_{n_0}^{\infty} \frac{\lambda^3}{n^3} g_{bf} e^{-\chi/kT} \\
&= \alpha_0 \sum_{n_0}^{\infty} \frac{\lambda^3}{n^3} g_{bf} 10^{-\theta\chi},
\end{aligned} \tag{8.7}$$

with $\theta = 5040/T$ as usual.

Unsöld (1955) has shown that the small contribution of the terms higher than $n_0 + 2$ can be replaced by an integral,

$$\sum_{n_0+3}^{\infty} \frac{1}{n^3} e^{-\chi/kT} = -\frac{1}{2} \int_{n_0+3}^{\infty} e^{-\chi/kT} d(1/n^2).$$

And according to Eq. (8.3), we have $\mathrm{d}\chi = -I\,\mathrm{d}(1/n^2)$ so that

$$\sum_{n_0+3}^{\infty} \frac{1}{n^3} \mathrm{e}^{-\chi/kT} = \tfrac{1}{2} \int_{\chi_3}^{I} \mathrm{e}^{-\chi/kT} \frac{\mathrm{d}\chi}{I}$$
$$= \frac{kT}{2I} (\mathrm{e}^{-\chi_3/kT} - \mathrm{e}^{-I/kT}),$$

where

$$\chi_3 = I\left[1 - \frac{1}{(n_0+3)^2}\right].$$

We are justified in neglecting the n dependence of g_{bf} because the integral is small compared to the first three terms that we retained explicitly in the summation. Finally, then Eq. (8.7) becomes

$$\kappa(\mathrm{H_{bf}}) = \alpha_0 \lambda^3 \left[\sum_{n_0}^{n_0+2} \frac{g_{\mathrm{bf}}}{n^3} 10^{-\theta\chi} + \frac{\log \mathrm{e}}{2\theta I} (10^{-\chi_3\theta} - 10^{-I\theta}) \right]. \tag{8.8}$$

Here $\log \mathrm{e} = 0.43429$ and α_0 is, again, 1.0449×10^{-26} when λ is in angstroms.

The free–free absorption of neutral hydrogen is much smaller. When a free electron collides with a proton, its orbit (unbound of course) is altered. A photon may be absorbed during such a collision, the orbital energy of the electron being increased by the photon energy. The strength of the absorption depends on the electron velocity, v, because a slow encounter increases the probability of a photon passing by during the collision. According to Kramers (1923), the atomic coefficient is

$$\mathrm{d}\alpha_{\mathrm{ff}}(\mathrm{H}) = \frac{2}{3^{3/2}} \frac{h^2 e^2 R}{\pi m_{\mathrm{e}}^3} \frac{1}{\nu^3 v} \mathrm{d}v,$$

which is the cross section in cm^2 per H atom for the fraction of electrons in the velocity interval v to $v + \mathrm{d}v$. The complete free–free absorption per electron is given by integrating this expression over velocity. In almost every case, it is reasonable to use a Maxwell–Boltzmann speed distribution (Eq. 1.12), which gives

$$\alpha_{\mathrm{ff}}(\mathrm{H}) = \frac{2}{3^{3/2}} \frac{h^2 e^2 R}{\pi m^3} \frac{1}{\nu^3} \int_0^{\infty} \left(\frac{2}{\pi}\right)^{1/2} \left(\frac{m}{kT}\right)^{3/2} v \mathrm{e}^{-(mv^2/kT)} \mathrm{d}v$$
$$= \frac{2}{3^{3/2}} \frac{h^2 e^2 R}{\pi m^3} \frac{1}{\nu^3} \left(\frac{2m}{\pi k T}\right)^{1/2}. \tag{8.9}$$

The quantum mechanical derivation by Gaunt (1930) leads to nearly the same expression, but modified by the correction factor $g_{\rm ff}$ given in Eq. (8.6).

The absorption coefficient in $\rm cm^2$ per neutral hydrogen atom is proportional to the number density of electrons, $N_{\rm e}$, and protons, $N_{\rm i}$, giving

$$\kappa(\mathrm{H_{ff}}) = \frac{\alpha_{\rm ff}(\mathrm{H}) g_{\rm ff} N_{\rm i} N_{\rm e}}{N_0}.$$

The number density of neutral hydrogen is denoted by N_0. Using Eq. (1.21) to obtain $N_{\rm i}N_{\rm e}/N_0$, we write

$$\kappa(\mathrm{H_{ff}}) = \alpha_0 \lambda^3 g_{\rm ff} \frac{\log \mathrm{e}}{2\theta I} 10^{-\theta I}. \tag{8.10}$$

The constant α_0 is defined in Eq. (8.4), and $I = hcR$ and $R = 2\pi^2 me^4/h^3c$ were used in the algebra leading to Eq. (8.10).

In stars of B, A, and F spectral types, neutral hydrogen is the dominant continuous absorber. The energy distributions of such stars show strong modulation at the edges of an increased absorption, e.g., the wavelengths of the ionization limits shown in Fig. 8.2. The major absorbers are compared in Fig. 8.6 below. Other examples of the neutral hydrogen signature can be seen in several of the figures in Chapter 10. But in cooler stars, the absorption from neutral hydrogen diminishes and the absorption from the negative hydrogen ion increases.

The negative hydrogen ion

The hydrogen atom is capable of holding a second electron in a bound state because the simple electron–proton combination is highly polarized. The extra electron is bound with 0.755 eV, so all photons with $\lambda < 16\,421$ Å have sufficient energy to ionize the $\mathrm{H^-}$ ion back to the neutral hydrogen atom and a free electron. The extra electrons needed to form $\mathrm{H^-}$ come from the ionized metals. The effectiveness of $\mathrm{H^-}$ therefore depends on the abundances of these electron donors as well as their state of ionization. In addition to bound–free absorption, free–free absorption can occur when there is a collision between a neutral hydrogen atom and an electron. Following the suggestion by Wildt (1939), calculations as well as laboratory measurements confirm the negative hydrogen ion as a major source of absorption in the Sun. A short review of the historical development is given by Massey (1969;

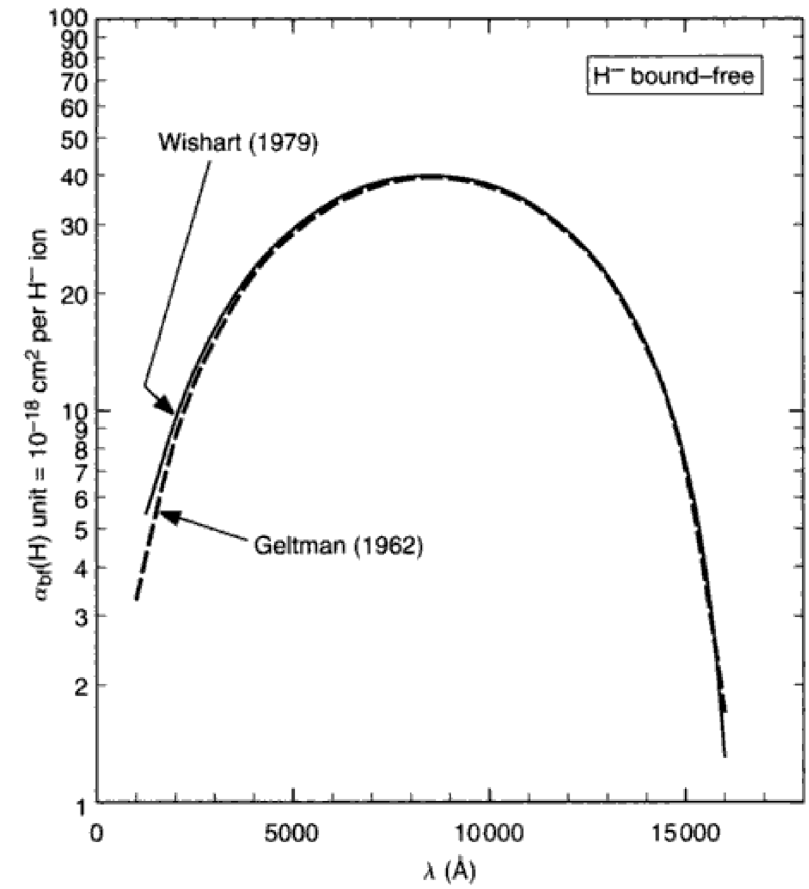


Fig. 8.3. The absorption coefficient of the negative hydrogen ion shows a maximum near 8500 Å. Two calculations are compared.

see also John 1989, 1991, 1992). At temperatures somewhat higher than those in the solar photosphere, not only does the absorption by neutral hydrogen take over, but H^- is ionized to such an extent that it ceases to be a strong absorber (see Fig. 8.5 later). For cooler stars, it is the dominant absorber but drops off rapidly in the very coolest stars because of the scarcity of free electrons.

The bound–free atomic absorption coefficients have been calculated by Geltman (1962), Doughty *et al.* (1966), Krogdahl and Miller (1967), Stewart (1978), Wishart (1979), and John (1988). The wavelength dependence is shown in Fig. 8.3. The maximum is near 8500 Å. The following polynomial fits Wishart's data to better than 0.2% between 2250 and 15 000 Å (about five times better than the expected precision of $\alpha_{bf}(H^-)$ itself).

$$\alpha_{bf}(H^-) = a_0 + a_1\lambda + a_2\lambda^2 + a_3\lambda^3 + a_4\lambda^4 + a_5\lambda^5 + a_6\lambda^6, \tag{8.11}$$

where $\alpha_{bf}(H^-)$ is in units of $10^{-18}\,cm^2/H^-$ ion; the stimulated emission factor is not included, and the constants have the values

$$a_0 = +1.996\,54$$
$$a_1 = -1.182\,67 \times 10^{-5}$$
$$a_2 = +2.642\,43 \times 10^{-6}$$
$$a_3 = -4.405\,24 \times 10^{-10}$$
$$a_4 = +3.239\,92 \times 10^{-14}$$
$$a_5 = -1.395\,68 \times 10^{-18}$$
$$a_6 = +2.787\,01 \times 10^{-23}$$

when λ is in Å. The H^- ionization is given by Eq. (1.19) with $u_0(T) = 1$ and $u_1(T) = 2$, namely,

$$\log \frac{N(H)}{N(H^-)} = -\log P_e - \frac{5040}{T} I + 2.5 \log T + 0.1248.$$

Then,

$$\kappa(H^-_{bf}) = 4.158 \times 10^{-10} \alpha_{bf} P_e \theta^{5/2} 10^{0.754\theta} \tag{8.12}$$

with units of cm^2 per neutral hydrogen atom.

The free–free component of the H^- absorption becomes dominant in the infrared. Calculations have been made by Geltman (1965), Doughty and Frazer (1966), John (1966), Stilley and Callaway (1970), and Bell and Berrington (1987). The behavior of the absorption coefficient with wavelength and temperature can be seen in Fig. 8.4. The following polynomials fit the data of Bell and Berrington to typically $\pm 1\%$ for θ in the interval 0.5–2.0 ($T = 10\,000$–$2\,500$ K) and for λ in the range 2600–113 900 Å. In units of cm^2 per neutral hydrogen atom,

$$\begin{aligned} \kappa(H^-_{ff}) &= \alpha_{ff}(H^-) P_e \\ &= 10^{-26} P_e 10^{f_0 + f_1 \log\theta + f_2 \log^2\theta}, \end{aligned} \tag{8.13}$$

where

$$f_0 = -2.2763 - 1.6850 \log\lambda + 0.766\,61 \log^2\lambda - 0.053\,346 \log^3\lambda$$
$$f_1 = +15.2827 - 9.2846 \log\lambda + 1.993\,81 \log^2\lambda - 0.142\,631 \log^3\lambda$$
$$f_2 = -197.789 + 190.266 \log\lambda - 67.9775 \log^2\lambda + 10.6913 \log^3\lambda - 0.625\,151 \log^4\lambda,$$

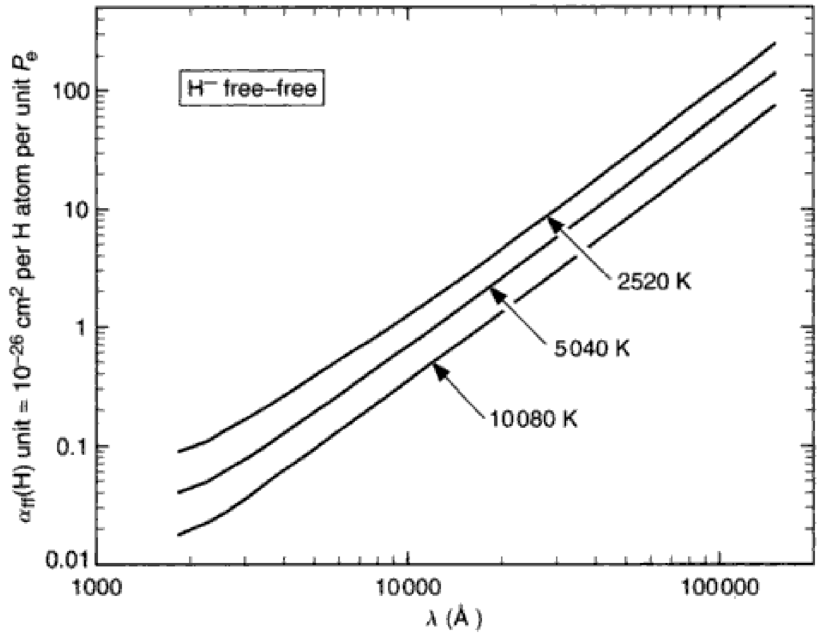


Fig. 8.4. The free–free absorption coefficient of the negative hydrogen ion increases rapidly with wavelength. The stimulated emission factor is included here.

when λ is in angstroms. Here the stimulated emission factor has been included in the polynomial.

The forbidden H^- continuum was found by Myerscough (1968) to be negligible with regard to its effect on the spectrum.

Other hydrogen continuous absorbers

Hydrogen molecules appear in large numbers in cool stars. The H_2 molecules are more numerous than atomic hydrogen for stars cooler than mid-M spectral type, and while the molecule itself does not absorb in the visible spectrum, various ions do. H_2^+ absorption has been studied by Bates (1951, 1952), Buckingham *et al.* (1952), Bates *et al.* (1954), Matsushima (1964), and Stancil (1994). It is a significant absorber in the ultraviolet, but quite generally is only a few per cent of the H^- absorption for $\lambda > 3800$ Å. Its absorption can be calculated from

$$\kappa(H_2^+) = \frac{16\pi^4 a_0^5 e^2}{3hc}\sigma(\lambda, T)N(H^+) \tag{8.14}$$

cm² per neutral hydrogen atom, in which $a_0 = 0.529\,17 \times 10^{-8}$ cm is the radius of the first Bohr orbit, $\sigma(\lambda, T)$ is the atomic cross section, and $N(H^+)$ is the number of protons per cm^3. Using the formulation given by Bates (1952), with the stimulated emission factor removed, one can separate σ into λ and T components and write

$$\begin{aligned}\kappa(H_2^+) &= \frac{16\pi^4 a_0^5 e^2}{3hc}\sigma_1(\lambda)\,10^{-U_1\theta}N(H^+) \\ &= 2.51\times10^{-42}\sigma_1(\lambda)\,10^{-U_1\theta}N(H^+) \\ &= 3.61\times10^{-30}\sigma_1(\lambda)\,10^{-U_1\theta}\theta P_e\frac{\Phi(H)/(1+\Phi(H)/P_e)}{\sum A_j\Phi_j/(1+\Phi_j/P_e)} \end{aligned}\qquad(8.15)$$

cm^2 per neutral hydrogen atom. In this expression, $\theta = 5040/T$, $N(H^+)$ has been replaced by $N_0\Phi(T)/P_e$, where N_0 is the number density of neutral hydrogen atoms, $\Phi(H)$ is the usual Saha ionization function for hydrogen given by Eq. (1.20), Φ_j is the same for the jth element, and A_j is the abundance of the jth element relative to hydrogen. (Casting $N(H^+)$ in terms of P_e is a computational expedient; also see Eq. (9.6)). Polynomial approximations for σ_1 and U_1 are

$$\sigma_1 = -1040.54 + 1345.71\log\lambda - 547.628\log^2\lambda + 71.9684\log^3\lambda,$$

and

$$U_1 = 54.0532 - 32.713\log\lambda + 6.6699\log^2\lambda - 0.4574\log^3\lambda.$$

These approximations match Bates' (1952) data to $\pm0.3\%$ over the wavelength range 3800–25 000 Å.

Although there are no stable bound states for H_2^- (Massey 1976), the free–free absorption becomes important in M and cooler stars, equaling a substantial fraction of the H^- absorption. The molecular equilibrium of H_2 is needed in order to compute this absorption (see Somerville 1964, Dalgarno & Lane 1966, John 1994).

Absorption by helium

In most stars, helium is the next most abundant element after hydrogen, but its electrons are so tightly bound ($\chi_1 = 19.72$ eV, $I_1 = 24.59$ eV, and $I_2 = 54.42$ eV) that its contribution to κ_ν becomes important only in O and early B spectral types, and primarily in the ultraviolet; I_1 corresponds to 504.19 Å, I_2 to 227.82 Å. Absorption edges closer to the visible region are of more interest in the context of stellar photospheres, and these come from the higher levels. Because χ_1 is so large, an excited electron will essentially see a net charge of +1 below it, and so the absorption will mimic that of hydrogen. Furthermore, an approximate treatment is often acceptable since the helium opacity rarely exceeds 10% of the hydrogen opacity. For $n \gtrsim 3$, Ueno (1954) suggested the approximate relation for scaling the hydrogen absorption,

$$\alpha(\text{He}) = 4e^{-10.92/kT}\alpha(\text{H}),$$

in which both $\alpha_{\text{bf}}(\text{H})$ and $\alpha_{\text{ff}}(\text{H})$ would be included in $\alpha(\text{H})$. A detailed treatment of the lower levels can be found in the work of Huang (1948), Underhill (1962), and Vardya (1964).

Because χ_1 is so close to I_1, by the time electrons start populating the excited levels, helium is beginning to ionize. Ionized helium, now a single-electron ion, again behaves much like hydrogen but the energies are scaled up, and the wavelengths scaled down, by a factor of 4.

Toward cool temperatures, the contribution of negative helium ion free–free transitions become increasingly significant. (The bound–free absorption is generally considered to be negligible for He^- because there is only one bound level with an ionization potential of 19 eV. The population of such a level is, under normal conditions, too small to warrant consideration.) The free–free absorption can be calculated from the data of McDowell *et al.* (1966) or John (1994); a useful approximation to John's tabulation is

$$\log\alpha(\text{He}^-_{\text{ff}}) = c_0 + c_1\log\lambda + c_2\log^2\lambda + c_3\log^3\lambda.$$

The coefficients, with $\theta = 5040/T$, are given by

$$c_0 = +9.66736 - 71.76242\theta + 105.29576\theta^2 - 56.49259\theta^3 + 10.69206\theta^4$$

$$c_1 = -10.50614 + 48.28802\theta - 70.43363\theta^2 + 37.80099\theta^3 - 7.15445\theta^4$$

$$c_2 = +2.74020 - 10.62144\theta + 15.50518\theta^2 - 8.33845\theta^3 + 1.57960\theta^4$$

$$c_3 = -0.19923 + 0.77485\theta - 1.13200\theta^2 + 0.60994\theta^3 - 0.11564\theta^4,$$

giving $\alpha(\text{He}^-_{\text{ff}})$ to a precision of 1% or better between 5000–150 000 Å and θ in the range of 0.5–2.0. The stimulated-emission factor has been included in these polynomials.

The absorption coefficient in cm^2 per hydrogen atom is then given by

$$\kappa(\text{He}^-_{\text{ff}}) = \frac{\alpha(\text{He}^-_{\text{ff}})A(\text{He})P_e}{1 + \Phi(\text{He})/P_e}. \tag{8.16}$$

The number abundance of helium relative to hydrogen is $A(\text{He})$, and the factor of $1 + \Phi(\text{He})/P_e$ in the denominator takes into account the ionization of

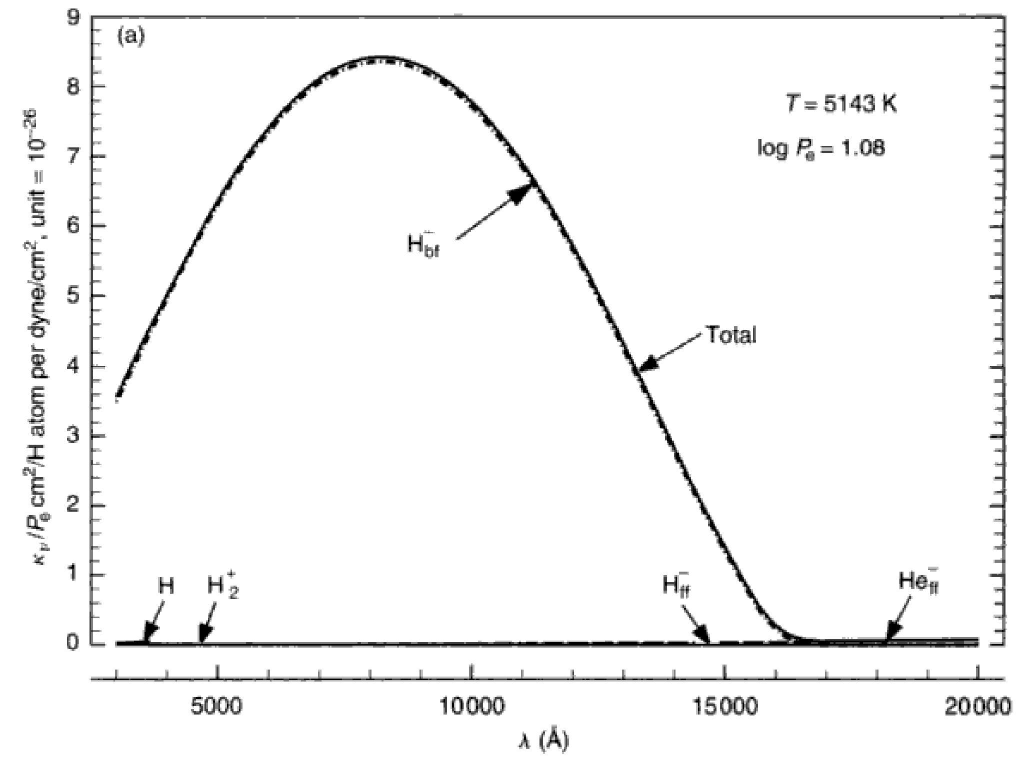


Fig. 8.5. (a) Absorption coefficients per unit electron pressure are compared for a temperature of 5143 K. The absorption is dominated by the H^-_{bf} ion. The stimulated-emission factor has been included in these comparisons (all panels). (b) The absorption coefficients per unit electron pressure are compared for a temperature of 6429 K. The absorption is dominated by the two components of the H^- ion. (c) The absorption coefficients per unit electron pressure are compared for a temperature of 7715 K. Here absorption from the H^- ion is dropping back compared to the cooler cases, while neutral hydrogen has grown with increasing temperature. (d) The absorption coefficients per unit electron pressure are compared for a temperature of 11 572 K. The absorption is now dominated by the neutral hydrogen component. Notice the large increase in the absolute scale of the ordinate compared to the previous panels.

helium, where Φ(He) is given by Eq. (1.20). As seen in Fig. 8.5, this ion is usually a small contributor to the total opacity.

Electron scattering

Free electrons scatter radiation with the same efficiency at all wavelengths. The absorption coefficient is

$$\begin{aligned}\alpha(\mathrm{e}) &= \frac{8\pi}{3}\left(\frac{e^2}{mc^2}\right)^2 \\ &= 0.6648 \times 10^{-24}\ \mathrm{cm^2/electron}.\end{aligned}$$

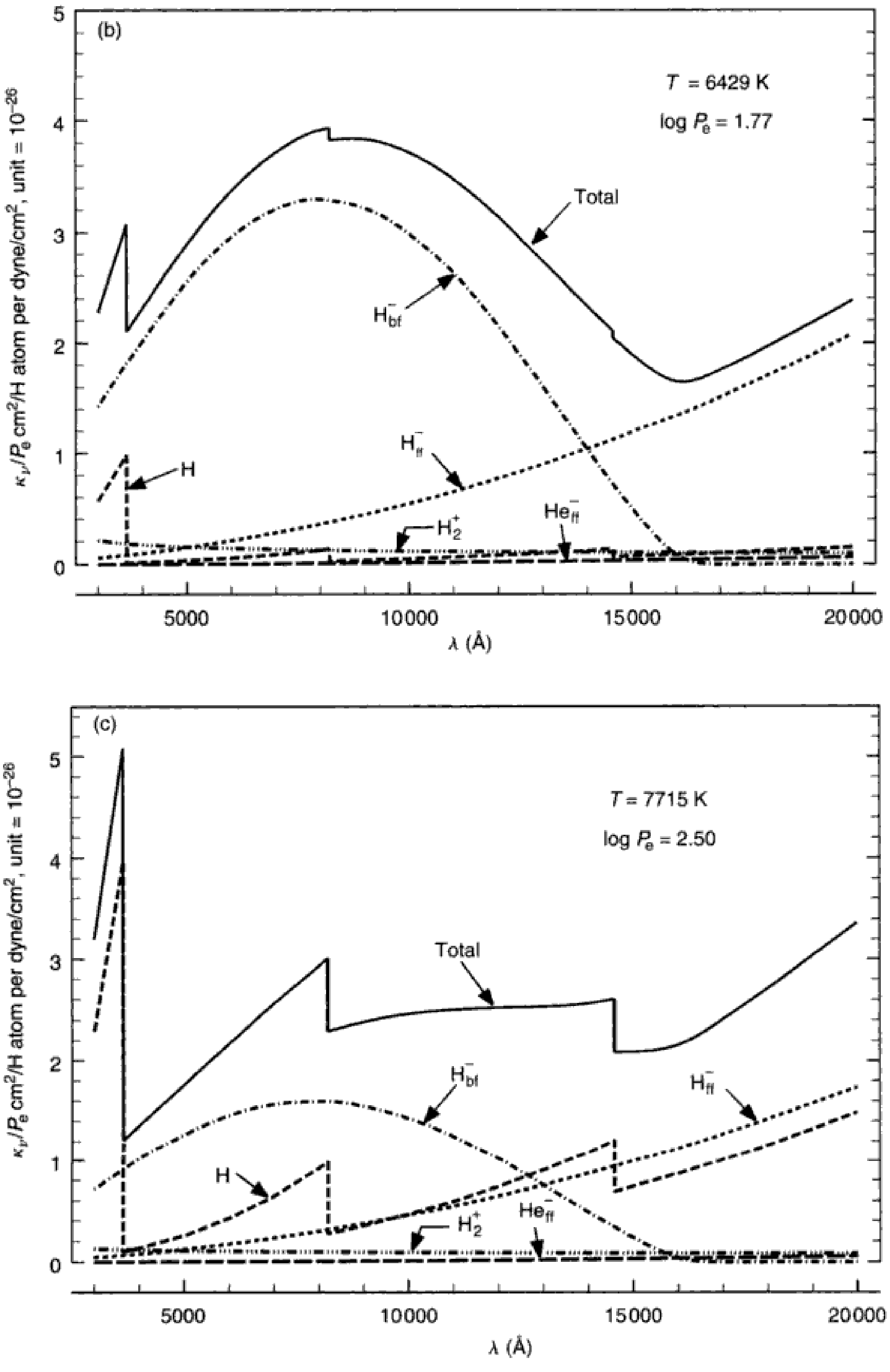

(b)
T = 6429 K
log P_e = 1.77
Total
H^-_{bf}
H^-_{ff}
H
H_2^+
He^-_{ff}
κ_ν/P_e cm^2/H atom per dyne/cm^2, unit = 10^{-26}
λ (Å)
(c)
T = 7715 K
log P_e = 2.50
Total
H^-_{bf}
H^-_{ff}
H
H_2^+
He^-_{ff}
κ_ν/P_e cm^2/H atom per dyne/cm^2, unit = 10^{-26}
λ (Å)

Fig. 8.5. (*cont.*)

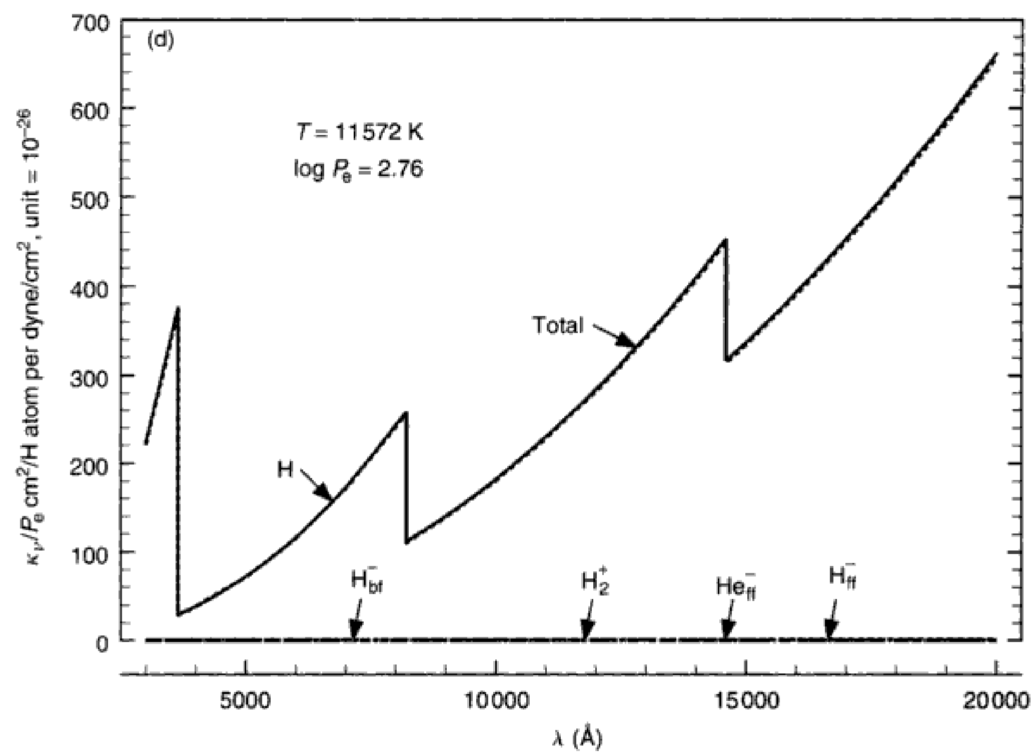


Fig. 8.5. (*cont.*)

It has generally been assumed that the intrinsic non-isotropy in the scattering process (phase function of the form $1+\cos^2\theta$) averages to zero over the symmetrical geometry of the star and many scatterings through the atmosphere. We write the absorption per hydrogen atom as

$$\kappa(\mathrm{e}) = \alpha(\mathrm{e})\frac{N_\mathrm{e}}{N_\mathrm{H}} = \alpha(\mathrm{e})\frac{P_\mathrm{e}}{P_\mathrm{H}},$$

where N_H is the number density of hydrogen and P_H is the partial pressure of hydrogen. P_H is related to the gas and electron pressure in the following way. If N is taken to represent the total number of particles per cubic centimeter, then

$$N = N_\mathrm{e} + \sum N_j = N_\mathrm{e} + N_\mathrm{H}\sum A_j$$

with N_j particles of the jth element per cubic centimeter, and $A_j = N_j/N_\mathrm{H}$. We solve for N_H and write

$$N_\mathrm{H} = \frac{N - N_\mathrm{e}}{\sum A_j} \qquad \text{or} \qquad P_\mathrm{H} = \frac{P_\mathrm{g} - P_\mathrm{e}}{\sum A_j},$$

which leads to

$$\kappa(\mathrm{e}) = \alpha(\mathrm{e})\frac{P_\mathrm{e}}{P_\mathrm{g} - P_\mathrm{e}}\sum A_j. \tag{8.17}$$

Again, $\kappa(\mathrm{e})$ has units of cm^2 per hydrogen particle. The stimulated-emission factor should not be used for $\kappa(\mathrm{e})$.

Electron scattering contributes significant absorption in the photospheres of O and early B stars. Since the hydrogen is ionized, $P_\mathrm{e} \approx 0.5P_\mathrm{g}$ and $\kappa(\mathrm{e}) \approx \alpha(\mathrm{e})\Sigma A_j \approx 7\times10^{-25}\,\mathrm{cm}^2$/hydrogen particle. To be consistent with the graphs in Fig. 8.5, we normalize to the electron pressure, typically 10^1–$10^3\,\mathrm{dyne/cm}^2$. Compared to the absorption shown in the figure, $\kappa(\mathrm{e})/P_\mathrm{e}$ is small unless the electron pressure becomes low, as in supergiant atmospheres. Toward still higher temperatures, electron scattering plays a stronger role as the absorption from other sources fades. In cool stars, $P_\mathrm{e} \ll P_\mathrm{g}$, and then $\kappa(\mathrm{e})/P_\mathrm{e} \approx 7\times10^{-25}/P_\mathrm{g}$. With $P_\mathrm{g} \approx 10^3$–$10^5\,\mathrm{dyne/cm}^2$, electron scattering can be neglected.

Rayleigh scattering is important in the ultraviolet in stars cool enough to have molecules in their atmospheres. The reader is referred to Vitense (1951), Gingerich (1964), Vardya (1970), or Allen (1973) for further details.

Other sources of continuous absorption

Elements such as carbon, silicon, aluminum, magnesium, and iron produce bound–free opacity which is important and often dominant in the ultraviolet, especially for $\lambda < 2500$ Å. However, data for the UV is incomplete. Many UV studies of solar-type and cooler stars refer to a "missing opacity" (see Bell *et al.* 2001, Allende Prieto *et al.* 2003, Short & Hauschildt 2003). In any case, the opacity increases so rapidly with decreasing wavelength that the atmospheric levels from which the radiation comes lie increasingly above the photosphere.

The proper way to handle the absorption is to combine well-determined experimental cross sections with the number of absorbers. This can actually be done in a few cases where the cross sections have been measured. Rich (1967), for example, measured silicon ($37\times10^{-18}\,\mathrm{cm}^2$ per absorber for the ground state) and his results were used by Gingerich and Rich (1966) to show that silicon dominates the metal opacity in the 1300–2000 Å region with absorption edges at 1526 Å (ground level ionization) and 1682 Å (first excited level ionization). Recent cross section results are often larger than the earlier ones, e.g., Seaton *et al.* (1994). A large upward revision in the continuous absorption of iron between 2000 and 4000 Å comes closer to explaining the UV spectrum of

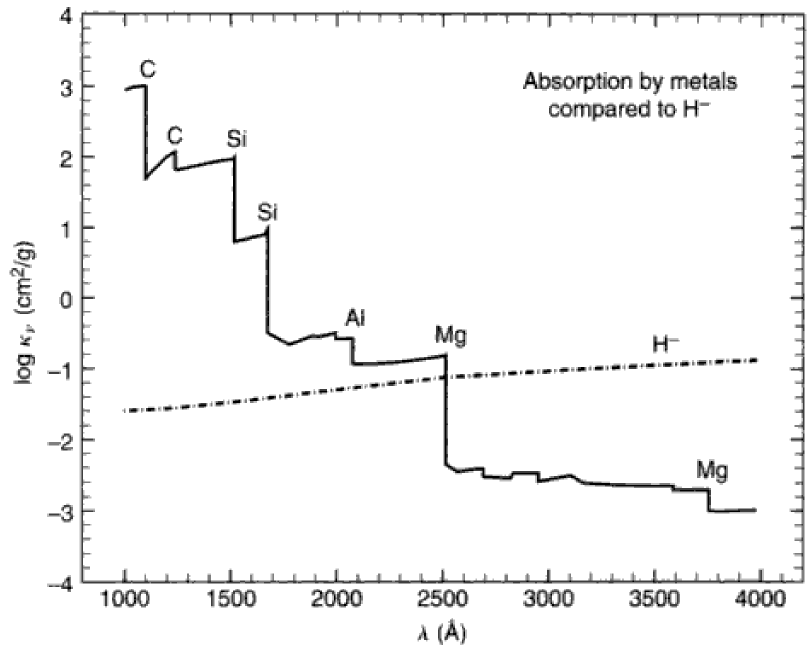


Fig. 8.6. This example of the absorption of metals in the UV is based on the work of Travis and Matsushima (1968).

the Sun as well as greatly weakening the conclusion that beryllium is depleted (Bell *et al.* 2001, Cowley & Bautista 2003).

But in many cases, less dependable estimates must be used. For those cases in which a single outer electron is involved in the transition, one might expect the ionization behavior to resemble that of hydrogen. An empirically adjusted nuclear charge is often used in the computations in place of the true charge. This method leads to a silicon absorption, for instance, which is only one-tenth the observed value mentioned above, a rather obvious danger sign. A better theoretical approach was given by Burgess and Seaton (1960) called the quantum defect method. It can only be applied to situations of LS coupling. Application has been made by Peach (1967, 1970) and Travis and Matsushima (1968), among others. Extensive tables of absorption coefficients have been published by the latter authors. The accuracy of the quantum defect method has been considered by Matsushima and Travis (1973).

The nature of the metal absorption coefficient is shown in Fig. 8.6 taken from the work of Travis and Matsushima (1968). A more recent tabulation is given by Dragon and Mutschlecner (1980), and Seaton *et al.* (1994) offer further improvements. The strength of the individual components obviously depends on the mix of the elements. The results of Fig. 8.6 are calculated using the abundances of Goldberg *et al.* (1960).

Several molecular ions such as CN^-, C_2^-, and H_2O^- are important continuous absorbers in late-type stars. A general approximation for free–free absorption coefficients of many atomic and molecular species is presented by Dalgarno and Lane (1966). They explicitly consider H, He, N, O, Ne, H_2, N_2, and O_2. Linsky (1969) considers molecular hydrogen and other absorbers expected to be at work in late-type stars. Water vapor opacity has been calculated by Auman (1966, 1967). A brief review of opacity in late-type stars is given by Vardya (1970). A more detailed discussion of some absorbers is presented by Massey (1969).

At temperatures of cool M types and below, opacity is contributed by numerous molecules such as TiO, TiO_2, CrH, FeH, and VO. MgH is computed by Weck *et al.* (2003b). Solid grains and liquids condense in the outer photospheres of very cool stars, from brown dwarfs to red giants. Grains of many types have been included in some models, e.g., Tsuji *et al.* (1996), Allard *et al.* (2001), and Ferguson *et al.* (2001).

Line opacity

Although line opacity is not strictly a part of the continuous opacity, the cumulative effect of many lines nevertheless can behave much like continuous opacity in the upper photosphere. This can be rather important when $T(\tau_\nu)$ is determined by imposing radiative equilibrium, as emphasized by Kurucz (1979). The problems associated with line opacity stem from the very large number of lines that can arise and the need to know their identification and atomic parameters. Data for millions of lines have been compiled for this purpose (Kurucz & Peytremann 1975, Kurucz 1979; but see Bell & Tripicco 1995). Line opacity depends on temperature, pressure, chemical composition, and, to a lesser extent, on microturbulence (see Chapter 13). It can also vary markedly from one spectral region to the next, although generally there are more atomic lines at shorter wavelengths within the visible window (refer to Fig. 10.12).

For the M stars and cooler, so many spectral lines, especially those of molecules, flood the whole spectrum that they become a major component in determining photospheric structure as well as the output spectrum. Ideally the line absorption coefficient should be computed at several points across each spectral line as a function of optical depth, based on proper transition probabilities and damping constants. The atomic and molecular data are often not available.

Simplifications have been introduced to expedite the numerical work. One technique is to generate an opacity distribution function for the lines. It then enters the model calculations much like a normal continuous absorption coefficient (see Kurucz 1979, for example). Another technique relies on a statistical sampling of the absorption at $\sim 10^3$ wavelengths to approximate the true picture (Peytremann 1974, Sneden *et al.* 1976, Johnson *et al.* 1977, Bessell *et al.* 1989). The relative merits of these schemes are discussed by Carbon (1979).

The more ambitious approach becomes more tractable as computing power increases. Examples can be seen in Alexander and Ferguson (1994), Jorgensen (1994), Allard *et al.* (2001), and Weck *et al.* (2003a). See also the discussion and references in Gustafsson and Jorgensen (1994).

The total absorption coefficient

We add the various absorption coefficients, reducing by the stimulated-emission factor those terms not implicitly including it,

$$\kappa_{\text{total}} = \{[\kappa(\text{H}_{\text{bf}}) + \kappa(\text{H}_{\text{ff}}) + \kappa(\text{H}^-_{\text{bf}}) + \kappa(\text{H}_2^+)](1-10^{-\chi_\lambda \theta}) + \kappa(\text{H}^-_{\text{ff}}) + \cdots\}$$
$$\times \frac{1}{1+\Phi(\text{H})/P_e} + \kappa(\text{metals}) + \kappa(\text{He}^-_{\text{ff}}) + \kappa(\text{e}) + \cdots. \tag{8.18}$$

The first few coefficients are reduced by the stimulated-emission factor in which $\chi_\lambda = 1.2398 \times 10^4/\lambda\,\text{eV}$ when λ is in angstroms; the factor of $1/[1+\Phi(T)/P_e]$ converts the absorption coefficients from normalization per neutral hydrogen atom to per hydrogen particle, thus taking into account the ionization of hydrogen at high temperatures (see Eq. 1.20).

The absorption coefficient in cm^2 per hydrogen particle as given above is then converted to units of cm^2/g by dividing by the number of grams per hydrogen particle,

$$\kappa_\nu = \frac{\kappa_{\text{total}}}{\sum A_j \mu_j}. \tag{8.19}$$

The mass, μ_j, of the jth element in the atmosphere is 1.6606×10^{-24}g times the atomic weight of the element. Again, A_j denotes the number abundance relative to hydrogen. A typical value of $\sum A_j \mu_j$ is 2.0×10^{-24}g/H particle.

Some investigators prefer to interpolate in tables of pre-computed absorption coefficients. One of the first comprehensive tabulations was done by Vitense (1951). A summary of these is given by Allen (1963). Bode (1965) has computed tables of absorption coefficients, which are again summarized by Allen (1973). The main difficulty with tabulations is their lack of flexibility, especially since the opacity depends on the chemical composition.

In the next chapter, we turn our attention to using these absorption coefficients in the construction of some model photospheres.

References

Alexander, D.R. & J.W. Ferguson 1994. *Ap. J.* **437**, 879.
Allard, F., P.H. Hauschildt, D.R. Alexander, A. Tamanai, & A. Schweitzer 2001. *Ap. J.* **556**, 357.
Allen, C.W. 1963. *Astrophysical Quantities*, 2nd edn. (London: Athlone), p. 96.
1973. *Astrophysical Quantities*, 3rd edn. (London: Athlone) p. 99.
Allende Preito, C., I. Hubeny, & D.L. Lambert 2003. *Ap. J.* **591**, 1192.
Auman, J.R., Jr. 1966. *Colloquium on Late Type Stars*, ed. M. Hack. (Trieste: Osservatorio Astronomico), p. 215.
1967. *Ap. J. Suppl.* **14**, 171.
Bates, D.R. 1951. *Mon. Not. Roy. Ast. Soc.* **111**, 303.
1952. *Mon. Not. Roy. Ast. Soc.* **112**, 40.
Bates, D.R., K. Ledsham, & A.L. Stewart 1954. *Phil. Trans. Roy. Soc. London*, Ser. A, **246**, 215.
Bell, K.L. & K.A. Berrington 1987. *J. Phys. B: Atomic Mol. Phys.* **20**, 801.
Bell, R.A. & M.J. Tripicco 1995. *Astrophysical Applications of Powerful New Databases*, ed. S. J. Adelman & W.L. Weise, Ast. Soc. Pacific Conf. Ser. **78**. (San Francisco: Ast. Soc. Pacific), p. 365.
Bell, R.A., S.C. Balachandran, & M. Bautista 2001. *Ap. J. Lett.* **546**, L65.
Bessell, M.S., J.M. Brett, P. R. Wood & M. Scholz 1989. *Ast. Ap.* **213**, 209.
Bode, G. 1965. Institutsbericht, Institut fur Theoretische Physik und Sternwarte der Universität Kiel, Keil, Germany.
Buckingham, R.A., S. Reid, & R. Spence 1952. *Mon. Not. Roy. Ast. Soc.* **112**, 382.
Burgess, A. & M.J. Seaton 1960. *Mon. Not. Roy. Ast. Soc.* **112**, 382.
Carbon, D.F. 1979. *Ann. Rev. Ast. Ap.* **17**, 513.
Cowley, C.R. & M. Bautista 2003. *Mon. Not. Roy. Ast. Soc.* **341**, 1226.
Dalgarno, A. & N.F. Lane 1966. *Ap. J.* **145**, 623.
Doughty, N.A. & P.A. Frazer 1966. *Mon. Not. Roy. Ast. Soc.* **132**, 267.
Doughty, N.A., P.A. Frazer, & R. P. McEachran 1966. *Mon. Not. Roy. Ast. Soc.* **132**, 255.
Dragon, J.N. & J.P. Mutschlecner 1980. *Ap. J.* **239**, 1045.
Ferguson, J.W., D.R. Alexander, F. Allard, & P. H. Hauschildt 2001. *Ap. J.* **557**, 798.
Gaunt, J.A. 1930. *Phil. Trans. Roy. Soc. London*, Ser. A, **229**, 163.
Geltman, S. 1962. *Ap. J.* **136**, 935.
1965. *Ap. J.* **141**, 376.
Gingerich, O. 1964. *Proc. First Harvard-Smithsonian Conf. on Stellar Atmospheres*, Smithsonian Special Report 167, Cambridge, MA, ed. E.H. Avrett, O. Gingerich, & C.A. Whitney, p. 17.
Gingerich, O. & J. C. Rich 1966. *Ast. J.* **71**, 161.
Goldberg, L., E.A. Müller, & L.H. Aller 1960. *Ap. J. Suppl.* **5**, 1.
Gustafsson, B. & U.G. Jorgensen 1994. *Ast. Ap. Review* **6**, 19.
Gustafsson, B., R.A. Bell, K. Eriksson, & A. Nordlund 1975. *Ast. Ap.* **42**, 407.
Huang, S.-S. 1948. *Ap. J.* **108**, 354.
John, T.L. 1966. *Mon. Not. Roy. Ast. Soc.* **131**, 315.
1988. *Ast. Ap.* **193**, 189.

1989. *Mon. Not. Roy. Ast. Soc.* **240**, 1.
1991. *Ast. Ap.* **244**, 511.
1992. *Mon. Not. Roy. Ast. Soc.* **256**, 69.
1994. *Mon. Not. Roy. Ast. Soc.* **269**, 865.
Johnson, H. R., J. G. Collins, B. Krupp, & R. A. Bell 1977. *Ap. J.* **212**, 760.
Jorgensen, U. G. 1994. *Ast. Ap.* **284**, 79.
Karzas, W. J., & R. Latter 1961. *Ap. J. Suppl.* **6**, 167.
Kramers, H. A. 1923. *Phil. Mag.* **46**, 836.
Krogdahl, W. S. & J. E. Miller 1967. *Ap. J.* **150**, 273.
Kurucz, R. L. 1979. *Ap. J. Suppl.* **40**, 1.
Kurucz, R. L. & E. Peytremann 1975. Smithsonian Inst. Ap. Obs. Res. Space Sci., report 362.
Linsky, J. L. 1969. *Ap. J.* **156**, 989.
Massey, H. S. W. 1969. *Electronic and Ionic Impact Phenomena*, vol. 2. (Oxford: Oxford University Press), p. 1255.
1976. *Negative Ions*, 3rd edn. (Cambridge: Cambridge University Press).
Matsushima, S. 1964. *Proc. First Harvard-Smithsonian Conf. Stellar Atmospheres*, Smithsonian Special Report 167, Cambridge, MA, p. 42.
Matsushima, S. & L. D. Travis 1973. *Ap. J.* **181**, 387.
McDowell, M. R. C., J. H. Williamson, & V. P. Myerscough 1966. *Ap. J.* **144**, 827.
Menzel, D. H. & C. L. Pekeris 1935. *Mon. Not. Roy. Ast. Soc.* **96**, 77.
Milne, E. A. 1924. *Phil. Mag.* **47**, 209.
Myerscough, V. P. 1968. *Ap. J.* **152**, 1115.
Peach, G. 1967. *Mem. Roy. Ast. Soc.* **71**, 1.
1970. *Mem. Roy. Soc.* **73**, 1.
Pecker, J. C. 1965. *Ann. Rev. Ast. Ap.* **3**, 135.
Peytremann, E. 1974. *Ast. Ap.* **33**, 203.
Rich, J. C. 1967. *Ap. J.* **148**, 275.
Seaton, M. J., Y. Yan, D. Mihalas, & A. K. Pradhan 1994. *Mon. Not. Roy. Ast. Soc.* **266**, 805.
Short, C. I. & P. H Hauschildt 2003. *Ap. J.* **596**, 501.
Sneden, C., H. R. Johnson, & B. M. Krupp 1976. *Ap. J.* **204**, 281.
Somerville, W. B. 1964. *Ap. J.* **139**, 192.
Stancil, P. C. 1994. *Ap. J.* **430**, 360.
Stewart, A. L. 1978. *J. Phys. B, Atom. Mol. Phys.* **11**, 3851.
Stilley, J. L. & J. Callaway 1970. *Ap. J.* **160**, 245.
Travis, L. D. & S. Matsushima 1968. *Ap. J.* **154**, 689.
Tsuji, T., K. Ohnaka, W. Aoki, & T. Nakajima 1996. *Ast. Ap.* **308**, L29.
Ueno, S. 1954. *Kyoto Contr.* Nr. **42**.
Underhill, A. B. 1962. *Pub. Dominion Ap. Obs. Victoria* **XI**, 467.
Unsöld, A. 1955. *Physik der Sternatmosphären*, 2nd edn. (Berlin: Springer-Verlag), p. 168.
Vardya, M. S. 1964. *Ap. J. Suppl.* **8**, 277.
1970. *Ann. Rev. Ast. Ap.* **8**, 87.
Vitense, E. 1951. *Z. f. Ap.* **28**, 81.
Weck, P. F., A. Schweitzer, P. C. Stancil, P. H. Hauschildt, & K. Kirby 2003a. *Ap. J.* **582**, 1059.
2003b. *Ap. J.* **584**, 459.
Wildt, R. 1939. *Ap. J.* **89**, 295.
Wishart, A. W. 1979. *Mon. Not. Roy. Ast. Soc.* **187**, 59P.

Questions and exercises

1. What is the dominant absorber in: (i) K dwarfs, (ii) K giants, (iii) A dwarfs, and (iv) O stars?
2. Why is it that the absorption of the negative hydrogen ion declines with increasing effective temperature?
3. Why does the absorption coefficient of neutral hydrogen show sharp edges?
4. Use Eq. (8.3) with Eq. (1.18) to find the relative populations of hydrogen atoms excited to the first few levels, e.g., $n = 1, 2, 3$, etc. Consider what these numbers mean in the context of bound–free and bound–bound absorption.
5. Look at Fig. 10.3. Explain, using your knowledge of black bodies and absorption coefficients, why Vega's continuous spectrum looks like it does. Be as specific as possible, relating specific wavelengths and atomic parameters to features you identify in the spectrum.

9

The model photosphere

What is a model photosphere, and why build one? It would seem logical to take our stellar observations and deduce from them the physical conditions existing in the atmosphere of the star – somewhat like a parallax measurement yields the distance to a star. Alas, the formation of the stellar spectrum involves many physical variables, and a rigorous deductive interpretation cannot generally be made. Instead we hypothesize a model through which we organize and relate the information conveyed in the starlight. The model resembles a scientific theory in that it is constructed on the basis of our observations and known physical laws. We may then gather additional observations to test our model much like we would test a theory. The model is modified and improved as information is added. When our model closely reproduces all the available observations, we begin to feel that our model is worthy of some trust. Then properties associated with the model, or deduced from further application of the model, are associated with the star, properties such as effective temperature, surface gravity, radius, chemical composition, or rate of rotation.

The model photosphere consists of a table of numbers giving the source function and the pressure as a function of optical depth for an assumed chemical composition. Additional columns may be added to the table depending on the use intended for the model. For example, one might also list the density, geometrical depth, absorption coefficients, electron pressure, mean intensity, velocity-field parameters, or magnetic field strength. An example can be seen in Table 9.2 below. From the model we often compute the surface flux, using for example, Eq. (7.15). Notice that this is the light escaping the model stellar surface *per unit area*. A model photosphere does not yield the luminosity or total power output until it is combined with an independently known stellar radius (see Eq. 10.5). A model photosphere differs fundamentally from a model interior on this point.

The material in this chapter is directed toward model photospheres that can be readily constructed. They are somewhat rudimentary compared to real stars. The reader should not jump to the conclusion that such models are useless – they are, in fact, indispensable tools for the analysis of stellar spectra, and they are certainly a good starting point for learning about model photospheres. We start with the usual simplifications.

1. Plane parallel geometry, making all physical variables a function of only one space coordinate.
2. Hydrostatic equilibrium, meaning that the photosphere is not undergoing large-scale accelerations comparable to the surface gravity; there is no dynamically significant mass loss.
3. Structures such as granulation or starspots are negligible, or at least can be adequately represented by mean values of the physical parameters.
4. Magnetic fields are excluded.

Later, in Chapters 17 and 18, we consider aspects of velocity fields and magnetic fields. It is also true that in the majority of cases we are thinking in terms of the thermodynamic equilibrium relations describing the excitation and ionization of the gas (Eqs. 1.18 and 1.19) and in terms of the black-body source function $B_\nu(T)$.The thermodynamic equilibrium concept is applied to relatively small volumes of the model's photosphere – volumes with dimensions of the order of unity in optical depth. For this reason, the name *local* thermodynamic equilibrium, usually abbreviated as LTE, was coined. The photosphere may then be characterized by one physical temperature at each depth. The excitation, ionization, source function, and thermal velocity distributions in the vicinity of one point are all described by this unique temperature. Progressing outward through the photosphere, each successive volume is assigned a lesser temperature so that in the LTE situation it is customary to replace the tabulation of the source function as it varies with depth by a tabulation of the temperature. The essential model then consists of temperature and pressure given as a function of optical depth. As we shall see, $B_\nu(T)$ makes a reasonable source function for many situations, but can fail badly for spectral lines.

The equation of hydrostatic equilibrium

The relation between pressure and optical depth is established from the assumption of hydrostatic equilibrium. Consider the small volume in Fig. 9.1. The difference in pressure between the top and bottom of the volume is the weight per unit area of the mass in the volume. The weight is the product of the density, ρ, volume, $\mathrm{d}A\,\mathrm{d}x$, and surface gravity, g. We let $\mathrm{d}F$ represent this weight,

$$\mathrm{d}F = \rho g\,\mathrm{d}A\,\mathrm{d}x.$$

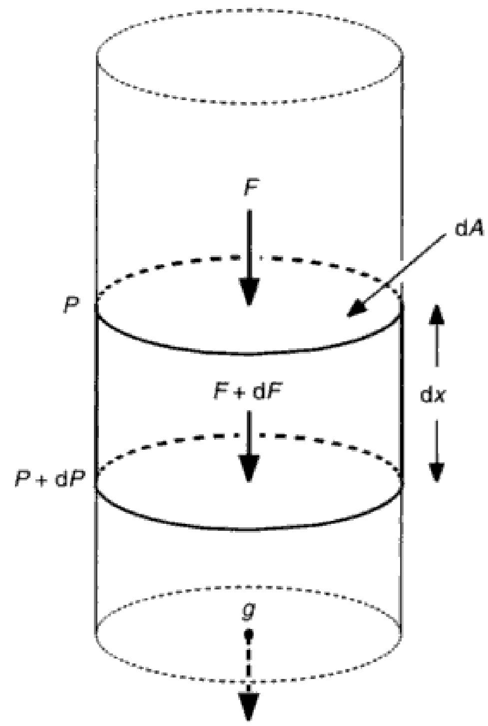


Fig. 9.1. The force F is exerted by the overlying gas on the area dA to give a pressure P. The weight of the material in the volume $dx\,dA$ adds a force dF in the depth dx, thereby increasing the pressure by dP.

Or, the increment in pressure over the distance dx is

$$dP = dF/dA = \rho g\, dx.$$

With x increasing inward as the pressure does, no negative sign appears in this relation. Now $d\tau_\nu = \kappa_\nu \rho\, dx$, according to Eq. (5.13), so

$$\frac{dP}{d\tau_\nu} = \frac{g}{\kappa_\nu}. \tag{9.1}$$

This is the hydrostatic equation we need.

The pressure in Eq. (9.1) is the total pressure supporting the small volume element. In most stars, the gas pressure accounts for the bulk of the total pressure. Only in extreme cases will other pressures be significant compared to P_g. Examples are (a) radiation pressure in very hot stars (refer to Eq. 5.11) with $P_R = (4\sigma/3c)T^4 = 2.52 \times 10^{-15}T^4$ dyne/cm^2, (b) magnetic pressure given by $B^2/4\pi$, and (c) turbulence pressure often assumed to be $\frac{1}{2}\rho v^2$, where ρ and v are the density and root-mean-square velocity of the turbulent elements,

Table 9.1. *Pressure tabulation.*

$T_{\rm eff}$ (K)	*Sp*	$P_{\rm g}$ (dyne/cm^2)	$P_{\rm R}$ (dyne/cm^2)	B (G)	v (km/s)
4000	K5 V	1×10^5	0.6	1121	7.5
8000	A6 V	1×10^4	10	354	10.6
12000	B8 V	3×10^3	52	194	13.0
16000	B3 V	3×10^3	165	194	15.0
20000	B0 V	5×10^3	403	251	16.7

respectively. Numerical examples are given for main sequence stars in Table 9.1. The magnetic field listed in the table is the value needed to have a magnetic pressure equal to $P_{\rm g}$. Similarly, turbulent elements would have to have the velocity listed to generate a pressure equal to $P_{\rm g}$. But in the rest of this chapter only $P_{\rm g}$ is retained in the hydrostatic equation.

Integration of the hydrostatic equation can be done in many ways, one of which is to write Eq. (9.1) as

$$P_{\rm g}^{1/2}\,{\rm d}P_{\rm g} = P_{\rm g}^{1/2}\frac{g}{\kappa_0}\,{\rm d}\tau_0,$$

where κ_0 is κ_ν at some reference wavelength which we choose to be 5000 Å. Integration gives

$$\begin{aligned} P_{\rm g}(\tau_0) &= \left(\tfrac{3}{2}g\int_0^{\tau_0}\frac{P_{\rm g}^{1/2}}{\kappa_0}\,{\rm d}t_0\right)^{2/3} \\ &= \left(\tfrac{3}{2}g\int_{-\infty}^{\log\tau_0}\frac{t_0^{1/2}P_{\rm g}^{1/2}}{\kappa_0\log e}\,{\rm d}\log t_0\right)^{2/3}. \end{aligned} \tag{9.2}$$

Integration on a logarithmic optical-depth scale gives better numerical precision – a detail that is obvious if you plot the integrand as a function of depth for a typical case. The procedure is to guess the function $P_{\rm g}(\tau_0)$ and perform the numerical integration. The new value of $P_{\rm g}(\tau_0)$ obtained from Eq. (9.2) can be used in the next iteration until convergence is obtained. With a reasonable guess at $P_{\rm g}(\tau_0)$, two to four iterations are typically needed. A poor guess will extend the iteration by a cycle or two.

Before the computation in Eq. (9.2) can proceed, we must know κ_0 as a function of τ_0 since κ_0 appears in the integrand. In Chapter 8 we saw that κ_ν was dependent on the temperature and the electron pressure. We conclude that we must first know how these two variables, T and $P_{\rm e}$, depend on τ_0. We first consider $T(\tau_0)$ and then $P_{\rm e}(\tau_0)$.

The temperature distribution in the solar photosphere

Basically we have two physical mechanisms to use as depth probes: limb darkening and the variation with wavelength of the absorption coefficient. The solar temperature distribution can be obtained from limb darkening measurements of the type shown in Fig. 9.2. Limb darkening exists because the continuum source function decreases outward. As we look toward the limb, we see systematically higher photospheric layers that are less bright. In fact, from Eq. (7.10), the specific intensity at the surface is

$$I_\nu(0) = \int_0^\infty S_\nu e^{-\tau_\nu \sec\theta} \sec\theta \, d\tau_\nu. \tag{9.3}$$

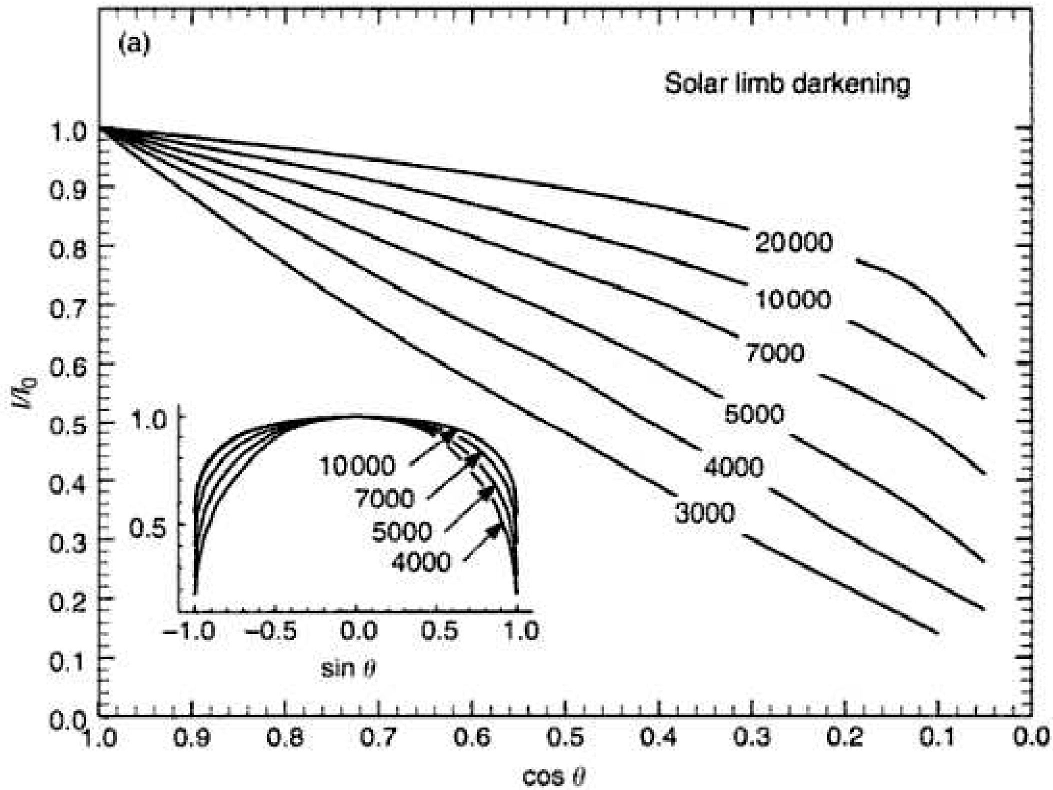


Fig. 9.2. (a) The limb darkening observations of the Sun are shown for several wavelengths. The darkening is nearly linear with $\cos\theta$. The small inset graph shows the limb darkening as a function of position across the disk, i.e., $\sin\theta$. Data from Pierce (2000). (b) A schematic illustration of the cause of limb darkening is shown. The top and bottom of the photosphere are indicated by the outer and inner circles. (The real solar photosphere is ~25 times thinner than the lines used to draw the circles!) Unit optical depth into the photosphere corresponds to different heights in the photosphere depending on θ. The dashed curve indicates the surface of unit optical depth from the top of the photosphere. Radiation seen at disk position $\sin\theta_2$ is characteristic of the higher cooler layers compared to the radiation seen at disk position $\sin\theta_1$.

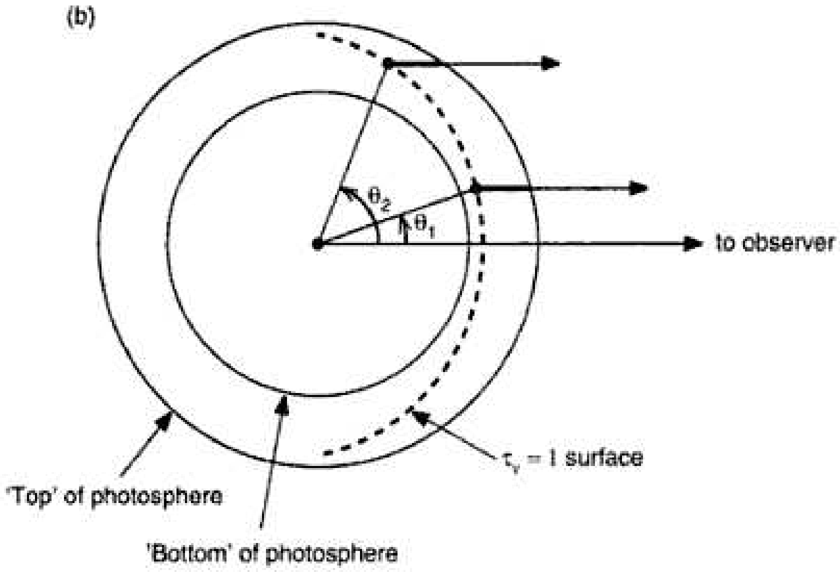


Fig. 9.2. (*cont.*)

We see that the exponential extinction varies as $\tau_\nu \sec\theta$, so the position of unit optical depth *along the line of sight* moves upward, i.e., to smaller τ_ν, as the line of sight is shifted from the center to the limb. In the approximate grey-case solution, we found a linear source function, Eq. (7.34), which we can write as

$$S_\nu = a + b\tau_\nu.$$

Then from Eq. (9.3), one obtains

$$I_\nu(0) = a + b\cos\theta,$$

which says that at $\cos\theta = \tau_\nu$, the specific intensity on the surface at the position θ equals the source function at the depth τ_ν. This is referred to as the Eddington–Barbier relation. Measurements of I_ν across the solar disk can be used to map the depth dependence of S_ν in this way. Slightly more complex analytical solutions for $S_\nu(\tau_\nu)$ have also been used (e.g., Mitchell 1959, Pierce & Waddell 1961). Alternatively, purely numerical solutions can be obtained by adjusting S_ν until the proper θ dependence is attained.

The concept of the depth of formation can be developed beyond the rudimentary relation described above by looking at the depth development of $I_\nu(\tau_\nu)$. More specifically, consider the integrand of Eq. (9.3), rewritten on a logarithmic scale,

$$I_\nu(0) = \int_0^\infty S_\nu(\tau_\nu) e^{-\tau_\nu \sec\theta} \tau_\nu \sec\theta \frac{d\log\tau_\nu}{\log e}. \tag{9.4}$$

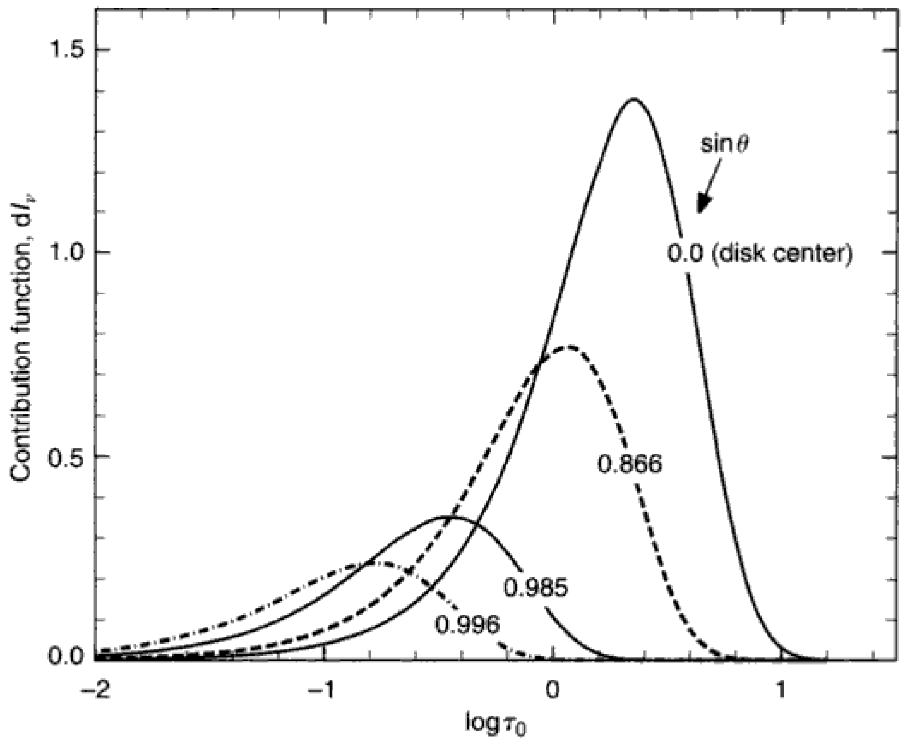


Fig. 9.3. The depth of formation at four limb distances shows quantitatively the decrease in intensity (area under the curve) and the shift to higher photospheric layers for lines of sight closer to the limb. This example is for $\lambda = 4000$ Å.

The integrand is called the *contribution function* since it specifies the contribution to $I_\nu(0)$ from each depth. Sample contribution functions are shown in Fig. 9.3. We see that no light comes from the very highest or the very lowest layers and that there is a simple smooth maximum. Comparing these three examples, we see that, on average, the surface intensity originates in higher layers for positions closer to the limb. By virtue of the considerable overlap in depth of these integrands, no fine structure can be discerned.

On the practical side of things, the relative limb darkening I_ν/I_ν^0 (where I_ν^0 is the intensity at the center of the disk) is measured as a function of θ, as in Fig. 9.2. Then Eq. (9.4) is used in a normalized form by dividing by I_ν^0, and $S_\nu(\tau_\nu)/I_\nu^0$ is obtained. In this way the *shape* of the source function is found from the *shape* of the limb darkening. A second set of measurements is made to obtain I_ν^0 itself. The *absolute* scale of the source function is obtained from the *absolute* specific intensity. Finally, the source function is set equal to the Planck function to obtain the temperature as a function of depth.

Our second handle on depth of formation is the wavelength variation of the absorption coefficient (see Fig. 8.6). We see deepest into the solar atmosphere near 1.6 μm. Toward shorter wavelengths, κ_ν systematically increases through the hump of H^-_{bf}, raising the depth of formation. For $\lambda \lesssim 2500$ Å, opacity of the metals lifts the depth of formation to the top of the photosphere and higher. Limb darkening measurements over a wide range in wavelength combine the information of both depth probes.

The variation in absorption across a strong spectral line produces a similar change in depth of formation (as discussed in Chapter 13). Numerous lines have contributed input to the solar models, especially the H and K lines of Ca II, the Mg b lines, etc. (See Appendix E for the strongest solar lines.)

A few of the results are compared in Fig. 9.4. The models of Pierce and Waddell, Elste, Holweger and Müller, Ayres and Testerman, and Maltby *et al.* depend heavily on empirical information, including the limb darkening and the behavior of strong lines. The other models (Carbon & Gingerich, Bell *et al.*, Kurucz) illustrate the results of radiative equilibrium calculations. Good agreement is seen in the layers around unit optical depth, where the observational information is strongest and least ambiguous. In the deepest layers, the uncertain effects of convection enter, leading to some divergence in the models (Pierce & Waddell's model has the radiative gradient). Increasing divergence is also seen in the higher layers. Part of this results from the breakdown of local thermodynamic equilibrium. Some models, such as that by Maltby *et al.*, include the non-LTE de-coupling of T from S_ν, whereas others, such as the Holweger–Müller model, assume the LTE $S_\nu = B_\nu$ to derive T. But Ayres and Testerman also find low temperatures from the CO lines, and for them the non-LTE effects are thought to be small (Ayres 1981, Ayres & Testerman 1981, Muchmore 1986, Ayres & Wiedeman 1989). This is taken as evidence for the heterogeneous structure of the real Sun, i.e., both hotter and cooler areas exist side-by-side, and the chromospheric temperature rise is delayed above these cooler pockets.

The more rigorous of the radiative equilibrium models, for example those by Kurucz (1979) and the updated version of his ATLAS code, the MARCS code (Gustafsson *et al.* 1975, Decin *et al.* 2003), Anderson (1989), or Allende Prieto *et al.* (2003) involve considerable computation, including many thousands of spectral lines, and in some cases, non-LTE effects as well. Their general agreement with the empirical models strengthens our confidence in the overall modeling picture.

The visible continuum and most of the spectral lines involve photospheric layers deeper than $\log \tau_0 = -4$. Table 9.2 at the end of the chapter gives a mean solar model to use as a reference photosphere. Its $T(\tau_0)$ values can be scaled to

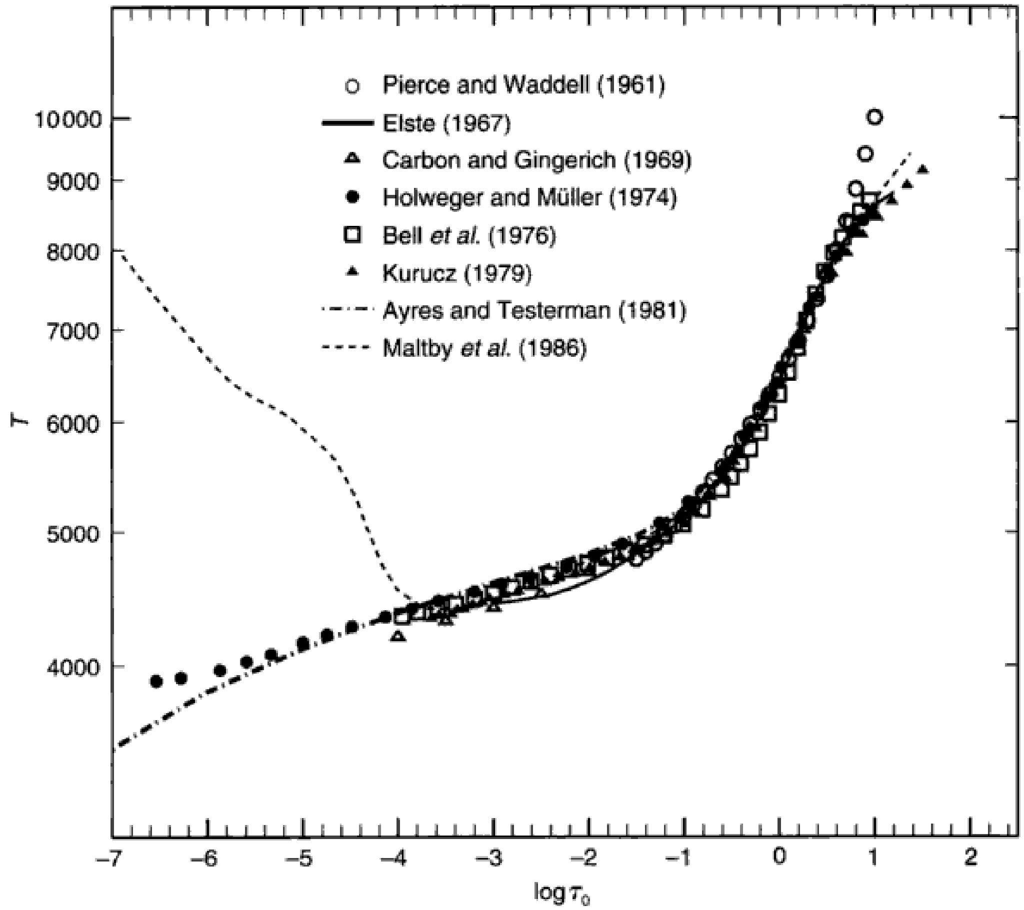


Fig. 9.4. Several solar temperature distributions are compared. The best agreement occurs in the depths between $\log \tau_0 = -1$ and 0.5, where the continuum is formed.

produce temperature distributions for other stars, as explained below. At the same time, we must not expect a single $T(\tau_0)$ to properly represent all aspects of a heterogeneous photosphere. That would be impossible. Temperature differences of several per cent are likely. Expressed on a geometrical scale, the temperature inhomogeneities can be substantially greater. For example, the hydrodynamical calculations of Nordlund (1985) show ≈25% variation in temperature. Differences between magnetic and non-magnetic regions have been modeled by Elste (1985).

Temperature distributions in other stars

Measurements of limb darkening for other stars using interferometers to resolve the stellar disks are close to happening. Meanwhile, the run of temperature with depth is obtained in one of two ways. The first is theoretical. Flux constancy with depth is imposed according to Eqs. (7.22), (7.28), or (7.30). In Chapter 7 we saw how this could be done analytically for the grey case using the Eddington approximation. In the non-grey models, numerical integrations over frequency are performed. Procedures for obtaining flux constancy are generally iterative, starting with some initial $S_\nu(\tau_0)$, computing $F_\nu(\tau_0)$ using Eq. (7.14), and finally correcting $S_\nu(\tau_0)$ in some way to meet the flux-constancy condition.

Good examples of the application of methods developed by Avrett and Krook (1963a, b) can be seen in the work of Carbon and Gingerich (1969). Linearization is discussed by Auer and Mihalas (1969). The more numerically efficient perturbation method (Cannon 1973a, b, Scharmer 1981) is the main topic of the compendium edited by Kalkofen (1987). Flux constancy is typically enforced to levels of between 0.1 and 1.0%. Imposing radiative equilibrium without line opacity is relatively straightforward because, except for the ionization edges, the positions of which are known, κ_ν varies slowly with wavelength. In other words, relatively few wavelengths need be dealt with to obtain sufficiently precise integrations. Including the spectral lines increases the level of computations by several orders of magnitude, especially for late-type stars, and has led to the development of approximations for the line opacity as mentioned in Chapter 8. The general result of including line opacity is to increase the temperature of the deeper layers (below the layers where most of the lines absorb) and cool the shallower layers. For this reason, the effects of line opacity are referred to as "line blanketing." A detailed discussion of line blanketing can be found in the literature, e.g., Böhm (1966), Athay and Skumanich (1969), Mihalas and Luebke (1971), Pecker (1973), Carbon (1979), Werner (1988), and Anderson (1989).

In very cool stars, late M and cooler, solid grains can form in the outer photospheric layers. The extra absorption produced by the grains acts to warm these outer layers (see Ferguson *et al.* 2001).

Errors in calculated flux constancy arise from inaccuracies in our treatment of κ_ν with depth and wavelength. Although the major continuous absorbers are thought to be reasonably well understood for most stars, the line opacity presents more of a challenge. A second point of uncertainty with the flux constancy method arises in the treatment of convection. According to the discussion in Chapter 7, the flux constancy condition should include the convective flux (Eq. 7.23). In the mixing-length formulation, the mixing length remains a free parameter, introducing a large uncertainty in the deep layers. Fortunately, only a small fraction of the surface flux arises in these layers and the spectrum is hardly affected. Unfortunately (for the simplest models) convection introduces significant temperature variations at a given depth, as noted above. These are frequently ignored as a matter of expedience. Although considerably more complex than the simple models discussed in this chapter, hydrodynamical models are designed to include convective flows from the beginning (see Chapter 17).

Radiative equilibrium calculations are complex enough to make customized modeling, star by star, rather laborious. Instead, grids of models spanning a range in effective temperature, surface gravity, and general chemical composition are often tabulated. One then interpolates in the grid as needed.

The second and far simpler method of obtaining $T(\tau_0)$ is to scale a standard temperature distribution, such as the solar one,

$$T(\tau_0) = S_0 T_{\odot}(\tau_0), \tag{9.5}$$

where S_0 is the scaling constant. The first indication that such a procedure might be possible was seen in the solution of the grey case, where we found (Eq. 7.36)

$$T(\tau) = \left\{\frac{3}{4}[\tau + q(\tau)]\right\}^{1/4} T_{\text{eff}}$$

or

$$T(\tau) = \frac{T_{\text{eff}}}{T_{\text{eff}}^{\odot}} T_{\odot}(\tau).$$

In the grey case, the scaling factor is the ratio of effective temperatures.

Theoretical temperature distributions computed on the basis of flux constancy differ by only a few per cent from the scaled solar $T(\tau_0)$, as seen in Figs. 9.5 and 9.6. These differences do not exceed by a large margin the uncertainties in the calculations themselves. Generally one finds that the temperature gradient is slightly steeper in giants compared to dwarfs (e.g., Ruland *et al.* 1980, Drake & Smith 1991), and one may wish to scale a "standard" giant model when analyzing giants.

Scaled models can also be used to advantage in customizing published grids of models by simply scaling the $T(\tau_0)$ tabulations of the grid models. In many applications, the scaling factor S_0 can be arbitrarily adjusted without asking what T_{eff} the model has. For most cases, using $T_{\text{eff}} = S_0 T_{\text{eff}}^{\odot}$ is a reasonable approximation. Scaled models have been used extensively (e.g., Gray 1984, 1989, Ruck & Smith 1995, Tripicchio *et al.* 1997). The tremendous practical advantage of using a scaled temperature distribution is the simplicity of the

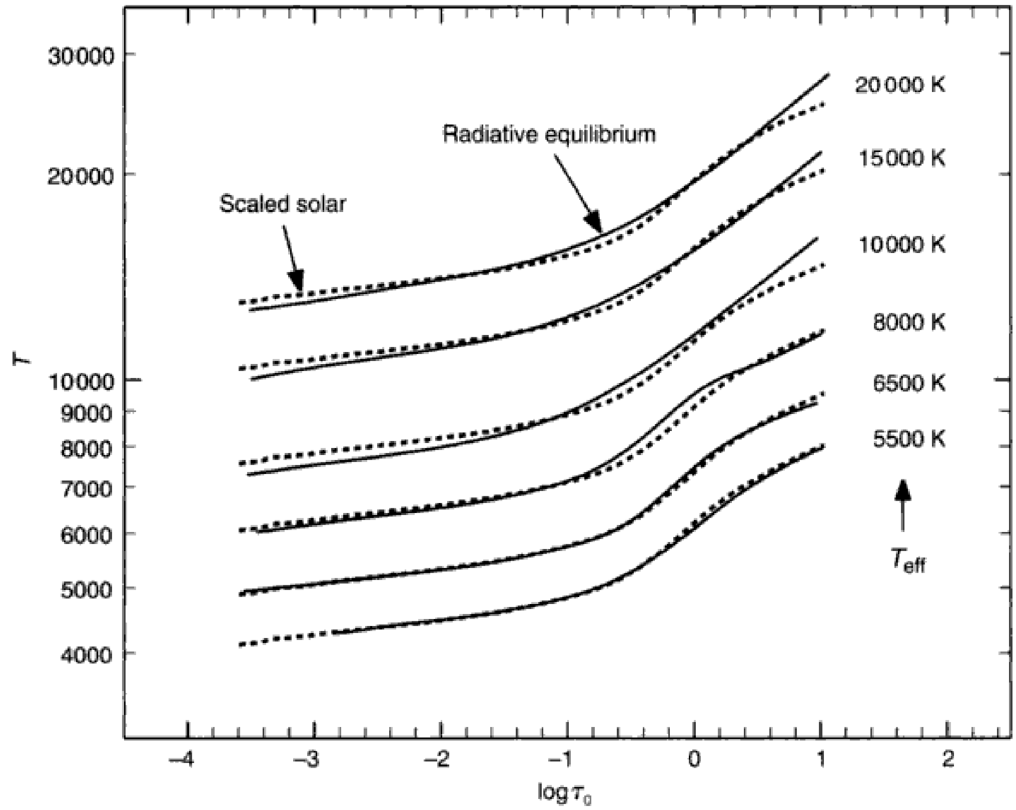


Fig. 9.5. Radiative-equilibrium models from the grid of Kurucz (1979) are compared to the scaled solar model from the same grid.

calculation. Whereas detailed radiative-equilibrium calculations require fast and powerful computers, scaled models can be compute with modest resources in a few seconds or less.

At this point we assume $T(\tau_0)$ has been determined, and we turn our attention to the calculation of electron pressure structure in the model.

The P_g–P_e–T relation

When solving Eq. (9.2) for the gas pressure, we start with an initial guess at $P_g(\tau_0)$. We then immediately require $P_e(\tau_0) = P_e(P_g)$ in order to find $\kappa_0(\tau_0) = \kappa_0(T, P_e)$ for the integrand. The electron pressure depends on the temperature and the chemical composition as follows. Let the number of ions per unit volume of the jth element be N_{1j} and the number of neutrals per unit volume be N_{0j}. Then the ionization Eq. (1.20) is

$$\frac{N_{1j}}{N_{0j}} = \frac{\Phi_j(T)}{P_e}.$$

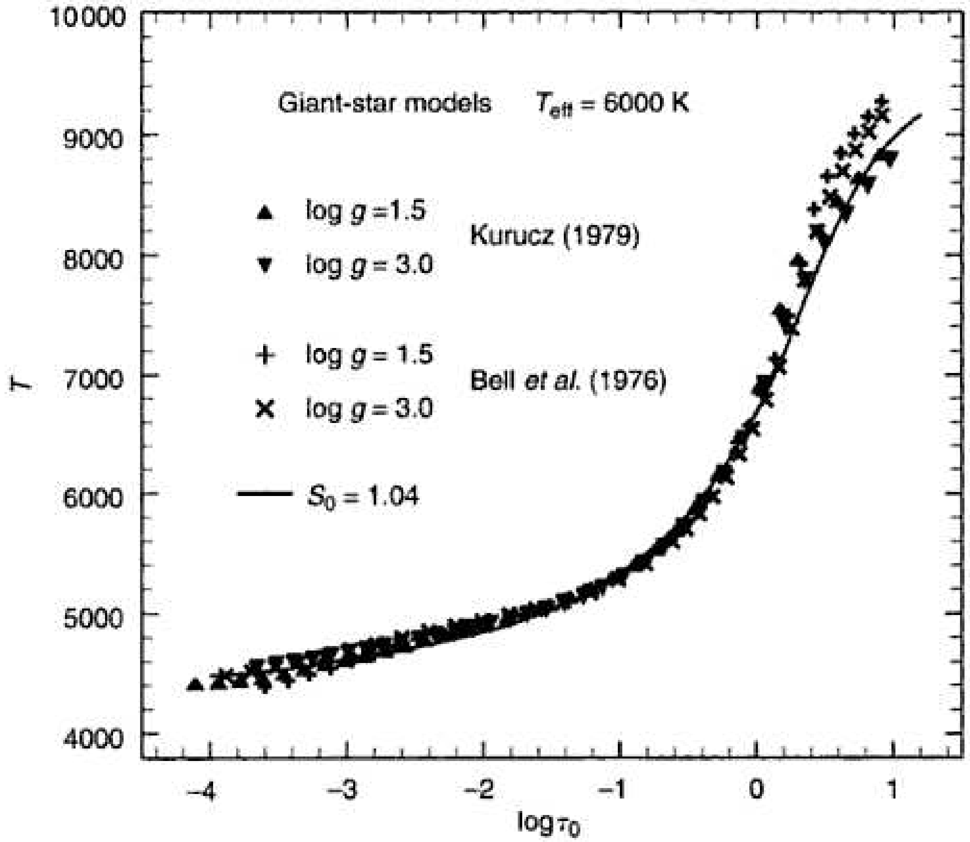


Fig. 9.6. The solar model of Table 9.2, when scaled up by 4%, is a reasonable match to radiative-equilibrium models computed for giants.

Neglect double ionizations and write $N_{1j} = N_{ej}$ which is the number of electrons per unit volume contributed by the jth element. Then

$$\frac{\Phi_j(T)}{P_e} = \frac{N_{ej}}{N_{0j}} = \frac{N_{ej}}{N_j - N_{ej}}.$$

The total number of jth element particles is $N_j = N_{1j} + N_{0j}$. We solve for N_{ej},

$$N_{ej} = N_j \frac{\Phi_j(T)/P_e}{1 + \Phi_j(T)/P_e}. \tag{9.6}$$

The pressures are

$$P_e = \sum_j N_{ej} kT$$
$$P_g = \sum_j (N_{ej} + N_j) kT. \tag{9.7}$$

which we ratio as follows.

$$\frac{P_e}{P_g} = \frac{\sum N_{ej}kT}{\sum (N_{ej} + N_j)kT}$$
$$= \frac{\sum N_j \left[\dfrac{\Phi_j(T)/P_e}{1 + \Phi_j(T)/P_e}\right]}{\sum N_j \left[1 + \dfrac{\Phi_j(T)/P_e}{1 + \Phi_j(T)/P_e}\right]}.$$

We introduce the number abundance of the element, $A_j = N_j/N_H$, where N_H is the number of hydrogen particles per unit volume. The relation we need is then

$$P_e = P_g \frac{\sum A_j \dfrac{\Phi_j(T)/P_e}{1 + \Phi_j(T)/P_e}}{\sum A_j \left[1 + \dfrac{\Phi_j(T)/P_e}{1 + \Phi_j(T)/P_e}\right]}. \tag{9.8}$$

The equation is transcendental in P_e, and is solved by iteration. Notice that the $\Phi_j(T)$ terms are constants for such an iteration. Since P_e and P_g are both functions of τ_0, Eq. (9.8) is solved at each depth, the first cycle of which uses the first guess of $P_g(\tau_0)$.

The results for the solar chemical composition are shown in Fig. 9.7. Hydrogen dominates at the high temperatures, and when it becomes essentially fully ionized, $P_g \approx 2P_e$. At cooler temperatures, P_e varies approximately as $P_g^{1/2}$. This can be seen as follows. Consider the photosphere to be composed of a single typical element (one which supplies electrons!) and reduce Eq. (9.8) to

$$P_e = P_g \frac{\Phi(T)/P_e}{1 + 2\Phi(T)/P_e}$$

or

$$P_e^2 = \Phi(T)P_g - 2\Phi(T)P_e = \Phi(T)(P_g - 2P_e).$$

Since $P_e \ll P_g$ in cooler stars, the expression reduces to

$$P_e^2 \approx \Phi(T)P_g. \tag{9.9}$$

That is, for a given temperature, the electron pressure is proportional to the square root of the gas pressure.

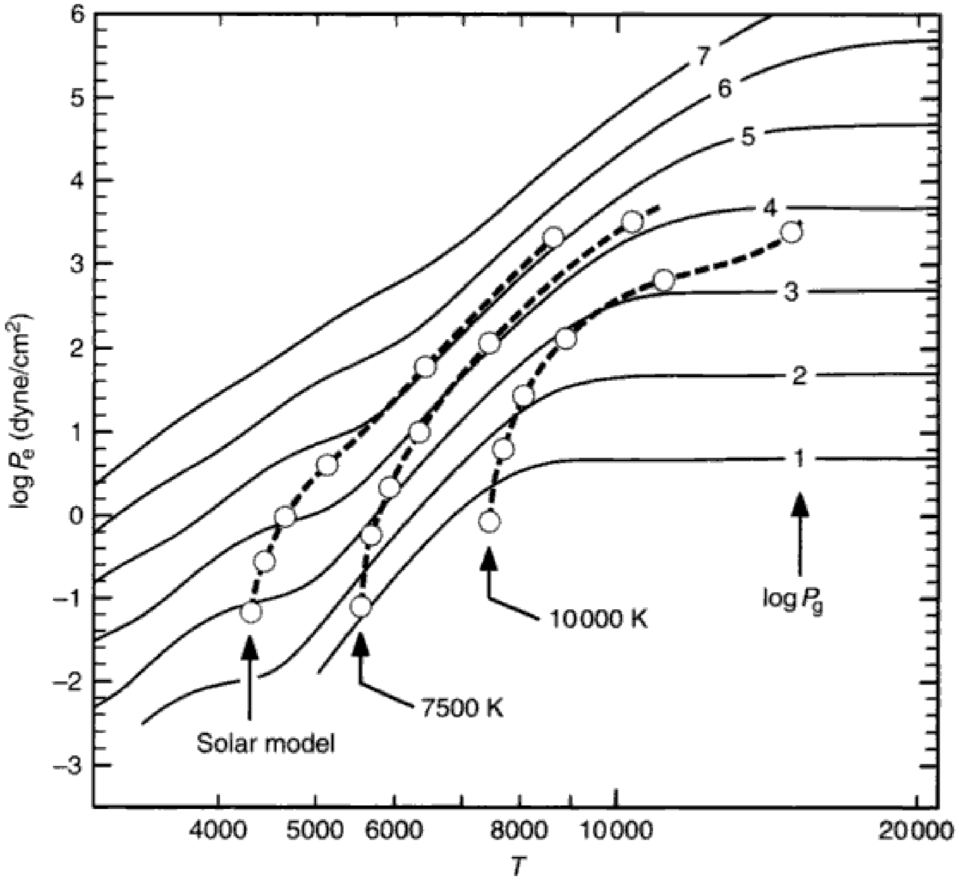


Fig. 9.7. Electron pressure is shown as a function of temperature for several values of gas pressure (dynes/cm^2). These computations use the solar chemical composition. The dashed lines and circles show the values in three different temperature model photospheres. The circles are placed one decade apart in optical depth, with the lowest at $\log \tau_0 = -4$. Notice the very steep slopes; electron pressure can be extremely sensitive to temperature.

Metals play a significant role in cooler photospheres where hydrogen is not ionized, and then the P_e–P_g–T relation is sensitive to the chemical makeup of the model. It is necessary to include the elements C, Si, Fe, Mg, Ni, Cr, Ca, Na, and K when dealing with the ionization equilibrium of models having normal chemical composition; these are the electron donors. We may find ourselves in a new iterative situation if the chemical composition of the stars we are modeling turns out, after analysis, to be different from what we assumed in the original P_e calculation.

Notice the enormous temperature sensitivity of electron pressure for the cooler models, with $d \log P_e / d \log T \approx 12$. Since the absorption coefficient, κ_ν, in such models is largely due to the negative hydrogen ion, which is proportional to P_e, the opacity increases very rapidly with depth.

Completion of the model

Now we are in a position to compute Eq. (9.2), the basic iteration for P_g. Taking $T(\tau_0)$ and our guess at $P_g(\tau_0)$, we compute $P_e(\tau_0)$ and then $\kappa_0(\tau_0)$ according to the details explained in Chapter 8. Then Eq. (9.2) gives a new $P_g(\tau_0)$ and we have completed the first cycle of the pressure iteration. At the outer boundary, when starting the integration, it is necessary to make some simple trapezoidal integrations for a step or two in $\log \tau_0$. After that, Simpson's rule or other standard integration formulae can be used. Usually a few extra steps are taken at the top of the model to ensure that the pressure at the depths where we need it is independent of how we start the integration. Iteration is continued to convergence, typically $\approx 1\%$, and we then have a complete model for the specified $T(\tau_0)$. If we are using a scaled solar temperature distribution, we are finished. The model has been made and we are ready to calculate auxiliary information from the model (such as the geometrical depth and surface flux). If we are determining $T(\tau_0)$ by the flux constancy condition, we are not finished, but must now calculate $\mathfrak{F}_\nu(\tau_\nu)$ using Eq. (7.14) for many frequencies across the spectrum so that $\int_0^\infty \mathfrak{F}_\nu \, d\nu$ can be properly evaluated as a function of depth. The temperature distribution is then corrected and the model is repeatedly re-computed until flux constancy is achieved.

The geometrical depth

The geometrical depth scale can be computed from $dx = d\tau_0/\kappa_0\rho$,

$$x(\tau_0) = \int_0^{\tau_0} \frac{1}{\kappa_0(t_0)\rho(t_0)} dt_0. \tag{9.10}$$

The density we calculate from the pressure and temperature using Eq. (1.8), namely,

$$\rho = N_H \text{ hydrogen particles/cm}^3 \times \sum A_j \mu_j \text{ grams/H particle}$$

in which μ_j represents the atomic weight of the jth element, and

$$N_H = \frac{N - N_e}{\sum A_j} = \frac{P_g - P_e}{kT \sum A_j}$$

in which N is the total particle density. This gives

$$x(\tau_0) = \int_{-\infty}^{\log \tau_0} \frac{\sum A_j k T(t_0)\, t_0}{\kappa_0(t_0) \sum A_j \mu_j [P_g(t_0) - P_e(t_0)]} \frac{d \log t_0}{\log e}. \tag{9.11}$$

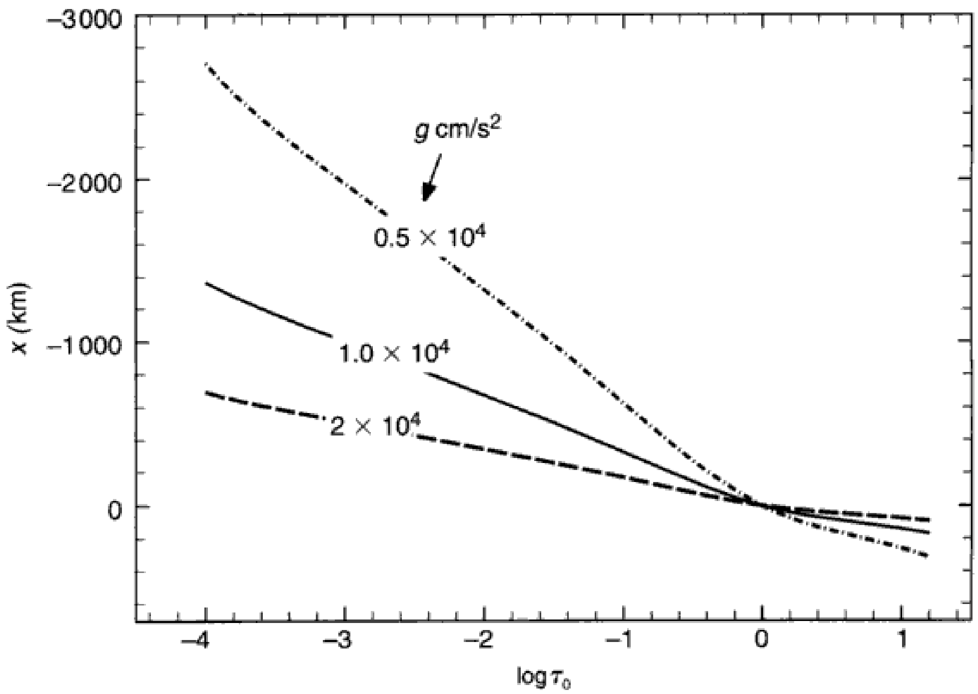


Fig. 9.8. These models show that the geometrical depth varies nearly linearly with log τ_0. Changing the surface gravity by a factor of 2 alters the thickness of the atmosphere inversely by the same factor.

An alternative method is to integrate on a P_g scale starting with $dP_g = \rho g\, dx$. Then

$$x(P_g) = \frac{1}{g}\int_0^{P_g} \frac{\sum A_j kT(p)}{\sum A_j \mu_j} \frac{dp}{p}. \tag{9.12}$$

This is an interesting form because it shows that the thickness of the atmosphere is inversely proportional to the surface gravity, since $T(P_g)$ depends only weakly on g. This can be seen in Fig. 9.8. We also see there that the geometrical depth is nearly linear with logarithmic optical depth.

Computation of the spectrum

The computation of the spectrum is usually the most important "result" of the modeling. We compute the spectrum using Eq. (7.15),

$$\mathfrak{F}_\nu = 2\pi \int_{-\infty}^{\infty} S_\nu(\tau_0) E_2(\tau_\nu) \frac{\kappa_\nu(\tau_0)\tau_0 \, d\log\tau_0}{\kappa_0(\tau_0)\log e} \tag{9.13}$$

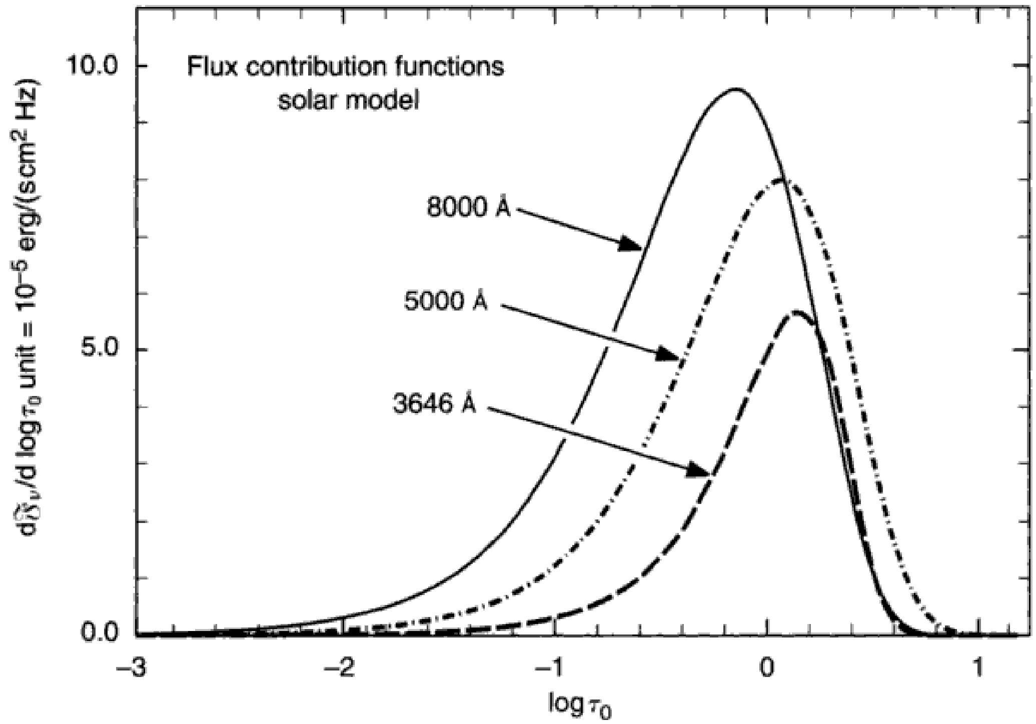


Fig. 9.9. Flux contribution functions show the relevant depths when computing the flux integral and they indicate the depths of formation, i.e., the layers from which the flux comes.

with $S_\nu(\tau_0) = B_\nu[T(\tau_0)]$ in the LTE case, and where the optical depth scales are related by

$$dx = \frac{d\tau_\nu}{\kappa_\nu \rho} = \frac{d\tau_0}{\kappa_0 \rho}$$

or

$$\tau_\nu = \int_0^{\tau_0} \frac{\kappa_\nu(t_0)}{\kappa_0(t_0)} \, dt_0 = \int_{-\infty}^{\log \tau_0} \frac{\kappa_\nu(t_0)}{\kappa_0(t_0)} \frac{t_0}{\log e} \, d\log t_0. \tag{9.14}$$

We must have τ_ν because it is the argument in E_2 in Eq. (9.13). The amount of light contributed to the surface flux as a function of depth in the atmosphere (on a $\log \tau_0$ scale) is specified by the integrand of Eq. (9.13). We call this integrand the *flux contribution function*. It is similar to the contribution function used with I_ν, but now includes an integration over the stellar disk. Figure 9.9 shows representative flux contribution functions. They tell us where in depth the surface flux originates and where, for example, $T(\tau_\nu)$ must be defined in order to compute the surface flux with sufficient precision. The contribution

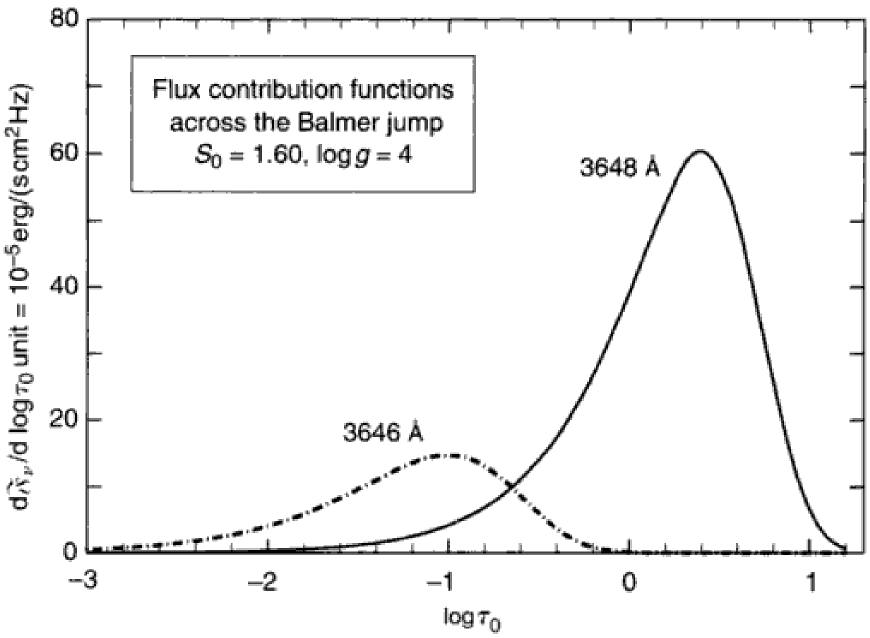


Fig. 9.10. Sharp differences in opacity can greatly shift the depth of formation, as illustrated here for a model where the Balmer discontinuity of κ_ν is large.

functions in Fig. 9.9 indicate that the radiation at 8000 Å is formed slightly higher than the radiation at 5000 Å. This reflects the larger κ_ν at 8000 Å compared to 5000 Å. If these contribution functions are plotted on their own $\log\tau_\nu$ scale instead of the $\log\tau_0$ scale, they all peak near $\tau_\nu = 1$.

A good illustration of the depth of formation concept occurs across the Balmer discontinuity. As shown in Fig. 9.10, the flux at 3646 Å arises from significantly deeper layers than the flux at 3648 Å

The depth of formation also dependends on the temperature of the model as the various contributions to κ_ν wax and wane with T causing κ_ν/κ_0 to vary. Changes such as those in Fig. 9.11 are typical. Surface gravity effects are much smaller, being barely noticeable in the hotter models and completely negligible in the cooler ones.

There are other techniques for computing the flux and they generally lead to other integrands. For example, suppose we integrate Eq. (7.15) by parts to obtain

$$\mathfrak{F}_\nu = \pi S_\nu(0) + 2\pi\int_0^\infty \frac{dS_\nu}{d\tau_\nu} E_3(\tau_\nu)\,d\tau_\nu. \tag{9.15}$$

Again, S_ν is replaced by B_ν in LTE, and $S_\nu(0) = B_\nu(T_0)$, where T_0 is the boundary temperature. The formulation in Eq. (9.15) is interesting because it

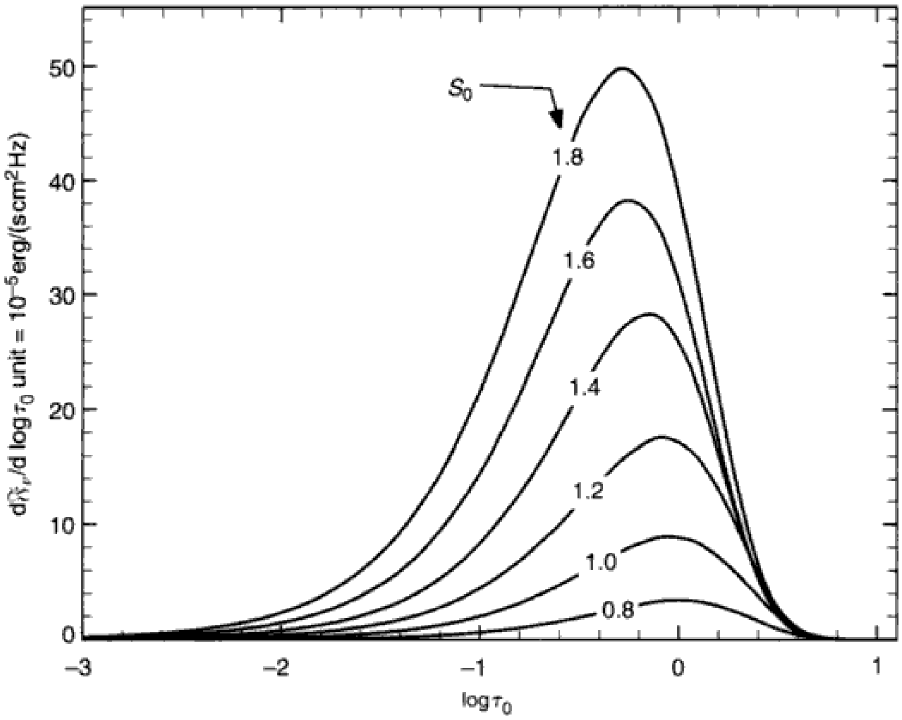


Fig. 9.11. Contribution functions at 6000 Å are shown for a run of models having different effective temperatures (S_0). The hotter models obviously produce more flux (area) and show a slight decrease in the depth of formation.

shows that the flux arises from the gradient of the source function. Greater contributions to the flux occur at depths where $dS_\nu/d\tau_\nu$ is larger. Notice, however, that in this formulation we have the curious situation of the flux depending on T_0 even though there may be no flux contributed from the boundary. Furthermore, the integrand $(dS_\nu/d\tau_\nu)E_3(\tau_\nu)$ is not itself an increment of flux. It follows that this integrand cannot be used as a contribution function in the usual sense of the term (see also Edmonds 1969).

Continuous spectra from model photospheres agree reasonably well with the observations. The details of model continua as they relate to real stars are discussed in Chapter 10.

Properties of models: pressure relations

The relation between temperature and gas pressure for four models is shown in Fig. 9.12. It is clear that $T(P_g)$ does not scale the way $T(\tau_0)$ does. Although P_g could have been used as an independent variable in the integration of the hydrostatic equation, the scaling procedure would then be inapplicable.

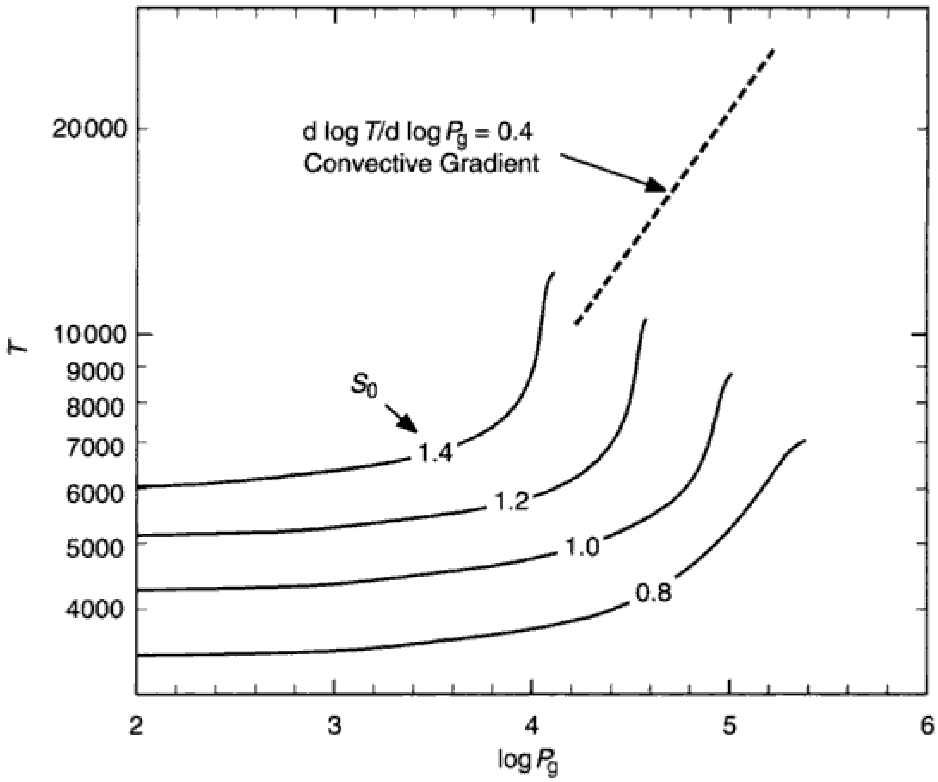


Fig. 9.12. The log T–log P_g plane is shown for four models having a range in effective temperature (S_0). The shape of $T(P_g)$ differs from one model to the next; one cannot scale $T(P_g)$ the way one can $T(\tau_0)$. The temperature gradient, d log T/d log P_g, beyond which convection occurs, is shown.

It is in the $\log T$–$\log P_g$ plane where the slope exceeding $1-1/\gamma \approx 0.4$ implies instability to convection (Eq. 7.39). Many models are convectively unstable by this criterion, and as indicated earlier in the chapter, convection should then be included in the energy transport. The $T(P_g)$ relation is altered by the convection, but the gradient does not normally reach the adiabatic value until below the photosphere because radiative transport is still competitive with convection at the relatively low densities of the photosphere.

The depth dependences of gas pressure and electron pressure are shown for two models in Figs. 9.13 and 9.14, respectively. In the cooler model, the electron pressure is orders of magnitude smaller than the gas pressure. Toward deeper layers, the increased temperatures boost the ionization and hence the electron pressure. Although hydrogen contributes to the electron pressure in the cooler model, it is very far from being completely ionized. By contrast, the hotter model has temperatures high enough to vigorously ionize hydrogen, the electron pressure is dominated by electrons from hydrogen, and $P_e \approx 0.5 P_g$.

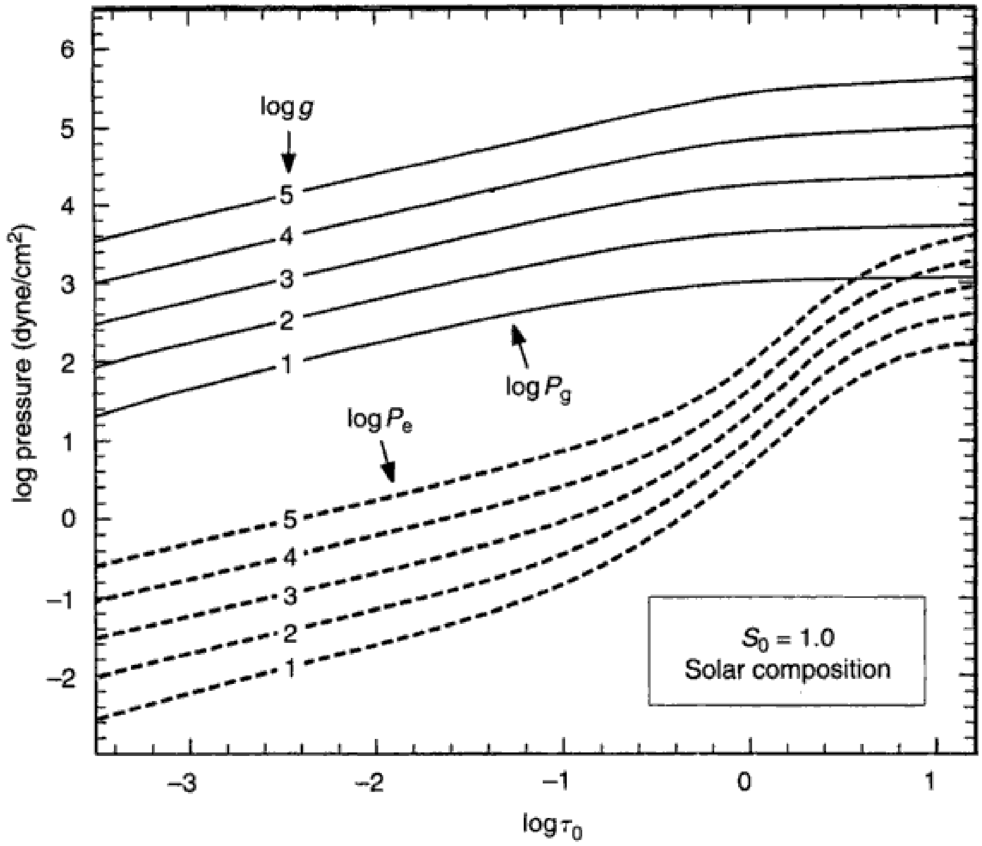


Fig. 9.13. The gas pressure varies nearly linearly with log τ_0, but levels off in the deeper layers. These models have $S_0 = 1.0$. The shape of the curves is nearly independent of the surface gravity.

Increasing the surface gravity compresses the photosphere, increasing all pressures. The variation with $\log g$ of any pressure, i.e., $\log P$, at any depth is nearly linear. We can therefore approximate these g dependences by writing

$$P_g \approx \text{constant } g^p, \tag{9.16}$$

where the power, p, ranges from 0.64 to 0.54 in going from deep to shallow layers in the cooler model, and 0.85 to 0.53 in the hotter model. Roughly speaking, $p = \frac{2}{3}$.

In a similar manner, the P_e dependence is

$$P_e \approx \text{constant } g^p, \tag{9.17}$$

where p ranges from 0.33 to 0.48 going from deep to shallow layers in the cooler model, and 0.82 to 0.53 in the hotter model. Approximately, $p = \frac{1}{3}$ and $\frac{2}{3}$ for the two regimes.

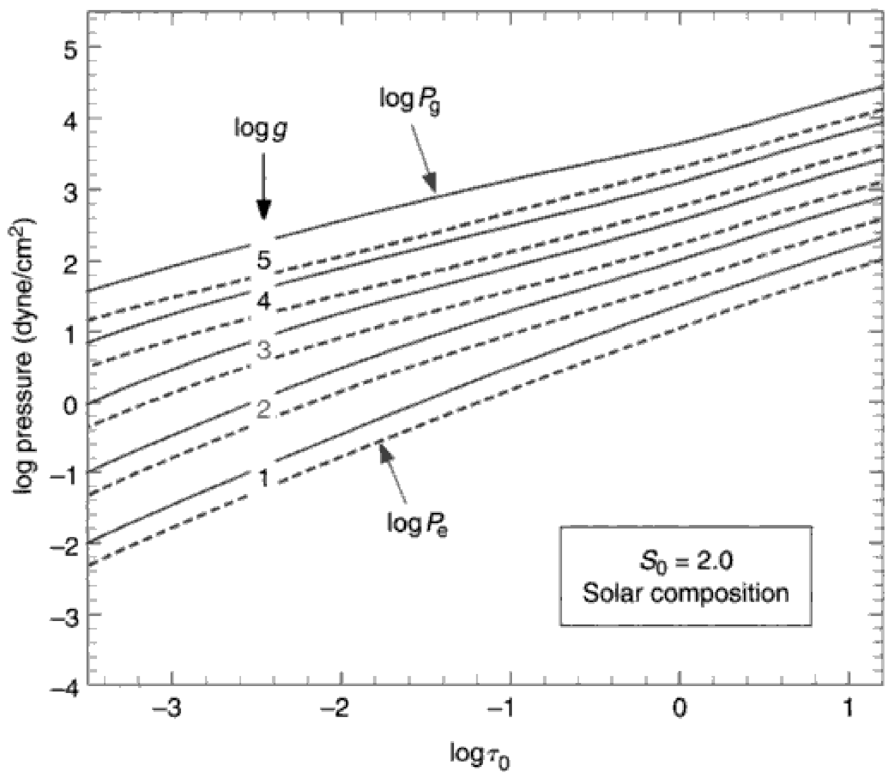


Fig. 9.14. Electron pressure is shown as a function of $\log \tau_0$ for the models used in Fig. 9.13. The shape is nearly independent of the surface gravity.

Some understanding of these gravity dependences can be obtained by considering Eq. (9.2) written as

$$P_g = g^{2/3}\left(\frac{3}{2}\int_{-\infty}^{\log \tau_0} \frac{t_0 P_g^{1/2}}{\kappa_0 \log e}\, d\log t_0\right)^{2/3}. \tag{9.18}$$

The pressure dependence inside the integral is weak and so

$$P_g \approx C(T) g^{2/3}. \tag{9.19}$$

That is, for a constant temperature, $C(T)$ is constant and P_g varies according to Eq. (9.16) with $p \approx \frac{2}{3}$, which is what is seen for the models. Then from Eq. (9.9), the electron pressure in cool models goes as

$$P_e \approx \text{constant } g^{1/3} \tag{9.20}$$

and in hot models, since $P_e \approx 0.5 P_g$, as

$$P_e \approx \text{constant } g^{2/3}, \tag{9.21}$$

again in approximate agreement with the numerical calculations.

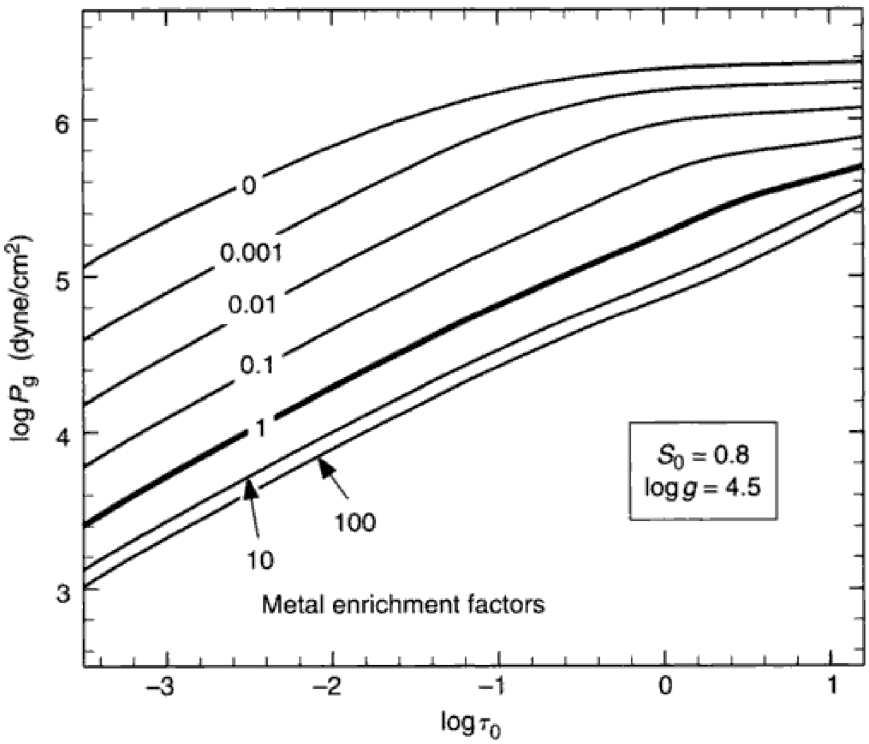


Fig. 9.15. The gas pressure as a function of depth shows a strong dependence on the metal abundance when the effective temperature is low. Increasing the metal abundance lowers the gas pressure at a given optical depth. The main change is actually in the optical depth, a result of enhanced absorption by the negative hydrogen ion with the increase in available electrons.

The effects of chemical composition

The pressure structure in a model can depends on the chemical composition. The nature of the dependence can be seen in Figs. 9.15 and 9.16, where a set of cool models is shown. (In hot models, hydrogen takes over as the electron donor, and the pressures become independent of chemical composition.) In the cooler models, increasing the metals increases the number of free electrons, resulting in a larger continuous absorption coefficient and a shorter geometrical penetration of the line of sight into the atmosphere. Therefore, the gas pressure at a given optical depth must decrease with increasing metal content. We can see the basics here as follows. Write, using ΣA_j for the sum of the metal abundances,

$$\frac{P_e}{P_g}\frac{1}{\Sigma A_j} = \frac{P_e}{P_g}\frac{N_H}{\Sigma N_j}\frac{kT}{kT} = \frac{P_e}{P_g}\frac{P_H}{\Sigma N_j kT} \approx \frac{P_e}{\Sigma N_j kT},$$

where P_H is the partial pressure of hydrogen and by virtue of its overwhelming abundance, dominates the gas pressure. But the sum of the element number

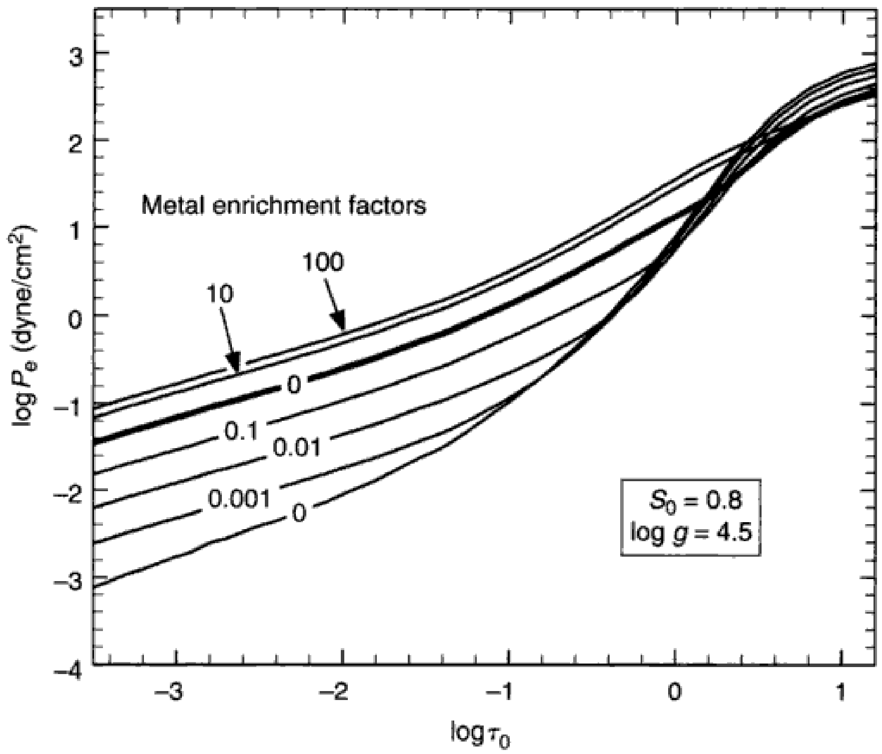


Fig. 9.16. The electron pressure for the same models used in Fig. 9.15 shows an increase with increasing metal abundance for all but the deepest layers.

densities is composed of the sum of both ions and neutrals,

$$\sum N_j = \sum (N_1 + N_0)_j \qquad \text{and} \qquad P_e = N_e kT = \sum N_{1j} kT,$$

assuming single ionizations. So the original expression becomes

$$\frac{P_e}{P_g}\frac{1}{\Sigma A_j} \approx \frac{\Sigma N_{1j}}{\Sigma (N_1 + N_0)_j}.$$

There are two obvious cases,

1. Metals ionized: $\Sigma N_{1j} \gg \Sigma N_{0j}$ giving

$$\frac{P_e}{P_g}\frac{1}{\Sigma A_j} \approx 1. \tag{9.22}$$

2. Metals neutral: $\Sigma N_{1j} \ll \Sigma (N_1 + N_0)_j$ giving

$$\frac{P_e}{P_g}\frac{1}{\Sigma A_j} \approx \frac{\Sigma N_{1j}}{\Sigma N_{0j}}. \tag{9.23}$$

For the Sun, case 1 is applicable, and we recast Eq. (9.1) as

$$\mathrm{d}P_g = \frac{g}{\kappa_0}\mathrm{d}\tau_0 = \frac{g}{P_e\kappa_0/P_e}\mathrm{d}\tau_0 = \frac{g}{P_g\sum A_j\kappa_0/P_e}\mathrm{d}\tau_0.$$

Since κ_0 is dominated by the negative hydrogen ion, κ_0/P_e is independent of P_e. Upon integration,

$$\tfrac{1}{2}P_g^2 = \frac{1}{\sum A_j}\int_0^{\tau_0}\frac{g}{\kappa_0/P_e}\mathrm{d}\tau_0,$$

and since g and T are being held constant,

$$P_g = c_0\left(\sum A_j\right)^{-1/2}, \tag{9.24}$$

where c_0 is a constant. Then from Eq. (9.22) it follows that

$$P_e = c_0\left(\sum A_j\right)^{1/2}. \tag{9.25}$$

While these expressions for the sensitivity of the pressures to composition are not rigorous, they do give the proper qualitative behavior when hydrogen is not an electron donor. In reality, hydrogen does take over in the deeper photospheric layers even for relatively cool models such as those shown in Figs. 9.15 and 9.16.

For Case 2 above, approximate the right-hand side of Eq. (9.23) using a single element so that it becomes $N_1/N_0 = \Phi(T)/P_e$. Following the same pattern with the hydrostatic equation then leads to

$$P_g = c_0\left(\sum A_j\right)^{-1/3}$$
$$P_e = c_0\left(\sum A_j\right)^{1/3},$$

where c_1 is another constant.

It is also instructive to compare the effects of g in Fig. 9.13 with the effects of ΣA_j in Fig. 9.16. In the higher layers, $\log\tau_\nu \le -2$, decreasing either of these variables produces similar results. In the deeper layers, where hydrogen is ionized, the chemical composition has only a small effect. We conclude that changes in chemical composition can mimic surface gravity effects (and vice versa) for those spectral features formed in the upper photosphere.

The helium abundance can alter the pressure structure by a small amount. The effect comes in through the sum of the atomic weights, $\Sigma A_j\mu_j$, used to reduce κ per hydrogen particle to κ_ν per mass. Using Eq. (8.19), we can write

$$\begin{aligned}\mathrm{d}P_g &= \frac{g}{\kappa_0}\mathrm{d}\tau_0\\ &= \frac{g}{\kappa}\sum A_j\mu_j\mathrm{d}\tau_0.\end{aligned}$$

where the unsubscripted κ is the area per hydrogen particle absorption coefficient. We see that g and $\Sigma A_j\mu_j$ play the same role. Since from Eq. (9.19) $P_g \approx$ constant $g^{2/3}$ we expect

$$P_g = \text{constant}\left(\sum A_j\mu_j\right)^{2/3}$$

But

$$\begin{aligned}\sum A_j\mu_j &= A(\mathrm{H})\mu_{\mathrm{H}} + A(\mathrm{He})\mu_{\mathrm{He}} + \cdots \\ &\approx 1.66\times 10^{-24}[1+4A(\mathrm{He})],\end{aligned}$$

grams since the other elements contribute only a very small mass. Therefore,

$$\begin{aligned}P_g &\approx \text{constant}\,[1+4A(\mathrm{He})]^{2/3} \\ P_e &\approx \text{constant}\,[1+4A(\mathrm{He})]^{1/3}\end{aligned} \qquad (9.26)$$

describe the dependence of the pressures on the helium abundance.

Changes with effective temperature

Finally, let us turn our attention to the effective temperature dependence of the pressure structure. The model data are presented in Figs. 9.17 and 9.18. The change of pressure at a given optical depth is a useful quantity, especially when we want to understand the behavior of spectral lines (Chapter 13). Gas pressure decreases at a given τ_0 as we move to hotter models until hydrogen starts to ionize. Refer back to Fig. 8.5. Notice the dramatic increase in opacity in going from the cool to the hotter cases. Larger opacity means less geometrical penetration to reach any given optical depth. We see less deeply into the photospheres of hotter stars; the gas pressure at a given optical depth is correspondingly less. On the other hand, the electron pressure increases with increasing temperature because of increasing ionization. For the cooler stars, we find it useful to write

$$P_e \approx \text{constant}\; e^{\Omega T} \qquad (9.27)$$

as an approximation to the curves in Fig. 9.18 (dashed lines). Values of Ω are shown on the graph, and more generally are in the range of 0.001–0.002/K. On the other hand, for $T_{\mathrm{eff}} \gtrsim 10\,000$ K, hydrogen has taken over as the electron donor and the curves of Fig. 9.18 level off.

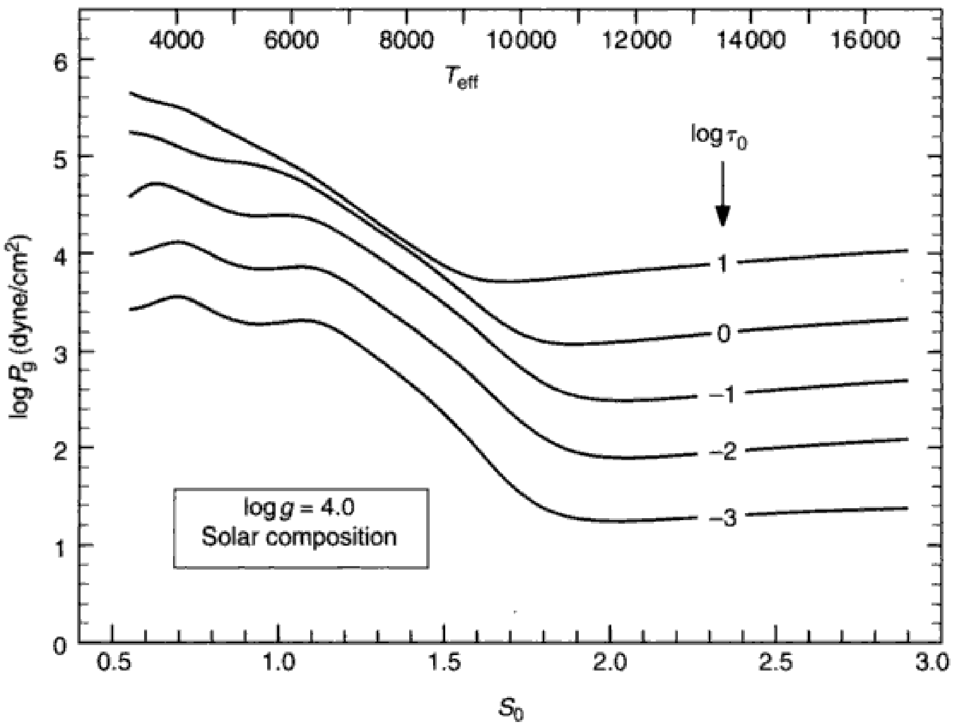


Fig. 9.17. Gas pressure is shown as a function of effective temperature for several depths in the model photospheres ($\log \tau_0$).

Tabulations of model photospheres

A small grid of models is given in Table 9.2. The solar $T(\tau_0)$ is scaled to produce the temperature distributions for the other models. From these tabulations and the graphs on the preceding pages, you can see what the numbers are like in model photospheres.

Many grids of models have been published. Some of the more informative ones are the O- to G-star models of Kurucz (1979) and the GK-giant series by Gustafsson *et al.* (1975) and Bell *et al.* (1976). Other tabulations include Mihalas (1969) and Carbon and Gingerich (1966). Models of hot stars can be found in the papers by Underhill (1968), Abbott and Hummer (1985), and Bohannan *et al.* (1986). Cool-star models have been tabulated by Gingerich *et al.* (1966), Tsuji (1966, 1978), Johnson *et al.* (1980), Schmid-Burgk *et al.* (1981), and Lester *et al.* (1982), for example. Models explicitly for giants have been computed by Parsons (1969) and Heasley *et al.* (1978), for supergiants by Groth (1961) and Osmer (1972); white-dwarf atmospheres are considered by Wesemael *et al.* (1980).

Some care should be used with older tabulations. Improvements in the hydrogen opacity sources over the last two decades will produce a few per cent

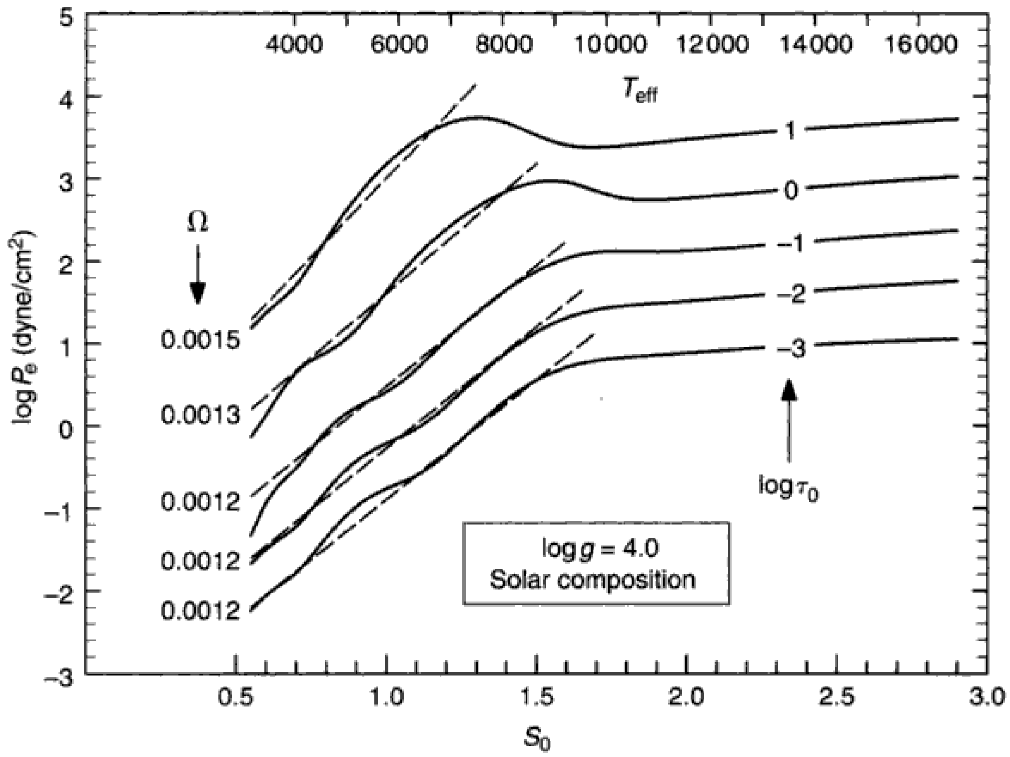


Fig. 9.18. Electron pressure increases exponentially with effective temperature: rapidly for the cool stars and slowly for the hotter ones. The exponent in Eq. (9.27) is shown on the left for the cool stars. The corresponding values for the hot stars range from 0.000 09 to 0.000 12.

change in the model parameters. Better continuous opacity for the metals can make significant differences, especially for $\lambda < 4000$ Å. And the improvements in treating line blanketing can amount to hundreds of degrees in $T(\tau_0)$, with dramatic alterations in the spectrum produced by the model. But in many cases you may find it more useful to compute your own models.

Reflection

The models we have considered here are simple but moderately realistic. They are completely determined when we specify the temperature parameter (i.e., S_0 or T_{eff}, for example), the pressure parameter (g), and the chemical composition. The original assumptions listed at the beginning of the chapter suggest some ways in which such models can be improved. Stars having extended atmospheres, as giants and supergiants do, may be better described in spherical rather than plane parallel geometry (see, for example, Scholz 1985, Bessell *et al.* 1989,

Table 9.2. *Model photospheres.*

$\log \tau_0$	T (K)	$\log P_g$ (dyne/cm^2)	$\log P_e$ (dyne/cm^2)	$\log \kappa_0/P_e$ (cm^2/g per dyne/cm^2)	x (km)
Solar model, $S_0 = 1.0$, $\log g = 4.438$ cm/s^2					
−4.0	4310	2.87	−1.16	−1.22	−509
−3.8	4325	3.03	−1.02	−1.23	−476
−3.6	4345	3.17	−0.89	−1.24	−448
−3.4	4370	3.29	−0.78	−1.25	−422
−3.2	4405	3.41	−0.66	−1.26	−397
−3.0	4445	3.52	−0.55	−1.28	−373
−2.8	4488	3.64	−0.44	−1.30	−349
−2.6	4524	3.75	−0.33	−1.32	−325
−2.4	4561	3.86	−0.23	−1.33	−301
−2.2	4608	3.97	−0.12	−1.35	−277
−2.0	4660	4.08	−0.01	−1.37	−252
−1.8	4720	4.19	0.10	−1.40	−228
−1.6	4800	4.30	0.22	−1.43	−203
−1.4	4878	4.41	0.34	−1.46	−177
−1.2	4995	4.52	0.47	−1.50	−151
−1.0	5132	4.63	0.61	−1.55	−124
−0.8	5294	4.74	0.76	−1.60	−97
−0.6	5490	4.85	0.93	−1.66	−70
−0.4	5733	4.95	1.15	−1.73	−43
−0.2	6043	5.03	1.43	−1.81	−19
0.0	6429	5.10	1.78	−1.91	0
0.2	6904	5.15	2.18	−2.01	15
0.4	7467	5.18	2.59	−2.11	27
0.6	7962	5.21	2.92	−2.18	37
0.8	8358	5.23	3.16	−2.23	46
1.0	8630	5.26	3.32	−2.25	56
1.2	8811	5.29	3.42	−2.27	68
$S_0 = 0.7$, $\log g = 4.6$, normal abundances					
−4.0	3017	3.22	−2.12	−0.46	−246
−3.0	3111	3.89	−1.51	−0.53	−179
−2.0	3262	4.45	−0.95	−0.64	−120
−1.0	3592	5.00	−0.26	−0.85	−59
0.0	4500	5.45	0.88	−1.31	0
1.0	6041	5.84	1.96	−1.82	68
$S_0 = 1.5$, $\log g = 4.0$, normal abundances					
−4.0	6465	1.32	−0.15	−1.40	−2273
−3.0	6667	2.35	0.56	−1.77	−1368
−2.0	6990	2.99	1.14	−1.89	−790
−1.0	7698	3.49	1.88	−1.97	−307
0.0	9643	3.76	2.96	−1.82	0
1.0	12945	3.88	3.53	−1.73	301

Table 9.2. (*cont.*)

$\log \tau_0$	T (K)	$\log P_g$ (dyne/cm^2)	$\log P_e$ (dyne/cm^2)	$\log \kappa_0/P_e$ (cm^2/g per dyne/cm^2)	x (km)
$S_0 = 1.0$, $\log g = 2.0$, normal abundances					
−4.0	4310	1.55	−2.38	−1.16	−129771
−3.0	4445	2.23	−1.72	−1.27	−90 459
−2.0	4660	2.79	−1.15	−1.37	−57 437
−1.0	5132	3.31	−0.44	−1.54	−24 170
0.0	6429	3.64	1.00	−1.88	0
1.0	8630	3.72	2.52	−1.96	8 103
$S_0 = 1.0$, $\log g = 4.438$, $0.1 \times$ normal abundances					
−4.0	4310	3.30	−1.60	−1.22	−449
−3.0	4445	3.95	−0.98	−1.28	−312
−2.0	4660	4.50	−0.41	−1.37	−194
−1.0	5132	4.99	0.35	−1.54	−80
0.0	6429	5.29	1.84	−1.91	0
1.0	8630	5.39	3.40	−2.27	34

Jorgenson *et al.* 1992, Plez *et al.* 1992). LTE seems to be a reasonable approximation for the continuum (Strom & Kalkofen 1966, Mihalas 1967, 1972, Smith & Strom 1969), and the LTE models adequately describe the Paschen continuum, for example. The Balmer and Paschen jumps may show non-LTE effects of a few per cent, and non-LTE effects in the line blanketing may alter the temperature gradient in the shallower layers (Anderson 1989, Rutten 1990). When mass loss and other aerodynamic phenomena occur, we question the validity of hydrostatic equilibrium, and so on down the list. In short, we need to extend these models as our research and understanding dictate. Some basic extensions of these models are made in later chapters.

References

Abbott, D.C. & D.G. Hummer 1985. *Ap. J.* **294**, 286.
Allende Preito, C., I. Hubeny, & D.L. Lambert 2003. *Ap. J.* **591**, 1192.
Anderson, L.S. 1989. *Ap. J.* **339**, 558.
Athay, R.G. & A. Skumanich 1969. *Ap. J.* **155**, 273.
Auer, L.H. & D. Mihalas 1969. *Ap. J.* **158**, 641.
Avrett, E.H. & M. Krook 1963a. *Ap. J.* **137**, 874.
1963b. *J. Quant. Spectr. Rad. Trans.* **3**, 107.
Ayres, T.R. 1981. *Ap. J.* **244**, 1064.
Ayres, T.R. & L. Testerman 1981. *Ap. J.* **245**, 1124.
Ayres, T.R. & G.R. Wiedeman 1989. *Ap. J.* **338**, 1033.
Bell, R.A., J. Eriksson, B. Gustafsson, & Å. Norlund 1976. *Ast. Ap. Suppl.* **23**, 37.

Bessell, M., J.M. Brett, M. Scholz, & P. R. Wood 1989. *Ast. Ap.* **213**, 209.
Bohannan, B., D.C. Abbott, S.A. Voels, & D.C. Hummer 1986. *Ap. J.* **308**, 728.
Böhm, K.H. 1966. *J. Quant. Spectr. Rad. Trans.* **6**, 534.
Cannon, C.J. 1973a. *J. Quant. Spectr. Rad. Trans.* **13**, 627.
1973b. *Ap. J.* **185**, 621.
Carbon, D.F. 1979. *Ann. Rev. Ast. Ap.* **17**, 513.
Carbon, D.F. & O. Gingerich 1969. *Theory and Observation of Normal Stellar Atmospheres*, ed. O. Gingerich. (Cambridge: MIT), p. 377.
Decin, L., B. Vandenbussche, C. Waelkens, *et al.* 2003. *Ast. Ap.* **400**, 709.
Drake, J.J. & G. Smith 1991. *Mon. Not. Roy. Ast. Soc.* **250**, 89.
Edmonds, F.N., Jr. 1969. *J. Quant. Spectr. Rad. Trans.* **9**, 1427.
Elste, G.H.E. 1967. *Ap. J.* **148**, 857.
1985. *Theoretical Problems in High Resolution Solar Physics*, ed. H. U. Schmidt. (München: Max Planck Inst. Ap.), p. 185.
Ferguson, J.W., D.R. Alexander, F. Allard, & P. Hauschildt 2001. *Ap. J.* **557**, 798.
Gingerich, O., D.W. Latham, J. Linsky, & S.S. Kumar 1966. *Coll. on Late-Type Stars*, ed. M. Hack. (Trieste: Osservatorio Astronomico), p. 291.
Gray, D.F. 1984. *Ap. J.* **281**, 719.
1989. *Ap. J* . **347**, 1021.
Groth, H.G. 1961. *Z.f. Ap.* **51**, 231.
Gustafsson, B., R. A. Bell, J. Eriksson, & Å. Nordlund 1975. *Ast. Ap.* **42**, 407.
Heasley, J.N., S.T. Ridgway, D.F. Carbon, R. W. Milkey, & D. N. Hall 1978. *Ap. J.* **219**, 970.
Holweger, H. & E.A. Müller 1974. *Solar Phys.* **39**, 19.
Johnson, H.R., A.P. Bernat, & B.M. Krupp 1980. *Ap. J. Suppl.* **42**, 501.
Jorgenson, U.G., H.R. Johnson, & Å. Nordlund 1992. *Ast. Ap.* **261**, 263.
Kalkofen, W. 1987. *Numerical Radiative Transfer*. (Cambridge: Cambridge University press).
Kurucz, R.L. 1979. *Ap. J. Suppl.* **40**, 1.
Lester, J.B., M.C. Lane, & R.L. Kurucz 1982. *Ap. J.* **260**, 272.
Maltby, P., E.H. Avrett, M. Carlsson, O. Kjeldseth-Moe, R.L. Kurucz, & R. Loeser 1986. *Ap. J.* **306**, 284.
Mihalas, D. 1966. *Ap. J. Suppl.* **13**, 1.
1967. *Ap. J.* **149**, 169.
1972. *Ap. J.* **176**, 139.
Mihalas, D.M. & W.R. Luebke 1971. *Mon. Not. Roy. Ast. Soc.* **153**, 229.
Mitchell, W.E., Jr. 1959. *Ap. J.* **129**, 93.
Muchmore, D. 1986. *Ast. Ap.* **155**, 172.
Nordlund, Å. 1985. *Theoretical Problems in High Resolution Solar Physics*, ed. H.U. Schmidt. (München: Max Planck Inst. Ap.), p. 19.
Osmer, P.S. 1972. *Ap. J. Suppl.* **24**, 255.
Parsons, S.B. 1969. *Ap. J. Suppl.* **18**, 127.
Pecker, J.C. 1973. *Problems of Calibration of Absolute Magnitudes and Temperature of Stars*, IAU Symp. 54, ed. B. Hauck & B. E. Westerlund. (Dordrecht: Reidel), p. 173.
Pierce, A.K. 2000. *Allen's Astrophysical Quantites*, 4th edn, ed. A. N. Cox. (London: Athlone), p. 355.
Pierce, A.K. & J.H. Waddell 1961. *Mem. Roy. Ast. Soc.* **68**, 89.
Plez, B., J.M. Brett, & Å. Nordlund 1992. *Ast. Ap.* **256**, 551.
Ruck, M.J. & G. Smith 1995. *Ast. Ap.* **304**, 449.
Ruland, F.D. Biehl, H. Holweger, R. Griffin, & R. Griffin 1980. *Ast. Ap.* **92**, 70.

Rutten, R.J. 1990. *Cool Stars, Stellar Systems, and the Sun*, 6th Cambridge Workshop, ed. G. Wallerstein. (San Francisco: Ast. Soc. Pacific), p. 91.
Scharmer, G.B. 1981. *Ap. J.* **249**, 720.
Schmid-Burgk, J., M. Scholz, & R. Wehrse 1981. *Mon. Not. Roy. Ast. Soc.* **194**, 383.
Scholz, M. 1985. *Ast. Ap.* **145**, 251.
Smith, M.A. & S. E. Strom 1969. *Ap. J.* **158**, 1161.
Strom, S.E. & W. Kalkofen 1966. *Ap. J.* **144**, 76.
Tripicchio, A., G. Severino, E. Covino, L. Terranegra, & R.J. Garcia Lopez 1997. *Ast. Ap.* **327**, 681.
Tsuji, T. 1966. *Colloquium on Late-Type Stars*, ed. M. Hack. (Trieste: Osservatorio Ast.), p. 260.
1978. *Ast. Ap.* **62**, 29.
Underhill, A. B. 1968. *Bull. Ast. Inst. Netherlands* **19**, 500.
Werner, K. 1988. *Ast. Ap.* **204**, 159.
Wesemael, F., H.M. van Horn, M.P. Savedoff, & L.H. Auer 1980. *Ap. J. Suppl.* **43**, 159.

Questions and exercises

1. Sketch qualitative limb darkening curves for a G8 dwarf as they might appear at 4000, 7000 Å, and at 10 cm. (The 10 cm radio emission comes from the chromosphere and lower corona.) Explain any differences you make in your curves.

2. For the case of a K4 supergiant, draw qualitative I_ν contribution functions for continuum radiation coming from (a) the center of the disk and (b) the extreme limb. If there are any differences between the two, suggest possible reasons for them.

3. Explain what we mean by the term "temperature distribution" in the context of stellar photospheres. Why do we need to know the temperature distribution of a star when computing a model photosphere? What methods are available for the determination of temperature distributions? Make a sketch of a typical temperature distribution in a stellar photosphere; be sure to include numerical scales. How is the temperature distribution related to the effective temperature?

4. Compute the electron pressure, P_e, at one point in a stellar atmosphere using Eq. (9.8). Assume $P_g = 1000\,\text{dyne/cm}^2$ and $T = 5040\,\text{K}$ at this point. Sum over the elements: H, C, Si, Fe, Mg, Ni, Cr, Ca, Na, K, and He. Use the solar abundances in Table 16.3 and the ionization potentials found in Appendix D. Iterate to get approximately two decimal places. Compare your value with what is shown in Fig. 9.7. Which of these elements contributes the most to the electron pressure? The least? Now re-do the calculations using the same gas pressure but raise the temperature to 12 000 K. Ask the same questions of your calculations.

5. Why does the metal abundance dramatically affect the electron pressure in the photosphere of a cool star, but not in a hot star?

6. Consider writing your own model photosphere program. Basically you will need to compute the hydrostatic integral in Eq. (9.2) and then the radiative transfer expressed in Eq. (9.13). To complete these computations, evaluate κ_ν according to Eqs. (8.18) and (8.19) and the optical depth scales through Eq. (9.14). As explained in the text, you will have to compute the electron pressure in order to evaluate κ_ν.

Use Eq. (9.8) to do this. For scaled models, get the temperature distribution from Eq. (9.5), and under the assumption of LTE, the source function is $B_\nu(T)$, i.e., Planck's law.

7. Use a model photosphere program (written by you or borrowed from a colleague or given to you by your instructor) to investigate how electron pressure and geometrical depth behave as a function of effective temperature (or S_0). Use one fixed surface gravity such as $\log g = 4$. Record the values of $\log P_g$ and $\log P_e$ at one depth, namely $\log \tau_0 = 0.0$, and the geometrical depth span, Δx, between $\log \tau_0 = 0.0$ and 1.0. Make plots of (i) P_e/P_g, (ii) $\log P_e/P_g$, and (iii) Δx as a function of effective temperature. What do you see happening? Why is it happening?

8. Continue to explore the geometrical depth with the model photosphere program. This time choose one temperature. Keep track of Δx, the geometrical depth span between $\log \tau_0 = 0.0$ and 1.0, as you run models over a wide range of surface gravity from, say, $\log g = 1$ to 8, i.e., supergiants down to white dwarfs. What kind of relation do you see between Δx and gravity?

10

The measurement of stellar continua

The primary aim of continuum measurements is to obtain the photospheric temperature scale, but numerous other uses range from gravity measurement, chemical composition studies, the detection of companion stars and disks, through properties of broad-band photometric systems, and bolometric corrections. In the hotter stars, the shape of the continuum is molded by the bound–free absorption of neutral hydrogen. From the ground, we can measure only a small part of the Balmer continuum (912–3647 Å) longward of the ozone cut-off ~3400 Å, the complete Paschen continuum (3647–8207 Å), and some of the Brackett continuum (8207–14 588 Å) which is badly cut up by terrestrial molecular absorption. The Balmer and Paschen discontinuities are useful as pressure diagnostics for late A and F stars, but they also depend on temperature. In cooler stars, where the negative hydrogen ion dominates, all continuum characteristics depend almost exclusively on temperature.

The spectrum measured with low resolution, e.g., 10–50 Å, is often called the "energy distribution." In such situations, the spectral lines are included, and then measurements of the fraction of light removed by spectral lines is needed to regain accurate information on the position of the continuum.

Our equipment must be capable of measuring radiation over a wide wavelength range, preferably in all wavelengths simultaneously. A star of interest is normally measured in four steps: (1) comparison of the program star to a standard star, (2) calibration of the shape of the standard star's energy distribution, (3) calibration of the absolute flux at one or more points in the energy distribution of the standard star, and (4) due allowance for the spectral lines in order to obtain true continuum values. Most observers will perform steps (1) and (4), while relying on specialists to accomplish steps (2) and (3).

Ultra-low-resolution spectrographs

An ultra-low-resolution spectrograph has been the instrument of choice for energy distribution work because the shape of the continuum can be completely specified with relatively few wavelength points. In addition, an ultra-low-resolution spectrograph will easily give wide wavelength coverage and the many-angstrom-wide bandpasses allow us to reach faint stars. Suppose for example we have a detector with 1000 pixels, and we wish to cover the 4000–10 000 Å range. A dispersion of ~6 Å/pixel is then called for. The smearing caused by a finite seeing disk and the optical aberrations would typically reduce the resolution to ~10–15 Å, or a resolving power of $\lambda/\Delta\lambda \sim 6000$ Å/10 Å $= 600$. Such low resolving power places this kind of work between multicolor photometry and normal high-resolution spectroscopy. But measuring the light from a few angstroms, compared to a few milli-angstroms for high-resolution spectroscopy, gives a thousand-fold gain in photon rate, so energy distributions of stars 1000 times fainter can be measured.

Figure 10.1 illustrates a low-resolution spectrograph suitable for energy distribution measurements. It is crucial to use a wide entrance slit or a circular diaphragm to ensure that the whole seeing disk enters the spectrograph. At the same time, the entrance aperture should be small enough to isolate the image of the star and exclude as much sky background light as possible. One might expect simultaneous measurement at all wavelengths to render the details of the entrance aperture immaterial. Alas, the size of the seeing disk varies with wavelength, and atmospheric refraction causes the seeing disk to lie in slightly different positions for different wavelengths. Too small an entrance aperture would occult some wavelengths more than others, causing systematic differences with wavelength, and ruin our energy distribution measurements.

The spectral resolution for the conventional spectrograph having an entrance slit width W' is given by Eq. (3.15),

$$\Delta\lambda = -W' \frac{\cos\alpha}{f_{\text{coll}}} \frac{d}{n},$$

where α is the angle of incidence on the grating, f_{coll} is the collimator focal length, and d is the grating ruling spacing for the nth-order spectrum. In the current case, W' is replaced by the size of the seeing disk. Suppose the seeing disk has an angular size $\Delta\phi$, and the telescope effective focal length is F_{eff}. Then $W' = \Delta\phi F_{\text{eff}}$, and the resolving power is

$$\Delta\lambda = -\cos\alpha \frac{F_{\text{eff}}}{f_{\text{coll}}} \frac{d}{n} \Delta\phi. \tag{10.1}$$

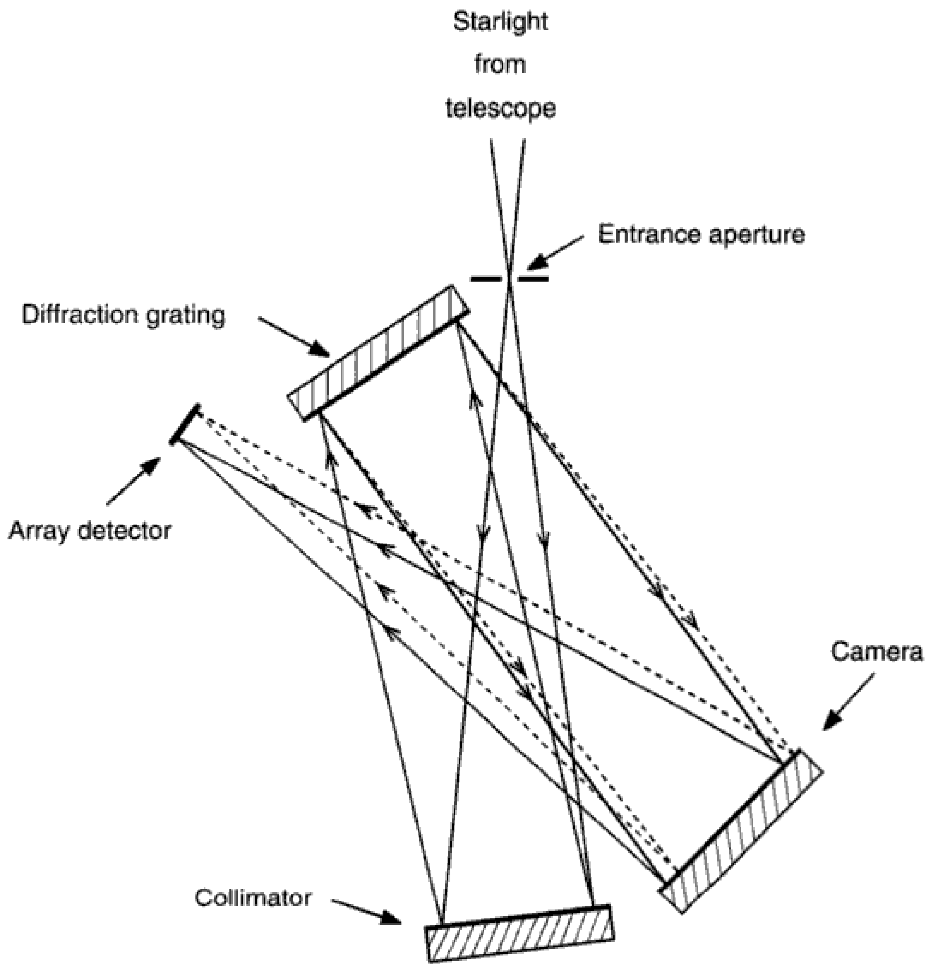


Fig. 10.1. This shows the optical layout for a typical low-dispersion spectrograph. A diffraction grating is used to disperse the light, and the extreme wavelength span recorded by the detector is shown by the solid and dashed lines from the grating to the detector.

We see that fluctuations in seeing result in fluctuations in resolving power, but the effect can be reduced by making the ratio $F_{\text{eff}}/f_{\text{coll}}$ small. Frequently the focal length of the camera is chosen to image several angstroms onto each pixel, in line with the concepts expressed at the beginning of this section.

Observations using standard stars

Step one in measuring the continuum of a star is to compare it with a standard star for which the continuum energy distribution has already been calibrated. The ratio of the output signals, after correction for sky plus dark signal and atmospheric extinction, is the ratio of fluxes for the two stars, assuming that we are blessed with linear behavior of the detector and its attendant circuitry.

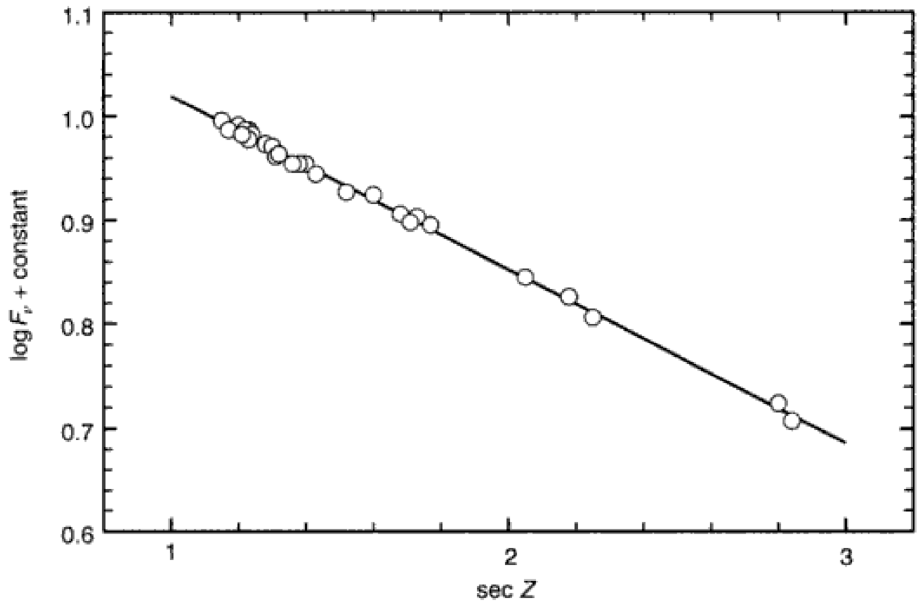


Fig. 10.2. Extinction curves like this one have to be measured to correct for the light lost in the terrestrial atmosphere. Either the program star or the reference stars should be measured through a range in zenith distance that encompasses zenith distances of the other star.

The sky correction is obtained by interspersing measurements of the nearby sky between the stellar measurements. One of the stars must be observed for a sufficiently long period to define the slope of the extinction curve of the type shown in Fig. 10.2. As long as the plane parallel approximation can be applied to the terrestrial atmosphere (see Hardie 1962 for curvature corrections), we expect the observed flux, F_ν, to vary with angle Z from the zenith according to

$$F_\nu = F_\nu^0 e^{-\tau_\nu \sec Z},$$

where F_ν^0 is the flux outside the atmosphere and τ_ν is the optical depth through the atmosphere toward the zenith. This is more commonly written as

$$\log F_\nu = \log F_\nu^0 - \tau_\nu \log e \sec Z, \tag{10.2}$$

The coefficient of $\sec Z$, $\tau_\nu \log e$, is often called an extinction coefficient, and is the slope of the line in Fig. 10.2. For comparative measurements, it is not necessary to know $\tau_\nu \log e$ or F_ν^0 explicitly, although they can be calculated if desired. Suppose we compare the two stars using Eq. (10.2),

$$\log\left[\frac{F_\nu(2)}{F_\nu(1)}\right] = \log\left[\frac{F_\nu^0(2)}{F_\nu^0(1)}\right] - \tau_\nu \log e(\sec Z_2 - \sec Z_1). \tag{10.3}$$

The left-hand side of the equation is equal to the ratio of data signals after allowance for any sky background and any amplifier gain differences. The flux ratio on the right-hand side of the equation is the answer we are after. It is desirable to have measurements of both stars near the same sec Z so the last term in Eq. (10.3) is small. Then Eq. (10.3) can be viewed as a simple interpolation along the extinction curve from which we find $F_\nu(1)$ at sec Z_2 or vice versa.

There are several assumptions incorporated in the discussion: (1) the sky is uniform and stable so that the extinction coefficient is the same for both stars and constant for the duration of the observations, (2) the star used for defining the extinction curve is not a variable star, and (3) the effective wavelength of the spectral band is not a function of sec$Z$. Difficulties with the first two assumptions are reasonably obvious if the observations are carefully made. A change in effective wavelength with sec$Z$ is a well-known phenomenon in broadband photometry (see Hardie 1962). It results from the change in the *detected* energy distribution across the photometric bandpass caused by the wavelength dependence of the extinction. Rayleigh scattering is the most pronounced extinction mechanism (the λ^{-4} term below). One can describe the extinction coefficient (see Hayes & Latham 1975, Cochran & Barnes 1981) in regions devoid of terrestrial molecular bands by

$$\tau_\nu \log e = A + B/\lambda + C/\lambda^4.$$

This steep wavelength dependence can cause the V band of the UBV system to change its effective wavelength by tens of angstroms when measurements are taken at large zenith angles. A very similar effect occurs when comparing stars of different intrinsic energy distributions such as a B and a K star. But for our purposes, where the stellar continuum is to be compared with model atmosphere calculations, we generally use much higher spectral resolution than afforded by the broadband photometric systems (even though it is "ultra-low"-resolution spectroscopy). Over a spectral band of 50 Å or less, the change in effective wavelength is negligible.

Absolute calibration of standard stars

Vega (α Lyr, HR 7001, HD 172167, A0 V, $V=0.03$) has been adopted as the primary photometric standard for the visible window. Secondary standard stars have been calibrated against Vega, as described in the next section. The *shape* of Vega's energy distribution is measured first by comparing the flux at various points in the continuum to the flux at λ5556. Then the actual number of photons comprising the flux at λ5556 is measured. Absolute measurements are more vulnerable to poor sky conditions because Eq. (10.2) rather than Eq. (10.3) must be used. Although we concentrate on Vega, the Sun has also been measured extensively (e.g., Neckel & Labs 1984, Lockwood *et al.* 1992, Burlov-Vasiljev *et al.* 1995; see below).

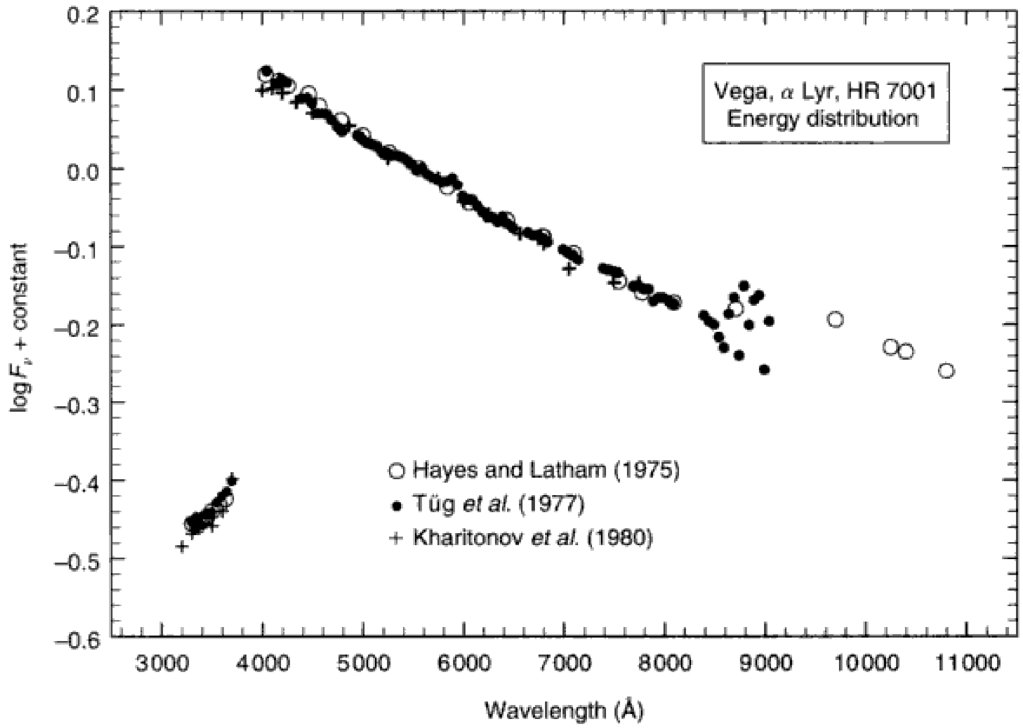


Fig. 10.3. Measurements of the energy distribution of Vega (A0 V) are compared. The large apparent scatter between 8500 and 9000 Å arises from the confluence of the Paschen lines.

Measuring the shape of Vega's continuum is difficult and meticulous work because the sensitivity of the instrument itself is a strong function of wavelength. The spectral response to the instrument must be removed using a standard light source, namely, a standard lamp, black body (see Chapter 6), or synchrotron radiator. Standard lamps, typically operating near 2800 K, are intermediate standards and require previous calibration, usually against a black body (see Bless *et al.* 1968). The most fundamental calibrations come from black bodies operated at the melting temperature of some metal such as platinum (2042 K), copper (1357.6 K), or gold (1337.58 K). The synchrotron radiation source has an advantage for ultraviolet wavelengths, where it is brighter than a typical black body (Code 1985).

Absolute calibrations of Vega have been given by Hayes and Latham (1975), and Hayes *et al.* (1975), Tüg *et al.* (1977), Terez and Terez (1979), Kharitonov *et al.* (1980), Knyazeva and Kharitonov (1990), among others. The comparison in Fig. 10.3 shows agreement to 1% between 5000 and 8000 Å, but errors are several times larger at shorter and longer wavelengths. Hayes (1985) has tabulated Vega's energy distribution in 25 Å steps continuously from 3300 Å to 10 500 Å. Table 10.1 is based on his tabulation. The column labeled "log F_ν" gives the logarithmic flux relative to the value at λ5556, and $m(\nu)$ is the same

Table 10.1. *The shape calibration of Vega.*

λ(Å)	$\log F_\nu$	$m(\nu)^a$	λ(Å)	$\log F_\nu$	$m(\nu)$
3300	−0.465	+1.163	5840	−0.022	+0.056
3400	−0.454	+1.135	6056	−0.044	+0.109
3500	−0.447	+1.117	6436	−0.068	+0.171
3571	−0.434	+1.086	6800	−0.090	+0.226
3636	−0.422	+1.056	7100	−0.113	+0.283
4036	+0.123	−0.307	7550	−0.143	+0.357
4167	+0.109	−0.273	7780	−0.159	+0.398
4255	+0.104	−0.260	8090	−0.174	+0.434
4464	+0.086	−0.216	8400	−0.197	+0.493
4566	+0.071	−0.177	8708	−0.181	+0.453
4785	+0.054	−0.134	9700	−0.195	+0.488
5000	+0.039	−0.097	10250	−0.231	+0.577
5263	+0.019	−0.047	10400	−0.238	+0.595
5556	0.000	0.000	10800	−0.260	+0.649

[a] Magnitude units: $-2.5 \log F_\nu$.

quantity expressed in magnitudes. Infrared measurements between 1 and 4.6 μm are discussed by Hayes (1985) and Mégessier (1995).

There is no consensus on how energy distributions should be plotted. Figure 10.4 shows some of the options and how the apparent shape changes with choice of units (frequency versus wavelength) and logarithmic versus linear scales. Although it might seem more consistent to use F_λ as the ordinate when using λ as the abscissa, since then equal areas under the curve represent equal power, once logarithmic scales are introduced, this is no longer true. Other combinations include logarithmic abscissae.

The absolute flux at 5556 Å is even more difficult to measure. The modern era started with the work of Code (1960). Mégessier (1995) has summarized recent measurements. Sometimes the values found in the literature are for Vega ($V = 0.03$) and sometimes for a star having a visual magnitude of $V = 0.00$. The difference amounts to 2.8%. The best estimate for Vega is

$$\begin{aligned} F_\lambda &= 3.46 \times 10^{-9}\,\mathrm{erg/(s\,cm^2\,Å)} \\ &= 3.46 \times 10^{-12}\,\mathrm{W/(m^2\,Å)} \end{aligned}$$

which is equivalent to

$$\begin{aligned} F_\lambda &= 3.56 \times 10^{-9}\,\mathrm{erg/(s\,cm^2\,Å)} \\ &= 3.56 \times 10^{-12}\,\mathrm{W/(m^2\,Å)} \\ &= 996\,\mathrm{photons/(s\,cm^2\,Å)} \end{aligned}$$

for a star with $V = 0.00$. The uncertainty is thought to be $\approx 0.7\%$.

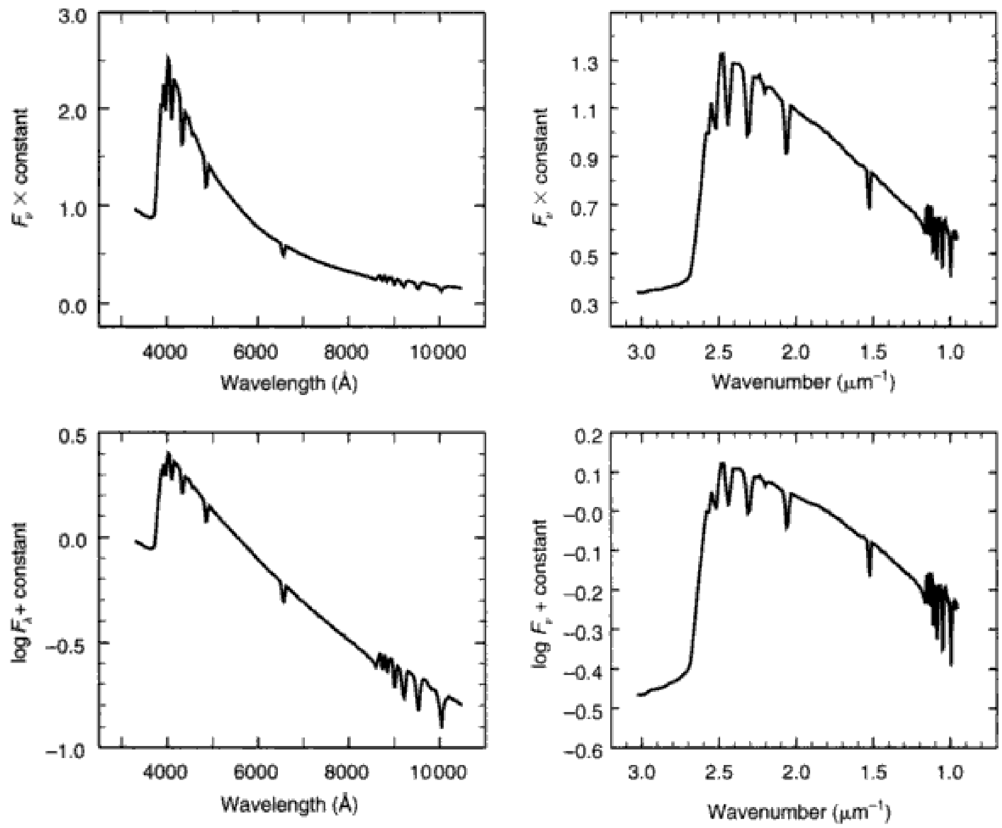


Fig. 10.4. The shape of a stellar energy distribution can look very different according to which coordinates are used. Those in Fig. 10.3 and the four shown here are the most common.

Most reports of possible variability ($\gtrsim$1%) of Vega are probably not to be believed, but we must bear in mind that all stars are likely to show variability if pushed to low enough levels. A surface feature carried across the apparent disk of the star as the star rotates, stochastic variations in large convective cells, and oscillations are all possible sources of low-level variation. The Sun, for example, can show variations of a few tenths of 1% when a large sunspot complex moves across its surface (see Fig. 10.17 below). The increase in flux arising from Vega's −14 km/s radial velocity amounts to only 0.04% per century.

The absolute flux at λ5556 for any other star can be obtained by scaling the fundamental calibration by its visual magnitude, using

$$\begin{aligned} \log F_\lambda &= -0.400V - 8.449\,\mathrm{erg/(s\,cm^2\,Å)} \\ &= -0.400V - 11.449\,\mathrm{W/(m^2\,Å)} \\ \log F_\nu &= -0.400V - 19.436\,\mathrm{erg/(s\,cm^2\,Hz)} \\ &= -0.400V - 22.436\,\mathrm{W/(m^2\,Hz)} \end{aligned} \tag{10.4}$$

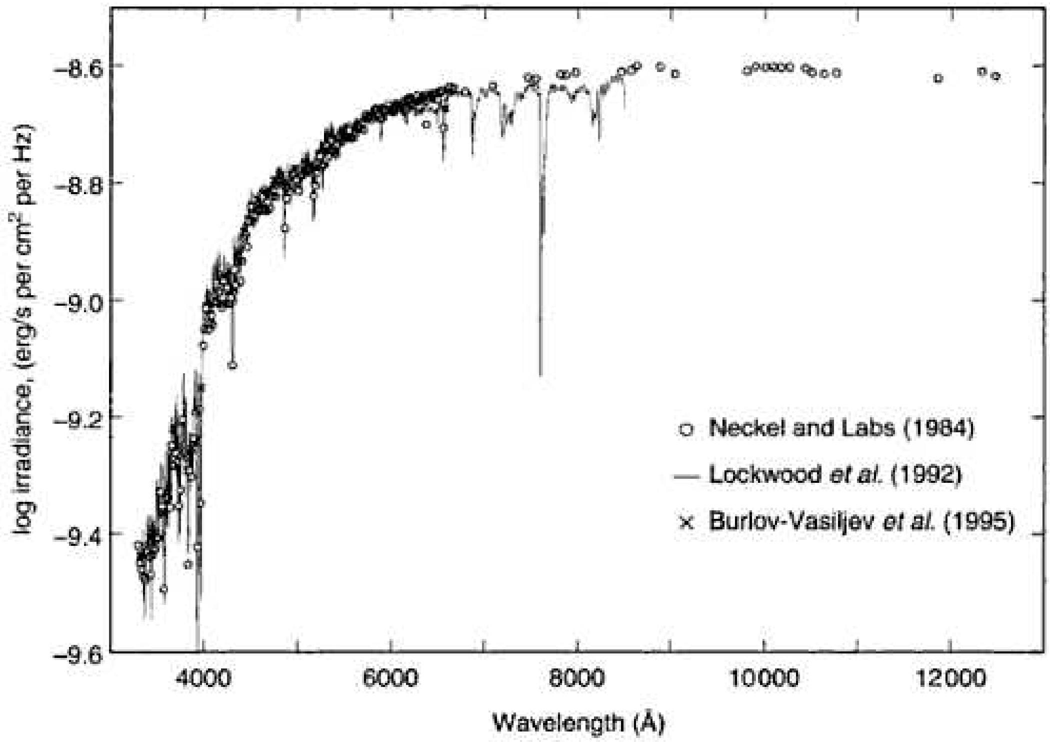


Fig. 10.5. Measurements of the solar flux energy distribution of the Sun are compared.

A small but systematic error is introduced when these expressions are used for stars of different temperatures because the effective wavelength of the V band is 5480 Å, that is, 76 Å shortward of the standard reference wavelength. From consideration of stellar energy distributions, one finds that adequate correction is given by a color term of the form

$$\log[F(\lambda 5556)/F(\lambda 5480)] = -0.006 + 0.018(B-V).$$

In the infrared, Vega appears to be overly bright compared to other A0 V stars, and brighter than models predict (Merezhin 1990, Mégessier 1995, Oudmaijer *et al.* 2001).

The solar energy distribution has been subject to extensive measurement. Figure 10.5 illustrates some of the visual-window observations. There are small but systematic differences among these observations, amounting to several per cent in some cases. The absolute flux at $\lambda 5556$ is uncertain by about $\pm 2\%$, and the apparent visual magnitude, $V_\odot = -26.75$, is uncertain by about ± 0.02 mag (Hayes 1985, Colina *et al.* 1996, Bessell *et al.* 1998). The corresponding absolute magnitude of the Sun is $M_v = 4.82$.

The color index of the Sun is also rather uncertain, with values ranging from $B-V = 0.619$ to 0.686 (Gray 1992). The best current value is: Johnson

$B-V=0.656\pm0.008$, Strömgren $b-y=0.414\pm0.005$, and Geneva $B2-V1=0.393\pm0.07$.

Photometric standard stars

Hot stars and metal-poor stars are chosen as secondary photometric standards for energy-distribution work because their spectra are the least cluttered by absorption lines, making the precise specification of the bandpass less demanding. The particular wavelength regions to be measured are selected to be as free from lines as possible. Commonly used regions are tabulated in the first column of Table 10.1 above. The grid of secondary standards chosen by Oke (1964) is well distributed in right ascension and has generally been used. More recently, Taylor (1984) has tied additional stars into the grid, and Tüg (1980a, b) has added 14 southern-hemisphere standards. The relative errors are in the 1–2% range. Additional stars suitable for use as secondary standards have been measured by Breger (1976), Kharitonov and Glushneva (1978), Kharitonov *et al.* (1980), Cochran (1981), Ipatov (1982), Glushneva and Ovchinnikov (1983), Hamuy *et al.* (1992), Glushneva *et al.* (1992), Pickles (1998), and Tereshchenko (2002).

Fainter standard stars are needed for work with large telescopes. Such standards have been set up by Stone (1977), Hayes and Philip (1983), Oke and Gunn (1983), Philip and Hayes (1983), Stone and Baldwin (1983), Baldwin and Stone (1984), Gutierrez-Moreno *et al.* (1988), Massey *et al.* (1988), and Oke (1990).

Integration over the bandpasses of a photometric system leads to absolute calibrations of the corresponding photometric magnitudes. For example, R. O. Gray (1998) gives for the Strömgren indices with effective wavelengths given in the second column,

$$
\begin{array}{lll}
u & \lambda 3491 & 11.72\times10^{-9}\pm0.12\,\mathrm{erg/(s\,cm^2\,\AA)} \\
v & \lambda 4111 & 8.66\times10^{-9}\pm0.09\,\mathrm{erg/(s\,cm^2\,\AA)} \\
b & \lambda 4662 & 5.89\times10^{-9}\pm0.04\,\mathrm{erg/(s\,cm^2\,\AA)} \\
y & \lambda 5456 & 3.73\times10^{-9}\pm0.01\,\mathrm{erg/(s\,cm^2\,\AA)}
\end{array}
$$

for a (fictitious) star having magnitude zero in the specified bandpasses, i.e., to find the absolute flux, multiply by $10^{-0.4m}$, where m is the magnitude u, v, b, or y. Again, for wider-band photometric systems, there will be a change in effective wavelength of the band with changing spectral type of the star.

Observations of stellar continua

The continuous spectrum is a strong function of temperature. Figure 10.6 displays a range of energy distributions as a function of spectral type. For stars

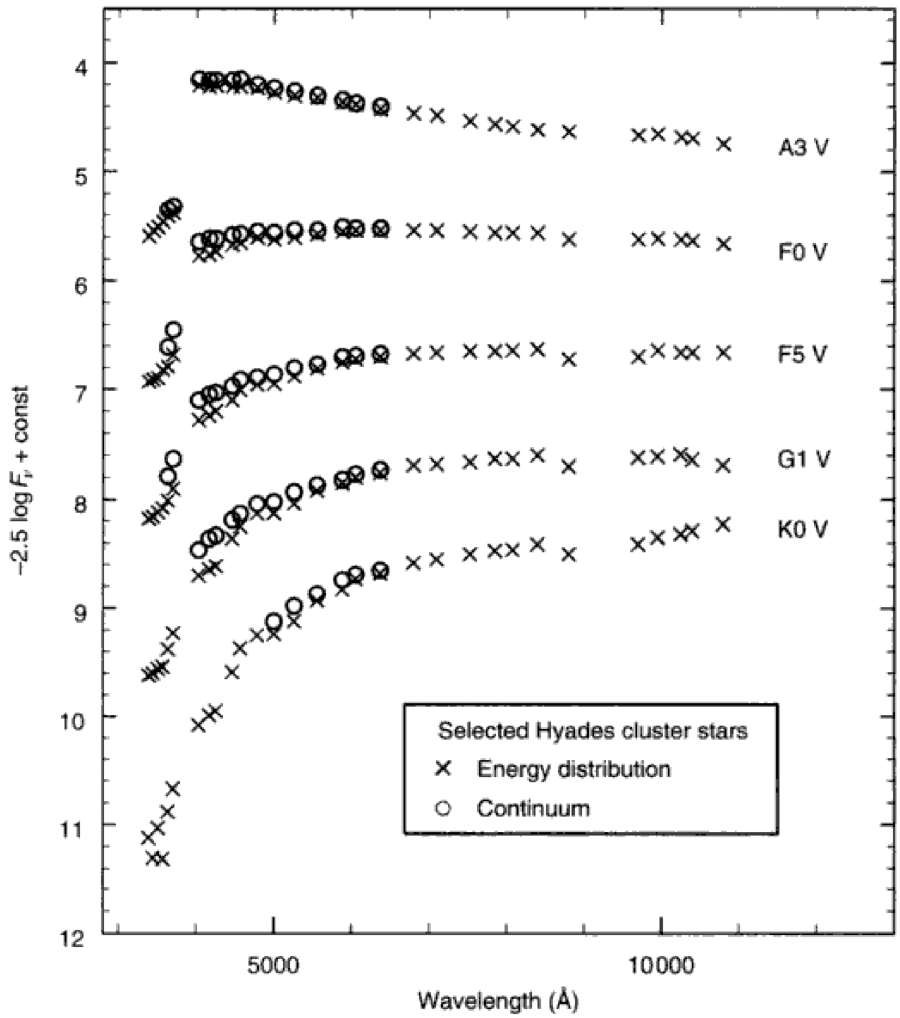


Fig. 10.6. The slopes of these Hyades-star continua show a large change with spectral type. Since the distance is nearly identical for all five stars, the measurements also give the correct absolute flux differences. Energy-distribution measurements are shown by crosses. The circles show the values after allowance for the absorption of the spectral lines. The A3 star is 68 Tau, the F0 star is 57 Tau, the F5 star is HD 27524, the G1 star is HD 28992, and the K0 star is BD + 17° 734. Based on data from Oke and Conti (1966).

hotter than about A0, the slope of the Paschen continuum becomes less sensitive to the temperature of the star because most of the flux is emitted in the Balmer continuum. Then ultraviolet measurements are called for as in Fig. 10.7, for example. The same happens in reverse for the cool stars, leading one to make infrared measurements.

Several compilations of energy distributions have been published, for example, Willstrop (1965), Breger (1976), Hardorp (1978), Ardeberg and Virdefors (1980), Cochran (1980), Gunn and Stryker (1983), Voloshina *et al.*

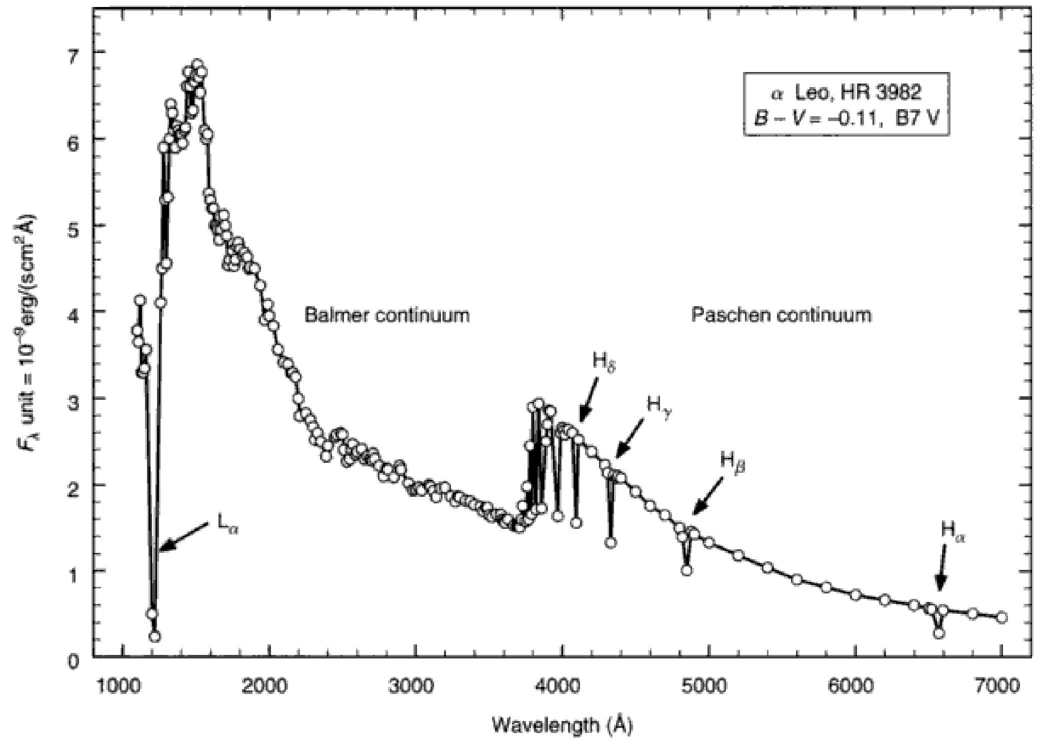


Fig. 10.7. The energy distribution of α Leo is strongest in the ultraviolet, and this portion of the spectrum is therefore very valuable in modeling the star. Even with low resolution, the hydrogen lines are prominent. Data from Code *et al.* (1976).

(1983), Jacoby *et al.* (1984), Pickles (1985), and Kiehling (1987). Mean energy distributions as a function of spectral type have been compiled by Spinrad and Taylor (1971) and by O'Connell (1973). Some care should be exercised before any of these data are applied to serious research because the adopted calibration of Vega changed with time and with the observer. In most cases, older data should be corrected to the latest calibration of Vega.

Anomalously large emission is not uncommon in the far infrared ($\lambda \gtrsim 5\ \mu m$). Such emission does not originate in the photosphere, but from shells, rings, or disks surrounding the stars (e.g., Russell *et al.* 1975, Auman *et al.* 1984, Beckwith *et al.* 1984, Harper *et al.* 1984, Odenwald 1986, Mariotti *et al.* 1987, Rose 1987).

Continua from photospheric models

The temperature and pressure sensitivity of the continuum can easily be explored using models of the type described in Chapter 9. Examples are shown in Fig. 10.8. Notice how the absolute flux is coupled to the change in slope as the model temperature changes. Except for the Balmer jump, the continua show

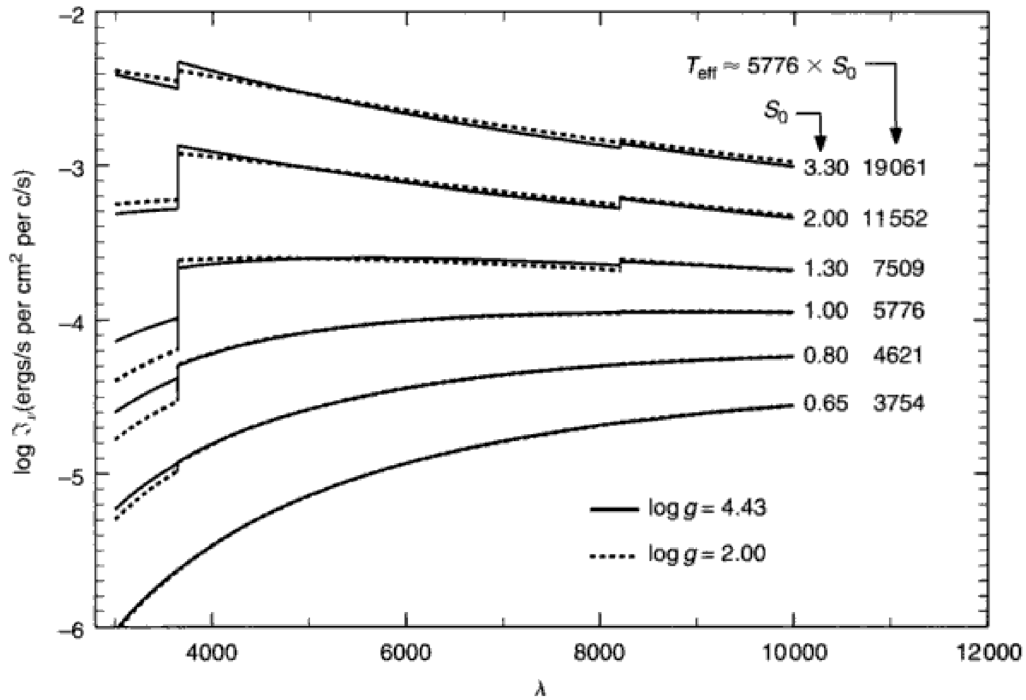


Fig. 10.8. Model-photosphere continua are shown for several scaled-solar models. They are plotted on an absolute flux scale, and so both the slope and the height of the graph change with temperature. The gravity dependence (solid versus dashed lines) is relatively weak.

only weak variation with surface gravity. The strength of the Balmer discontinuity is plotted in Fig. 10.9 as a function of temperature for different gravity values. The rise and fall of the Balmer jump corresponds to the temperature range over which neutral hydrogen dominates the continuous opacity. At lower temperatures, the negative hydrogen ion grows, while the neutral hydrogen diminishes from lack of excitation. At the higher temperatures, the Balmer jump becomes smaller as the ratio of Balmer to Paschen absorption decreases and as the hydrogen absorption fades with increasing ionization of hydrogen.

The observed Balmer jump according to Barbier (1958) is also shown in Fig. 10.9. The excellent agreement should not be taken too seriously since there are basic uncertainties in the models and the numerical size of the discontinuity depends on the details of how it is measured. Specifically, the top of the Balmer jump is obliterated by the Balmer lines, as they reach the series limit, so that extrapolation of the observations is required. Figure 10.10 illustrates this problem in the energy distribution of Sirius. It is clear that the hydrogen lines overlap shortward of H_δ and the true Paschen continuum lies higher than any of these measurements. Rather than work with the Balmer jump as such, it is better to model the Paschen and Balmer continua together.

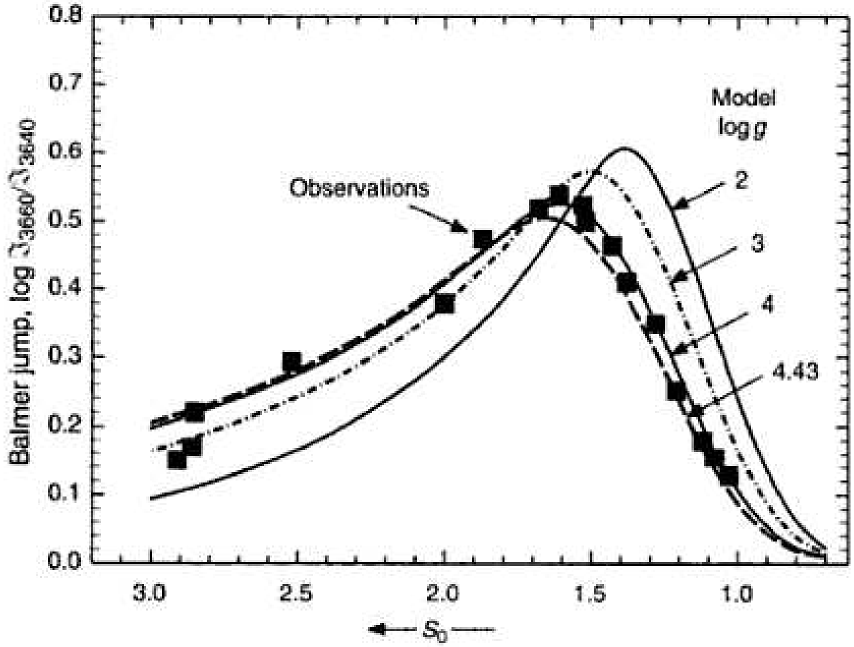


Fig. 10.9. The Balmer jump depends on both temperature (S_0) and surface gravity (g). Solid lines are from scaled-solar models. The squares show the observations tabulated by Barbier (1958).

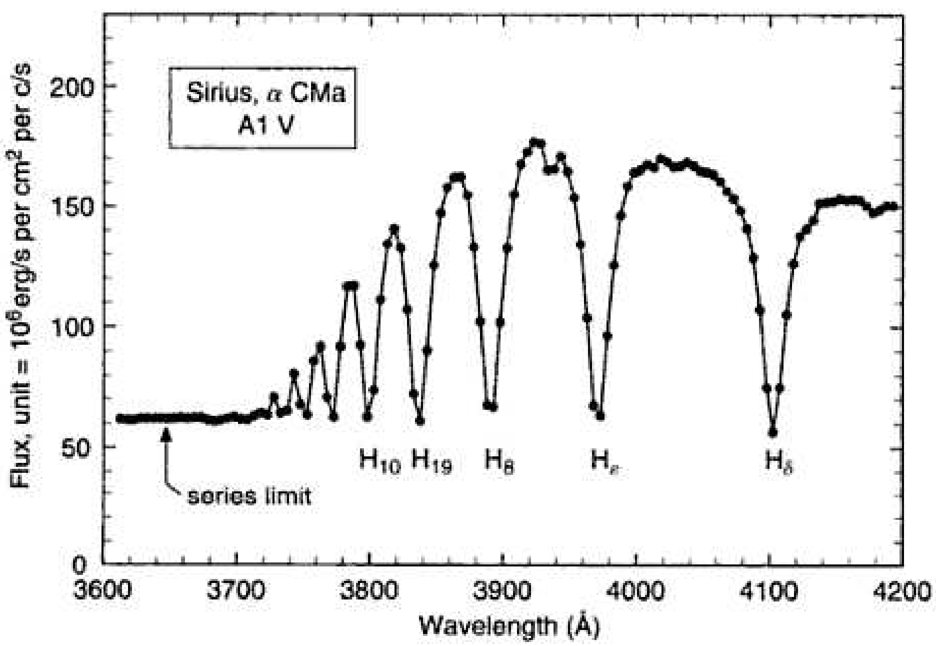


Fig. 10.10. When the Balmer jump is strong, so are the Balmer lines. The lines obliterate the Paschen continuum near the series limit. At 5 Å resolution for the data, and with the relatively high surface gravity of Sirius, the lines disappear well above the series limit. Based on data from Panek (1977).

Theoretical color indices are often computed by integrating the computed spectra multiplied by the bandpass of the photometric system. Examples can be seen in VandenBerg and Bell (1985), Edvardsson and Bell (1989), and Buser and Kurucz (1992). To be successful, careful inclusion of spectral lines in the computations is needed. We will not expand on this topic, but return to the direct comparison of energy distributions and continuous spectra.

Line absorption

Prior to matching continuum models to observed energy distributions, it is necessary to consider the absorption caused by the spectral lines. When ultra-low-resolution measurements are made of a stellar spectrum, the absorption of the lines is, of course, included. Necessarily then the measured values lie below the true continuum. One tack is to include in the computation all the lines within each measured spectral band and integrate over the band. The result of this integration can then be compared directly with the observations. In cases where there are only a few lines, this is easy enough to do, although the amount of calculation is increased somewhat because several points per line profile are then needed. With spectra of cooler stars, where there are thousands of lines, one is faced with a major computational effort. Nevertheless, some success has been achieved in this way (see Figs. 16.8, 16.9, and Bell & Gustafsson 1978, Piskunov *et al.* 1995, Fontenla *et al.* 1999, Kupka *et al.* 2000). Unfortunately the amount of absorption arising in computed spectral lines requires the complete and proper tabulation of those lines actually present in the stellar spectrum, the correct f-values for those lines, the value of the microturbulence dispersion (see Chapter 18), and damping constants for the stronger lines (see Chapter 11). These requirements are difficult to meet.

An alternative approach is to measure the actual line absorption in the stellar spectrum within selected regions using high enough spectral resolution to resolve the lines. Only relative flux measurements, F_ν/F_{cont}, are needed for such line absorption work, so no absolute calibration is involved here. Examples are shown in Fig. 10.11, where the typical line absorption amounts to a few per cent in the orange region of the spectrum of cool stars. The observed value in the energy-distribution measurement for these wavelength bands can then be increased by 1.1234 and 1.0345 times, respectively, before they are compared to a (lineless) model continuum.

The wavelength distribution of lines can be seen in Fig. 10.12, where the spectra of the Sun and Arcturus are displayed. The shape of the continuum has been removed by normalizing the continuum to unity at all wavelengths. We see that the line absorption is dramatically greater toward shorter wave-

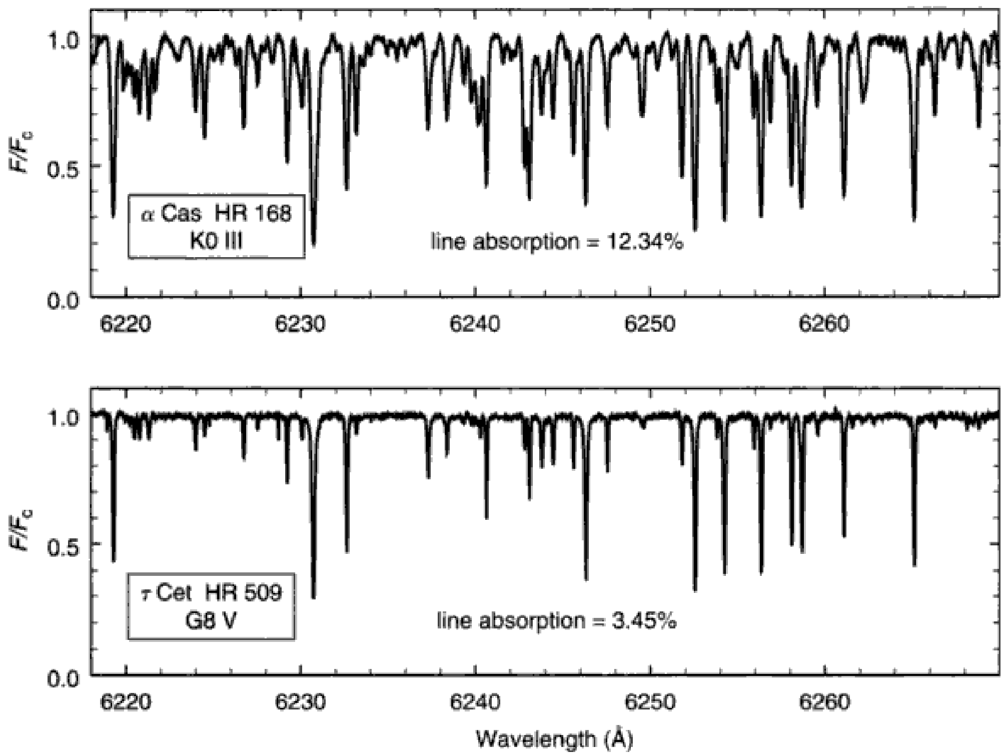


Fig. 10.11. Line absorption is the fraction of light removed from the spectrum. The values shown in the plots above have been determined by direct integration of these spectra. The cooler K giant has considerably more line absorption than the warmer dwarf.

lengths. It is also greater for cooler stars and for metal-rich stars. When the corrections become large, more care must be exercised in their measurement to maintain overall accuracy. Extensive tabulations have been published for β Gem by Ruland *et al.* (1980).

Comparison of model to stellar continua

The energy distribution of a star can be compared to a grid of model continua, like those shown in Fig. 10.8 for example, to select an appropriate model for the star. In the examples shown here, line absorption is handled by empirical measurement as outlined above, but the same principles are used if the lines are explicitly included in the models instead. New models can be calculated, or an interpolation can be performed within a grid, until the observations are matched to within the errors. Such a match is shown in Fig. 10.13. The star is relatively cool and the continuum is insensitive to surface gravity; temperature remains the lone adjustable parameter. For the F star in Fig. 10.14, the Paschen continuum is fitted by temperature adjustment, and the Balmer jump is then matched by

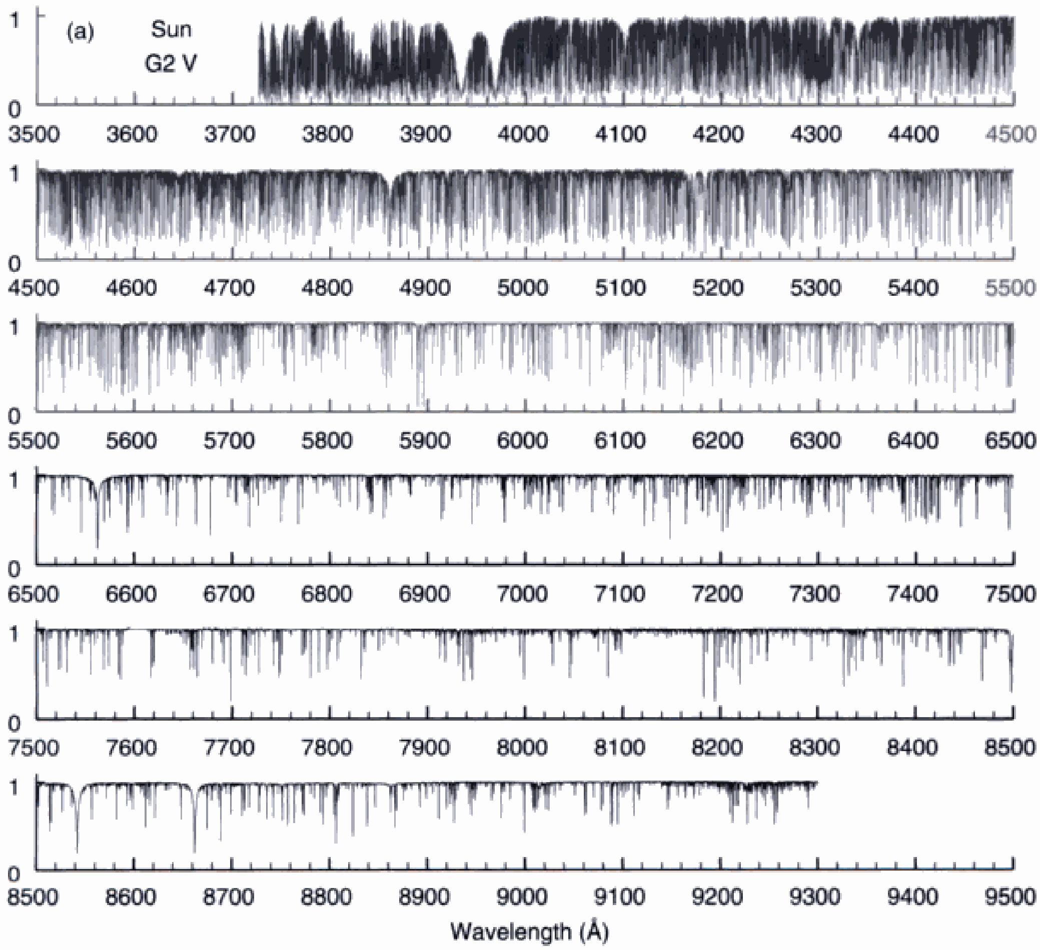


Fig. 10.12. The visible window spectra of (a) the Sun and (b) Arcturus are shown here with the continuum normalized to unity, i.e., the shape of the energy distribution has been removed. The line absorption increases dramatically toward shorter wavelengths. Data from Hinkle *et al.* (2000).

adjustment of the surface gravity used in the model. But, recall (Fig. 9.13) that the modeled Balmer jumps are dependent on $T(\tau_0)$, so some caution should be exercised. In still hotter stars, such as Vega (A0 V) shown in Fig. 10.15, the Balmer jump has become nearly independent of surface gravity again.

In most cases, the stellar continua can be fitted within errors by the models. The fundamental sources of uncertainty are: (1) the accuracy of the absolute flux calibration, (2) the realism of the model, especially the shape of $T(\tau_0)$, and (3) interstellar reddening. The interstellar reddening can be a problem for stars more distant than $\sim$100 pc. If the star is reddened, its energy distribution can no longer be used to select a model (although sometimes reddening corrections are applied to salvage it), and then alternate criteria such as spectral lines and the Balmer jump must be relied on.

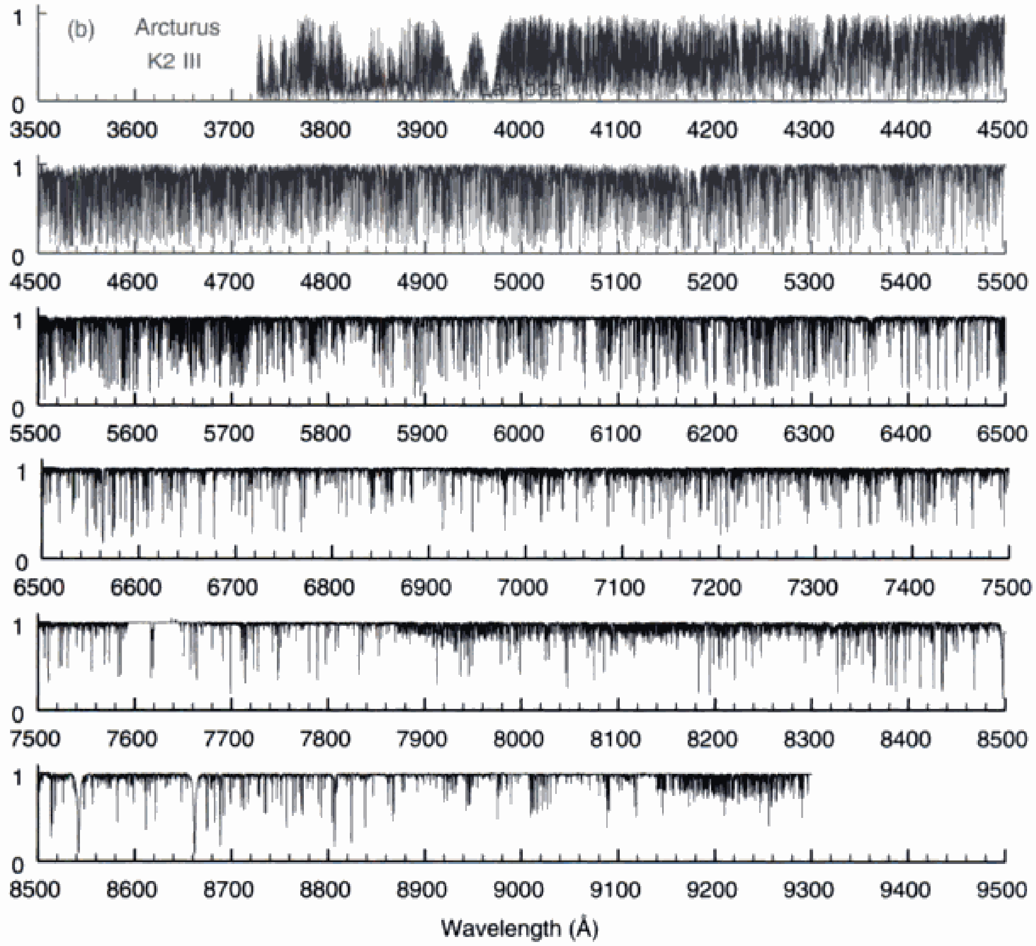


Fig. 10.12. (*cont.*)

Since the line absorption is stronger toward shorter wavelengths, it is sometimes also possible to extract the chemical-composition index called metallicity (see Chapter 16) from the energy distributions. An example can be seen in Morossi *et al.* (2002).

A different kind of comparison between models and stars is that of "synthetic photometry." Theoretical color indices are often computed by integrating the computed spectra multiplied by the bandpass of the photometric system. To be successful, careful inclusion of spectral lines in the computations is needed, as explained above. The computed color indices are then used to infer temperatures, surface gravities, and metallicity from the photometric measurements of stars. Examples of synthetic photometry can be seen in the works of Bell and Gustafsson (1978, 1989), Buser and Kurucz (1985, 1992), VandenBerg and Bell (1985), Buser (1986), Lester *et al.* (1986), and Edvardsson and Bell (1989).

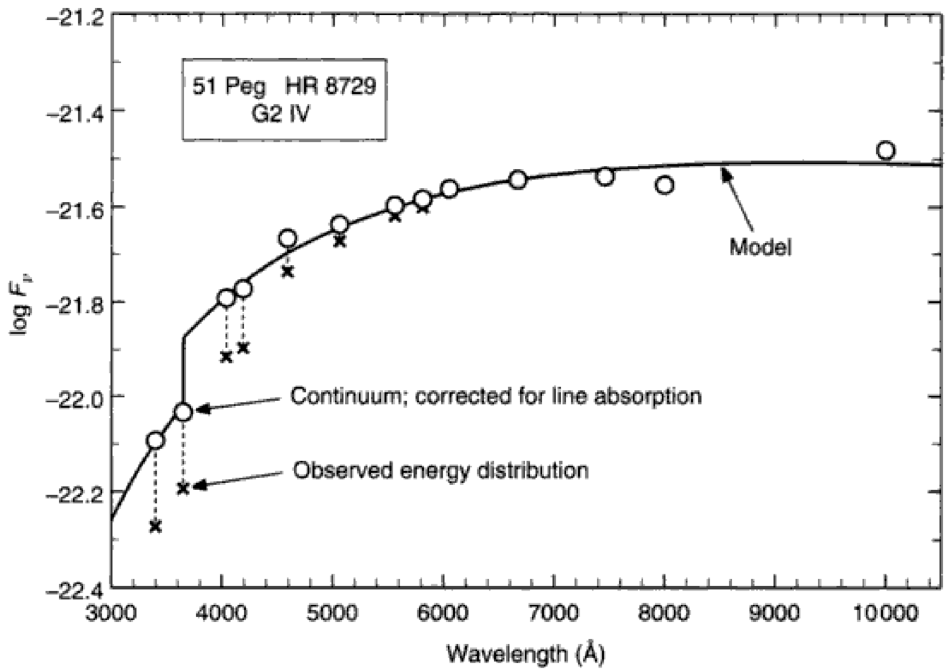


Fig. 10.13. The observed energy distribution of 51 Peg is shown by the × symbols (based on data by Melbourne 1960 and Code 1960). Correction for line absorption gives the continuum (circles), which are then compared to a model ($S_0 = 0.98$, $\log g = 3.0$).

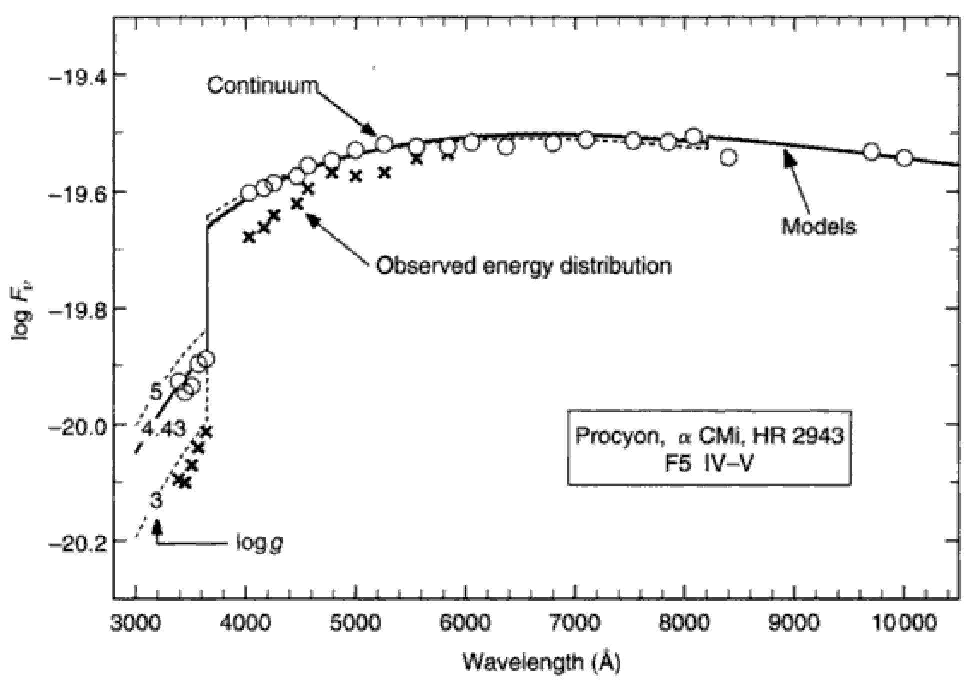


Fig. 10.14. The observed energy distribution (×) of Procyon is corrected for line absorption (○), and compared to models (lines). Based on data by Bessell (1967). The models have $S_0 = 1.18$ and $\log g$ as labeled.

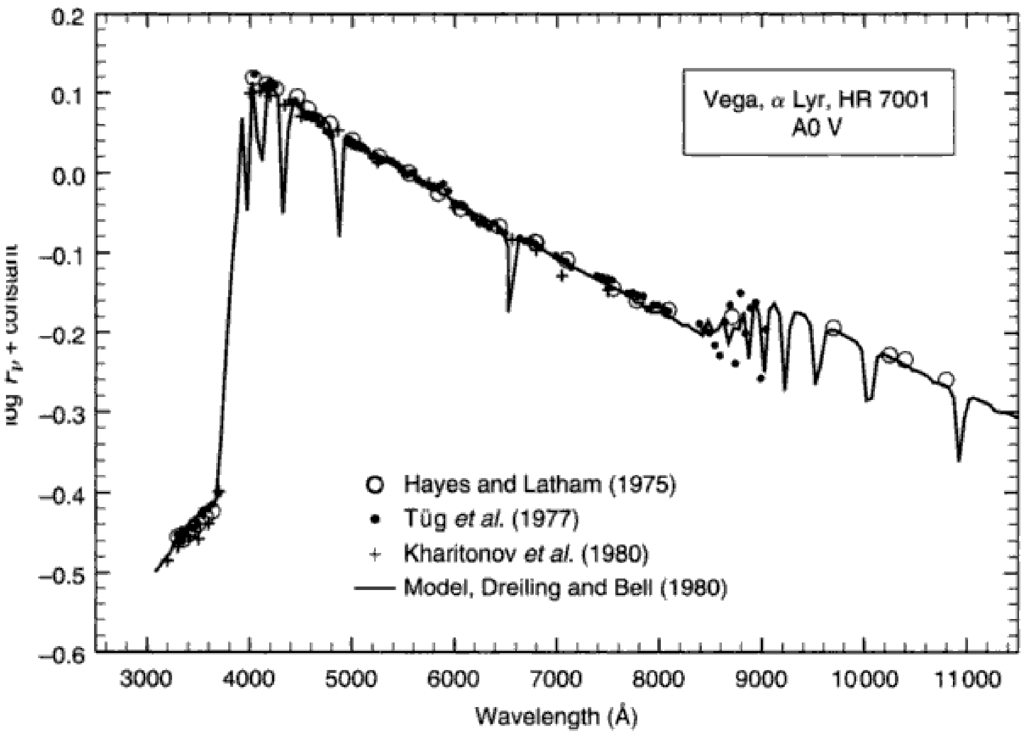


Fig. 10.15. The observations of Vega from Fig. 10.3 are compared here to a model having $T_{\text{eff}} = 9650$ K, $\log g = 4.0$ computed by Dreiling and Bell (1980). See also Castelli and Kurucz (1994).

Luminosity and bolometric flux

Through the visible window we measure part of the flux emitted by the star. All of the flux is called the bolometric flux, while the luminosity includes the surface area,

$$L = 4\pi R^2 \int_0^\infty \mathfrak{F}_\nu \mathrm{d}\nu = 4\pi R^2 \mathfrak{F}_{\text{bol}}, \tag{10.5}$$

where L is the luminosity, R is the stellar radius, $\mathfrak{F}_\nu$ is the frequency-dependent flux, and $\mathfrak{F}_{\text{bol}}$ is the bolometric flux. Bolometric flux is important for studies of the effective temperature scale (Chapter 14) and in linking stellar atmosphere to stellar interior calculations.

Flux outside the visible window can be measured using rockets, satellites, and high-altitude balloons. A large number of ultraviolet observations have been made in this way (Davis & Webb 1970, Bless & Code 1972, Code 1973, Beeckmans *et al.* 1974, Code *et al.* 1976, Malagnini *et al.* 1986). A less direct method is to calculate the flux in the non-visible portions of the spectrum using a model photosphere (e.g., Oke & Conti 1966, Kurucz 1979, Bessell *et al.*

1998, Houdashelt *et al.* 2000). Either way, one obtains the ratio, $a_{\rm bol}$, of the bolometric flux to the flux in some spectral window

$$a_{\rm bol} = \frac{\int_0^\infty F_\nu \, d\nu}{\int_{-\infty}^\infty W(\nu) F_\nu \, d\nu} = \frac{\int_0^\infty \mathfrak{F}_\nu \, d\nu}{\int_{-\infty}^\infty W(\nu) \mathfrak{F}_\nu \, d\nu}, \tag{10.6}$$

where $W(\nu)$ is the spectral window through which we record the more limited data, F_ν is the flux at Earth, and $\mathfrak{F}_\nu$ is the flux at the stellar surface; the distance factor is the same in the numerator and the denominator. If, for example, the bolometric flux is compared to the flux seen in the visual band of the UBV system, then $W(\nu)$ is the response function for V given in Fig. 1.4.

Bolometric corrections are defined in magnitude units by

$$\begin{aligned} BC &= 2.5 \log a_{\rm bol} + \text{constant} \\ &= m_V - m_{\rm bol} \\ &= M_V - M_{\rm bol}. \end{aligned} \tag{10.7}$$

The additive constant arises because the zero point of the magnitude scale is arbitrary and set by adoption. Typically, the minimum in the relation between the bolometric correction and the spectral type or color index was set to zero. This resulted in the solar bolometric correction wandering between ≈ 0.07 and 0.19 depending on whose study was used. In fact, the proper way to use bolometric corrections is to normalize to the Sun, i.e.,

$$-2.5 \log(L/L_\odot) = M_V - M_V^\odot - (BC - BC_\odot)$$

or (10.8)

$$M_{\rm bol} - M_{\rm bol}^\odot = M_V - M_V^\odot - (BC - BC_\odot).$$

As noted above, $m_V^\odot = V_\odot = -26.75$, giving $M_V^\odot = 4.82$. Notice that the difference in bolometric corrections in Eq. (10.8) eliminates the zero point of the scale. Consistency here can be important because of the many different zero points found in the literature.

The large change in bolometric correction with color index and effective temperature is shown in Fig. 10.16. As stated above, the Sun is at $B - V = 0.656 \pm 0.008$ in this plot. For stars hotter than the Sun, the bolometric corrections are independent of the luminosity class except for a very small deviation in $B - V$ for supergiants. Since the sample for stars cooler than the Sun is almost exclusively giants, it is not possible to make any definitive statement about their differences with luminosity class. Bolometric

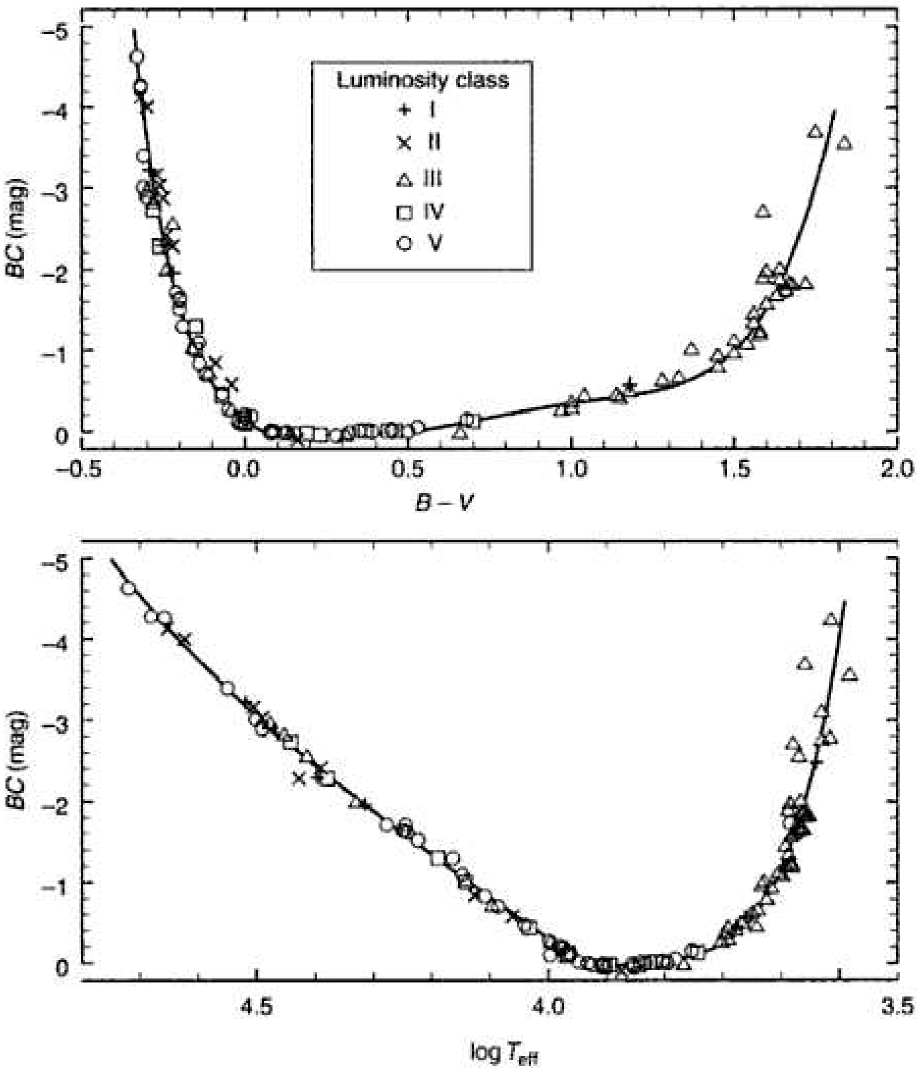


Fig. 10.16. Bolometric corrections are shown as a function of $B-V$ color index in the top panel and as a function of the logarithm of the effective temperature in the lower panel. Based on data summarized by Flower (1996).

corrections relative to the infrared I and K magnitudes are given by Bessel *et al.* (1999). The curves fitted to the data in Fig. 10.16 are given by the following polynomials:

$$\begin{aligned} BC &= -0.14 + 3.1(B-V) - 12.0(B-V)^2 \\ &\quad + 5.5(B-V)^3 - 163(B-V)^4 \qquad \text{for} \quad B-V \leq 0.1 \\ BC &= 0.06 - 0.86(B-V) + 5.44(B-V)^2 - 12.39(B-V)^3 \\ &\quad + 10.410(B-V)^4 - 3.015(B-V)^5 \qquad \text{for} \quad B-V \geq 0.1 \end{aligned} \tag{10.9}$$

and

$$\begin{aligned}
BC &= -888.82 + 884.69\log T - 326.850\log^2 T \\
&\quad + 53.400\log^3 T - 3.2744\log^4 T \qquad \text{for} \quad \log T \geq 4 \\
BC &= -64\,741.46 + 67\,717.00\log T - 26\,566.141\log^2 T \\
&\quad + 4632.926\log^3 T - 303.0307\log^4 T \qquad \text{for} \quad \log T \leq 4
\end{aligned} \tag{10.10}$$

The luminosity of the Sun follows from direct measurements of the solar constant, now often called the (total) *irradiance*: $\ell = 1.367 \times 10^6 \,\mathrm{erg/(s\,cm^2)} = 1367\,\mathrm{W/m^2} \pm 0.2\%$ (based on Neckel & Labs 1973, Willson & Hudson 1981, Newkirk 1983, Foukal & Lean 1986, Fröhlich & Lean 2002),

$$\begin{aligned}
L_\odot &= 4\pi r^2 \ell \\
&= 3.845 \times 10^{33}\,\mathrm{erg/s} \\
&= 3.845 \times 10^{26}\,\mathrm{W},
\end{aligned}$$

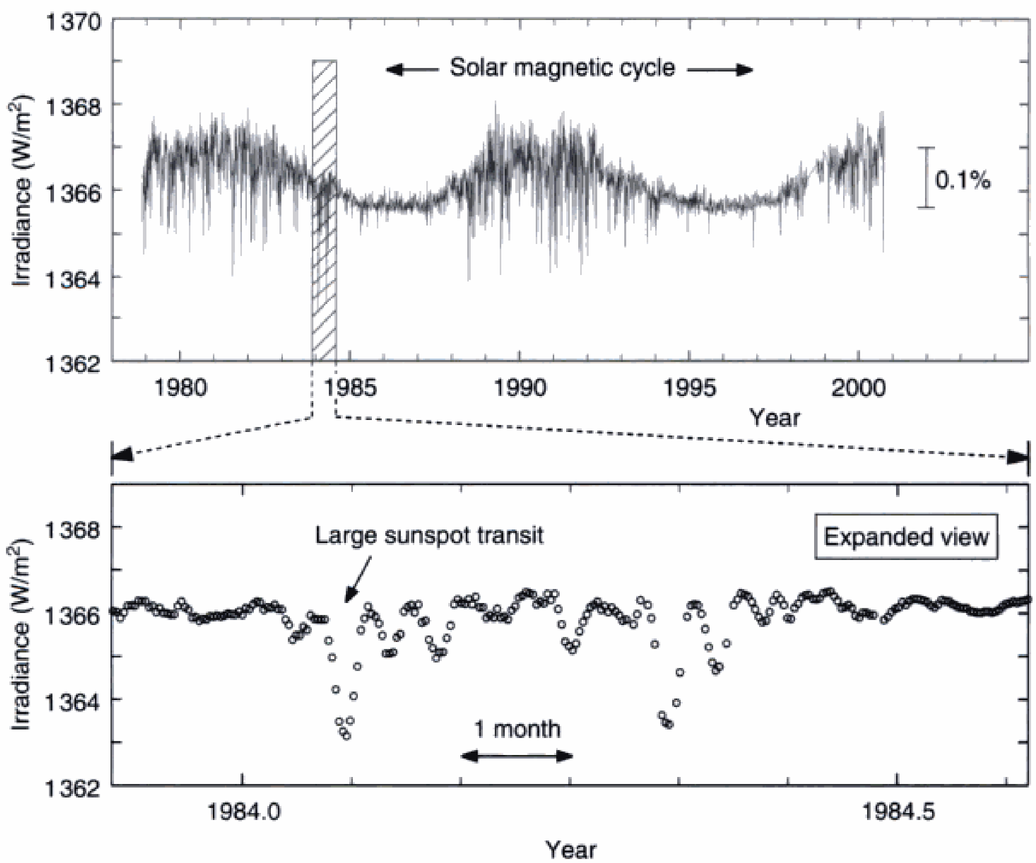


Fig. 10.17. Satellite observations of the solar flux, or irradiance, show a variation ≈0.1% across a magnetic cycle (upper panel). The Sun is brighter when it is more magnetically active. The individual dips on the 2 week time scale (lower panel) correspond to a (dark) sunspot being carried across the disk of the Sun by the Sun's rotation. Data: refer to Fröhlich and Lean (1998).

with $r = 1$ a.u. $= 1.495\,979 \times 10^{13}$ cm (10^{11} m). Using the solar radius of 6.9598×10^{10} cm (10^{8} m), this gives an effective temperature of $T_{\rm eff}^{\odot} = 5777$ K.

The solar power reaching Earth is known to have varied over recent magnetic cycles, as shown in Fig. 10.17. The variation is much smaller than the errors discussed above, but is of great interest in the context of global warming and solar physics.

References

Ardeberg, A. & B. Virdefors 1980. *Ast. Ap. Suppl.* **40**, 307.
Auman, H.H., F.C. Gillett, C.A. Beichman, *et al.* 1984. *Ap. J. Lett.* **278**, L23.
Baldwin, J.A. & R.P.S. Stone 1984. *Mon. Not. Roy. Ast. Soc.* **206**, 241.
Barbier, D. 1958. *Handb. d. Phys.*, Vol. 50. (Heidelberg: Springer) p. 274.
Beckwith, S., B. Zuckerman, M.F. Skrutskie, & H.M. Dyck 1984. *Ap. J.* **287**, 793.
Beeckmans, F., D. Macau, & D. Malaise 1974. *Ast. Ap.* **33**, 93.
Bell, R.A. & B. Gustafsson 1978. *Ast. Ap. Suppl.* **34**, 229.
1989. *Mon. Not. Roy. Ast. Soc.* **236**, 653.
Bessell, M.S. 1967. *Ap. J. Lett.* **149**, L67.
Bessell, M.S., F. Castelli, & B. Plez 1998. *Ast. Ap.* **333**, 231.
Bless, R.C. & A.D. Code 1972. *Ann. Rev. Ast. Ap.* **10**, 197.
Bless, R.C., A.D. Code, & D.J. Schroeder 1968. *Ap. J.* **153**, 545.
Breger, M. 1976. *Ap. J. Suppl.* **32**, 7.
Burlov-Vasiljev, K.A., E.A. Gurtovenko, & Yu. B. Matvejev 1995. *Solar Phys.* **157**, 51.
Buser, R. 1986. *Highlights of Astronomy*, Vol. 7, ed. J. P. Swings. (Dordrecht: Reidel) p. 799.
Buser, R. & R.L. Kurucz 1985. *Calibration of Fundamental Stellar Quantities*, IAU Symposium 111, ed. D.S. Hayes, L.E. Pasinetti, & A. G. D. Philip. (Dordrecht: Reidel), p. 513.
1992. *Ast. Ap.* **264**, 557.
Castelli, F. & R.L. Kurucz 1994. *Ast. Ap.* **281**, 817.
Cochran, A.L. 1980. *Spectrophotometry with a Self-Scanned Silicon Diode Array. III. A Catalogue of Energy Distributions*. (Austin: Univ. Texas), Pub. No. 16.
1981. *Ap. J. Suppl.* **45**, 83.
Cochran, A.L. & T.G. Barnes, III 1981. *Ap. J. Suppl.* **45**, 73.
Code, A.D. 1960. *Stellar Atmospheres*, ed. J. L. Greenstein (Chicago: Univ. Chicago), Chapter 2.
1973. *Problems of Calibration of Absolute Magnitudes and Temperatures of Stars*, IAU Symp. 54, ed. B. Hauck & B.E. Westerlund. (Dordrecht: Reidel), p. 131.
1985. *Calibration of Fundamental Stellar Quantities*, IAU Symposium 111, ed. D.S. Hayes, L.E. Pasinetti, & A.G.D. Philip. (Dordrecht: Reidel), p. 209.
Code, A.D., J. Davis, R.C. Bless, & R. Hanbury Brown 1976. *Ap. J.* **203**, 417.
Colina, L., R.C. Bohlin, & F. Castelli 1996. *Ast. J.* **112**, 307.
Davis, J. & R.J. Webb 1970. *Ap. J.* **159**, 551.
Dreiling, L.A. & R.A. Bell 1980. *Ap. J.* **241**, 736.
Edvardsson, B. & R.A. Bell 1989. *Mon. Not. Roy. Ast. Soc.* **238**, 1121.
Flower, P.J. 1996. *Ap. J.* **469**, 355.
Fontenla, J., O.R. White, P.A. Fox, E.H. Avrett, & R.L. Kurucz 1999. *Ap. J.* **518**, 480.

Foukal, P. & J. Lean 1986. *Ap. J.* **302**, 826.
Fröhlich, C. & J. Lean 1998. *Geophys. Research Lett.* **25**, 4377.
2002. *Astr. Nachr.* **323**, 203.
Glushneva, I. N. & S. L. Ovchinnikov 1982. *Soviet Ast.* **26**, 548.
Glushneva, I. N., A. V. Kharitonov, L. N. Knyazeva, & V. I. Shenavrin 1992. *Ast. Ap. Suppl.* **92**, 1.
Gray, D. F. 1992, *Pub. Ast. Soc. Pacific* **104**, 1035.
Gray, R. O. 1998. *Ast. J.* **116**, 482.
Gunn, J. E. & L. L. Stryker 1983. *Ap. J. Suppl.* **52**, 121.
Gutierrez-Moreno, A., H. Moreno, G. Cortes, & E. Wenderoth 1988, *Pub. Ast. Soc. Pacific* **100**, 973.
Hamuy, M., A. R. Walker, N. B. Suntzeff, P. Gigoux, S. R. Heathcote, & M. M Phillips 1992. *Pub. Ast. Soc. Pacific* **104**, 533.
Hardie, R. H. 1962. *Astronomical Techniques*, ed. W. A. Hiltner. (Chicago: Univ. Chicago), p. 178.
Hardorp, J. 1978. *Ast. Ap.* **63**, 383.
Harper, D. A., R. F. Loewenstein, & J. A. Davidson 1984. *Ap. J.* **285**, 808.
Hayes, D. S. 1985. *Calibration of Fundamental Stellar Quantities*, IAU Symposium 111, ed. D. S. Hayes, L. E. Pasinetti, & A. G. D. Philip. (Dordrecht: Reidel), p. 225.
Hayes, D. S. & D. W. Latham 1975. *Ap. J.* **197**, 593.
Hayes, D. S., D. W. Latham, & S. H. Hayes 1975. *Ap. J.* **197**, 587.
Hayes, D. S. & A. G. D. Philip 1983. *Ap. J. Suppl.* **53**, 759.
Hinkle, K., Ll. Wallace, J. Valenti, & D. Harmer 2000. *Visible and Near Infrared Atlas of the Arcturus Spectrum, 3727–9300Å*. (San Francisco: Ast. Soc. Pacific)
Houdashelt, M. L., R. A. Bell, & A. V. Sweigart 2000. *Ast. J.* **119**, 1448.
Ipatov, A. P. 1982. *Soviet Ast.* **26**, 368.
Jacoby, G. H., D. A. Hunter, & C. A. Christian 1984. *Ap. J. Suppl.* **56**, 257.
Kharitonov, A. V. & I. N. Glushneva 1978. *Sov. Ast.* **22**, 284.
Kharitonov, A. V., V. M. Tereshchenko, L. N. Knyazeva, & P. N. Boiko 1980. *Sov. Ast.* **24**, 417.
Kiehling, R. 1987. *Ast. Ap. Suppl.* **69**, 465.
Knyazeva, L. N. & A. V. Kharitonov 1990. *Sov. Ast.* **34**, 626.
Kupka, F. G., T. A. Ryabchikova, N. E. Piskunov, H. C. Stempels, & W. W. Weiss 2000. *Baltic Ast.* **9**, 590.
Kurucz, R. L. 1979. *Ap. J. Suppl.* **40**, 1.
Lester, J. B., R. O. Gray, & R. L. Kurucz 1986. *Ap. J. Suppl.* **61**, 509.
Lockwood, G. W., H. Tüg, & N. M. White 1992. *Ap. J.* **390**, 668.
Malagnini, M. L., C. Morossi, L. Rossi, & R. L. Kurucz 1986. *Ast. Ap.* **162**, 140.
Mariotti, J.-M., C. Perrier, & F. Lacombe 1987. *Ast. Ap.* **182**, L11.
Massey, P., K. Strobel, J. V. Barnes, & E. Anderson 1988. *Ap. J.* **328**, 315.
Mégessier, C. 1995. *Ast. Ap.* **296**, 771.
Melbourne, W. G. 1960. *Ap. J.* **132**, 101.
Merezhin, V. P. 1990. *Ap. Space Sci.* **174**, 181.
Morossi, C., P. Di Marcantonio, M. Franchini, M. L. Malagnini, & M. Chavez, 2002. *Ap. J.* **577**, 377.
Neckel, H. & D. Labs 1973. In *Problems of Calibration of Absolute Magnitudes and Temperatures of Stars*, I.A.U. Symp. No. 54, ed. B. Hauck & B. E. Westerlund. (Dordrecht: Reidel), p. 149.
1984. *Solar Phys.* **90**, 205.
Newkirk, G., Jr. 1983. *Ann. Rev. Ast. Ap.* **21**, 429.
O'Connell, R. W. 1973. *Ast. J.* **78**, 1074.

Odenwald, S.F. 1986. *Ap. J.* **307**, 711.
Oke, J.B. 1964. *Ap. J.* **140**, 689.
1990. *Ast. J.* **99**, 1621.
Oke, J.B. & P.S. Conti 1966. *Ap. J.* **143**, 134.
Oke, J.B. & J.E. Gunn 1983. *Ap. J.* **266**, 713.
Oudmaijer, R.D., J. Palacios, C. Eiroa, *et al.* 2001. *Ast. Ap.* **379**, 564.
Panek, R.J. 1977. *Ap. J.* **217**, 749.
Philip, A.G.D. & D.S. Hayes 1983. *Ap. J. Suppl.* **53**, 751.
Pickles, A.J. 1985. *Ap. J. Suppl.* **59**, 33.
1998. *Ast. Ap.* **92**, 1.
Piskunov, N.E., F. Kupka, T.A. Ryabchikova, W.W. Weiss, & C.S. Jeffery 1995. *Ast. Ap. Suppl.* **112**, 525.
Rose, W.K. 1987. *Ap. J.* **312**, 284.
Ruland, F., R. Griffin, R.F. Griffin, D. Biehl, & H. Holweger 1980. *Ast. Ap. Suppl.* **42**, 391.
Russell, R.W., B.T. Soifer, & W.J. Forrest 1975. *Ap. J. Lett.* **198**, L41.
Spinrad, H. & B.J. Taylor 1971. *Ap. J. Suppl.* **22**, 445.
Stone, R.P.S. 1977. *Ap. J.* **218**, 767.
Stone, R.P.S. & J.A. Baldwin 1983. *Mon. Not. Roy. Ast. Soc.* **204**, 347.
Taylor, B.J. 1984. *Ap. J. Suppl.* **54**, 259.
Tereshchenko, V.M. 2002. *Ast. Zh.* **79**, 249.
Terez, G.A. & E.I. Terez 1979. *Soviet Ast.* **23**, 449.
Tüg, H. 1980a. *Ast. Ap.* **82**, 195.
1980b. *Ast. Ap. Suppl.* **39**, 67.
Tüg, H., N.M. White, & G.W. Lockwood 1977. *Ast. Ap.* **61**, 679.
VandenBerg, D.A. & R.A. Bell 1985. *Ap. J. Suppl.* **58**, 561.
Voloshina, I.B., I.N. Glushneva, V.T. Doroshenko, L.V. Mossakovskaya, S.L. Ovchinnikov, & T.S. Khruzina 1983. *Tr. Gos. Astron. Inst. Shternberg*, Tom **53**, p. 50 (in Russian).
Willson, R.C. & H.S. Hudson 1981. *Ap. J. Lett.* **244**, L185.
Willstrop, R.V. 1965. *Mem. Royal Ast. Soc.* **69**, 83.

Questions and exercises

1. If one is making extinction corrections, as in Fig. 10.2, why not just use Eq. (10.2) directly and extrapolate to zero air mass instead of Eq. (10.3) as recommended?

2. The people who perform absolute calibrations of energy distributions have several difficult problems to overcome. Think about what needs to be done to: (i) get the shape of the energy distribution correct and (ii) get the absolute flux at the reference 6656 Å position. List all the factors you can think of and discuss why (ii) is more difficult than (i).

3. In this exercise, we learn about the formation of the continuum. Run your model photosphere program to generate two or more models. For example, choose a hot model with an effective temperature of $\sim$15 000 K, a middle temperature model $\sim$7500 K, and a cool model $\sim$5000 K, all with $\log g \sim 4.0$ or 4.5. For these models, compute the flux at 15–20 wavelengths across the visible window. Plot these values against wavelength to delineate the continua. What differences do you see among these spectra? Compare your results with Figs. 10.6 and 10.8. Why do these differ-

ences occur? Would you say that the slope of the spectra is a good temperature indicator? Can you visualize the connection between a continuum slope and the $B-V$ color index? Plot and compare the contribution functions, i.e., the integrand of Eq. (9.13), at 3640, 3660, 5000, and 7000 Å. Contrast your results with Figs. 9.9 and 9.10. What do we learn from these contribution functions?

4. Compare the solar energy distribution shown in Fig. 10.5 with a black-body spectrum having the solar effective temperature of 5777 K. Make of copy of Fig. 10.5 and plot your calculated black-body values on it. Where do the greatest deviations occur? Do the same for α Leo, shown in Fig. 10.7. You can estimate the effective temperature of α Leo from its color index or its spectral type through Fig. 14.6, Eq. (14.16), or the tables in Appendix B.

11

The line absorption coefficient

The strengths and shapes of spectral lines contain a great deal of information about the stars, and the line absorption coefficient plays a fundamental role here. The situation in this regard is similar to the effect the continuous absorption coefficient has on the shape of the continuum. Lines are more interesting, however, because several different physical effects can enter the structuring of the final absorption coefficient. Each of these has its own variation with wavelength across the line, that is, its own absorption coefficient. The main processes we consider in this chapter are: (1) natural atomic absorption, (2) pressure broadening, of which there are several, and (3) thermal Doppler broadening. The final combined absorption coefficient is the multiple convolution of these individual absorption coefficients. One of the remarkable results of these studies is that the natural atomic absorption and all the significant pressure broadenings have the same wavelength dependence in their individual absorption coefficients (with the notable exception of the hydrogen lines), namely the dispersion profile. This leads to a tremendous simplification, as we shall see, since the convolution of dispersion profiles is a dispersion profile with a half-width equal to the sum of the individual half-widths of the contributing processes. The thermal broadening, on the other hand, reflects the Maxwellian velocity distribution of the absorbing atoms and ions via the Doppler effect, so its wavelength shape is a Gaussian. The final absorption coefficient is therefore a convolution of the dispersion profile with the Gaussian profile.

The atomic absorption coefficient with units of area per absorber we call α (the same notation as in Chapter 8). Here, as in Chapter 8, α itself is not a distribution, but $\alpha\, d\nu$ is, and refers to the amount of power abstracted from a unit I_ν beam. Most physical processes deal with energy, thereby making frequency units appropriate. The linear Stark effect in hydrogen is an important exception, and there the equations are expressed in wavelength units with explicit conversion by λ^2/c when necessary. In order to avoid writing many of the equations twice, the traditional cgs formulation will generally be given, but in a notation easily translated to mks units by dividing by $4\pi\varepsilon_0$, where ε_0 is the permittivity of free space and has a value of $8.842\times 10^{-12}\,\mathrm{C}^2/(\mathrm{N\,m}^2)$ or $1/(4\pi\varepsilon_0) = 9.000\times 10^9\,\mathrm{N\,m}^2/\mathrm{C}^2$.

For discussions of helium lines, which we do not consider here, see Griem (1968), Barnard *et al.* (1974), Mihalas *et al.* (1974), Milosavljević and Djeniže (2001), or Pérez *et al.* (2003).

The natural atomic absorption

The simplest classical model for the interaction of light with atoms is that of a plane electromagnetic wave interacting with dipoles. This results in absorption or attenuation of the wave. Imagine an incident electromagnetic wave, E, as it interacts with an ensemble of dipole-charge oscillators representing the absorbing atoms in the gas. Choose the coordinates so E is traveling in the x direction. E must satisfy the wave equation

$$\frac{\partial^2 E}{\partial t^2} = v^2 \frac{\partial E}{\partial x^2},$$

where v is the velocity. Any function of the form $E = f(x - vt)$ is a solution.

We can allow E to remain general by considering it to be a Fourier composition of sinusoidals. Because the electric field can be linearly decomposed, the manipulations performed on any one component can be performed on the others as well – the final answer being the sum of the manipulated components. This one Fourier component we write as a sinusoidal

$$E = E_0 e^{-i2\pi(x/\lambda - t/P)} = E_0 e^{-2\pi i(x - vt)/\lambda} = E_0 e^{-i\omega(x/v - t)}, \qquad (11.1)$$

where P is the period of the oscillation or the reciprocal of the frequency, ν, and $\omega = 2\pi\nu$ is the angular frequency.

For light, Maxwell's equations tell us (e.g., Stone 1963) that the wave velocity is related to the electrical and magnetic properties of the medium through which it is passing,

$$v = c\left(\frac{\varepsilon_0\mu_0}{\varepsilon\mu}\right)^{1/2}.$$

Here ε is the permittivity, μ is the magnetic permeability, ε_0 and μ_0 are the same for free space. Further, since we are dealing with gases and they have negligible magnetic permeability, $\mu = \mu_0$. We can evaluate $\varepsilon/\varepsilon_0$ by going back to the basic reason why ε differs from ε_0, which is the charge separation of the dipoles induced by the imposed field, E. The total electrical field is the sum of E and the field of the separated charges. The ratio of this total field to the field of the wave is

$$\frac{\varepsilon}{\varepsilon_0} = \frac{E + 4\pi Nqz}{E} = 1 + \frac{4\pi Nqz}{E}. \tag{11.2}$$

The number of dipoles per unit volume is N, the size of the dipole charge is q, and their induced separation is z. In most stellar spectra, where we deal with electronic transitions, replace q with $e = 4.803 \times 10^{-10}$ e.s.u. (in mks units, divide the last term by $4\pi\varepsilon_0$ and use $e = 1.602 \times 10^{-19}$ C).

The displacement z can be obtained by considering a single oscillator at $x = 0$ as it is forced into oscillation by the imposed field, E. Adding the acceleration, damping, and restoring force terms in that order, we obtain the usual harmonic oscillator equation with the driving term on the right,

$$\frac{\mathrm{d}^2 z}{\mathrm{d}t^2} + \gamma\frac{\mathrm{d}z}{\mathrm{d}t} + \omega_0^2 z = \frac{e}{m}E_0 \mathrm{e}^{\mathrm{i}\omega t}, \tag{11.3}$$

where $m = 9.1094 \times 10^{-28}$ g is the mass of the electron. The damping constant, γ, is of some importance, and is discussed in detail below. The driving wave on the right-hand side of the equation is the same E as in Eq. (11.1), but with $x = 0$. The solution is $z = z_0 \mathrm{e}^{\mathrm{i}\omega t}$. Then $\mathrm{d}z/\mathrm{d}t = \mathrm{i}\omega z$ and $\mathrm{d}^2 z/\mathrm{d}t^2 = -\omega^2 z$, all of which, when placed back into Eq. (11.3), gives

$$-\omega^2 z + \mathrm{i}\gamma\omega z + \omega_0^2 z = \frac{e}{m}E_0 \mathrm{e}^{\mathrm{i}\omega t}$$

or

$$z = \frac{e}{m}\frac{E_0 \mathrm{e}^{\mathrm{i}\omega t}}{\omega_0^2 - \omega^2 + \mathrm{i}\gamma\omega} = \frac{e}{m}\frac{E}{\omega_0^2 - \omega^2 + \mathrm{i}\gamma\omega}.$$

The amplitude z reaches its largest value at $\omega = \omega_0$, and so ω_0 is called the resonant frequency. In our model, this is the position of the spectral line. For Eq. (11.2), where we need z/E, we obtain

$$\frac{\varepsilon}{\varepsilon_0} = 1 + \frac{4\pi Ne^2}{m}\frac{1}{\omega_0^2 - \omega^2 + \mathrm{i}\gamma\omega}. \tag{11.4}$$

If we further realize that in a gas $\varepsilon \approx \varepsilon_0$, we can write the wave velocity as

$$\frac{c}{v} = \left(\frac{\varepsilon}{\varepsilon_0}\right)^{1/2} \approx 1 + \frac{1}{2}\frac{4\pi Ne^2}{m}\frac{1}{\omega_0^2 - \omega^2 + \mathrm{i}\gamma\omega}$$

$$= 1 + \frac{2\pi Ne^2}{m}\left[\frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2} - \mathrm{i}\frac{\gamma\omega}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2}\right].$$

For convenience, write this as $c/v = n - \mathrm{i}k$, a complex refractive index. When $\mathrm{i}k$ is combined with the $\mathrm{i}\omega x/v$ term in Eq. (11.1) a real term is produced, corresponding to exponential extinction, $\mathrm{e}^{-k\omega x/c}$. Since the intensity is proportional to EE^*, where the asterisk denotes a complex conjugate, the light extinction can be expressed as

$$I = I_0 \mathrm{e}^{-2k\omega x/c} = I_0 \mathrm{e}^{-\ell_\nu \rho x},$$

where standard notation has been used in the last step; the mass absorption coefficient for the line is ℓ_ν. Upon introducing the full expression for k, we have

$$\ell_\nu \rho = \frac{4\pi N e^2}{mc} \frac{\gamma\omega^2}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2}. \tag{11.5}$$

Since this function peaks sharply, giving non-zero values only when $\omega \approx \omega_0$ we write

$$\omega_0^2 - \omega^2 = (\omega_0 - \omega)(\omega_0 + \omega) \approx (\omega_0 - \omega)2\omega = 2\omega\Delta\omega.$$

In this way, the basic form of the absorption coefficient becomes

$$\ell_\nu \rho = N \frac{\pi e^2}{mc} \frac{\gamma}{\Delta\omega^2 + (\gamma/2)^2}. \tag{11.6}$$

We recognize this as a dispersion profile (refer to Eq. 2.8); also called a damping profile, a Lorentzian profile, a Cauchy curve, and, by some, the Witch of Agnesi.

If we now consider the absorption coefficient per atom, α, we write for Eq. (11.6) $\ell_\nu \rho = N\alpha$ and then

$$\begin{aligned} \alpha &= \frac{2\pi e^2}{mc} \frac{\gamma/2}{\Delta\omega^2 + (\gamma/2)^2} \\ &= \frac{e^2}{mc} \frac{\gamma/4\pi}{\Delta\nu^2 + (\gamma/4\pi)^2} \\ &= \frac{e^2}{mc} \frac{\lambda^2}{c} \frac{\gamma\lambda^2/4\pi c}{\Delta\lambda^2 + (\gamma\lambda^2/4\pi c)^2}. \end{aligned} \tag{11.7}$$

The distance from the center of the spectral lines is $\Delta\omega$, $\Delta\nu$, or $\Delta\lambda$ according to the scale selected. (Divide by $4\pi\varepsilon_0$ mks units.) Equation (11.7) is the basic result; the shape of the absorption coefficient for natural atomic broadening is a dispersion profile with its width determined by a damping constant, γ. This damping constant is called the "natural" broadening, also known as "radiation damping," and we shall evaluate it shortly. But first we consider a quantum mechanical factor of some importance.

Direct integration of the dispersion profile in Eq. (11.7), or a comparison with the unit area form in Eq. (2.8), shows that

$$\int_0^\infty \alpha \, d\nu = \int_{-\infty}^{\infty} \alpha \, d\Delta\nu = \frac{\pi e^2}{mc}. \tag{11.8}$$

This is the energy per second per atom per square radian absorbed by the total line from the unit I_ν beam. The total energy taken from an I_λ beam is

$$\int_0^\infty \alpha \, d\lambda = \int_{-\infty}^{\infty} \alpha \, d\Delta\lambda = \frac{\pi e^2}{mc} \frac{\lambda^2}{c}. \tag{11.9}$$

Actual measurements of this total absorption are usually smaller than indicated by Eq. (11.8) or (11.9), and by an amount that differs from one spectral line to another. A quantum mechanical treatment leads one to introduce a quantity called the oscillator strength, f, such that

$$\int_0^\infty \alpha d\nu = \frac{\pi e^2}{mc} f. \tag{11.10}$$

The oscillator strength, also called the f value, is different for each atomic level and is related to the atomic transition probability B_{lu} (defined near the end of Chapter 5) as follows. The integral in Eq. (11.10) must be equivalent to the probability of absorption times the unit energy for the transition, i.e.,

$$\int_0^\infty \alpha d\nu = B_{lu} h\nu$$

which means that

$$f = \frac{mc}{\pi e^2} B_{lu} h\nu = 7.484 \times 10^{-7} \frac{B_{lu}}{\lambda} \tag{11.11}$$

for λ in angstroms. Using Eq. (6.8), we also see that

$$f = \frac{mc^3}{2\pi e^2 \nu^2} \frac{g_u}{g_l} A_{ul} = 1.884 \times 10^{-15} \lambda^2 \frac{g_u}{g_l} A_{lu}, \tag{11.12}$$

where g_u and g_l are the statistical weights of the upper and lower levels, respectively, and λ is again in angstroms. Recall in passing that the Einstein probability coefficients defined in Chapter 5 are per unit solid angle and not over the full 4π square radians of a unit sphere, as is sometimes done.

It is also possible to define an f value for emission. The two f values are related by $g_u f_{em} = g_l f_{abs}$. Tabulations of measured f values give gf so that no ambiguity arises. Most f values are determined from empirical laboratory measurements, although in less complicated cases calculations can be done. For example, the expression for hydrogen is

$$f=\frac{2^5}{3^{3/2}\pi}\frac{g_{\mathrm{bb}}}{l^5u^3}\left(\frac{1}{l^2}-\frac{1}{u^2}\right)^{-3},$$

where u and l are the quantum numbers of the upper and lower levels as usual, and g_{bb} is the bound–bound Gaunt factor tabulated by Baker and Menzel (1938). The g_{bb} values for the first eight Balmer lines are given by $g_{\mathrm{bb}}=0.869-3.00/u^3$ to better than 0.35%. Numerically $f=0.6407$ for H_α, 0.1193 for H_β, 0.0447 for H_γ, and so on (see Table 11.4 later in this chapter).

Damping constant for natural broadening

The damping constant appearing in Eq. (11.7) can be calculated from the classical dipole emission theory by taking a time average of the acceleration (Menzel 1961). An equation of the form

$$\frac{\mathrm{d}W}{\mathrm{d}t}=-\frac{2}{3}\frac{e^2\omega^2}{mc^3}W=-\gamma W \tag{11.13}$$

results. The solution is $W=W_0e^{-\gamma t}$, where $\gamma=2e^2\omega^2/(3mc^3)=0.22/\lambda^2$ for λ in centimeters for both systems of units. The half-half width of α is $\gamma/2$, $\gamma/4\pi$, or $\gamma\lambda^2/(4\pi c)$, depending on which of the forms in Eq. (11.7) is used. In the $\Delta\lambda$ form, the half-half width is 0.59×10^{-4}Å for all lines. This classical damping constant, also called radiation damping, is usually smaller by an order of magnitude compared to the observations, and again a quantum-mechanical interpretation must be made to reconcile the difference.

A simple phenomenological way to introduce the quantum aspects is to view the energy W in Eq. (11.13) as quantized and write $W=N_uh\nu$, where N_u is the population of the upper level for the transition. Then Eq. (11.13) can be rewritten as $\mathrm{d}N_u/\mathrm{d}t=-\gamma N_u$. But we already know from Chapters 5 and 6 that $\mathrm{d}N_u/\mathrm{d}t=-4\pi A_{ul}N_u$ for each transition starting from the level u, so the total downward rate due to spontaneous emission is

$$\frac{\mathrm{d}N_u}{\mathrm{d}t}=\sum_{l<u}\frac{\mathrm{d}N_{ul}}{\mathrm{d}t}=-4\pi N_u\sum_{l<u}A_{ul}. \tag{11.14}$$

Comparing these equations, we see that

$$\gamma_u=4\pi\sum_{l<u}A_{ul}. \tag{11.15}$$

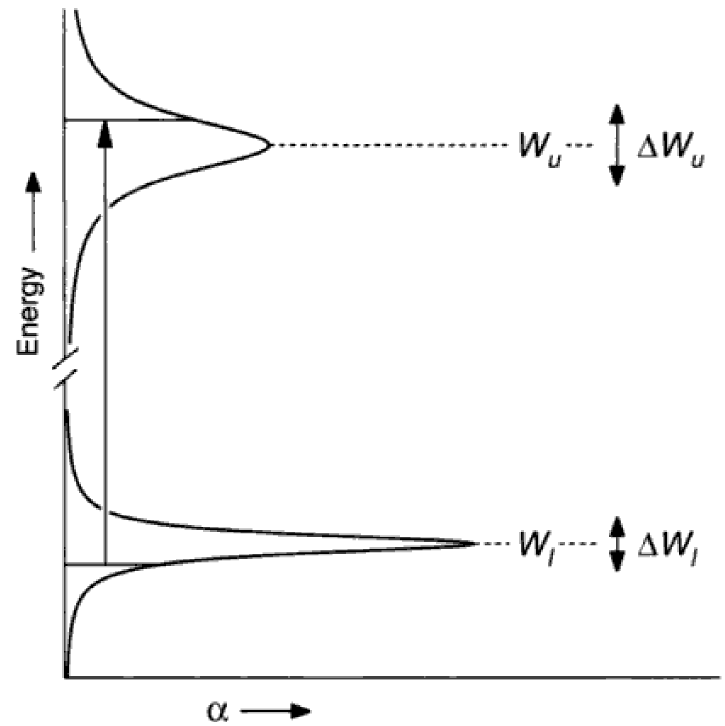


Fig. 11.1. This schematic energy-level diagram illustrates the width of the atomic levels. An absorption-line transition starts somewhere within the lower level with a probability of a specified energy given by the α for the lower level. The transition ends in the upper level with a terminal energy probability given by α for the upper level. The characteristic energy widths of the levels is indicated by the ΔW values on the right. Note the cut in the ordinate axis; the separation of the levels is much larger than their widths.

The physical meaning of this γ can be found by viewing it in terms of the lifetime of the level. Since A_{ul} has units of reciprocal seconds and specifies the probability that the electron will come down from the upper level u in 1 s, the quantity $\Delta t = 1/\Sigma 4\pi A_{ul}$ corresponds to the probable time interval over which the electron will, if left alone, stay in the upper level.

This concept in turn can be coupled to the Heisenberg uncertainty principle. Define the uncertainty of the electron energy in the level u to be ΔW. Then $\Delta W_u \Delta t \gtrsim h/2\pi$ or $\Delta W_u \gtrsim 2h\Sigma A_{ul}$. In other words, the energy level u has an intrinsic energy width set by nature at ΔW_u. The lower level shows a similar broadening, ΔW_l and γ_l. Consider the absorption process depicted in Fig. 11.1. The electron leaves the level l from the energy range ΔW_l. The probability of the electron being at some point in this energy band is α from Eq. (11.7) with $\gamma = \gamma_l$. By a similar token, the probability of the electron being at some energy within ΔW_u in the final state is α with $\gamma = \gamma_u$. Since any starting energy within the lower level can correspond to the full range in energy in the upper level, we see that the full absorption coefficient is a *convolution* between α_u and α_l. In Chapter 2 we saw that a convolution between dispersion profiles results in a new dispersion profile having a larger γ given by

$$\gamma = \gamma_u + \gamma_l.$$

It is this expression that is used for the natural broadening component in the computation of $\ell_\nu \rho$, although in the case of a strong radiation field, γ_u is sometimes expanded to

$$\gamma_u = 4\pi \sum A_{ul} + 4\pi \sum I_\nu B_{ul} + 4\pi \sum I_\nu B_{uk}.$$

The second term allows for stimulated emission and the third for electrons leaving the level u for higher levels by absorption of a photon. A similar expression is written for γ_l.

The observed widths of stellar spectral lines are always larger than given by the natural atomic damping, sometimes much larger. We go on to discuss some of these other broadening mechanisms.

Pressure broadening

The term pressure broadening implies a collisional interaction between the atoms absorbing the light and other particles. The other particles can be ions, electrons, or atoms of the same element as the absorbers or another type, or in cool stars, they might be molecules. One imagines the atomic levels of the transition of interest to be disturbed such that their energy is altered. The distortion is a function of the separation, R, between the absorber and the perturbing particle. We expect the upper level u to be more strongly altered than the lower level l. Figure 11.2 shows a schematic presentation of this situation where the potential energy of the level is plotted against the distance to the perturber. Transitions of type 1 have the undisturbed energy. Transitions of type 2 have lower energy, while type 3 shows an increased energy. Depending on the distribution of encounter separations, and the shape of the energy curves, the net effect of all the absorbers along a line through the stellar photosphere can result in line shifts, asymmetries, and broadening. Line asymmetries and shifts due to collisions have been studied very little in stellar spectra. In keeping with the level of treatment, we shall concentrate on the broadening and refer the reader to the references for a discussion of pressure shifts and asymmetries.

The change in energy induced by the collision when plotted against R can often be approximated by a power law of the form

Table 11.1. *Types of pressure broadening.*

n	Type	Lines affected	Perturber
2	Linear Stark	Hydrogen	Protons, electrons
4	Quadratic Stark	Most lines, especially in hot stars	Ions, electrons
6	van der Waals	Most lines, especially in cool stars	Neutral hydrogen

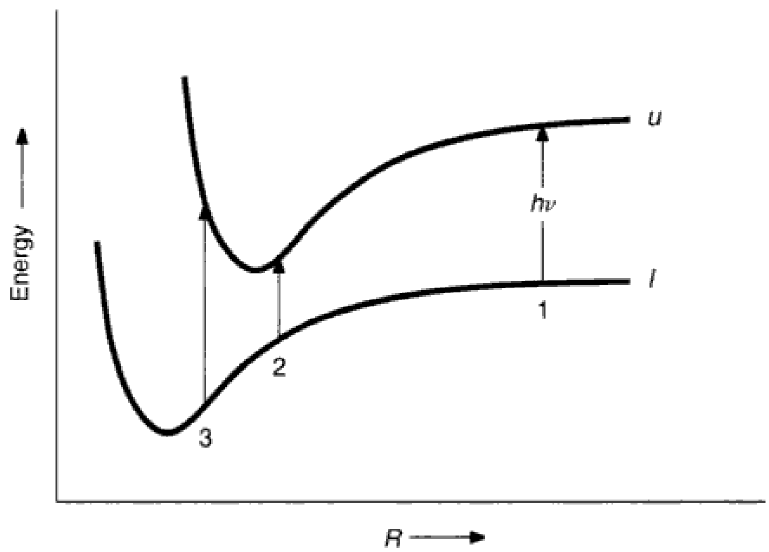


Fig. 11.2. The energies associated with the upper and lower atomic levels on a transition depend on the distance R to the perturber. The transition energy can be either less (2) or greater (3) than the unperturbed value (1).

$$\Delta W = \text{constant}/R^n, \tag{11.16}$$

where the integer n depends on the type of interaction. For example, the dipole coupling force between pairs of atoms of the same kind is proportional to R^{-4}, making ΔW proportional to R^{-3}, so $n = 3$ is used for this interaction. Dipole coupling perturbations are called resonance broadening. As a second example, we cite the result of London (1930a, b) who found theoretically that when the perturber is of a different species, the perturbation energy is proportional to R^{-6}, so $n = 6$. Table 11.1 gives a summary of the types of interactions that are most important in stars.

We can convert the energy change given in Eq. (11.16) to a change in frequency in the spectrum by subtracting the equation for the lower level from the equation for the upper level,

$$\Delta\nu = C_n/R^n. \qquad (11.17)$$

The interaction constant C_n must be measured or calculated for each transition and type of interaction. It is known for only a few lines.

In most of what we consider, it is assumed that no atomic transitions result from the collision itself. This assumption is called the adiabatic assumption. Baranger (1958a, b) has given a generalized theory in which this limitation is not imposed. A review of the progress made in understanding pressure broadening has been given by Van Regemorter (1965). Additional material and references can be obtained from Fuhr *et al.* (1973), Van Regemorter (1973), Griem (1974, 1997), and Lesage *et al.* (1999).

The impact approximation

The temperatures of most stellar atmospheres lead to relatively high particle velocities. This is especially true for hydrogen and helium atoms and for electrons. Furthermore, for stars on or above the main sequence, the atmospheric pressures are modest ($\lesssim$1 atm). Under these conditions, one expects the duration of the collision to be small compared to the time between collisions. In this way one is led to consider the impact approximation as the starting point for a theoretical treatment. The impact approximation is also referred to as the interruption or phase shift approximation.

A simple form of the impact treatment was proposed by Lorentz (1906) in which he assumed the electromagnetic wave was terminated by the impact, with the remaining photon energy being converted to kinetic energy. This simple treatment was refined under the hands of Lenz (1924), Kallmann and London (1929), and Weisskopf (1932). In place of a termination of the electromagnetic wave, they showed how a phase change in the oscillation could occur resulting in broadening. Suppose we have a photon of time duration Δt, typically about 10^{-9} s, that we view as a product of an infinite sinusoid and a box. The spectrum of the electric field E is then a sinc $\pi\Delta t(\nu-\nu_0)$ function centered on the frequency of the sinusoid and of characteristic width $\Delta\nu = 1/\Delta t$. Of course the intensity spectrum is EE^* or $\text{sinc}^2\pi\Delta t(\nu-\nu_0)$. If the same sinusoid is being absorbed or emitted by an atom that suffers several impacts from perturbers, one can view the original wave as having been cut into pieces, the phase between segments having been changed by the impacts. The original box of length W is now replaced with a sum of boxes with lengths Δt_j. In other words, the original photon has been broken down into two or more shorter pieces. The Fourier transform of such a sum is the sum of the transforms. But since each Δt_j is smaller than the original Δt, the line is broadened with each $\Delta\nu_j = 1/\Delta t_j$ being greater than $\Delta\nu = 1/\Delta t$.

The distribution of the Δt_j values along with the sinc function determines the shape of the absorption coefficient. The distribution P of Δt_j is found to be

$$\mathrm{d}P(\Delta t_j) = \mathrm{e}^{-\Delta t_j/\Delta t_0}\mathrm{d}\Delta t_j/\Delta t_0, \tag{11.18}$$

where Δt_0 is the average length of uninterrupted segment. (The derivation of this distribution is as follows. Let $P(\Delta t)$ be the probability that there is, and $p(\Delta t)$ that there is not, an impact collision in the interval of time Δt. Then $p(\Delta t) = 1 - P(\Delta t)$. Now in the interval of time between Δt and $\mathrm{d}\Delta t$, the probability of collision is $\mathrm{d}\Delta t/\Delta t_0$, where Δt_0 is the average collision time for the gas. Therefore the probability of no collision in $\mathrm{d}\Delta t$ is $p(\mathrm{d}\Delta t) = 1 - \mathrm{d}\Delta t/\Delta t_0$. Probability theory tells us that $p(\Delta t + \mathrm{d}\Delta t) = p(\Delta t)p(\mathrm{d}\Delta t) = p(\Delta t)(1 - \mathrm{d}\Delta t/\Delta t_0)$ or $p(\Delta t + \mathrm{d}\Delta t) = p(\Delta t) - p(\Delta t)\,\mathrm{d}\Delta t/\Delta t_0$. However, from a Taylor expansion we have $p(\Delta t + \mathrm{d}\Delta t) = p(\Delta t) + (\mathrm{d}p/\mathrm{d}\Delta t)\mathrm{d}\Delta t + \cdots$, so that we are led to equate $\mathrm{d}p/\mathrm{d}\Delta t$ with $-p/\Delta t_0$, which upon integration gives $p(\Delta t) = \exp(-\Delta t/\Delta t_0)$. Finally, $P(\Delta t) = 1 - p(\Delta t) = 1 - \exp(-\Delta t/\Delta t_0)$, and $\mathrm{d}P(\Delta t) = \exp(-\Delta t/\Delta t_0)\,\mathrm{d}\Delta t/\Delta t_0$.) The atomic line absorption coefficient is then proportional to the weighted average given by

$$\int_0^\infty \Delta t^2 \left(\frac{\sin \pi\Delta t(\nu - \nu_0)}{\pi\Delta t(\nu - \nu_0)}\right)^2 \mathrm{e}^{-\Delta t/\Delta t_0}\frac{\mathrm{d}\Delta t}{\Delta t_0}.$$

This integral can be evaluated analytically (see Petit Bois 1961, for example). The result is

$$\alpha = \frac{\text{constant}}{4\pi^2(\nu - \nu_0)^2 + (1/\Delta t_0)^2},$$

which can be re-written as

$$\alpha = \text{ constant}\frac{\gamma_n/4\pi}{(\nu - \nu_0)^2 + (\gamma_n/4\pi)^2}. \tag{11.19}$$

This has the form of Eq. (11.7). This is a remarkable result because we end up with a dispersion profile similar to that for natural damping. Furthermore, the impact approach led to this result quite independent of the type of collisional interaction.

In order to use Eq. (11.19) in a line profile calculation, it is necessary to evaluate $\gamma_n = 2/\Delta t_0$, including its pressure and temperature dependences. Unlike radiation damping, collisional damping is a function of depth in the stellar photosphere.

Theoretical evaluation of the collisional damping constant

The simplest approach is to assume all encounters can be placed into one of two groups depending on the strength of the encounter. We measure strength by the size of the phase shift. If we find the phase is shifted by, say 1 rad or more, we count it, but if the shift is less, we neglect it. Clearly this is an order of magnitude estimate, for there is no reason to choose 1 rad versus, say, π rad. The chosen value simply represents what is considered to be a significant phase shift. The frequency change is given by Eq. (11.17). The cumulative effect of this change in frequency is the phase shift we seek, namely,

$$\phi = 2\pi\int_0^\infty \Delta\nu\,\mathrm{d}t = 2\pi\int_0^\infty C_n R^{-n}\,\mathrm{d}t. \tag{11.20}$$

We evaluate this integral by assuming the perturber moves past the atom on a straight line trajectory and maintains a constant velocity v, as shown in Fig. 11.3. The impact parameter, ρ, and the angle θ are related by $\rho = R\cos\theta$, so that Eq. (11.20) becomes

$$\phi = 2\pi\int_0^\infty C_n \frac{\cos^n\theta}{\rho^n}\,\mathrm{d}t.$$

Using the fact that $v = \mathrm{d}y/\mathrm{d}t = (\rho/\cos^2\theta)\,\mathrm{d}\theta/\mathrm{d}t$ gives $\mathrm{d}t = (\rho/v)\,\mathrm{d}\theta/\cos^2\theta$ and so

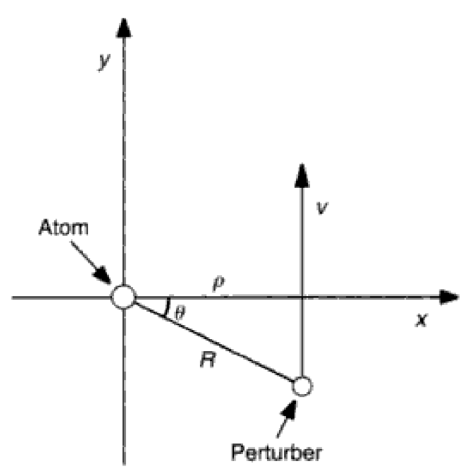


Fig. 11.3. The perturber passes the atom at an impact distance ρ with a velocity v. The y axis is chosen to be parallel to v.

Table 11.2. *Values of the integral.*

n	$\int_{-\pi/2}^{\pi/2} \cos^{n-2}\theta \, d\theta$
2	π
3	2
4	$\pi/2$
5	4/3
6	$3\pi/8$

$$\phi = \frac{2\pi C_n}{v\rho^{n-1}} \int_{-\pi/2}^{\pi/2} \cos^{n-2}\theta \, d\theta. \tag{11.21}$$

The value of the integral is given in Table 11.2 for several values of n. At this point, we use our definition of a significant phase shift and set $\phi = 1$ rad, thereby defining a limiting impact parameter,

$$\rho_0 = \left[\frac{2\pi C_n}{v} \int_{-\pi/2}^{\pi/2} \cos^{n-2}\theta \, d\theta \right]^{1/(n-1)}. \tag{11.22}$$

As stated at the beginning of the section, we count only those collisions that have $\phi \geq 1$ rad, or $\rho \leq \rho_0$. The number of such collisions is given by $\pi\rho_0^2 vN\Delta t_1$, where $\pi\rho_0^2 v$ is the volume swept out per second, Δt_1 is the time interval over which we count the collisions, and N is the number of perturbers per unit volume. If we set $\Delta t_1 = \Delta t_0$, from Eq. (11.18), then on average the number of collisions must be one, and $\pi\rho_0^2 vN\Delta t_0 = 1$. In this way

$$\gamma_n = \frac{2}{\Delta t_0} = 2\pi\rho_0^2 vN \tag{11.23}$$

angular frequency units, with ρ_0 given by Eq. (11.22). The average relative velocity v for the atom of mass m_A and perturber of mass m_p is, according to Eq. (1.14),

$$v = \left[\frac{8kT}{\pi} \left(\frac{1}{m_A} + \frac{1}{m_p} \right) \right]^{1/2}. \tag{11.24}$$

The number density of perturbers, N, is given by the model photosphere.

A more general and accurate statement of the damping constant is

$$\gamma_n = N \int_{-\infty}^{\infty} vf(v)\sigma_n(v) \, dv, \tag{11.25}$$

Table 11.3. *Sample Stark interaction constants.*

Line	Wavelength	log C_4
Na I D_2	λ5890	−15.17
D_1	λ5896	−15.33
Mg I b_2	λ5172	−14.52
b_1	λ5183	−14.52
Mg I	λ5528	−13.12

where σ_n is the broadening cross section and $f(v)$ is the velocity distribution of the perturbers, a Gaussian in most cases. The challenge is then to measure this cross section experimentally or to compute it theoretically.

Numerical values for collisions with charged perturbers

We now obtain some numerical expressions for the quadratic Stark effect. The quadratic Stark effect ($n = 4$) arises from perturbations by charged particles. In a stellar atmosphere these are electrons and ions. From Table 11.2 and Eq. (11.22), we have $\rho_0 = (\pi^2 C_4/v)^{1/2}$. Then with N perturbers per unit volume, Eq. (11.23) gives us

$$\gamma_4 = 2\pi v N\left(\frac{\pi^2 C_4}{v}\right)^{2/3} \approx 29 v^{1/3} C_4^{2/3} N.$$

A slightly more complicated theory (e.g., Lindholm 1945, Foley 1946, Unsöld 1955) gives

$$\gamma_4 \approx 39 v^{1/3} C_4^{2/3} N, \tag{11.26}$$

which, considering the approximate nature of the calculation, is the same as the previous equation. Nevertheless, to be numerically consistent with material in the literature, we adopt Eq. (11.26). Now we take this equation and, using Eq. (11.24) for the velocity, write for the combined perturbations of electrons and ions

$$\gamma_4 \approx 39 C_4^{2/3}\left[\frac{8kT}{\pi}\left(\frac{1}{m_A}+\frac{1}{m_e}\right)\right]^{1/6} N_e + 39 C_4^{2/3}\left[\frac{8kT}{\pi}\left(\frac{1}{m_A}+\frac{1}{m_i}\right)\right]^{1/6} N_{ion}.$$

We can take $N_{ion} = N_e$ without serious error. Further, the mass of ions is so large compared to the electron mass that they hardly matter. Numerically the expression becomes

$$\log \gamma_4 \approx 19 + \tfrac{2}{3}\log C_4 + \log P_e - \tfrac{5}{6}\log T, \tag{11.27}$$

where P_e is expressed in dyne/cm^2, and we must remember that the constant 19 is quite uncertain.

Values of the interaction constant C_4 are known from laboratory measurements for a few lines. Examples are cited in Table 11.3, and additional values can be found in Griem (1964), Landolt-Börnstein (1950), Allen (1973), Moity *et al.* (1975), González *et al.* (2002), and Milosavljević and Djeniže (2003). Calculation of C_4 can also be done. Hunger (1960), Mugglestone and O'Mara (1966), Freudenstein and Cooper (1978), Dimitrijevic and Konjevic (1986), Lesage *et al.* (1990), Popovic *et al.* (1999, 2001), and Dimitrijevic *et al.* (2003) give examples of such computations.

Numerical values for collisions with neutral perturbers

We now turn our attention to collisions with neutral perturbers. Neutral hydrogen is by far the dominant perturber for stars cooler than about 10 000 K. Although the classic van der Waals formulation with $n = 6$ in Eq. (11.17) has now been largely superceded, it is useful to review it before considering recent improvements. Typically, the van der Waals formulation gives damping constants that are about a factor of 1.5–2 too small, but the error varies substantially from one line to the next, running the gamut from ≈0.1 to ≈5 (see, for example, Ryan 1998).

As with the quadratic Stark case, start with Eq. (11.26), using ρ_0 given by Eq. (11.22) and Table 11.2 where $n = 6$. We are led to

$$\gamma_6 \approx 17 v^{3/5} C_6^{2/5} N, \tag{11.28}$$

where, as we did with Eq. (11.26), the constant has been increased from 14 to 17. Using v_H to denote the velocity of hydrogen atoms,

$$\gamma_6 \approx 17 v_H^{3/5} C_6^{2/5} 3 \left[\frac{1}{1 + \Phi_H(T)/P_e} \right] N_H$$

in which N_H represents the total number of hydrogen particles, neutrals and ions, and the factor in parentheses takes into account the ionization of hydrogen since only the neutrals cause the perturbations. ($\Phi_H(T)/P_e = N_1/N_0$ is the ionization equation, Eq. (1.21).) Using typical masses in Eq. (11.24), the expression reduces to

$$\log \gamma_6 \approx 20 + 0.4 \log C_6 + \log P_g - 0.7 \log T \tag{11.29}$$

for P_g in dyne/cm^2. Unsöld (1955) showed how to evaluate the perturbation energy in an approximate way. The difference between the energies for the two levels of the transition leads to the expression

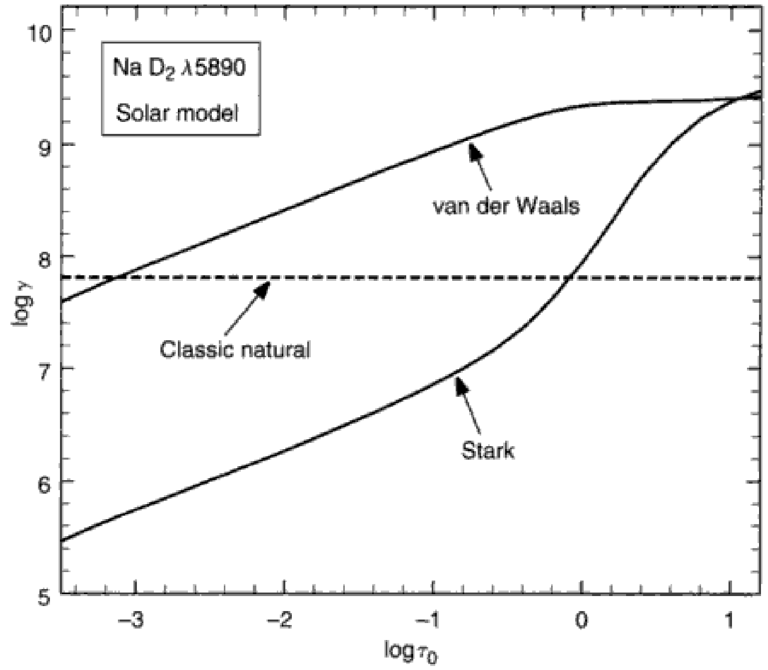


Fig. 11.4. Damping constants for the Na I D_2 line are shown as a function of depth in a solar model. The van der Waals damping constant is computed using Eq. (11.9). The Stark damping constant comes from Eq. (11.27). For omparison, the natural radiation damping according to Eq. (11.13) is shown.

$$C_6 = 0.3 \times 10^{-30}\left[\frac{1}{(I-\chi-\chi_\lambda)^2} - \frac{1}{(I-\chi)^2}\right], \tag{11.30}$$

where I is the ionization potential and χ is the excitation potential of the lower level in electron volts for the atom of interest. The symbol $\chi_\lambda = h\nu = 1.2398 \times 10^4/\lambda$ with λ in angstroms, and is the energy of a photon in the line.

An illustration of the van der Waals and quadratic Stark damping is given for the Na D line in Fig. 11.4. Here log C_6 has a value of -31.7. The damping constants are shown as a function of optical depth in a solar-type model, and it can be seen how the Stark broadening is much smaller than the van der Waals broadening over most of the photosphere. For other lines, such as the Mg b lines, the difference is less and the Stark broadening dominates in the deeper layers.

Equations like (11.27) and (11.29) have been used in a great many line profile calculations. In most cases, the damping constant, notably the γ_6 component, is found to be too small and an empirical "enhancement factor" is used. Examples can be seen in Cowley *et al.* (1969), Burgess and Grindlay (1970), Fullerton and Cowley (1971), Holweger (1971, 1972), Ayres (1977), and Lambert and Luck (1978). Criticisms, revisions, and improvements in handling collisions with neutral hydrogen have been made by Kusch (1958), Roueff and Van Regemorter (1969, 1971), Brueckner (1971), Lewis *et al.* (1971, 1972), Blackwell *et al.* (1972a, b), Lwin *et al.* (1977), and more recently by Anstee and O'Mara (1991, 1995).

In the theory of Anstee and O'Mara, the σ_6 cross sections in Eq. (11.25) are calculated and tabulated as a function of *effective* principal quantum numbers, n^*, for the upper and lower levels of neutral species. Barklem *et al.* (1998a) give examples of how these are computed. (Note that we continue to use the subscript 6 even though the approximation embodied in Eq. (11.16) has been superceded.) Armed with these two quantum numbers, one enters their tables to obtain $\sigma_6(v_0)$ and p to use in the following expression for γ_6:

$$\gamma_6 = N_H\left(\frac{4}{\pi}\right)^{p/2}\Gamma\left(\frac{4-p}{2}\right)\left(\frac{v}{v_0}\right)^{-p} v\sigma_6(v_0),$$

where Γ is the gamma function, $v_0 = 10\,\text{km/s}$, and v is given by Eq. (11.24). Additional tables for other atomic configurations are given by Barklem and O'Mara (1997) and Barklem *et al.* (1998b).

This formulation reproduces the broadening seen in the wings of stronger solar lines for at least a few significant cases. It is relatively easy to use and can be applied to many lines. More accurate calculations can be done, but on a tedious line-by-line basis. Barklem and O'Mara (1998, 2000) show how line-by-line calculations are needed to evaluate the broadening in the lines of ions, and they present results for selected lines of several ions. Tabulated values of γ_6 entered into the Vienna Atomic Line Database can be found through Barklem *et al.* (2000a).

Hydrogen line broadening

Hydrogen lines are different from other spectral lines because the atomic structure of hydrogen is particularly subject to the linear Stark effect. The corresponding wavelength shifts can be enormously larger than is typical of "normal" spectral lines, as illustrated in Fig. 11.5. It was Struve (1929) who gave a convincing argument that the great widths of the hydrogen lines in early-type stars are due to the linear Stark effect. A decade earlier, Holtsmark (1919) began the studies of theoretical distributions of ions in a plasma and the subsequent broadening caused in removing the degeneracy of the hydrogen energy levels. The splitting of the energy levels can be expressed as the wavelength shift of the spectral components,

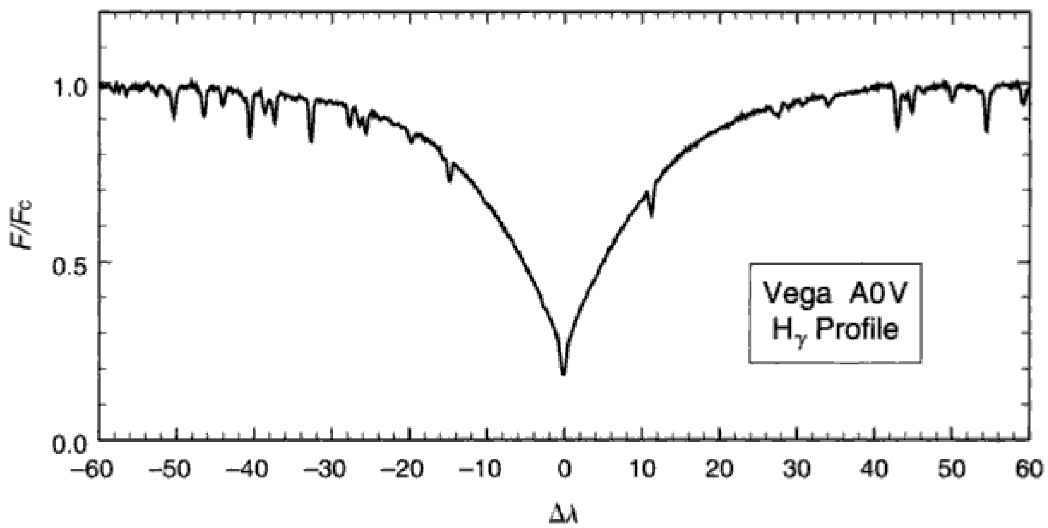


Fig. 11.5. The hydrogen lines in hot stars like Vega are about two orders of magnitude wider than typical metal lines. Taken at Observatoire de Haute-Provence; resolving power of ~16 000. Courtesy of F. Royer.

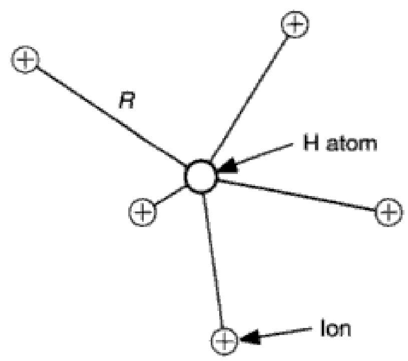


Fig. 11.6. The distribution of ions in space gives a non-zero electric field at the position of the atom. This field causes a perturbation in the energy levels of the atom.

$$\Delta\lambda_j = c_j E \tag{11.31}$$

in which E is the electric field at the hydrogen atom, and c_j is the constant of proportionality for the jth component. Historically, the electric field was at first thought to depend only on the electric fields of ions. One imagined the situation to look like Fig. 11.6. If we let R be the separation of some ion from the hydrogen atom, then the field from that ion is $E = e/R^2$, where we have assumed the perturbing ion to be singly ionized. This form is consistent with Eq. (11.17).

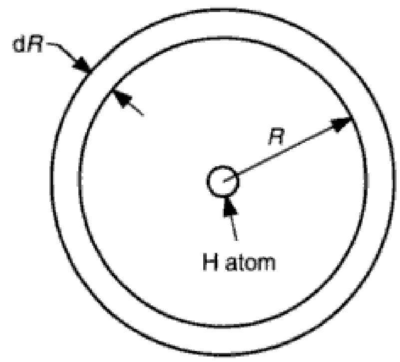


Fig. 11.7. The shell of radius R and thickness $\mathrm{d}R$ around the hydrogen atom has a volume $4\pi R^2 \mathrm{d}R$.

Since the strongest interaction occurs between the atom and the closest perturber, the lowest-order approximation is given by completely neglecting the lesser fields contributed by the other ions slightly farther away. This approach is know as the nearest-neighbor approximation. We take N to be the number density of ions. The probability of finding an ion in the shell bounded by R and $R + \mathrm{d}R$ is proportional to the number of ions expected to be in that size volume, namely, $4\pi R^2 \mathrm{d}R\, N$, as in Fig. 11.7. We seek the ion that is closest to the atom, so we combine the probability that the ion is *not* in the sphere of radius R but *is* in the shell of thickness $\mathrm{d}R$. The probability, $P(R)$, of both these conditions occurring at the same time is the product of their individual probabilities,

$$P(R)\,\mathrm{d}R = P_n(R) 4\pi R^2\,\mathrm{d}R\; N, \tag{11.32}$$

where $P_n(R)$ denotes the probability of no ion being in the sphere. We know that as R increases, $P_n(R)$ must diminish according to the increased likelihood of encompassing some ion. This leads us to write

$$\begin{aligned} P_n(R + \mathrm{d}R) &= P_n(R)(1 - 4\pi R^2\,\mathrm{d}R\, N) \\ &= P_n(R) - 4\pi R^2 N P_n(R)\,\mathrm{d}R. \end{aligned}$$

But since quite generally $P_n(R + \mathrm{d}R) = P_n(R) + (\mathrm{d}P_n/\mathrm{d}R)\mathrm{d}R$, we see that $\mathrm{d}P_n/\mathrm{d}R = -4\pi R^2 N P_n(R)$, which upon integration gives

$$P_n(R) = P_n(0)\,\mathrm{e}^{-4\pi R^3 N/3}.$$

When $R = 0$, $P_n = 1$ on physical grounds, and using the mean distance between ions, $R_0 = (4\pi N/3)^{-1/3}$, as the unit of length, the probability of no ion being as close as R is $P_n(R) = \mathrm{e}^{-(R/R_0)^3}$. The electric field corresponding to R_0 is

$$E_0 = (4\pi/3)^{2/3} e N^{2/3}$$
$$= 1.2481 \times 10^{-9} N^{2/3}. \tag{11.33}$$

Then from Eq. (11.32) the probability that the ion is in the shell dR is

$$P(R)\,dR = 3\frac{R^2}{R_0^3} e^{-(R/R_0)^3} dR.$$

This distribution in R can be translated into the distribution of electric fields by using $(R/R_0)^2 = E_0/E$ and $|dR/R_0| = 0.5(E_0/E)^{3/2}$. Thus

$$P(E)\,dE = \frac{3}{2}\left(\frac{E_0}{E}\right)^{5/2} e^{-(E_0/E)^{3/2}} \frac{dE}{E_0},$$

or

$$P(\beta)\,d\beta = \tfrac{3}{2}\beta^{-5/2} e^{-\beta^{-3/2}} d\beta, \tag{11.34}$$

where $\beta = E/E_0$ is the field strength in units of E_0. This is the simplified distribution based on the nearest-neighbor approximation that explicitly avoids the vector addition of E from several perturbers. More elaborate calculations take into account the full distribution of perturbers, and so one obtains a somewhat more accurate result for $P(\beta)$ called the Holtsmark distribution. For example, Underhill and Waddell (1959) give the Holtsmark distribution as the sum over all spectral components,

$$P(\beta)\,d\beta = \frac{4}{3\pi}\sum_{j=0}^{\infty}(-1)^j \frac{\Gamma[(4j+6)/3]}{2j+1}\beta^{2j+2} d\beta \tag{11.35}$$

in which Γ is the gamma function. Figure 11.8 shows the perturber distribution according to Eqs. (11.34) and (11.35). The two distributions give the same result for large fields, as one would expect based on the simple argument that strong fields arise from close encounters, in which case one ion dominates and the nearest-neighbor approximation is fulfilled.

The complete hydrogen Stark broadening, arising from many atoms but normalized to the f value for the line, is the weighted sum of the individual Stark components with each showing a $P(\beta)$ or $P(\Delta\lambda)$ distribution according to Eq. (11.35). Following the customary definition (Underhill 1951), we denote the values of the unshifted Stark components by f_0 and define the f values of the shifted components as

$$f_{\pm} = \frac{1}{l^2}\sum_j f_j,$$

the summation being carried out over all shifted components. The principal quantum number of the lower level has been called l. The Stark profile is then

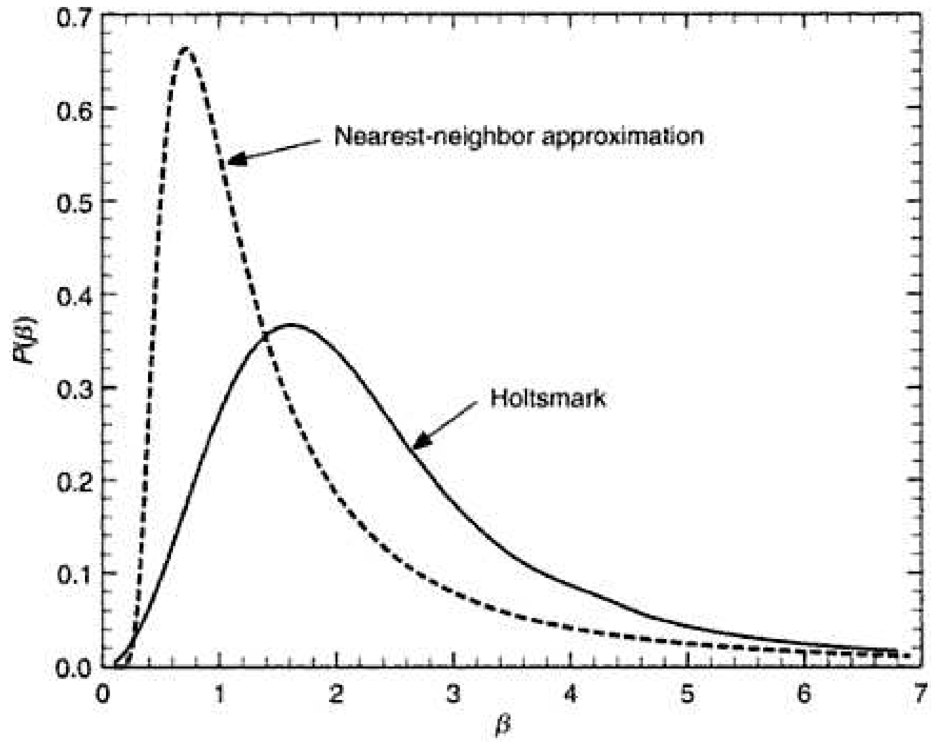


Fig. 11.8. The Holtsmark Stark-broadening distribution (———), according to Eq. (11.35), is compared to the nearest-neighbor approximation (---------) given by Eq. (11.34).

defined as the weighted sum

$$S(\Delta\lambda/E_0)\frac{d\Delta\lambda}{E_0} = \sum \frac{f_j}{\sum f_j} P(\beta)d\beta$$
$$= \sum \frac{f_j}{l^2 f_\pm} P\left(\frac{\Delta\lambda}{c_j E_0}\right)\frac{d\Delta\lambda}{c_j E_0}. \qquad (11.36)$$

The oscillator strengths are given in Table 11.4. The sum S is normalized to unity according to

$$\int_{-\infty}^{\infty} S(\Delta\lambda/E_0)\frac{d\Delta\lambda}{E_0} = 1.$$

The energy subtracted from a unit-strength I_λ beam can now be written as

$$\alpha d\Delta\lambda = \frac{\pi e^2}{mc}\frac{\lambda^2}{c}\left[\frac{f_\pm S(\Delta\lambda/E_0)}{E_0} + f_0\delta(\Delta\lambda)\right]d\Delta\lambda. \qquad (11.37)$$

The unshifted component is represented by the delta-function, $\delta(\Delta\lambda)$. This expression for α is typically evaluated numerically by interpolating in tables of

Table 11.4. *Hydrogen oscillator strengths.*

Line	u	f_0	$f_\pm$	f
H_α	3	0.248 689	0.392 058	0.640 742
H_β	4	0.00	0.119 321	0.119 321
H_γ	5	0.007 014	0.037 656	0.044 670
H_δ	6	0.00	0.022 093	0.022 093
H_ε	7	0.001 310	0.011 394	0.012 704
H_ζ	8	0.00	0.008 036	0.008 036

$S(\Delta\lambda/E_0)$.

The original Stark sums given by Underhill and Waddell (1959) neglected the impact broadening of electrons, but this turns out to be important. In the wings of the lines, it amounts to a factor of 2. Developments by Griem (1954, 1962, 1967), Edmonds *et al.* (1967), Brissaud and Frisch (1971), Vidal *et al.* (1971, 1973), Stehlé *et al.* (1988), and Stehlé (1994) have moved the theory forward. Tabulations of $S(\Delta\lambda/E_0)$ have been given by Edmonds *et al.* (1967), Vidal *et al.* (1973), Griem (1974), Clausset *et al.* (1994), Lemke (1997), and Stehlé and Hutcheon (1999) among others.

In Fig. 11.9, $S(\Delta\lambda/E_0)$ is shown for the first few Balmer lines. The asymptotic wavelength dependence in the wing goes as $\Delta\lambda^{-5/2}$ and several of the references above give proportionality constants. The main differences among the different formulations appear in the cores of the lines and here the Doppler broadening from thermal (see below) and mass motions in a stellar atmosphere blurs them away. It is also in these core and inner-wing regions where qualitative disagreements still exist with laboratory measurements.

Further broadening of the hydrogen lines occurs from the perturbation of the hydrogen absorber by other hydrogen atoms, i.e., self-broadening. This is a van der Waals-type interaction. In hotter stars, where hydrogen is at least partially ionized, the usual Stark effect is much larger. But in solar-type stars and cooler, the self-broadening is competitive because the numbers of ions and electrons has diminished by four or five orders (as in Fig. 9.13). Barklem *et al.* (2000b) model the Balmer lines of the Sun using the "overlapping-line" theory (Baranger 1958a, b, Peach 1981) combined with the usual Stark broadening. Inclusion of self-broadening improves the fit to the observations, but some discrepancies remain (see also Bergeron *et al.* 1991). It is often difficult to disentangle inadequacies in the model photospheres from problems with the line absorption coefficient.

We have now covered the major aspects of pressure broadening. Other wavelength shifts result from the Doppler effect. The thermal component falls in this category, and we now turn our attention to it.

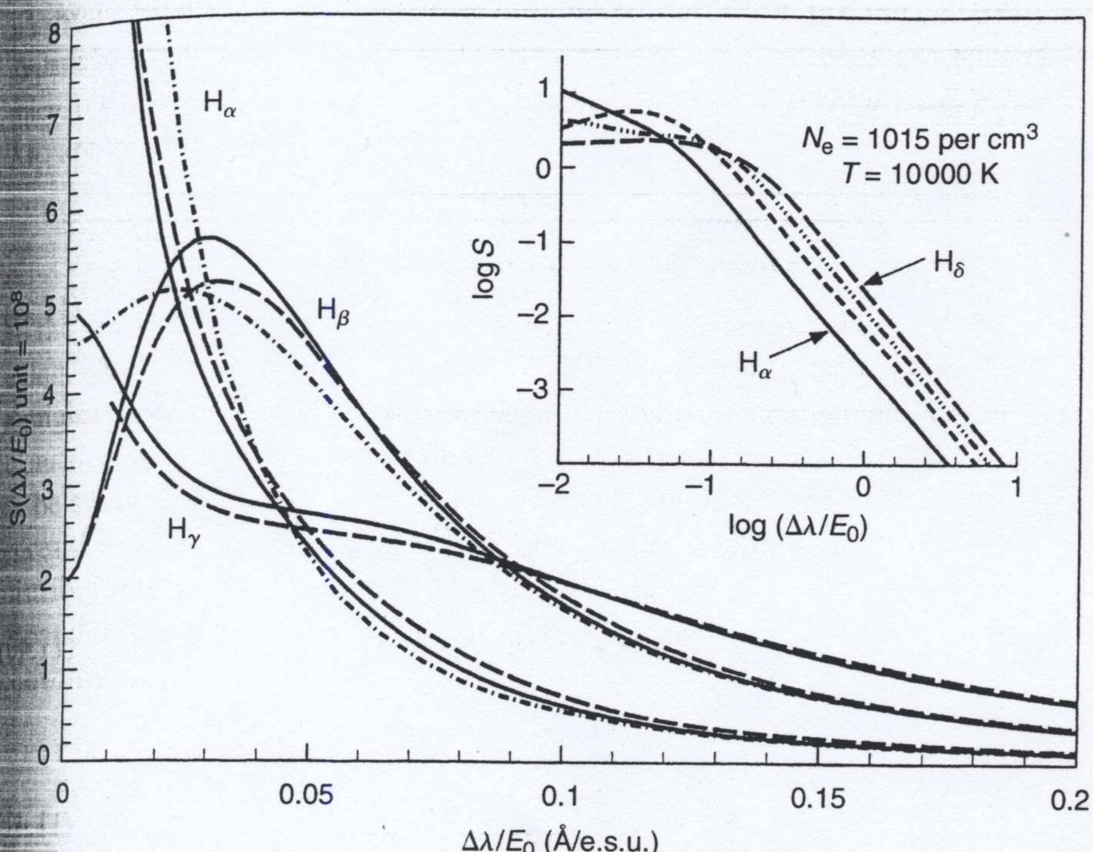


Fig. 11.9. Stark profiles, $S(\Delta\lambda/E_0)$, are shown for the first few Balmer lines. Only half of the profiles are shown since they are essentially symmetric. The results for three independent computations are shown: —, Vidal *et al.* (1973); – – –, Griem (1974); -·-·-·-, Stehlé (1994). The inset shows the Vidal *et al.* functions with logarithmic coordinates, where the $\Delta\lambda^{-5/2}$ behavior in the wing is apparent.

Thermal broadening

Each atom has a component of velocity along our line of sight to the star due to its thermal motion. Denote this velocity, the radial velocity, by v_R. Then the Doppler shift of the line emitted or absorbed by the atom is given by

$$\frac{\Delta\lambda}{\lambda} = \frac{\Delta\nu}{\nu} = \frac{v_R}{c}, \tag{11.38}$$

where c is the velocity of light. The distribution of $\Delta\lambda$ values gives us the shape of the absorption coefficient, but since $\Delta\lambda$ and v_R are proportional, we may simply use the distribution of atomic velocities according to Eq. (1.11),

$$\frac{\mathrm{d}N}{N} = \frac{1}{\pi^{1/2}v_0}\mathrm{e}^{-(v_R/v_0)^2}\mathrm{d}v_R,$$

where the variance v_0 is related to the temperature by $v_0^2 = 2kT/m$. Here m is the mass of the atom and k is Boltzmann's constant. The Doppler wavelength shift associated with v_0 is

$$\begin{aligned}\Delta\lambda_D &= \frac{v_0}{c}\lambda_0 = \frac{\lambda_0}{c}\left(\frac{2kT}{m}\right)^{1/2} \\ &= 4.301 \times 10^{-7}\lambda_0(T/\mu)^{1/2} \qquad (11.39)\end{aligned}$$

in which λ_0 is the central wavelength of the line and μ is the atomic weight in atomic mass units. The corresponding frequency shift is

$$\begin{aligned}\Delta\nu_D &= \frac{v_0}{c}\nu_0 = \frac{\nu_0}{c}\left(\frac{2kT}{m}\right)^{1/2} \\ &= 4.301 \times 10^{-7}\nu_0(T/\mu)^{1/2}. \qquad (11.40)\end{aligned}$$

The distribution of $\Delta\lambda$ values is then

$$\frac{dN}{N} = \frac{1}{\pi^{1/2}\Delta\lambda_D} e^{-(\Delta\lambda/\Delta\lambda_D)^2} d\Delta\lambda. \qquad (11.41)$$

The energy removed from a unit intensity beam is $(\lambda^2/c)\pi e^2 f/(mc)$ for wavelength units and $\pi e^2/(mc)$ for frequency units times dN/N or

$$\alpha d\lambda = \frac{\pi^{1/2}e^2}{mc} f \frac{\lambda_0^2}{c} \frac{1}{\Delta\lambda_D} e^{-(\Delta\lambda/\Delta\lambda_D)^2} d\lambda, \qquad (11.42)$$

and

$$\alpha d\nu = \frac{\pi^{1/2}e^2}{mc} f \frac{1}{\Delta\nu_D} e^{-(\Delta\nu/\Delta\nu_D)^2} d\nu. \qquad (11.43)$$

These expressions define α for the thermally broadened absorption coefficient.

Microturbulence

Small-scale mass motions, where the characteristic dimensions of the moving material are small compared to the unit optical depth, have traditionally been dubbed "microturbulence." Velocity fields are well documented in studies of stellar spectral lines, and we take up the subject in more detail in Chapter 17. In the classical model of microturbulence, one postulates an isotropic Gaussian velocity distribution of dispersion ξ. These small-scale velocities produce Doppler shifts that are completely analogous to those arising from thermal motions. One therefore has an absorption coefficient associated with microturbulence that is identical to Eqs. (11.42) and (11.43) in which $\Delta\lambda_D$ and $\Delta\nu_D$ are replaced by ξ.

Combining absorption coefficients

In the preceding pages, we saw that there are many processes: natural broadening, Stark broadening, van der Waals' broadening, thermal broadening, and microturbulence broadening all of which take place simultaneously. Consider any two of these absorption coefficients, and imagine the first one to be synthesized by a series of δ functions. Each of the δ-functions experiences the broadening of the second absorption coefficient. In short, the two distributions must be convolved together to obtain the combined result. If we expand this to include all the above-mentioned processes, we obtain a multiple convolution,

$$\begin{aligned}\alpha(\text{total}) = \alpha(\text{natural}) * \alpha(\text{Stark})\ \alpha(\text{v.d. Waals})\\ * \alpha(\text{thermal}) * \alpha(\text{micro}),\end{aligned} \tag{11.44}$$

and $\int_0^\infty \alpha(\text{total})\,d\nu$ is normalized to $\pi e^2 f/(mc)$. For normal metal lines, the first three absorption coefficients each have a dispersion profile according to the impact approximation (hydrogen lines are treated separately below). In Chapter 2 we saw that such convolutions result in a new dispersion profile with $\gamma = \gamma_{\text{natural}} + \gamma_4 + \gamma_6$. Similarly, the last two absorption coefficients produce a new Gaussian with dispersion

$$\Delta\lambda_D = \frac{\lambda_0}{c}\left(\frac{2kT}{m} + \xi^2\right)^{1/2} \quad \text{or} \quad \Delta\nu_D = \frac{\nu_0}{c}\left(\frac{2kT}{m} + \xi^2\right)^{1/2}. \tag{11.45}$$

This is an extremely powerful result because of the great computational simplification involved. In essence, we are left with a single convolution between a dispersion function and a Gaussian,

$$\begin{aligned}\alpha &= \frac{\pi e^2}{mc} f \frac{\gamma/4\pi^2}{\Delta\nu^2 + (\gamma/4\pi)^2} * \frac{1}{\pi^{1/2}\Delta\nu_D}\, e^{-(\Delta\nu/\Delta\nu_D)^2}\\ &= \frac{\pi^{1/2} e^2}{mc}\frac{f}{\Delta\nu_D} H(u, a)\end{aligned}$$

Table 11.5. *Hjerting function components.*

u	$H_0(u)$	$H_1(u)$	$H_2(u)$	$H_3(u)$	$H_4(u)$
0.0	+1.000000	−1.12838	+1.0000	−0.752	+0.50
0.1	+0.990050	−1.10596	+0.9702	−0.722	+0.48
0.2	+0.960789	−1.04048	+0.8839	−0.637	+0.40
0.3	+0.913931	−0.93703	+0.7494	−0.505	+0.30
0.4	+0.852144	−0.80346	+0.5795	−0.342	+0.17
0.5	+0.778801	−0.64945	+0.3894	−0.165	+0.03
0.6	+0.697676	−0.48582	+0.1953	+0.007	−0.09
0.7	+0.612626	−0.32192	+0.0123	+0.159	−0.20
0.8	+0.527292	−0.16772	−0.1476	+0.280	−0.27
0.9	+0.444858	−0.03012	−0.2758	+0.362	−0.30
1.0	+0.367879	+0.08594	−0.3679	+0.405	−0.31
1.1	+0.298197	+0.17789	−0.4234	+0.411	−0.28
1.2	+0.236928	+0.24537	−0.4454	+0.386	−0.24
1.3	+0.184520	+0.28981	−0.4392	+0.339	−0.18
1.4	+0.140858	+0.31394	−0.4113	+0.280	−0.12
1.5	+0.105399	+0.32130	−0.3689	+0.215	−0.07
1.6	+0.077305	+0.31573	−0.3185	+0.153	−0.02
1.7	+0.055576	+0.30094	−0.2657	+0.097	+0.02
1.8	+0.039164	+0.28027	−0.2146	+0.051	+0.04
1.9	+0.027052	+0.25648	−0.1683	+0.015	+0.05
2.0	+0.0183156	+0.231726	−0.12821	−0.0101	+0.058
2.1	+0.0121552	+0.207528	−0.09505	−0.0265	+0.056
2.2	+0.0079071	+0.184882	−0.06863	−0.0355	+0.051
2.3	+0.0050418	+0.164341	−0.04830	−0.0391	+0.043
2.4	+0.0031511	+0.146128	−0.03315	−0.0389	+0.035
2.5	+0.0019305	+0.130236	−0.02220	−0.0363	+0.027
2.6	+0.0011592	+0.116515	−0.01451	−0.0325	+0.020
2.7	+0.0006823	+0.104739	−0.00927	−0.0282	+0.015
2.8	+0.0003937	+0.094653	−0.00578	−0.0239	+0.010
2.9	+0.0002226	+0.086005	−0.00352	−0.0201	+0.007
3.0	+0.0001234	+0.078565	−0.00210	−0.0167	+0.005
3.1	+0.0000671	+0.072129	−0.00122	−0.0138	+0.003
3.2	+0.0000357	+0.066526	−0.00070	−0.0115	+0.002
3.3	+0.0000186	+0.061615	−0.00039	−0.0096	+0.001
3.4	+0.0000095	+0.057281	−0.00021	−0.0080	+0.001
3.5	+0.0000048	+0.053430	−0.00011	−0.0068	0.000
3.6	+0.0000024	+0.049988	−0.00006	−0.0058	0.000
3.7	+0.0000011	+0.046894	−0.00003	−0.0050	0.000
3.8	+0.0000005	+0.044098	−0.00001	−0.0043	0.000
3.9	+0.0000002	+0.041561	−0.00001	−0.0037	0.000
4.0	+0.000000	+0.039250	0.00000	−0.0033	0.000

Table 11.5. (*cont.*)

u	$H_1(u)$	$H_3(u)$	u	$H_1(u)$	$H_3(u)$
4.0	+0.039 250	−0.003 29	8.0	+0.009 0306	−0.000 15
4.2	+0.035 195	−0.002 57	8.2	+0.008 5852	−0.000 13
4.4	+0.031 762	−0.002 05	8.4	+0.008 1722	−0.000 12
4.6	+0.028 824	−0.001 66	8.6	+0.007 7885	−0.000 11
4.8	+0.026 288	−0.001 37	8.8	+0.007 4314	−0.000 10
5.0	+0.024 081	−0.001 13	9.0	+0.007 0985	−0.000 09
5.2	+0.022 146	−0.000 95	9.2	+0.006 7875	−0.000 08
5.4	+0.020 441	−0.000 80	9.4	+0.006 4967	−0.000 08
5.6	+0.018 929	−0.000 68	9.6	+0.006 2243	−0.000 07
5.8	+0.017 582	−0.000 59	9.8	+0.005 9688	−0.000 07
6.0	+0.016 375	−0.000 51	10.0	+0.005 7287	−0.000 06
6.2	+0.015 291	−0.000 44	10.2	+0.005 5030	−0.000 06
6.4	+0.014 312	−0.000 38	10.4	+0.005 2903	−0.000 05
6.6	+0.013 426	−0.000 34	10.6	+0.005 0898	−0.000 05
6.8	+0.012 620	−0.000 30	10.8	+0.004 9006	−0.000 04
7.0	+0.011 8860	−0.000 26	11.0	+0.004 7217	−0.000 04
7.2	+0.011 2145	−0.000 23	11.2	+0.004 5526	−0.000 04
7.4	+0.010 5990	−0.000 21	11.4	+0.004 3924	−0.000 03
7.6	+0.010 0332	−0.000 19	11.6	+0.004 2405	−0.000 03
7.8	+0.009 5119	−0.000 17	11.8	+0.004 0964	−0.000 03
8.0	+0.009 0306	−0.000 15	12.0	+0.003 9595	−0.000 03

$$= \frac{\pi^{1/2} e^2}{mc^2} \frac{\lambda_0^2 f}{\Delta\lambda_D} H(u, a). \tag{11.46}$$

Here $H(u, a)$ is called the Hjerting function (Hjerting 1938); its arguments are defined by $u = \Delta\nu/\Delta\nu_D = \Delta\lambda/\Delta\lambda_D$ and

$$\begin{aligned} a &= \frac{\gamma}{4\pi} \frac{1}{\Delta\nu_D} \\ &= \frac{\gamma}{4\pi} \frac{\lambda_0^2}{c} \frac{1}{\Delta\lambda_D}. \end{aligned} \tag{11.47}$$

The H function itself is the convolution

$$\begin{aligned} H(u, a) &= \int_{-\infty}^{\infty} \frac{\gamma/4\pi^2}{(\Delta\nu - \Delta\nu_1)^2 + (\gamma/4\pi)^2} \mathrm{e}^{-(\Delta\nu_1/\Delta\nu_D)^2} \mathrm{d}\nu_1. \\ &= \frac{a}{\pi} \int_{-\infty}^{\infty} \frac{\mathrm{e}^{-u_1^2}}{(u - u_1)^2 + a^2} \mathrm{d}u_1. \end{aligned}$$

Closely allied to the Hjerting function is the Voigt function, $V(u, a) = H(u, a)/(\pi^{1/2}\Delta\nu_D)$, which is used equally often.

The absorption coefficient can be calculate using the simple series expansion

$$H(a, u) = H_0(u) + aH_1(u) + a^2H_2(u) + a^3H_3(u) + a^4H_4(u) + \cdots. \quad (11.48)$$

Interpolation in Table 11.5, taken from Harris (1948), gives the numerical values. The first term, $H_0(u)$, is a Gaussian. In the far wing, one can use the asymptotic expressions

$$H_1(u) = \frac{0.564\,19}{u^2} + \frac{0.846}{u^4}$$

and

$$H_3(u) = \frac{-0.56}{u^4}.$$

In a power expansion like Eq. (11.48), a smaller a means faster convergence. Generally a is small. For example, if $\gamma = 10^9$/s, $T = 6000$ K, $\lambda = 6000$ Å $\times 10^{-8}$ cm/Å, and $m = 56 \times 1.66 \times 10^{-24}$ g for Fe, we have $a = 0.035$.

Figure 11.10 shows how the central portion of $H(u, a)$ is dominated by the

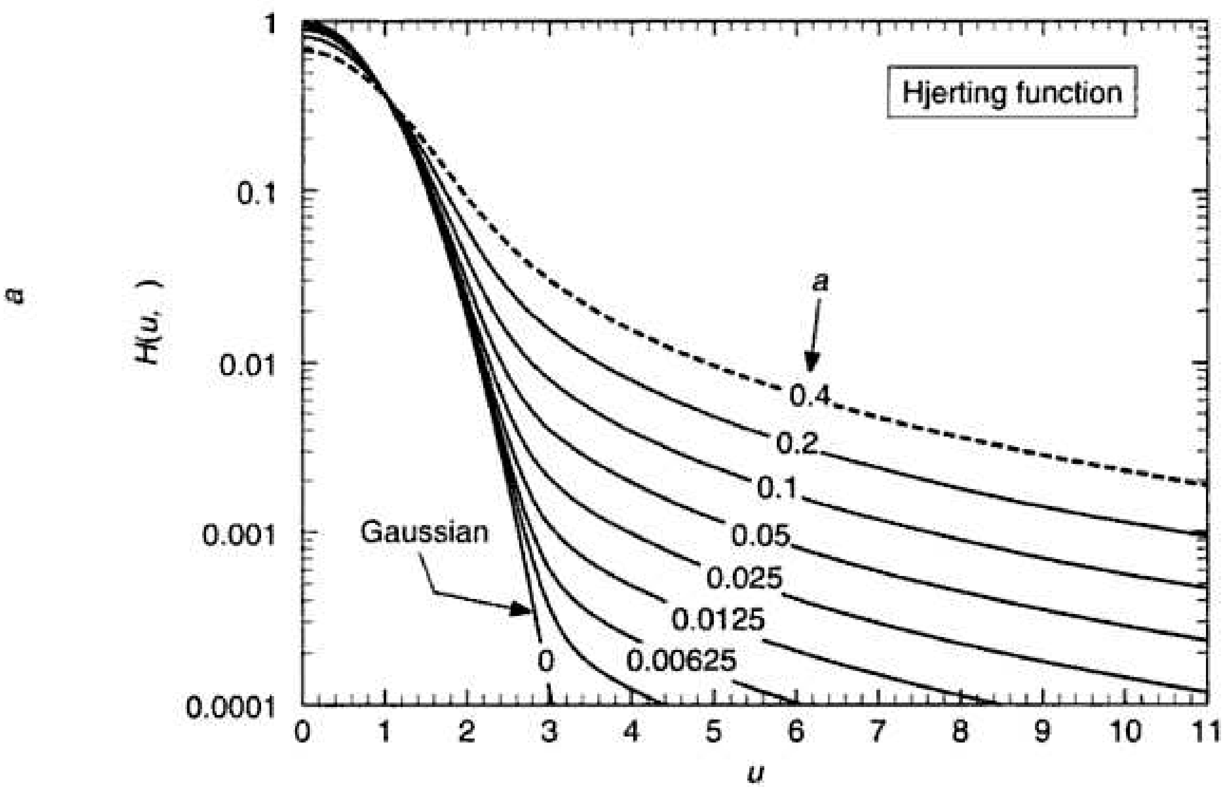


Fig. 11.10. Hjerting functions are plotted with a logarithmic ordinate to emphasize the growth of the wings with increasing damping parameter a.

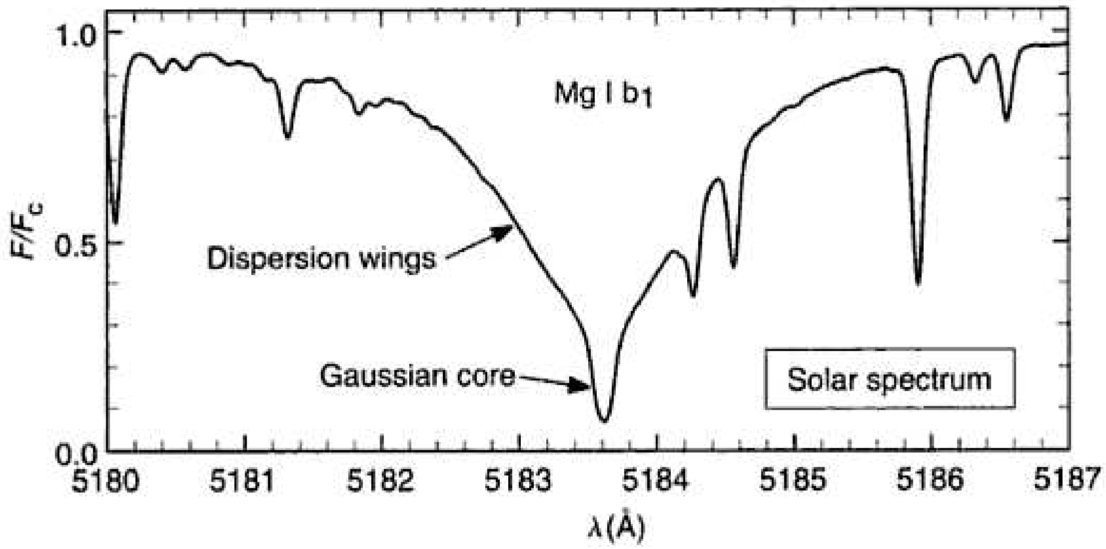


Fig. 11.11. In a few cases spectral lines, like this magnesium line in the solar spectrum, clearly show the Gaussian core and the dispersion wings with a relatively sharp transition between the two near $F/F_c \approx 0.3$.

Gaussian term and how the wings of the dispersion component grow with increasing a. The shaping of stellar line profiles by the line absorption coefficient is illustrated in Fig. 11.11.

The absorption coefficient for hydrogen is also given by Eq. (11.44) or (11.46), but with the Stark broadening term specified by Eq. (11.37). Denoting the convolution of all the α values except the Stark term by the Hjerting function, we write

$$\alpha = \frac{\pi^{1/2}e^2}{mc^2}\frac{\lambda_0^2}{\Delta\lambda_D}\left[f_\pm \frac{S(\Delta\lambda)}{E_0} * H(u, a) + f_0 H(u, a)\right]. \tag{11.49}$$

In stars of early spectral type, the Stark broadening is much greater than the width of $H(u,a)$ so that the convolution of $S(\Delta\lambda)$ with $H(u, a)$ essentially returns $S(\Delta\lambda)$ back again. In such cases, we can write

$$\alpha = \frac{\pi e^2 \lambda_0^2}{mc^2}\left[f_\pm \frac{S(\Delta\lambda)}{E_0} + f_0 \frac{H(u, a)}{\pi^{1/2}\Delta\lambda_D}\right]. \tag{11.50}$$

This simplified form is often used in model atmosphere computations.

Hyperfine and isotopic splitting

Hyperfine structure stems from the interaction of nuclear spin with the angular momentum vector of the rest of the atom, splitting the energy levels of odd atomic-number elements. Isotopic splitting of similar magnitude occurs with the different nuclear mass of the different isotopes. Zeeman splitting requires the absorbers to be in a magnetic field and that is the case in many stars. To properly handle these effects, one needs to know the positions and relative strengths of the components of the spectral line (e.g., Wahlgren *et al.* 1997). Then the absorption coefficients of each of them, in the form of Eq. (11.46), are added together to obtain the complete absorption coefficient for the "line" that now is really several closely spaced lines.

Splitting can alter the shapes of narrow spectral lines and reduce the amount of saturation in strong lines. In these ways one can be misled when high-resolution stellar spectra are analyzed. Some aspects of hyperfine structure are discussed and examples given by Abt (1952), Bely (1966), Holweger and Oertel (1971), Arnesen *et al.* (1982), Bohlender *et al.* (1998), Dworetsky *et al.* (1998), Jomaron *et al.* (1999), Ryabchikova *et al.* (1999), Nielsen *et al.* (2000), Prochaska and McWilliam (2000), and Wahlgren *et al.* (2001). Rare earths show particularly extensive hyperfine structure. Isotopic splitting is often discussed in the context of nuclear processing of chemicals, and some of its interesting story can be seen in Woolf and Lambert (1999), Nielsen *et al.* (2000), Sneden *et al.* (2002), Aoki *et al.* (2003), Dolk *et al.* (2003) Johansson and *et al.* (2003).

The mass absorption coefficient for lines

Once we have established α, the absorption coefficient per atom, we calculate the mass absorption coefficient needed in the radiative transfer equation and its solutions. To do this we write $\ell_\nu \rho = N\alpha$ as with the discussion leading to Eq. (11.7). Write N/ρ, which is the number of absorbers per unit mass as

$$\frac{N}{\rho} = \frac{N}{N_E}\frac{N_E}{N_\mathrm{H}}\frac{N_\mathrm{H}}{\rho}, \tag{11.51}$$

in which N/N_E is the fraction of the element E that is capable of absorbing the line of interest; this term takes into account the ionization and excitation of the element E. The ratio N_E/N_H is the number abundance, A, of the element E, and N_H/ρ is the number of hydrogen atoms per unit mass of stellar material, as given by the reciprocal of

$$\frac{\rho}{N_\mathrm{H}} = \sum_j \frac{N_j}{N_\mathrm{H}}\mu_j = \sum_j A_j \mu_j \,. \tag{11.52}$$

The summation is carried out over all the elements in the stellar material – each having an abundance $A_j = N_j/N_\mathrm{H}$ and an atomic mass of μ_j. The summations in Eqs. (11.52) and (8.19) are the same.

The mass absorption coefficient, expressed in area per unit mass, for a typical metal line can now be written in frequency units as

$$\ell_\nu = \frac{\pi^{1/2}e^2}{mc} f \frac{H(u,a)}{\Delta\nu_{\rm D}} \frac{N}{N_E} \frac{A}{\Sigma A_j \mu_j}(1 - 10^{-\chi_\lambda \theta})$$

$$= 1.497 \times 10^{-2} \frac{H(u,a)}{\Delta\nu_{\rm D}} \frac{Af}{\Sigma A_j \mu_j} \frac{N}{N_E}(1 - 10^{-\chi_\lambda \theta}) \tag{11.53}$$

in units of $\mathrm{cm^2/g}$.

In wavelength units the expression becomes

$$\ell_\nu = 4.995 \times 10^{-21} \frac{H(u,a)}{\Delta\lambda_{\rm D}} \frac{\lambda_0^2 Af}{\sum A_j \mu_j} \frac{N}{N_E}(1 - 10^{-\chi_\lambda \theta}) \tag{11.54}$$

with λ_0 and $\Delta\lambda_{\rm D}$ in Å. The stimulated-emission factor is $1 - 10^{-\chi\lambda\theta}$. The damping parameter a is given by

$$a = \frac{\gamma}{4\pi} \frac{1}{\Delta\nu_{\rm D}} = 7.96 \times 10^{-2} \frac{\gamma}{\Delta\nu_{\rm D}}$$

$$= \frac{\gamma}{4\pi} \frac{\lambda_0^2}{c} \frac{1}{\Delta\lambda_{\rm D}} = 2.65 \times 10^{-20} \frac{\gamma\lambda^2}{\Delta\lambda_{\rm D}} \tag{11.55}$$

with λ_0 and $\Delta\lambda_{\rm D}$ in Å. In LTE, N/N_E is calculated from the excitation and ionization Eqs. (1.18) and (1.20).

A similar expression for ℓ_ν can be written for hydrogen lines using Eq. (11.50).

Comments

Armed with the line absorption coefficient, we are now in a position to compute theoretical spectral lines. The line absorption coefficient, as expressed in Eq. (11.54) for example, is added to the continuous absorption coefficient when solving the radiative transfer equation. This is discussed in Chapter 13, where we deal with the behavior of spectral lines. The shape of the line absorption coefficient is reflected in the shape of spectral lines, but so are more macroscopic processes such as large-scale velocity fields in the photospheric gas (macroturbulence) and the rotation of the star. Analysis of the shapes of spectral lines is a powerful tool for understanding the physics of stellar photospheres and the behavior of stars.

On the observational side, spectral lines are narrow; high-resolution spectrographs are therefore required in order to see the shapes of these lines.

Chapter 12 takes us into this arena.

References

Abt, A. 1952. *Ap. J.* **115**, 199.

Allen, C. W. 1973. *Astrophysical Quantities*, 3rd edn. (London: Athlone), pp. 84, 89.

Anstee, S. D. & B. J. O'Mara 1991. *Mon. Not. Roy. Ast. Soc.* **253**, 549.

1995, *Mon. Not. Roy. Ast. Soc.* **276**, 859.

Aoki, W., S. G. Ryan, N. Iwamoto, *et al.* 2003. *Ap. J.* **592**, 67.

Arnesen, A., R. Hallin, C. Nordling, *et al.* 1982. *Ast. Ap.* **106**, 327.

Ayres, T. R. 1977. *Ap. J.* **213**, 296.

Baker, J. G. & D. H. Menzel 1938. *Ap. J.* **88**, 52.

Baranger, M. 1958a. *Phys. Rev.* **111**, 481, 494.

1958b. *Phys. Rev.* **112**, 855.

Barklem, P. S. & B. J. O'Mara 1997. *Mon. Not. Roy. Ast. Soc.* **290**, 102.

1998. *Mon. Not. Roy. Ast. Soc.* **300**, 863.

2000. *Mon. Not. Roy. Ast. Soc.* **311**, 535.

Barklem, P. S., S. D. Anstee, & B. J. O'Mara 1998a. *Pub. Ast. Soc. Australia* **15**, 336.

Barklem, P. S., B. J. O'Mara, & J. E. Ross 1998b. *Mon. Not. Roy. Ast. Soc.* **296**, 1057.

Barklem, P. S., N. Piskunov, & B. J. O'Mara 2000a. *Ast. Ap. Suppl.* **142**, 467.

2000b. *Ast. Ap.* **363**, 1091.

Barnard, A. J., J. Cooper, & E. W. Smith 1974. *J. Quant. Spec. Rad. Trans.* **14**, 1025.

Bely, F. 1966. *Abundance Determinations in Stellar Spectra*, I.A.U. Symp. 26, ed. H. Hubenet. (London: Academic), p. 254.

Bergeron, P. F. Wesemael, & G. Fontaine 1991. *Ap. J.* **367**, 253.

Blackwell, D. E., G. Calamai, & R. B. Willis 1972a. *Mon. Not. Roy. Ast. Soc.* **160**, 121.

Blackwell, D. E., J. H. Kirby, & G. Smith 1972b. *Mon. Not. Roy. Ast. Soc.* **160**, 189.

Bohlender, D. A., M. M. Dworetsky, & C. M. Jomaron 1998. *Ap. J.* **504**, 533.

Brissaud, A. & U. Frisch 1971. *J. Quant. Spec. Rad. Trans.* **11**, 1767.

Brueckner, K. A. 1971. *Ap. J.* **169**, 621.

Burgess, D. D. & J. E. Grindlay 1970. *Ap. J.* **161**, 343.

Clausset, F., C. Stehlé, & M.-C. Artru 1994. *Ast. Ap.* **287**, 666.

Cowley, C. R., G. H. Elste, & H. Allen 1969. *Ap. J.* **158**, 1177.

Dimitrijevic, M. S. & N. Konjevic 1986. *Ast. Ap.* **163**, 297.

Dimitrijevic, M. S., T. Ryabchikova, L. C. Popovic, D. Shulyak, & V. Tsymbal 2003. *Ast. Ap.* **404**, 1099.

Dolk L., G. M. Wahlgren, & S. Hubrig 2003. *Ast. Ap.* **402**, 299.

Dworetsky, M. M., C. M. Jomaron, & C. A. Smith 1998. *Ast. Ap.* **333**, 665.

Edmonds, F. N., Jr., H. Schlüter, & D. C. Wells, III 1967. *Mem. Roy. Ast. Soc.* **71**, 271.

Foley, H. M. 1946. *Phys. Rev.* **69**, 616.

Freudenstein, S. A. & J. Cooper 1978. *Ap. J.* **224**, 1079.

Fuhr, J. R., W. L. Wiese, & L. J. Roszman 1973. *Bibliography on Atomic Line Shapes and Shifts*, Nat. Bur. Standards Publ. No. 366. (Washington, DC: National Bureau of Standards).

Fullerton, W. & C. R. Cowley 1971. *Ap. J.* **165**, 643.

González, V. R., J. A. Aparicio, J. A. del Val, & S. Mar 2002. *J. Phys.* B **35**, 3557.

Griem, H. R. 1954. *Z. Phys.* **137**, 280.

1962. *Ap. J.* **136**, 422.

1964. *Plasma Spectroscopy*. (New York: McGraw-Hill).

1967. *Ap. J.* **147**, 1092.
1968. *Ap. J.* **154**, 1111.
1974. *Spectral Line Broadening by Plasmas*. (New York: Academic).
1997. *Principles of Plasma Spectroscopy*. (Cambridge: Cambridge University Press).
Harris, D. L., III 1948. *Ap. J.* **108**, 112.
Hjerting, F. 1938. *Ap. J.* **88**, 508.
Holtsmark, P. J. 1919. *Phys. Z.* **20**, 162.
Holweger, H. 1971. *Ast. Ap.* **10**, 128.
1972. *Solar Phys.* **25**, 14.
Holweger, H. & K. B. Oertel 1971. *Ast. Ap.* **10**, 434.
Hunger, K. 1960. *Z. f. Ap.* **49**, 129.
Johansson, S., U. Litzén, H. Lundberg, & Z. Zhang 2003. *Ap. J. Lett.* **584**, L107.
Jomaron, C. M., M. M., Dworetsky, & C. S. Allen 1999. *Mon. Not. Roy. Ast. Soc.* **303**, 555.
Kallmann, H. & F. London 1929. *Z. Phys. Chem.* **B2**, 207.
Kusch, H. J. 1958. *Z. f. Ap.* **45**, 1.
Lambert, D. L. & R. E. Luck 1978. *Mon. Not. Roy. Ast. Soc.* **183**, 79.
Landolt-Börnstein 1950. *Zahlenwerte und Funktionen aus Physik, Chemie, Astronomie, Geophysik und Technik*, Vol. 1. (Berlin: Springer-Verlag), p. 246.
Lemke, M. 1997. *Ast. Ap. Suppl.* **122**, 285.
Lenz, W. 1924. *Z. Phys.* **25**, 299.
Lesage, A., J. L. Lebrun, & J. Richou 1990. *Ap. J.* **360**, 737.
Lesage, A., N. Konjevic, & J. R. Fuhr 1999. *Spectral Line Shapes 10*, AIP Conference Proceedings, Vol. 467, ed. R. M. Herman. (New York: AIP), p. 27.
Lewis, E. L., L. F. McNamara, & H. H. Michels 1971. *Phys. Rev.* A **3**, 1939.
1972. *Solar Phys.* **23**, 287.
Lindholm, E. 1945. *Ark. Mat. Astron. Fys.* **32A**, No. 17.
London, F. 1930a. *Z. Phys.* **63**, 245.
1930b. *Z. Phys. Chem.* **B11**, 222.
Lorentz, H. A. 1906. *Proc. Amst. Acad.* **8**, 591.
Lwin N., D. G. McCartan, & E. L. Lewis 1977. *Ap. J.* **213**, 599.
Menzel, D. H. 1961. *Mathematical Physics*. (New York: Dover).
Mihalas, D., A. J. Barnard, J. Cooper, & E. W. Smith 1974. *Ap. J.* **190**, 315.
Milosavljević V. & S. Djeniže 2001. *European Phys. J.* D **15**, 99.
2003. *Ast. Ap.* **405**, 397.
Moity, J., P. E. Pieri, & J. Richou 1975. *Ast. Ap.* **45**, 417.
Mugglestone, D. & B. J. O'Mara 1966. *Mon. Not. Roy. Ast. Soc.* **132**, 87.
Nielsen, K., H. Karlsson, & G. M. Wahlgren 2000. *Ast. Ap.* **363**, 815.
Peach, G. 1981. *Advances Phys.* **30**, 367.
Pérez, C., R. Santamarta, M. I. de la Rosa, & S. Mar 2003. *European Phys. J.* D **27**, 73.
Petit Bois, G. 1961. *Tables of Indefinite Integrals*. (New York: Dover), p. 148.
Popovic, L. C., M. S. Dimitrijevic, & T. Ryabchikova 1999. *Ast. Ap.* **350**, 719.
Popovic, L. C., S. Simic, N. Milosavljevič, & M. S. Dimitrijevic 2001. *Ap. J. Suppl.* **135**, 109.
Prochaska, J. X. & A. McWilliam 2000. *Ap. J. Lett.* **537**, L57.
Roueff, E. & H. Van Regemorter 1969. *Ast. Ap.* **1**, 69.
1971. *Ast. Ap.* **12**, 317.
Ryabchikova, T., N. Piskunov, I. Savanov, F. Kupka, & V. Malanushenko 1999. *Ast. Ap.* **343**, 229.
Ryan, S. G. 1998. *Ast. Ap.* **331**, 1051.
Sneden, C., J. J. Cowan, J. E. Lawler, *et al.* 2002. *Ap. J. Lett.* **566**, L25.

Stehlé, C. 1994. *Ast. Ap. Suppl.* **104**, 509.
Stehlé, C. & R. Hutcheon 1999. *Ast. Ap. Suppl.* **140**, 93.
Stehlé, C., A. Mazure, G. Nollez, & N. Feautrier 1988. *Ast. Ap.* **205**, 368.
Stone, J. M. 1963. *Radiation and Optics*. (New York: McGraw-Hill), p. 73.
Struve, O. 1929. *Ap. J.* **69**, 173.
Underhill, A. B. 1951. *Pub. Dom. Ap. Obs. Victoria* **8**, 385.
Underhill, A. B. & J. H. Waddell 1959. *Nat. Bur Standards Circ.* **603**
Unsöld, A. 1955. *Physik der Sternatmosphären*, 2nd edn. (Berlin: Springer-Verlag), pp. 326, 331.
Van Regemorter, H. 1965. *Ann. Rev. Ast. Ap.* **3**, 71.
1973. *Reports on Astronomy*, Trans. Intern. Astron. Union **15A**, ed. C. De Jager. (Reidel: Dordrecht), p. 155.
Vidal, C. R., J. Cooper, & E. W. Smith 1971. *J. Quant. Spectr. Rad. Trans.* **11**, 263.
1973. *Ap. J. Suppl.* **25**, 37.
Wahlgren, G. M., S. G. Sveneric, U. Litzen, *et al.* 1997. *Ap. J.* **475**, 380.
Wahlgren, G. M., T. Brage, J. C. Brandt, *et al.* 2001. *Ap. J.* **551**, 520.
Weisskopf, V. 1932. *Z. Phys.* **75**, 287.
Woolf, V. M. & D. L. Lambert 1999. *Ap. J.* **521**, 414.

Questions and exercises

1. What is "pressure broadening" in the context of spectral lines? Briefly explain what causes pressure broadening. List the main types of pressure broadening and indicate the perturber for each type.

2. Describe the Hjerting function. Explain how we come to need this function in our studies of stellar photospheres. Since the wing portion of the Hjerting function is extremely tiny for weak spectral lines, would it be reasonable to expect weak spectral lines formed in a stellar photosphere to have a Gaussian shape? Support your answer with arguments.

3. Use Fig. 11.4 to find an approximate value of γ for the sodium D line. Then calculate the parameter $a = \lambda^2\gamma/(4\pi c\Delta\lambda_D)$ in the Hjerting function. Is this value small enough so that the series in Eq. (11.48) is convergent?